

The Astrans

Part I The Spark

Chapter 1: The Tokaplex

Price of ^3He	\$20,000,000/kg
Current Production	20 kg/year
Current Consumption	12 kg/year
Time Since Start	0 to 1 day

Dr. Elin Sarai braced against the cold wind as she checked the pressure readings on her slate. *Ilha Fusão* sprawled across an artificial island fifty miles off Lisbon. *D* would meet ^3He , releasing clean fusion energy. The tokamak had worked before, but today wasn't a test. This was a live connection to Portugal's grid. One hundred megawatts wasn't enough to change the world. But it would prove fusion could feed a commercial grid.

The wind shifted, blowing her hair across her face. When she turned back to go inside, she saw Gonalo Silva coming out.

He joined her on the walkway and looked across the island. "*Bom dia, doutora.*" He looked like he had been up all night. "The ^3He tank is installed. We are ready."

Elin looked out to sea. "That tank nearly bankrupted us. It's half the world's production." She looked back at Gonalo. "Are we ready?"

"She's an angry old *senhora* who's waited a long time to feel important."

Gonalo's remark lingered in her mind. She knew he meant the plant, but his words fit her, too. Today, this plant would show the world what it could do.

She pulled up the system checklist on her slate. Two weeks ago, it had been full of red and yellow flags. Now there was only a single yellow flag: a pressure differential that was outside expectations but within tolerance. She kept her eyes on the slate. A red warning flickered briefly as the deuterium extractor started up, then turned to green when the flow became steady. Another yellow flickered on her slate, then went green again. "*Valha-nos Deus,*" she muttered. Working this long near Lisbon had affected the curses she used.

A soft rumble passed beneath them. Deuterium tanks were hissing. Trace amounts of ^3He were being let out in tiny bursts. Somewhere far below, in the containment

torus, the $D\text{-}^3\text{He}$ reaction chamber was beginning its warm-up cycle. The corridor curved, descending in a spiral toward the turbine hall. Elin led the way. They passed a crew of techs hunched over an access panel. In front of them, three massive units stood silent, waiting for steam. Elin paused at the observation window. These were conventional technology, superheated steam driving generator rotors, maybe forty percent efficient if they were lucky. Functional, proven, but wasteful.

“This is the bottleneck,” she said. “We’re extracting thermal energy with nineteenth-century technology.”

Gonalo shrugged. “Efficiency was never the point. It works. And for a hundred megawatts, it’s enough.”

“For now,” Elin replied.

Each level had its own voice, its own smell. Salt and metal. Polymer and heat. And everywhere vibration, small, subsonic, and constant. Like the plant was holding its breath.

By the time they reached the observation ring around the containment torus, Elin was sweating through her collar. Gonalo keyed the access door.

Inside, the world changed. The torus was not quiet, but it was sacred. A ring of magnet housings loomed around a core of dense shielding. Inside, a reaction chamber poised to hold a plasma at six hundred million degrees. Cameras fed real-time diagnostics to the translucent wall panels.

The field was up. The magnets were humming.

“I never get used to it,” Gonalo said.

Elin shook her head. “You shouldn’t.”

She took a last look at the checklist. The yellow flag was still there, but the pressure had not moved. She signed off on it. “*Está pronta.*”

Gonalo looked down at his slate, now showing her approval. “You know what Raines told me yesterday?”

Elin sighed. “Do I want to know?”

“He said the press has to see the helium loading into the reactor live on camera. He’s planning to let them into the control room.”

She laughed. It was sharp and brief. “I talked to him about that; there is no way the press or anyone else is going into the control room. I mounted a large twin-blade knife switch on the wall so they can all watch him throw it. The engineers will know what to do. I also wired an Edison bulb to the generator, up above the switch. As the turbine spins up, the bulb will glow.”

After the inspection, they found Raines near the conference room, surrounded by investors, dignitaries, and the Minister of Energy from Argentina, who seemed excited and had already begun hinting at the possibility of a second plant, maybe closer to the tropics. Raines was in his element. Laughing, clasp shoulders, turning charm into sales contracts.

He saw Elin and immediately excused himself, cutting the circle short with a gesture and a few well-oiled words. The group didn’t mind.

“Elin,” he said, voice warm and certain, like he already knew the answer. “She’s ready, isn’t she?”

She didn’t nod. Just looked him in the eye. “She’ll hold.”

He exhaled and grinned, letting the tension slide off him like a coat. “That’s all I need to hear.”

Raines clapped her gently on the shoulder. “I know how much you hate this day. But thank you.”

“Don’t thank me until she makes it through the week.”

“Deal.”

He turned, buoyant again, already slipping back into his public face, back into the performance of selling the dream to the world.

They came by tiltrotor and by ferries. Premiers and presidents in motorcades threaded the island’s narrow ring road. The King of England, dressed with understated dignity, arrived by helicopter. Billionaires who had staked fortunes were there, and a small press corps trimmed to just twenty, each one combed, curated, and cleared.

They all gathered in the Assembly Hall. Polished steel and glass under a vaulted ceiling. There was seating for two hundred. Everyone had a view of the oversized screen, and beyond it, the observation pane that stared down to the reactor.

Tobias Raines stepped up to the podium, relaxed, with a smile on his lips.

“When I created Helion Dynamics,” he said, stepping out from behind the podium, “fusion was a fantasy told by physicists. I found a few who believed that fantasy. We needed someplace to build, but the fear of unproven nuclear power was strong. Lisbon agreed to let us build off her shores. We built this island with our own hands. Nothing worked. We’ve replaced the tokamak five times. But now we have tamed the reaction. We have made something real. The world is hungry for energy. Today, this marvel will feed the world’s hunger.”

When he finished, the screen retracted into the ceiling, revealing a switch and a bulb. The audience began murmuring and whispering to each other. Raines held up his hand, and the crowd went silent. He grabbed the green glass handle firmly, lifting it. The twin blades of the knife switch were a foot long, heavy, polished copper. Bracing with his legs, he drove it home.

An indicator light came on behind the glass in the control room. The supervisor called out. “Raines is ready.”

A technician set down his *bica* and pulled gently on a lever. Engineers moved like ghosts across the command tier. The reactor hissed. Somewhere deep in her belly, valves opened. *D* met ^3He .

There was no explosion. No flash. Just the warm, unremarkable glow of a single, antique Edison bulb wired to a ceremonial loop.

It came to life slowly, dimly at first, then glowing brilliantly.

The room erupted in applause.

Hours later, after the reception emptied, Elin and Raines stood in the observation room, watching the Edison bulb glow. The plant hummed steadily beneath them.

Elin pulled herself onto a chair, resting her arms on the table, bowing her head.

Raines turned, gesturing broadly with his arms. “Argentina’s in,” he said. “They claim fusion is central to their national industrialization plans. They plan to have an artificial island off the coast ready in six months, large enough for a ten-gigawatt tokaplex. How soon can we have it built?”

She slowly turned to him. “That scale would require a ton of ^3He every year. We don’t have that. We’ve talked about this before.”

“Elin, I know it is expensive, but you have no idea how much money they will give, now that it is running. I have already scheduled calls with suppliers. They are charging about twenty thousand dollars per gram.”

“There are no suppliers at the scale you’re talking about. The nuclear industry only produces about twenty kilograms per year. Total. We need tons.”

Her eyes moved back to the numbers on her slate, tracking the plasma and first-wall temperatures.

“That’s fine. That’s what the industry is doing today. But we can breed tritium. It decays to become ^3He .”

“Yes, that’s viable,” Elin answered. “But it’s a few kilograms, and depends on scraping traces from fission systems that were not built for this. And tritium has a half-life of a decade. That trickle is enough to keep this tokaplex running, but not a larger tokaplex.”

“Tritium is only a part of the picture. I know a supplier pulling helium from the Minnesota Topaz site. He can get a thousand tons for the right price.”

“The Topaz site is very rich in ^3He . But that thousand tons might hold a hundred grams of it,” she said. “The rest is helium ash. That’s not helpful for us. Tobias, *Ilha Fusão* was built to prove fusion can power electric generation. We have proved that. But if we don’t open up more ^3He reserves, this tokaplex is nothing but a curio.”

Tobias Raines said nothing. He watched the Edison bulb powered by their steady reactor, then turned toward the window.

Elin was watching the gibbous Moon rise.

Chapter 2: Helium Fever

Price of ^3He	\$8,000,000/kg
Current Production	1 ton/year
Current Consumption	12 kg/year
Time Since Start	1 week to 2 months

Li Yun Tan was alone in her apartment, watching the news. Locally, a traffic accident in downtown Phoenix resulted in multiple injuries. The ambulances moved freely to the intersection, lights flashing, traffic moving aside. In Kuala Lumpur, when she was twelve, the streets had been gridlocked for hours.

In financial news, the story opened with canned footage of the Edison bulb in Ilha Fusão. Helion Dynamics was offering a two-year price guarantee of eight million dollars per kilogram for lunar ^3He , and a loan of up to one million dollars for transit.

She looked around her apartment; no one to call, no debt, no lease, just a photo of her and her parents when she was twelve. She opened her accounts and worked through the numbers. Her savings, plus accrued vacation time from Arizona Public Service, were enough to buy the necessary gear. Getting that gear into space was the real cost. Heavy lift from Texas was the cheapest way to get up, at twenty dollars per kilogram. There was a Russian ferry operator that could haul up to twenty-five tons from low Earth orbit to the Moon, but he was charging as much as heavy lift. The transit loan terms were steep; twenty percent interest, but they let her pay it back in ^3He . With a steady hand, she hit the “APPLY” button.

She waited as they processed her application. Every day, more people were headed to the Moon. Big corporations were building advanced equipment and sending well-outfitted crews. Others were cobbling together used equipment and going on their own, all of them driven by the fever for ^3He .

Li didn't tell anyone at work. Every day when she got home, she checked the claim map, watching as clusters of claims grew. The ^3He distribution was nearly uniform; she knew it didn't matter. Still, she wondered what they knew that she didn't. She hoped the good sites wouldn't all be gone by the time she got the loan. If she got the loan.

She was in the office at her desk when she got the notification on her slate. Work suddenly meant nothing. She stopped by HR and quit, leaving immediately. On the way home, she began checking Craigslist, under Resumes/Moonwork for an equipment operator. She found Darrin Brooks maintaining offshore turbines in the North Sea. For years, he had logged every inspection and checked each seal twice. That was enough. She offered him a contract, passage included. Darrin looked up at the Moon and said yes within an hour.

Li went through a lawyer to get the claim. Each claim was limited to one square mile per person, which, on average, could yield just over one ton of ^3He . By including Darrin Brooks on her paperwork, she was able to file a claim for two square miles in the southwest corner of Ball Crater. She got the paperwork back three days later. Her name was written as Tan, Li Yun. Fixing it would take a week; new claims were being filed every day. Her name was now Tan. She sold what remained of her assets and met Darrin in Tucson, at the market that had grown beside the Air Force Boneyard.

The vendors at the Lunar Supply Market traded in surplus and salvage, ex-military and scientific equipment, and a few new components from specialty manufacturers. They started at the aisle of helium extractors.

Li found large ovens with ten-ton tumblers and one-minute cycle times. The power rating was hundreds of megawatts.

Darrin called her across the aisle. “We can’t run those. That’s for industrial operations. RF extractors don’t heat the rock, just the gas.”

Li joined him. “How does that work?”

Darrin was reading a brochure. “It uses something called ‘Larmor resonance.’ The magnets adjust it, and the RF generator excites the ^3He atoms more than it excites the other atoms.”

“So we get pure ^3He ?”

“We’ll be lucky to get five percent, but it is still a huge savings in bottling and shipping.”

Darrin was looking for trusted brands. Li kept her eye on the power requirements and the shipped weight.

They started looking at one extractor because it mounted on stilts to let them dump processed regolith easily, and had a large hopper so collection could be independent of operation.

The vendor talked them through the operation. “On the moon, regolith isn’t all the same. You’ve got ilmenite and other magnetic minerals mixed unevenly with anorthosite and other non-magnetic minerals. You can see sometimes it’s darker, sometimes lighter. When you run it, you adjust the magnetic field strength to compensate. That’s what the levers are for.”

That made no sense to Li. “How would I know where to set the levers? Test the load before it goes in?”

“If only it were that simple. You can’t see directly what the right field strength is, but you can see the effect it is having. You watch this screen. A mass spec is sampling just a few atoms at a time every few seconds. The screen will show peaks representing how much ^3He , ^4He , and hydrogen you are pulling.”

“So I keep the levers where I am getting more ^3He than hydrogen.”

“Yes and no. ^3He is rare. You will always get more hydrogen and ^4He than you get ^3He . You maximize the ^3He ratio. Keep it cold, watch the hydrogen and ^4He . If you let the field get too weak, hydrogen gets released faster. If it is too strong, more of the energy heats the rock, and your ^4He extraction increases. You want to live in the sweet spot between them. Don’t just set the levers. You keep adjusting them the entire time. Generally, your purity will start to fall after you extract about half the ^3He . That’s a good time to stop and process another batch.”

Li pulled out her slate and took a few notes. She and Darrin stepped a few stalls away. “Darrin, I know you like this one, but it weighs twelve hundred pounds more than some others, and it will cramp our budget.”

“It weighs more, mostly because of the regolith feeding equipment. That will make mining much more productive. The other ones that have hoppers and stilts weigh as much.”

“Lunar regolith eats everything regardless of reliability.”

“This one is built for easy repair.”

Li looked at a few more extractors, but Darrin was right. She returned and bought the one he recommended.

Darrin had started looking at living quarters while Li finished the extractor purchase. He found one he liked and called Li over. “Hey, what do you think of this? Dual Bosch air recyclers and a top-quality airlock.”

“Are you kidding? Other huts have much bigger airlocks.”

“I know, but this is from a top manufacturer, durable and solid. And small is better. Every time we cycle the airlock, we lose oxygen. The smaller the volume, the less we lose.”

Li studied the layout. It was a tiny Quonset hut, nine feet square, the size of a four-man tent. The bed was a three-quarter mattress, four feet across. The bathroom was just an alcove, without a door or even a curtain, and the shower had a hose with a low-volume mist emitter. The kitchen cabinet supported a small microwave oven with an electric heating coil on top.

Li pointed to the eighteen-inch window, which was packed separately. “What’s this for?”

The vendor looked at it, a hint of amusement in his eyes. “People on the Moon like to look at Earth. When you anchor the hut, you can cut a small hole in the roof, and this will fit in and seal up the hole. You align it to frame Earth. Because the Moon is tidally locked, Earth doesn’t move much. Pretty much every hut on the near side has an Earth window.”

She checked the price. It was higher than most huts of comparable size, but the compact dimensions and low mass made launch affordable.

It came with a heavy crate containing the life support core. The core was a column of cylindrically wound polypropylene tubing with embedded LEDs, configured as a closed algal loop. It took in wet and dry waste and stale air, returning oxygen, heat, light, and edible biomass.

“The hand grinder and impeller are more work, but it’s one less motor to maintain,” Li said, checking the heat welds. “This closes the nutrient loop.”

Darrin stared at the emerald-stained plastic, his face paling. “Nutrient loop? We have to eat that?”

Li wrinkled her nose. “Not appealing, but we are a long way from Earth. I’m also buying a seed package so we can grow microgreens and sprouts. We’ll manage.”

They looked at rovers and crawlers next. Li liked the faster, lighter rovers for covering ground quickly. But Darrin shook his head. “We need to haul twenty tons

of gear to the site, then move hundreds of tons of regolith once we're there. The crawlers are slower, but they can handle the load."

Li bought the crawler, but wasn't completely satisfied. "We won't always have a full load."

They got to the comms section. Li was looking at the radios. "Life up there will be strange, always saying 'copy' and 'over'."

A vendor stepped up. "People don't do that much. They use Local Band. Here's a headset, try it out. Don't worry about etiquette, just talk naturally."

Li and Darrin both put one on. Talking felt normal. They could both talk at once, hear other people clearly when close, while voices faded with distance.

She listened carefully, hearing two other conversations at the same time. "So natural."

"It is an AM band. Multiple signals can add together, just like talking here on Earth. The one odd thing is that sometimes you get a heterodyne squeal when two systems get close for the first time. Just a short chirp."

"So everyone can hear everything we say?"

"Most radio sets have a few encoded FM bands for restricted conversations. Then you have to use terms like 'copy'."

They collected all their gear and weighed it. Fifteen tons. Li bought ten tons of solar panels to reach the twenty-five-ton limit. They loaded everything into a semi-trailer and hauled it to Texas.

The launch team packed it alongside cargo from a few other crews. Li was perfectly calm until the elevator ride to the passenger compartment. The platform was higher than she expected. Below her, everything that had been important looked so small. She had been too busy to say goodbye. Again. As they strapped her into her seat, she closed her eyes.

The lift to orbit took twelve minutes at three G. Li's vision narrowed, and her chest compressed. Every breath was a fight. Darrin's knuckles were white against the armrest beside her. Then the pressure vanished. Weightlessness. They had made it.

In low Earth orbit, they transferred to a lunar ferry. It was a retrofitted module with Cyrillic lettering and a Russian pilot who did not waste words. The trip to the Moon took three days, cramped and vibrating the whole way.

A news alert flickered on her slate: three miners dead at Mare Serenitatis. Suit failures. The preliminary report cited “substandard seals and counterfeit pressure bladders.” The company issued a statement: “We are cooperating fully with the investigation.”

Li closed the feed and looked at the suits Darrin had bought. She’d found suits for a third of the price and thought they’d be adequate, but Darrin had insisted on these expensive suits. He bought decommissioned Indian Space Research Organization stock and insisted on inspecting every valve himself. At the time, she called it paranoia. Now she was grateful. She tightened her grip on the gear straps. Soon, they’d see how well every choice held up.

The ferry set them down three miles west of Ball Crater. Just regolith fines and glare, with the crater wall rising a thousand feet to the east. No cloud of dust on arrival. Nothing frightening or welcoming. The horizon was closer than on Earth, but it didn’t feel small; it felt empty.

Darrin and the pilot pulled the loaded crawler out of the cargo bay. Li noticed where the pilot’s gaze fell. It was always on people or equipment, never the Moon. They loaded their gear on the crawler and set off. Li was fine with the slow movement of the crawler. She was fine with absolutely everything.

She followed the lunar claim map on her slate. Darrin sat beside her, watching the terrain roll past. “How far?” he asked.

“Three miles to the rim. Our claim is just inside, on the southwestern floor.”

They’d spent the entire ferry ride planning where to position the solar panels for optimal sun exposure, how to route the power lines, and which section to start extraction from first. “First thing we do is anchor the hut,” Darrin said. “Get life support stable before we even think about extraction.”

Li looked up at the Earth, judging its phase. “We have about ten days left before lunar night. I’ll set up the panels, then I’ll run a site survey. Map the concentrations before we start sweeping.”

“You think we can pull an entire bottle before lunar night?”

“No, but even a half bottle is probably worth shipping back.”

Darrin laughed. “Then what?”

“Then we do it again. And again.” She glanced at him. “You ready for this?”

“I’ve been ready since you called me in the North Sea.” He looked up at the Earth. “Waxing gibbous. The best phase to start something new.”

Li looked at the claim. She considered where she would post it when they set up camp. The rim of Ball Crater rose ahead of them, a wall of ancient rock and debris about a kilometer high. The slope looked to be maybe twenty degrees at the approach point Li had mapped. “There’s our door,” she said, pointing to a pass where the rim had degraded into a rough ramp of rubble.

The crawler moved easily even as it began the climb. The treads bit into loose regolith, found purchase on harder substrate beneath. They climbed slowly, steadily, the wall rising around them. They reached the rim twenty minutes later. Before they could see into the crater, Local Band gave a loud heterodyne squeal. Less than a second, and it was gone.

Li looked at Darrin.

Darrin said nothing. The crawler pulled over the lip and onto the relatively flat ring of the crater’s edge.

“Li, something is wrong here. Verify our claim on the map.” Darrin leaned forward.

Li stopped the crawler. The coordinates matched. She checked twice. This was their claim, filed three months ago, survey boundaries confirmed. But someone was already there. The rig was no temporary setup. Solar mats stretched in neat rows, cables buried under regolith berms. An RF extractor hummed over processed material. Two huts sat leveled on foundation pads, their airlocks sealed, power lines running between them. This crew had been here for weeks. Maybe longer.

“That’s our claim,” Darrin said quietly.

“I know, Darrin.”

“So what do we do?”

She stared at the operation below them. She pulled up the claim on her slate. Morrison & Chen, certified attorneys. Tucson filing. All the seals and signatures. Everything proper. “We show them the paperwork,” she said.

Darrin looked at her, then back at the established rig below. “You think that’ll work?”

“It has to.” She started the crawler down the interior slope toward their claim.

When they arrived, Darrin climbed down, holding their paperwork. “This sector of Ball Crater is Li Yun Tan’s claim. You and your team need to pack up and move.”

The other miners had started gathering. From the side, one used his survey laser, cutting the paper in half while Darrin held it.

The leader looked at Darrin. “No.”

Li recoiled.

“We can’t just leave,” Darrin said. “We have no other claim. No way to get home.”

“We’ll get a surveyor in here to settle this,” Li said.

The leader turned his gaze to Li. “You need to leave now.”

One of the miners leveled a survey laser at Darrin’s chest. Just enough power to see the green dot. Another miner etched a warning line into the dust, a foot from Darrin’s boots.

Li’s throat went dry. The world collapsed into that green, speckled dot. A gentle squeeze and the laser would cut right through his pressure suit.

She understood these men would kill them. These men would keep the equipment she and Darrin had so carefully chosen, they would work the claim, and sleep fine. Darrin and Li would simply stop existing, and no one would ever know or care.

The paperwork meant nothing. The signatures, the seals, and the lawyer’s certification were all worthless. No one was coming. No one would ever come.

She looked at her claim, cut in half, on the crater floor. “Darrin, get back in the crawler.” Her voice sounded distant, like it belonged to someone else.

Darrin climbed back in. Li turned and moved the crawler as fast as it would go. She wished they had a rover. The claim-jumpers kept a green dot on Darrin until they made it over the pass.

Coming out of the pass, they continued straight southwest. Her hands were shaking. She’d believed the paperwork mattered. She’d paid lawyers, filed claims, done everything right. None of it meant anything.

They’d figure out the rest once they had distance. The Moon felt hostile, but there was nowhere to go but the Moon.

A few ridges later, they passed a wrecked crawler. Li slowed. No markings, no beacon. Solar panels shattered. The cabin hatch hung open, dark inside.

“Should we check it?” Darrin asked.

Li stopped, and they got out. “Salvagers already did. Look at the hatch. It was forced open from the outside.”

“What happened to the crew?”

“No obvious damage. Darkside, maybe. Ran too long on batteries, didn’t make it back before their battery died. Two weeks of night out here with no power...” She let the thought hang. “Their suits didn’t last.”

Darrin was quiet. The crawler was intact except where it had been pried open. Darrin found two oxygen bottles in the cargo space on the back. Empty. He tossed them over to their crawler. “Li, we don’t know when we may need another oxygen bottle. These folks don’t need theirs anymore. What do you say?”

“Sure. Let’s see if we can pull their battery pack, too. Our crawler can handle the load.”

Li peered inside the broken crawler, then recoiled. The driver lay in the back, where salvagers had tossed him after stripping his suit. Mummified. Face dark as jerky, eyes deflated. Li backed away.

“Darrin, forget about the battery.”

“I’ve almost got it, Li. Just a few more bolts.”

“Darrin. Let’s just go. Scavengers took their suits.”

They got back in, and Li drove without stopping. Someone else’s mistake. Someone else’s claim now. It wasn’t hostility. The Moon didn’t care.

They kept moving until there were no more tracks. No one else had gone this far. The ground got rough, with frequent small craters. No more claims showed on the map. The crawler moved slowly, its tracks whispering across the dust.

Then Li saw it.

A wall a mile high, maybe more. White in the sunlight, rising from the curve of the Moon like a broken edge of the world. She stopped the crawler.

“What is it?” Darrin asked.

She didn’t answer. She stared at the wall, at the way it climbed into the black sky, massive and silent and utterly indifferent. Navigation showed the crater ahead was Tycho. She’d heard of it. Everyone had. Ancient. Huge. Empty.

She started the crawler again, steering directly toward it.

Chapter 3: The Moon is Hard

Price of ^3He	\$8,000,000/kg
Current Production	6 tons/year
Current Consumption	12 kg/year
Time Since Start	2 months to 7 months

The wall pulled her. It looked like a fortress. She needed to see it up close. They took turns driving. There were eight days left before lunar night when they finally arrived. The slope towered over them, jagged and geologically young.

Li climbed down from the cab, boots sinking into the fines. She tilted her head back, following the wall up until it met the stars.

“Li, that is too dangerous to climb, and it will take more than a day,” Darrin said.

She looked at him. Then back at the wall.

“We have over a week until sundown. I want to climb this.”

They stopped at the base and ran the first anchors. The slope wasn’t vertical, but it was steep enough that the crawler’s drive alone wouldn’t hold. The fine regolith was more like silted glass than soil.

Li checked the line tension, then engaged the front winches. The cables drew tight, whining as the crawler began to pull itself upward. The treads fought for grip. The whole frame shuddered, metal straining against rock.

“Keep the pace slow,” Darrin said over the intercom. “Too fast and we’ll shear a line.”

Li didn’t answer. The crawler inched forward, steady but tense.

A hundred yards up, the left tread slipped, sending regolith out in an arc, scattering on the slope below. The crawler lurched sideways until the cables caught, jerking them to a stop. Darrin cursed, unstrapped, and climbed out to reset the anchor. He moved carefully, checking every clasp before locking it again.

Back in the cab, Li feathered the winch controls until the cables vibrated like bowstrings. She slowly engaged the treads, and the machine crawled forward another few meters.

They guided the crawler up the wall with winches and determination. The treads slipped and bit again, throwing regolith with each surge. They fell into a rhythm of stopping, driving new pitons into the slope, and setting fresh lines every twenty meters. They spoke little. There was nothing to say except what the equipment already told them.

Halfway up, Darrin's voice came through again. "We're off alignment. It's pulling sideways in the rear, and the track is close to coming off. If we keep this path, we'll twist the frame."

Li paused the winches. "Then we reset."

"That's another hour."

"Then it's an hour."

They reanchored, realigned, and started again.

The last few meters of the climb were smooth. The treads engaged well, pulling them onto the rim with a final surge of power. They still had six days before lunar night.

Li sat back, breathing hard. Her heart hammered. Sweat slicked her suit liner. She looked out across the crater.

The floor spread below them, gray, smooth, and silent, fifty miles to the far wall. A central peak rose in the middle distance, stark against the black. Nothing moved. Nothing had moved here in a hundred million years.

She climbed out of the cab slowly, leaving the first footprints on the crater. She walked to the inner edge of the rim where the crater opened at her feet. It was vast, empty, and utterly still.

Something tightened in her chest. Not fear. Not relief. Something harder.

Hers.

"The rest of the Moon can keep its dust," Li said. "This crater is ours. That wall is our claim, and anyone who wants to contest it will have to climb it first."

She raised her survey laser and aimed it across the crater. A bright green pulse struck the far wall, and a scatter of regolith tumbled down, leaving a clear scar.

“And then face us.”

She didn’t look at him. She looked at the scar she’d made, at the crater spread below, at the walls rising in every direction.

Darrin said nothing for a long moment. Then, quietly: “What if someone does come? What if they make the climb?”

Her lips smiled, but it didn’t reach her eyes.

“Then those cheechakos will learn here what we learned at Ball Crater.”

Darrin looked at Li and saw the quiet hardness around her mouth. It sent a chill through him.

He changed the subject. “Li, if we are staying here, let’s set up the panels on the rim. Inside, we will get more shadow.”

“Does it really matter?”

“An extra ten hours per cycle, compared to the plain, but an extra twenty hours compared to the crater floor. I think it matters.”

They set them on a prominence and ran electrical cables down into the crater. On the Moon, fuel didn’t work. Nuclear would work, but was impossible to license, so solar was king: cheap, predictable, and reliable, with batteries for lunar night.

Then they anchored fresh lines and began their descent into the interior. The ratchet and pawl clicked in slow rhythm as they descended. They stopped on each terrace until the brakes were cool enough to proceed. The descent was steeper than the climb had been, and went down much further, but the winches did almost all the work, lowering them from terrace to terrace until they reached the floor.

Li looked across the floor of the crater. “That’s where we’ll live,” she said, pointing to the central prominence.

Darrin sighed. He would have preferred someplace closer, but the prominence would be fine. They unloaded their gear on the north side of the prominence and set up their hut. It looked so tiny in the crater. Darrin installed the Earth window so they could see Earth when lying down in bed. He set up a tube to protect the Earth view, then buried the hut in several feet of regolith, shielding it from temperature

swings, micrometeoroids, and radiation. Inside, earthshine cast a blue glow, a steady reminder of home.

The first survey sweep crushed Li's optimism. 3.2 parts per billion on the western floor. Four point eight near the central peak. She'd hoped for fifteen, more typical of the lunar surface, but Tycho was young and hadn't had time to accumulate that much helium.

Darrin saw it in her face. "How bad?"

"One cycle to fill one bottle. Maybe. We have to work hard and smart." She showed him the numbers. "We skip the first rendezvous, catch the second one. If the yields hold, we can ship a full bottle then. That should be worth enough to start making improvements."

Darrin filled the chamber, and Li began their first extraction. She monitored the residual gas analyzer, narrowing the resonance window until the ^3He nuclei's Larmor frequency matched the RF's output. A microtesla drift, and the energy would bleed into the ^4He . She balanced maximum extraction with purity, settling on ten percent ^3He . She triggered the pulse, feeling the hum shift as the RF energy coupled directly into the target atoms, vibrating them loose from the mineral lattice while the surrounding regolith stayed cold and silent.

The extraction process was relentless trial and error. The RF extractor hummed beneath her boots, pulsing energy through regolith. First load: 4.7 milligrams. Not enough. Not nearly enough. She adjusted the magnetic field and extended the timing. Load by load, yields improved.

"We hit eight!" Li exclaimed. "That's the target. If we maintain this level, we can fill a bottle per cycle. We are moving a lot more regolith than we planned, but we can do it."

Darrin smiled weakly. "OK, Li." He knew he would have to be as efficient as she was when it was his shift on the RF. The RF extractor defined them. It had to run constantly during the lunar day, or production would lag. Sweeping was much simpler. They could pile up an Earth day's worth of regolith in six to ten hours. But they took shifts sleeping out of sequence, so someone was always up, ensuring the RF extractor never stopped.

The sun dropped below Tycho's western rim. There was no dusk; the world became blue with earthshine. The temperature outside plunged to a hundred seventy below zero Celsius within hours. The solar panels up on the rim gathered power for a few more hours.

Inside the buried hut, the heaters for the algae tank kept them warm. In the last twenty minutes of solar power, they had both taken a brief shower before shutting off the water heater. Li was last. There was no door or privacy. The water was all redistilled for reuse. When Li stepped out of her brief shower, she looked at the name tag on her pressure suit. TAN, LI YUN. Because of the mix-up, everyone on the Moon called her Tan. Except Darrin. He got it right. He'd always gotten it right.

She dried off and put on her inner layer. They wouldn't wear their pressure suits until they were ready to go out again, after lunar night. She looked over at Darrin, who had done the same. He was inspecting his suit. She followed his lead.

Darrin caught her looking at her name badge. "Why don't you get that corrected?"

"Wasn't important enough to delay our launch then; don't care what people here call me now."

In spite of the cold, heat loss through the regolith was slow. Li and Darrin shut down everything but life support. The only light was coming from the algae system. Just air recycling and warmth, and a little heat for cooking.

She sat at the console, legs folded under her, eyes half-closed. Darrin handed her a bowl of algae broth seasoned by a flavor packet with the vague taste of shrimp. She took it without thanks. The packets from Earth made it nutritionally complete and more palatable. It was not so much that it was bad as it was monotonous.

"I miss the flavors I grew up with in Kuala Lumpur," Li said.

"Maybe next time we order, we can get some curry fish head flavor packets."

Li kept an eye on the power meters. She had calculated how much battery capacity they would need and added twenty percent. More batteries would have meant fewer solar panels.

Day six, Darrin was watching Li check the monitors. "Are we going to live through it?"

"I've worked in power since they died. If the city had power, the ambulances could have gotten through in time. We'll have enough."

Day twelve of darkness, battery at twenty-six percent. They had let the CO₂ climb to four thousand PPM, making it harder to concentrate, but reducing the electrical demands on the algae system. By day thirteen, the batteries had dwindled to nineteen percent.

“Twenty-six hours,” Li said, looking at the phase of the Earth.

On day fourteen, the sun returned. It hit the panels on the rim ten hours earlier than it hit the interior of the crater. The batteries began charging. They’d made it through their first lunar night. Li thought back to the wrecked rover. Not everyone did.

They set the CO₂ levels back down to eight hundred PPM. Li took one long breath, then got back to work.

In the second cycle, they were faster. They had a better strategy for moving regolith, they knew how to manage the magnetic field on the extractor, and had better routing for the crawler to minimize time between loads.

Darrin learned to anticipate failures before they happened. A slight vibration in the RF extractor housing meant a mounting bolt had worked loose. A half-degree temperature drift meant a seal was starting to degrade.

By the end of the second cycle, they’d pulled almost a hundred grams. They had started on their second bottle.

The next night, they stopped with four hours of sunlight on the panels left, making sure to fully charge the crawler. They rested as before for ten days, then they loaded the bottle into the crawler along with every blanket they had and made the climb out of Tycho. The climb out was easier because they had left the cables mounted on the rim. The winches pulled them up and lowered them down. They kept it slow and cautious. When they got to the rendezvous site, about fifty miners were there waiting, quiet and watchful. This was their first time around other miners since Ball Crater, and Li was uneasy. They stayed in their crawler, covered in blankets, keeping warm.

The cargo ship landed twenty hours before dawn. Li read the manifest. It listed the pilot as Andrei Volkov. The name didn’t ring a bell.

When Andrei climbed out, he recognized Darrin and Li immediately. “Li, Darrin, how are you?”

Li recognized the voice. He was the pilot who had ferried them here. She looked at his face. Different in a helmet, but definitely the same man. “Hi, Andrei!”

Andrei broke into a wide grin. “Come, see my ship.” He brought them over to his ship, where he had stenciled “LUNAR TUG” on the hull. “I incorporated in Miami. This is a growing business.”

“Smart move. Are you expanding?” she asked.

“As fast as I can hire pilots. Rendezvous are a huge business, even bigger than fares.”

Most of the miners were here to ship helium to Earth. Andrei weighed each bottle, with no concern for what was in it. His only concern was the cost of shipping. Assay and helium purchase were handled by Helion Dynamics after the bottles got to Earth. Li shipped their first bottle to Helion Dynamics. The QR code for her account was engraved on the bottle.

After he loaded their bottle, Li and Darrin said their goodbyes.

“Andrei, nice to see you again. We will see you at the next rendezvous.”

“No, don’t go, wait until the business is done. I have something for you.”

Li looked at Darrin. They negotiated with their eyes. “Sure, we can stay for a bit.”

Andrei continued chatting as he processed everyone’s bottles. “So you two were headed for Ball Crater, right?”

“Yeah, well, change of plans. We ended up in Tycho.”

“Tycho! That’s too far. Why would you go there?”

“It was available.”

“Da, everything on the Moon is available.”

With all the bottles processed, Andrei started distributing packages that people had bought. Water and air dominated, but replacement parts and other enhancements were also common.

When everything had been delivered and all the other miners had left, Andrei showed them what he had for them. “See these? Extra water, oxygen bottles, even some batteries this time. People paid for them, so I brought them. But the buyers of these packages darksided before I landed. It happens on every rendezvous.”

“You are giving them to us?”

“Da, I don’t have any reason to haul them back to LEO. You keep them.”

Li picked up one of the water containers. A family photo and inscription were laminated to the side. She set it down in the crawler, facing inward. They never got to say goodbye, either.

“Thank you, Andrei. This will be very helpful.”

The sun hit some peaks around them. They were burning daylight. They needed to get back to extracting to fill the next bottle. To repeat the cycle.

Tycho was too far from the rendezvous for most miners. The climb in and out was hard, and the yields were not good. Those challenges gave them security. No one else would want what they had.

It was thirty hours into the lunar day by the time they made it to their hut. They started the RF extractor as soon as they got back. They had left the hopper half full. They both went straight to work, collecting regolith, extracting helium, and bottling it.

It would be a week before their shipment was assayed and sold. A week before, they'd know what it was worth, before they'd have money to improve anything. For now, there was only the work.

When the deposit finally hit, they cleared \$730,000. Helion Dynamics took \$146,000 to service the loan, leaving her with \$584,000. Li stared at the number. “Darrin, it's shopping time. If we order now, they will deliver it at the next rendezvous. Besides water and oxygen, what do we need?”

“Real food. I hate algae biscuit soup.”

“It's nutritious, and it recycles our waste,” she chided. “Everyone on the Moon is boiling those biscuits.”

“Doesn't make it good.”

“No. But it keeps us alive.” She pulled up the logistics map. “We will need air, water, and food. Every time out of the airlock, we lose a little air and water, even though we pump it down. And our suits are well sealed, but some escapes while we are outside. My second priority is getting the equipment we need to make this all more efficient. First, batteries. They didn't last as long as I expected last time.”

The rhythm became a relentless, exhausting routine. Mine, extract, bottle, sell. Only one of them would go to the rendezvous, so the other could start the RF extractor as soon as there was enough power. It was a hard, brutish existence.

Four months in, Li tallied their production. Two hundred eighty-three grams so far. They were making it. They could afford real food now, not just algae. They filled a bottle every cycle and had enough profit for improvements: more batteries, more panels, and a fractional distiller, separating pure ^3He from the ^4He and other

gases. Purified, they could put ten times as much ^3He in a bottle. Li calculated that shipping costs were over 15% of their income. Distillation cut shipping to less than 2%. She hated making the run every lunar night, anyway, time away, needless risks, power consumption. Li smiled to herself.

Darrin was doing maintenance on the RF. He pulled the housing cover and cursed. Dust again. The regolith was fine as smoke and sharp as broken glass. The jagged microscopic shards had never known the smoothing effect of wind or water. They crept into every gap, abraded all the moving parts, and shorted contacts. Constant maintenance was the only way to keep the equipment running well.

Then she read the new claim filings. “Darrin,” she called.

He came over, reading over her shoulder. “Ocean State filed a claim west of Tycho crater. That’s fine by me. But the plat is rectangular, and it contains ten acres of our crater floor. We have to defend it.”

“They filed from Earth. Haven’t even landed yet.” He zoomed in on the overlap. “Probably don’t know we’re here.”

“We have to defend our crater, but they outnumber us. We can’t win a fight. We need to find another way.”

“It’s their legal claim, Li. Let it go,” Darrin suggested, but knew that suggestion would be rejected as soon as he said it.

Li looked at the calendar. Twenty-one days until Ocean State arrived. Seven days of light, then two weeks of night.

“We strip it,” she said. “We pull every gram we can before they arrive. We clean them out, then put the stripped regolith back, making them think it was never touched.”

Darrin stared at her. “Li, that’s...”

“What?” She met his eyes. “Illegal? They filed paperwork from Earth for a crater they’ve never seen. We’re here. We hold this ground. That’s the only law that matters here.”

He remembered Ball Crater. The green laser dot on his chest. He didn’t answer.

“We’ve got seven days before lunar night,” she continued. “Double shifts. When they land, there won’t be anything left worth taking. This is our crater.”

They were fast and efficient. Sweeping was their priority job. This cycle, they would not hit their extraction goals, but their claim had to be defended. They worked in shifts. Li would sweep for sixteen hours straight. Halfway through, Darrin would wake up and run as much of it through the RF extractor as possible. When she was done, she would head to bed, and he would start sweeping, continuing the long, methodical sweeps. The regolith piled up, coming in faster than they could process it.

They had the power to keep running the crawler through the lunar night, but not the RF. They knew their margins pretty well. They went more slowly, saving power, but still sweeping. A week into the night, she climbed out to clear a jammed valve. Her glove came back with a shallow cut across the outer layer. She checked it. The suit held pressure. Any rational person would have stopped. Li taped over it and continued. A day before lunar dawn, they'd stripped about eight acres of the overlap. Darrin began quickly covering the rest with processed tailings. A shallow assay would show this area had no helium. At the same time, Li kept extracting from what remained, pulling every gram she could.

Suddenly, Darrin's voice cracked over comms. "Li... incoming. West ridge."

Li froze, the sweeper still whirring beside her.

One last bag. The scoop arm dipped, dragging another scoop into the sack.

She reversed the crawler, slow and deliberate, the bag less than half full.

"How many?"

"Three rovers. Heavy-frame. Ocean State livery."

A tone hit her headset. "Unidentified vehicle, you are inside a mineral claim held by Ocean State Resources. You are in violation. Acknowledge."

She keyed the mic. "Tan, Tycho Outpost. Leaving now."

Static. Then: "Leaving with our regolith."

Three rovers crested the ridge, moving faster than her crawler could retreat. They stopped thirty meters out. Three figures stepped out of the lead rover, suits marked in Ocean State blue. The other two vehicles waited behind.

She pulled her survey laser from its mount, not pointing it, just holding it, ready.

"Tan," the lead miner said. "You're actively collecting inside our claim."

“Leaving your claim,” she corrected. The crawler kept backing away. Five meters. Ten.

One of them raised his survey laser, aiming toward her cab.

Li’s laser snapped up, centering on the leader’s helmet. Her arm was steady. The crawler continued to reverse, engine humming.

“This is an Ocean State claim,” the leader said. “You can’t take our ore.”

A fourth figure stepped from the rear rover and began taking samples.

“This crater is mine,” she said. “Not some of it. All of it.”

“It’s not yours, Tan. It’s ours. Leave the bag.”

The fourth figure crouched again, dropping fragments into his mass spectrometer.

“This area doesn’t make sense for you anyway,” Li said, still backing away. “Most of your claim is on the far side of the crater. You planning to haul your loads over that wall every shift?”

A bright green dot appeared on the second man’s chest. Perfectly round.

Everyone froze.

Li recognized it. Darrin had climbed the ridge behind the hut. He was dug in under regolith, safe from any return laser shots.

The Ocean State team stopped moving. Li kept backing away, her laser still trained. Forty meters. Forty-five.

“You can work this ridge if you want,” she said evenly. “But every time you leave, I’ll be back. Every time you’re over that wall, I’ll strip it clean. You’ll burn more fuel climbing than you’ll ever pull in helium.”

The fourth figure said something over a private channel. She couldn’t hear it, but she knew what he’d seen. Darrin had covered that section with stripped regolith. It would look worthless.

“This crater is mine,” she said. “I’m here. You’re not. Stay west of the wall. You’ll find more helium there anyway, and you won’t have to climb a mile every shift. Stay out of Tycho.”

Silence. Then the lead miner tilted his helmet, turned, and the three of them climbed back into their rovers. They headed west, over the ridge and out of sight.

Li cut the channel. When she could no longer see them, she jumped out of the rig and secured the bag properly. Her hands shook. Once the seals held, she turned the crawler around. Forward wasn't much faster than reverse, but she wanted distance.

No one followed.

Back at the hut, the processors hummed, pulling scraps of ^3He from the dust. Darrin stood by the airlock, arms folded.

"You didn't have to," she said.

He shrugged. "Didn't seem like they wanted to talk."

She looked at him for a long moment, then turned to the gauges. "They'll be back."

"Maybe. That ridge slows them down. Easier to work their western sites. And that sample they pulled showed no helium."

"They'll remember your name now," Darrin said quietly.

"They already did." She checked the pressure readings.

They were preparing for the next cycle when Darrin pulled up a relay feed. "Another darkside incident, Li. Just discovered."

Li glanced at the slate. Four miners. They'd pushed their extraction run too long, trying to fill one more load after nightfall. Batteries died two days before dawn. Their hut lost heat, then oxygen. By the time the sun came up, it was too late.

"They were gambling on one more load," Darrin said quietly. "They thought they knew their capacity, but they overplayed it."

"They lost." Li stared at the screen. Four names, four families somewhere on Earth who'd get a notification. "That's what the solar power does. It makes you choose between shutting down for two weeks or gambling with your life."

She closed the feed.

"We can't keep shutting down for two weeks every month," she said. "We can't continue depending on sunlight."

She paused, then added, “We need a reactor. Something that runs through nightfall.”

He hesitated. “Nukes are banned, Li.”

She gave him a long look. “So is pulling helium from someone else’s claim.”

“That’s not the same...”

“The law only matters if someone shows up to enforce it. If I can get one, it changes everything. No more blackout periods. No more power rationing. Just work, all cycle long.”

“And if they find out?”

“They will. Eventually. But if we’re smart, we’ll be too useful to stop.” She paused, then added, “Out here, production keeps the peace.”

Chapter 4: Cold Storage

Price of ^3He	\$8,000,000/kg
Current Production	18 tons/year
Current Consumption	12 kg/year
Time Since Start	18 months

Tobias Raines sat in his office at Helion Dynamics headquarters in Geneva, looking out over Jet d'Eau in Lac Léman. The bells at Saint-Pierre marked the hour. His policy was no meetings before nine, but today he had a visitor from China, where it was already noon.

Energy Minister Chen Zhengrong walked in with three assistants just as the sixth bell pealed.

“Good morning, Mr. Raines,” Chen said. “Our premier sends his greetings. He wished to attend personally, but was caught up in internal affairs.”

“Good morning, Mr. Chen. What can I do for you?” Raines asked.

“Fusion power is an enticing concept, but our concern is stability. We have been watching the Lisbon performance data.”

“We publish the Ilha Fusão logs specifically to allow verification.”

“Yes, of course, Mr. Raines. We have reviewed your extraordinary logs. We can see routine maintenance ongoing, without a flicker in voltage or power output since operations began eighteen months ago. This stability is what has piqued our interest.”

“Fantastic. You’ve done your homework. Several countries have purchased our ten-gigawatt tokaplex. The first one, in Argentina, will go live soon. Would you like to purchase one of those? We have strict requirements on infrastructure, which we can go over...”

“No, Mr. Raines. We do not wish to purchase a ten-gigawatt tokaplex.” Chen folded his hands. “We have read your documentation on required infrastructure, and for China, we find that to be inadequate. Instead, we would like to purchase a one-terawatt tokaplex.”

“Mr. Chen, our production of the ten-gigawatt tokaplexes is deeply streamlined. All major components are produced on an assembly line in Biloxi. We can get one set up for you quickly.”

“No, Mr. Raines. You do not understand our needs. We wish to centralize production, freeing our country to use electricity in new and better ways. We call it hyperelectrification. With your terawatt tokaplex, we will be able to provide the power needed to enable that step forward.”

“That’s... forward-thinking.”

“It is necessary. Ten-gigawatt facilities are insufficient for our scale. We would begin with one terawatt installation. Operational validation. Then expand. Eventually, we would expect several such plants.”

“Mr. Chen, our terawatt facilities have not proven as popular yet with other nations. I’m not certain the infrastructure specifications have yet been made public.”

“We would appreciate your alacrity in this matter,” Chen said. “China will do what we can to help. Our engineers can draft infrastructure proposals to move the power out across the country. And we can assure you that you will not face any bureaucratic difficulties.”

“Mr. Chen, as I’m sure you’re aware, we have not yet made a final determination on the price of such facilities. I’ll get back to you with that...”

“Yes, of course, Mr. Raines. However, our premier wishes to assure you of our commitment to the future. He has sent a note with me.”

Chen handed Raines a red envelope. Inside was a card with an account number with a Hong Kong bank, a PIN, and a balance.

\$500,000,000,000.

He kept his hand steady as he folded it and slipped it into his jacket pocket.

“Please tell the premier that we appreciate his commitment and will put this together now. I’m sure you realize we have a significant backlog...”

“The premier wishes to tell you this is not a down payment, but a gift for your consideration in this matter.”

Chen rose, they shook hands, and his assistants filed out behind him.

Raines stood at the window, looking out over the lake. His slate was already buzzing. Meetings, messages, a dozen fires needing attention.

He called Elin.

Elin's answer cut right through all the pleasantries. "Raines, why are you calling so late? It's almost midnight. I am at my wits' end! We're out of molybdenum. Production stops in two days without a resupply."

"Elin, I am so glad you are still up. I will be arriving in Biloxi tomorrow at noon. I would love to see you while I am there. Molybdenum? Let me check into that."

He pulled up his calendar. A meeting with India in three hours. Egypt that afternoon. Poland and South Africa tomorrow.

He canceled them all.

A half hour later, his Gulfstream was still climbing when he reached Miguel Flores, a metals broker in Lima.

"Miguel, I need molybdenum."

"Tobias, it's so nice to hear from you. Unfortunately, you have caught us at a bad moment. We have eighteen tons here, but it's already been sold to a buyer in Shanghai."

Raines said nothing.

"Let me make a call."

Raines touched down at the Gulfport airport. He waited for the molybdenum to arrive, got it loaded in a semi, and went to find Elin in Back Bay. She was giving a talk to newly hired welders when he arrived.

"Elin, wonderful to see you. I found some moly to tide you over. Eighteen tons," he said, gesturing to the truck.

Elin got a crew to unload the truck.

"I appreciate this, Tobias. We have five hundred tons coming from South Africa next week, but we have improved our production cadence. We're building faster than the supply contracts anticipated."

"That's the third time this month."

“I know. Every time we improve the process, every time we get faster, we burn through materials faster. The contracts can’t keep up. Production here has been a tornado.”

“That tornado has been a miracle, Elin. I can’t believe how much you have done. Eighteen months ago, this was bare ground.”

Tobias looked around at her manufacturing facility. It looked nothing like a factory. The grounds stretched for miles in every direction. Six cooling towers stood in various stages of completion. The cranes moved overhead, welders beneath sending up showers of sparks, heat rising through the salt air. In the distance, they could see the metal storage yards, large Quonset huts filled with refractory and base metals.

“Argentina’s core is being loaded onto a barge now,” she said. “Japan’s will leave in a month. We’re moving, Tobias.”

They stepped inside one of the buildings. Raines watched as electricians wired up control panels.

“We can build every major component in about a year. With pipelining, our goal was to complete one per month. I am trying to catch up with orders, but the faster we move, the further we are behind. Each step forward brings its own headaches.”

Raines responded, “From your side.”

She stopped. “What does that mean?”

“You’re improving the process continually. Each improvement reveals new constraints, new risks, and new obstacles. From my side, it’s a cascade. Every plant we sell triggers demand for ten more. Interest becomes orders before we can prepare. I couldn’t slow this if I tried.”

“Maybe you should try,” she said quietly. “Let’s get some lunch.”

They took a car to the Beau Rivage.

The doorman greeted Elin by name when they arrived. Elin brought Raines to her favorite restaurant. “Coralline’s. They have great seafood and a private dining room.”

Raines sat down. “How do you like staying here?”

“It is close enough to the manufacturing site.”

The waiter left with their orders. Raines waited for the door to close. “I spoke with China yesterday. They want a terawatt tokaplex.”

Elin put her fork down. “Seriously?”

“Minister Chen was serious. They call it hyperelectrification. Centralized power production for the entire country.”

“That’s...” She pulled out her slate, already running calculations. “Tobias, that’s exactly what this technology is meant for. Economies of scale kick in at the terawatt level. The current design is a compromise. I favored simplicity and speed in construction to meet the volume. Moving up, though, I will be able to use direct energy capture. Efficiency improves dramatically, and non-fuel expenses will be similar to the current design. You could power...” She stopped, staring at the numbers on her slate. “Oh.”

“What?”

“Tungsten requirements.” She turned the slate toward him. “Eighty thousand tons per plant. Global annual production is under one hundred thousand.”

Raines nodded slowly. “I see.”

“And that’s just one material. Molybdenum, tantalum, and beryllium all scale the same way. Square-cube problem: a hundred times the power needs a thousand times the materials.”

“The manufacturing facility is designed for ten-gigawatt tokaplexes. We’re good at it. But terawatt? That’s not just bigger. It requires materials that don’t exist.”

“I understand.”

“I want to build it, Tobias. This is what fusion should be. But I can’t.”

“What if you didn’t have to worry about materials?”

She looked at him. “What does that mean?”

“China is working on infrastructure design. When they’re done, I want you to review them. Can you do that much? Not the real plant, but how they plan to distribute electricity.”

“Of course. But it won’t matter if there is no tokaplex to power it all.”

“In the meantime, I would like you to consider what a terawatt structure would look like. Not detailed, but more than an overview.”

“Design for when the materials magically appear?”

“When I figure out how to get them.”

She took a slow drink of carbonated water. “Okay,” she said finally. “I’ll indulge your fantasy. I will check China’s specs, and I will daydream about a terawatt facility. Is that enough?”

“That’s all I am asking.”

“Until then, I have a backlog of twenty tokaplexes. Before we are done with those, I expect you will drop another thirty on us. I have real work to do.”

After lunch, Raines returned to the airport. He had one more stop.

Dutch Harbor was gray and cold when his jet touched down. The working port stretched along the waterfront where fishing boats were unloading king crab, processing plants were running overtime, and warehouses were stacked with pallets of frozen seafood.

His facility looked like all the others. A low warehouse, maybe eight hundred square feet, with weathered siding, no signage. Cryogenic containers came off the boats, got wheeled inside, and others came back out empty and were reloaded on boats. Routine.

Today, Raines needed solitude. He walked in. The floor was ordinary. A few rows of 150 liter ^3He cryobottles using ^4He as a shield to prevent loss. Two workers in coveralls were moving helium tanks.

He went to the back office and closed the door.

This was a cold storage facility for bottles of ^3He . Behind a partition, there was a compressor takeoff feeding an armored pipe. That pipe ran through bedrock into the North Pacific to the edge of the continental shelf. From there, it descended eight thousand meters into the Aleutian Trench, where the pressure reached twelve thousand psi. Three spheres, each the size of a city block, were filling with ^3He , their flexible bladders displacing seawater as they filled.

Current level: 12.3 tons. Capacity: 600 tons.

Enough buffer capacity to run multiple terawatt plants for a year while lunar production scaled.

He opened a secure line to his purchasing team in Geneva.

“I need contracts,” he said. “Tungsten, tantalum, beryllium, titanium. Long-term agreements. I’m authorizing up to double the current spot prices. Our manufacturing facility can’t stop.”

“Understood. I’ll have proposals by end of week.”

He ended the call and pulled up Elin’s materials list.

For the ten-gigawatt plants, the contracts would work. Premium prices, aggressive terms, but the metals existed. He could buy them.

For terawatt-scale... he scrolled through the numbers. Tungsten requirements: 80,000 tons per plant. Global annual production: 95,000 tons. China wanted one now, but would probably want three more as soon as they got that one. Then all the major economies. These were no longer replacing old generators or expanding generation. Hyperelectrification would change everything.

But the math didn’t work.

He could stockpile helium. He could lock in metal contracts for current production. He could keep Elin supplied, keep the ten-gigawatt plants rolling out, keep the orders coming.

But terawatt? He pulled Chen’s *hongbao* from his jacket.

He had no way to deliver what they wanted.

He sat there for a long time, staring at the numbers.

Then he walked outside. The wind off the Bering Sea was sharp and cold. The dock was empty. Just the ocean, the cold, and the problem he couldn’t fix.

He stood there, hands in his pockets, until the light began to fade. There was no moon tonight.

He walked back to the airport. There was no way on Earth, and the moon didn’t have the refractory metals he needed. That would not stop him.

Chapter 5: Argentina

Price of ^3He	\$8,000,000/kg
Current Production	19 tons/year
Current Consumption	11 tons/year
Time Since Start	19 months

Elin's Gulfstream descended into San Fernando International Airport, just north of Buenos Aires. While the aircraft taxied into the hangar, she reviewed the inspection briefs. She stepped out with only an overnight bag and her slate.

The Ministry of Energy representative approached and extended his hand. "Dr. Elin Sarai?" he asked. "Welcome. I'm Álvaro Luis, Director of Strategic Projects. The preparations for this tokaplex have been my responsibility. Would you like to inspect our infrastructure?"

She shook his hand. "No worries. Show me what you want me to see."

Álvaro smiled and gestured toward the car. "We'll take the river drive. You'll see the northern substations on the way."

They pulled away from the airfield in silence, the black sedan's windows sealed tight against the outside heat. Cranes loomed behind transit depots, where new transformers were being craned into place. There were Ministry placards posted on utility poles and a banner stretched across an overpass. *"LA NUEVA RED – PARA UNA ARGENTINA ELÉCTRICA."*

He glanced at her. "You'll find we're ready."

She didn't reply.

Past the Retiro yards and the old port, they came to a hardened facility surrounded by armored fencing and a ring of security. Álvaro leaned forward. "This is Subestación Belgrano Norte. One of eight new HVDC hubs. It's part of the megavolt redundancy mesh."

Elin checked the converters and cabling. She had already seen the testing documentation. She had verified test data. Now she was verifying build quality.

After passing a few more sites, Elin noticed a generator truck. “Commissioning under load?” she asked.

“Yes. We are connecting a mountain village that was previously off-grid. They are installing a step-down transformer.”

Elin and Álvaro got out and donned hard hats. The familiar scent of ozone and mineral oil told her testing was underway. She inspected the wiring, noting the runs were sharply dressed and tied down. She talked to the foreman. “They never had power?”

“Pero claro. They have a diesel generator, but no road. Mules bring the fuel.”

Elin kept the helmet with her when they got back in the car. She gazed at the transmission towers crossing the countryside. “You’ve done the hard part. The routing, the redundancy. You built for more than ten gigawatts.”

“We built for forty. Our president is looking to the future,” Álvaro said.

Elin gave a short nod. “Let’s go see the first one.”

The car pulled back into traffic. Elin watched the transmission lines stretch south as they drove to Balneario Monte Hermoso airfield. From there, they boarded a tiltrotor for the fifty-mile flight to the tokaplex.

In the air, she could see three large black lines converging near the sea, connecting to submarine cables. Twenty minutes later, they arrived at the artificial island, Central de Fusión de Monte Hermoso. It was larger but simpler than Ilha Fusão. Álvaro led her off the landing pad to a low-rise brick building with rounded corners and large glass windows on the second floor.

Once inside, Álvaro led her to the N.O.C. door. “Ready to see the beast?”

The room’s walls were lined with screens showing actual and predicted demand, voltages, maintenance schedules, and crew assignments.

“This is the mesh,” Álvaro said, gesturing to the largest screen, which displayed a map of nested loops and arterial corridors. “We’re the first in Latin America to attempt a full megavolt HVDC overlay. Drop any node, and it automatically reroutes without cascade failure.”

Elin studied the model. Interlocking rings in the north, deep ties south to the mining zones, fresh connectors across the pampas where before there had been

only thin, century-old transmission lines. Every connection lit in a low amber that pulsed as loads shifted in real time.

“You designed this quickly,” she said, with a matter-of-fact tone.

“We didn’t design it alone,” Álvaro replied, tapping the screen. “ABB handled the HVDC overlay architecture. Our team integrated them with national policy. This isn’t a local tie-in. It’s a national backbone. If a section goes offline, every node has failover paths without loss of synchronism.”

He tapped a zone northwest of Córdoba.

“Ilha Fusão was smaller,” he said. “A hundred megawatts. Simulations were sufficient to ensure smooth, safe operations. Here, simulations aren’t enough. Every component has been pushed to its limit.”

He pointed to a secondary display. Three heavy black lines radiated from the tokaplex, each connecting to a different node of the mesh. She recognized them from the flight in.

“These are the HVDC links. Fully isolated, with separate substations and protection. If one fails, the others carry the load.”

Elin looked at the links. “Meets the requirements. The real question is what happens when it goes live.” She looked again at the projected grid, breathing, humming, ready to distribute enormous amounts of energy. With Ilha Fusão, she had been there every day for five years before it went live. For Monte Hermoso, all she got was some reports and less than a week for inspections. The stakes were much higher, the risks were much greater, and she had no time to work with the installation.

They didn’t linger at the control center. Álvaro had a schedule, and the schedule moved fast. He brought her to a conference room down the hall. Elin checked her itinerary, and this meeting wasn’t on it.

Elin had designed the situation room for engineers, with hard lines instead of wireless and integrated displays. Today, it held several ministers, envoys, and one uniformed military observer who said nothing and made no introductions. Elin took her seat near the far end. A plate with chipas and facturas and a glass of sparkling water sat at each place. She broke a chipa open, releasing the steam and the aroma of toasted Reggianito cheese before popping it in her mouth.

The Minister of Energy opened the session briskly. “We’re in final integration for Monte Hermoso,” he said, tapping a projection to his right. “The island is on schedule. The mesh is ready. We would like to discuss future commitments.”

A secondary map appeared with three red outlines across the country, one near Buenos Aires, one near Mar del Plata, and one further south near Comodoro Rivadavia, each marked *SOLICITADO*, all of them with planned tie-ins to the mesh.

“These are the follow-ons,” he said. “This will support development of transportation, industry, and mining operations around the country and balance our mesh.”

Elin didn’t answer at first. Looked around the room. Not a single engineer. She read the names, the grid ties, the siting logic. “You’ve assumed Helion Dynamics will be able to schedule these deployments,” she said finally. “Who confirmed those deployments?”

The Trade Minister answered. “We were your first paying customer, and we have paid in full for Monte Hermoso. We expect priority.”

Elin nodded slowly. “I’ll relay that, but it’s not my contract to make. You were first, but there have been many orders since, all fully paid.”

Another official, this one from the Ministry of Industry, leaned in. “Aluminum and lithium plants have been built already. The demand is high, and they can employ several thousand, but they need power.”

“We know you have Raines’ ear,” said the Energy Minister. “And we know you understand what this kind of energy makes possible. We are not just talking about supply. We are talking about national transformation.” He slid an envelope to Elin.

Elin placed her hand flat on the envelope, stopping it before it reached her. Blushing hotly, she looked at each person in the room.

“I am in manufacturing. I built this place, even this room. Raines decides where they go.”

Elin’s eyes moved from minister to minister. “If you need the power to arrive sooner, build the other islands quickly and connect them to the mesh. You know the requirements. Then talk to Raines.”

After the meeting, Álvaro found Elin. “The tiltrotor is ready, Dr. Sarai. We’ve reserved a room in town and dinner is arranged.”

Elin shook her head. “I’ll stay on the island tonight.”

Álvaro paused. “The apartments are very basic.”

“I know. I designed them.”

He smiled slightly. “The vending machines may not satisfy you. My wife packed a dinner for me in case things went long. It is in the refrigerator. It’s yours if you want it.”

“Thank you, Álvaro. That sounds perfect. I would just like some time alone now.”

“*Buenas noches*, Dr. Sarai. I will see you in the morning.”

She found the milanesa a la napolitana con puré in the refrigerator and warmed it. The solitude was perfect, just her and the quiet machine. After dinner, she walked the corridors alone, listening, connecting. The tokaplex hummed softly on standby: cooling systems, auxiliary pumps, metal contracting in the evening air.

The apartments were simple with just a cot, a sink, and a shared restroom at the end of the hall. Elin had designed them herself, years ago, for overnight crews. She’d never expected to use one. But staying close to the machine was grounding. This is why she was here. She slept well.

The next morning, Álvaro brought her medialunas from home. Elin thanked him as she ate.

Her first meeting that morning was with the general manager, Gabriel Mansilla. She found him windblown, looking across the floor.

“Elin,” he said, nodding once.

She skipped the pleasantries. “How’s the floor?”

“Cleared, cleaned, and ready,” he said. “We pulled the last crates two nights ago. Gantries are programmed. Power couplings are aligned and tested.”

She glanced down at her notes. “Turbines?”

“Hot,” he confirmed. “Running on simulation heat. We’re venting most of it, but efficiency is higher than spec. The sCO₂ units stabilize faster than we expected.”

“Any anomalies?”

He hesitated. “Minor fouling on line B. There was some grit in the condensing loop, probably from the cleaning process. Flushed and rebalanced. They are running it again.”

She made a mark on her slate. “And the core?”

“A few hours out. I expect it around noon, plus or minus.”

“Assembly team is ready?”

“Staggered shifts. They’re tuning the robotic frames and walking through dry runs. They’re ready.”

“And you?”

Gabriel gave a tired half-smile. “I’ve got three hours of sleep and a cooling jacket strapped to my lower back. So yes. Ready.”

Her eyes laughed, but her voice stayed steady. “Stay on comms. Notify me when the core clears port.”

Elin headed to the tiltrotor pad. Gonçalo Silva was coming. The tiltrotor dropped onto the eastern pad. Elin was already there when the rotors began to slow. Gonçalo stepped out with his usual pace, unhurried, steady. He wore the same field jacket he had worn in Ilha Fusão, light, practical, unbranded. His bag was slung crosswise, heavy with tools and notebooks. Before he made his way to her, he paused to take in the east gantry, then the core floor scaffolding. She let him walk the first few meters alone.

When they met, they didn’t shake hands. She tilted her head slightly. He smiled.

“Looks smooth,” he said.

“More streamlined than Ilha Fusão,” Elin replied. “We learned from all the mistakes we made there.”

Gonçalo was looking at the skyline. “I can see that. It’s much larger, but it feels less cluttered.”

They started walking toward the operations mezzanine. Between them, the control tower loomed in prefabricated concrete and fiber panels, with broad window slits set under sun-shielding eaves.

They climbed a short flight of metal steps that led up to the observation level. Below them, the control room staff ran diagnostics in near silence, each screen tracing voltages and heat gradients across systems not yet powered. Gabriel stood with one of the interface engineers, looking up at a wall display of turbine feedback loops.

Gonçalo gestured at the far end of the floor. “Gabriel’s getting there. He’s detail-forward, but he’ll learn to step back. You can’t fly a thing this size like you’re tuning it by hand.”

“He respects the systems,” Elin said. “He doesn’t try to outthink them.”

“Is the emergency cooling loop really that simple?” he asked.

Elin brought up a diagram on her slate. “The core is below sea level now. If containment fails or thermal load runs away, gravity pulls seawater straight through. No pumps, no heat exchangers. Just open the valves. Four separate valve sets. The computer controls the automated valve set. If it detects a thermal runaway, it opens the valve.”

“That’s nice if everything works perfectly. What if the computer shuts down or the sensors fail?”

“If the computer fails for any reason, there are two manual valves, accessible from different locations. And even if no one notices, there is one final failsafe, a thermal valve with a fusible plug. If the temperature gets too high, the solder will melt, opening the valve.”

He leaned on the railing. “That’s the difference, isn’t it. Ilha Fusão is safe, but it needs babysitting. This one’s safer because it doesn’t.”

Elin and Gonçalo were eating lunch when the core arrived. Elin stayed in the cafeteria, but Gonçalo got up to watch Mansilla’s crew install it.

The gantry cranes lowered it into position, where a team of a dozen technicians immediately swarmed the core, connecting the feed pipes, monitors, and cooling loops.

After installation was complete, Elin found Gonçalo. “How did they do?”

“Where did you get these guys, a Formula One team? Mansilla has trained them very well. Installation went smoothly and much faster than I expected.”

“At Ilha Fusão, we didn’t design for installation. This time around, ease of assembly was a key constraint. It’s supposed to be smooth.”

With the core secured, the site shifted tempo. The Argentine crews moved quickly, hands practiced, expressions focused. The last hookups were made and checked.

Heavy conduit arms were swung into place and locked down. The sCO₂ turbine, already hot-tested, was mated to the flow channel. A hiss and pressure check marked the moment when the CO₂ storage tank came online, dumping its contents into the sCO₂ turbine.

At the perimeter of the island, a low-power carbon collector clicked back to life, drawing in atmospheric CO₂ to replenish the storage tank. It would refill the tank in about a month. In theory, they would not need more CO₂, but if there were a leak developed, they would need to be fixed and completely recharged.

Power lines were checked, then rechecked. The HVDC terminals blinked status green. Routing software loaded. Backup systems synced. The mesh recognized the plant’s endpoints. For the first time, the island registered as a node in Argentina’s electrical system.

Assembly was declared complete at 14:42 local time. Just a line item in the log, signed by the work crew chief, seconded by Gabriel.

Gabriel found Elin near the mezzanine, hands in his jacket pockets, boots dusty from the floor. “Dr. Sarai, *ya está listo*.”

Elin notified her contact, a US Navy lieutenant on the amphibious assault ship offshore. It had brought the container down from Dutch Harbor with a destroyer escort around the horn. The US government had asked to secure the delivery due to the extraordinary value. The ³He in the container was enough to power this plant for a year. It had been scraped from the Moon at enormous cost in labor, equipment, lives, and eight billion dollars.

The cryotank came in low, under fighter escort. Two F-35s bracketed the CH-53E as it descended on its pad. Propwash kept everyone at a distance.

From the operations platform, Elin watched as the rear ramp unfolded. Marines deployed first, fanning out with unhurried precision. Then the cargo came down, matte silver, heavily insulated, marked with stenciled black codes, and frost blooming around the seals. The cryotank held a year’s supply of ³He, one metric ton.

A Marine captain held his salute until Elin acknowledged him. He passed over the digital manifest. Elin inspected it, then signed.

Installation was already underway. The hauler maneuvered the container to the sealed platform near the reactor housing. Hydraulic arms lifted it into position. The cradle clicked into its locks. Engineers verified coolant flow and containment pressure. Readouts held steady.

The Marine escort didn't linger. Before departing, their commander turned to Gabriel and saluted sharply. Gabriel crisply returned the salute. Then the Marines were back aboard the helicopter and into the air.

She walked with Gabriel to the control floor for the test run. The control room was already full. Gabriel was at the operator's station, Gonçalo standing behind him, arms crossed, leaning slightly forward. Elin watched from one of the side consoles, monitoring the turbine feedback loops.

On screen, the core came to life in steps. Systems primed. Vacuum held. Field coils charged. Then the plasma ignited briefly, controlled, a soft event visually, just a line of green across the diagnostic pane. Not even enough to raise the turbine feed pressure, but the superconducting ring registered the field.

Gonçalo leaned in and called for another pulse, this one longer. Seconds, not milliseconds. Heat output rose. The turbine responded, barely. A ripple in the pressure. Cooling held steady. The sCO₂ cycle hadn't fully ramped, but it was alive. The storage buffers reported a charge. Grid output was still zero, as expected.

They ran five more bursts. In each, the response was tighter. The plasma shape held. The pressure gradients behaved. Thermal gradients stabilized faster than the simulations had suggested. The reactor wasn't just stable, it was predictable and controllable.

Gabriel turned and gave Elin a nod. "This thing listens."

She checked the timestamp. The whole break-in run had lasted under twenty minutes. Long enough to show continuity across fusion, routing, recovery, and control systems, but not enough to prove efficiency. The turbine had barely started to turn. Just a machine confirming it was what they had built.

Gonçalo stepped back from the console. "The next run will be for power, Gabriel."

The control room was silent in the seconds before ignition. Gabriel keyed the final command. The reactor ramped up in a controlled ascent. Plasma formed, and ignition held. The display confirmed a stable burn in under three seconds. Turbine readouts spiked. Heat flooded the sCO₂ loop, and pressure hit threshold. Grid output climbed, six, eight, ten gigawatts. Steady. Smooth.

For a moment, it all held.

Then one of the HVDC lines dropped. Just a blink and the connection was lost. Power continued to route smoothly as the mesh's reroute protocols kicked in. Load spilled to the southern loop and northern loops, then back across the mesh. Power rebalanced.

Gabriel swore under his breath. Gonalo stared at the numbers, expression flat.

Elin leaned forward and logged the fault. The mesh had responded exactly as designed. But the line had failed under load, and that meant something had slipped.

She didn't say anything yet.

Instead, they left the control center and followed the long service corridor toward the power conditioning chamber. This was the part no one showed on press tours. The CL filters filled the space like machinery in a cathedral, vast, industrial, sculptural. The capacitors stood fifteen feet tall, six feet across, ribbed in black insulation. Inductors coiled behind caged housings, humming faintly. Each unit shaped and stabilized the rectified current, converting raw internal power into clean HVDC output.

Gonalo paused in the aisle between them. He thought of Ilha Fuso, where the output had been AC, conditioned by standard hardware. It had been elegant, minimal. Nothing like this. He felt like a mouse in the land of giants.

He turned to Elin. "Isto   outra escala."

She turned to him. "Another kind of responsibility."

They continued walking, passing into the thyristor hall, where the problem had occurred. Too much feedback on the western line had caused multiple commutation failures as it tried to ramp up. The valves were blocked to prevent them from melting. Gabriel restarted the process, letting the western line climb slowly. A thousand volts, ten thousand, a hundred thousand. Gabriel touched one

more button on his slate, and full power was restored. There was no friction to the power now. It moved outwards with a strange ease.

Back in the control room, the grid displays showed full delivery restored. Argentina was receiving power. Gabriel gave his diagnosis. “It was a harmonic echo. The western line’s fundamental resonance matched our switching frequency. The circuit couldn’t provide sufficient reverse-bias duration to allow the thyristors to shut off between pulses. From now on, we ramp the western line on a staggered curve to stay below that resonant peak.”

Gonçalo beamed. He didn’t understand DC power nearly as well as Gabriel, but it was clear the plant was in good hands.

They had 58.3% thermal conversion efficiency. Elin made a note.

The system had worked, and so had the failure. The recovery system had proven itself at full power.

The tiltrotor lifted in a smooth vertical climb, skimming above the breakers before banking east. Through the cabin window, Elin watched the artificial island shrink into geometry, cranes stilled, intake arms folded, the blunt white dome of the reactor glinting in sea haze. A line of pylons faded into the mainland.

She turned back to the slate in her lap. Uptime has held steady since the full ignition. Grid telemetry showed stable dispatch at ten-point-four gigawatts, even with the reroute. Cooling margins were conservative. Substation frequency held. Forecasted demand peaks from the southern industrial corridor had already started to adjust upward.

There were six more contracts in early-phase development. Different soils. Different governments. Different tolerances for delay.

She tapped through Gabriel’s final site report. Nothing unexpected. Turbine efficiency was holding.

The plant would run. But more importantly, it could be repeated.

She closed the report and looked back out the window, but the island was already gone. Just sea, now, layered and flat. On the horizon, faint in the morning light, the HVDC towers on the island fed the submarine cables that went to the mainland. She could see where the submarine cables joined the aerial cable towers on land. Argentina was powered. Elin had work to do.

Chapter 6: Beyond Helium

Price of ^3He	\$8,000,000/kg
Current Production	30 tons/month
Current Consumption	30 tons/year
Time Since Start	22 months

Li liked the Moon. The silence, the slow drift of shadow, the way boots felt against ground. The Earth was beautiful from the crater.

The Earth was slightly gibbous. She had only a few hours left before the floor turned blue, and another couple of hours before the solar panels on the ridge stopped producing power. She had added to her pack every rendezvous. This lunar night, they would have twenty-five percent more power than they needed.

Every rendezvous, she learned of people who had tried just a little too hard and darksided. She took their orders home. She would pay the next of kin when she could find them.

As they got into bed, Li looked at Darrin. “I’ve been working on a way out of this for almost a year, you know.”

“You will never get enough batteries for us to operate during the lunar night.”

At breakfast the next morning, Darrin prepared the algae biscuit soup, with freeze-dried beef and vegetables. Li took a sip. “Andrei has a friend, Belov. They grew up together. Belov is salvaging old Soviet submarines.”

“And?”

“He was tearing down an Alfa-class submarine.”

Darrin frowned. “You bought a submarine?”

“Just the reactor. An OK-550.”

“You’re kidding.”

Five days in, they were going to bed, watching the Earth through the Earth window. “It’s perfect for us, Darrin. One hundred fifty megawatts thermal. Compact enough to fit in a lander frame if you strip the shielding. Belov will clean it up, replace the fuel rods, and check every weld. When we get it, it will be solid.”

“We barely use one megawatt. What are you planning to do with that much power?”

“There aren’t a lot of tiny reactors lying around. This one meets our needs. We will run it low.”

Li was going over their extraction charts on day ten when Darrin spoke up. “You know nukes are banned here.”

“The law only matters if someone can enforce it. In the meantime, this is power when the sun disappears, and the cold tries to kill us.”

“And if they check?”

“I filed it as a high-output RF extractor. It will come wrapped in tungsten foil and a three-millimeter steel box. Safe enough while it’s cold. And I talked to Andrei Volkov. I gave him an extra ten million to not ask any questions. He can use the cash while he is getting Lunar Tug on its feet.”

Day fourteen, their last night. Earth was almost at its last quarter. “When is this coming?”

“Belov said it would take six months to prepare before he could ship it. He is using Russian heavy lift.”

“Won’t that cost more than getting it to Texas and using SpaceX?”

“Fewer people will handle it. Fewer questions.”

Darrin set the pit near the north crater wall. The wall would shade the radiators from the sun. He dug ten meters deep, using the rock to shield them and provide distance.

“What if this thing decides to go critical on us?” he asked, leveling the excavator.

Li knelt at the edge, scooping up a handful of regolith and letting it sift through her gloves. “Then we won’t have time to worry about it.”

“That’s comforting.”

“Belov said it’s engineered to fail safely. Control rods drop, reaction stops. It was made for military use, so it’s built to operate under duress.”

Darrin’s helmet turned toward her. “Water?”

“Lead-bismuth coolant, but same principle.” She stood, brushing dust off her knees. “Hot liquid in, hotter liquid out. We tap the heat, run it through generators, ovens, and anything else.”

He went back to digging. “You make it sound like ordinary plumbing.”

“Ordinary, but with molybdenum pipes.”

For the next six months, every time they went to rendezvous, there were extra tungsten plates and lead-bismuth alloy pellets and TZM pipe sections. Li had cut a deal with Lunar Tug to ship them as ballast whenever they had spare capacity.

Darrin routed an ammonia cooling loop to radiators on the permanently shaded interior face of the crater’s northern wall. Set in reflective troughs, they would radiate heat into space smoothly.

Lunar Tug actually landed inside the crater to deliver the reactor. It was a cube about three meters to a side. It looked like just plain carbon steel plates, scuffed from the journey. Li watched the Lunar Tug crew offload it with the kind of care reserved for live ordnance. No one questioned the claim that it was an oversized RF. Belov had stenciled “RF Extractor” and “УСТАНОВКА РАДИОЧАСТОТНОЙ ЭКСТРАКЦИИ” on the steel panels. Li wondered how much the Lunar Tug pilots knew.

“Christ, it’s heavy,” Darrin muttered as they guided it toward the pit.

“Be glad Belov removed the depleted uranium shielding before shipping this. It would have weighed four times as much with that in place. What’s left is just enough for transport.” Li ran her hand along its surface.

They spent the next two days preparing it. First came the tungsten plates Li had been accumulating for real shielding. Each plate was five hundred kilograms. Even on the moon, they weighed too much for Darrin to lift. They bolted them to the reactor’s sides and top, piece by heavy piece, several layers deep.

Only then did they lower it into the pit and connect the molybdenum piping. The hot loop ran out a hundred yards from the reactor, then doubled back. Li had already set the pipe stanchions, giving the pipe about one foot of clearance above the exposed rock. Every four yards, Darrin had installed a tap where equipment

could draw heat from the flowing metal. Between the taps, silicon aerogel insulation and protective covers kept the heat contained. When it was completed, they buried it in regolith.

Darrin dropped lead-bismuth pellets in a feeder pipe until it was full. Then he looked to Li, who was seated at the console. “Ready.”

Li turned the handle, lifting the control rods slowly, watching temperatures climb degree by degree. As it warmed, the pellets melted and started flowing deeper. Darrin kept feeding in more. They melted quickly, filling the piping and beginning their circulation as a river of liquid metal.

“Temperature’s climbing,” Darrin called from his console.

Li checked her readouts. “The reactor’s warming smoothly. The metal is at operating temperature. How’s the flow?”

Darrin scanned the regolith with an infrared imager. “Steady. No visible leaks.”

The real test came when they started up the supercritical CO_2 turbine. Li had connected it to the first tap on the hot loop, where the liquid metal was hottest. The turbine itself was compact, the size of a car engine, but rated for ten megawatts electric. She let the current into the generator, using it as a motor to turn the turbine slowly, making sure that the blisks were not warped by the density difference between the liquid phase and gas phase of the CO_2 charge. As the liquid entered the heat exchanger on the tap, it heated fast, pushing back against the turbine. She monitored the temperature. It wasn’t moving fast enough. The CO_2 was heating above its dissociation temperature, chemically attacking the exchanger. She increased the power slightly, and the velocity brought the temperature back down. Pressure and temperature were rising. The CO_2 passed the critical point. She kept pushing it deeper into the supercritical range, then opened the valve to the chilled ammonia line and stopped driving the turbine. It kept moving. Li and Darrin watched as the output rose, driving it all the way to ten megawatts before throttling it back down to one.

“We’ve got power,” she announced. “We could run every hut on the Moon with this thing. Let’s set up the ovens.”

Darrin had connected two regolith processors into the second and third taps on the hot loop two days ago. Now, he wired in the electric power to control the rotating drums and the gas extraction. Each oven could hold twenty tons of raw regolith, industrial-scale processors designed to crack rock and extract every trace gas locked inside. Unlike the RF extractor, which gently coaxed helium atoms to

vibrate free, these ovens hit the regolith with raw thermal energy from the liquid metal. No finesse. No precision. Just extreme heat.

He loaded a test batch into the first oven, and they watched the internal temperature climb. 700°C is the generally accepted extraction temperature. Darrin let it climb to 1,000°C.

“Jesus,” Li whispered, watching the mass spec data.

The regolith didn’t just release helium. It gave up everything: hydrogen, neon, argon, water vapor, gases that had been locked in the rock for billions of years. Her old distillation equipment, designed for the flow from RF extraction, couldn’t keep up with the torrent of mixed gases pouring out of the load.

“We’re going to need a bigger distillation column,” Darrin said, imitating Brody.

Li laughed. She was already pulling up catalogs on her slate. “Two bigger columns. And a pre-separator to strip out the non-helium gases. We just pulled out 1.8 times more ^3He than our old rig could manage. And that’s not counting all the other gases.”

She showed him the readout. Beyond the helium, they were recovering significant quantities of water vapor and noble gases. On a world where everything had to be shipped from Earth, those byproducts were valuable.

When lunar night came, for the first time since coming to the Moon, Li ran their hut at full power. Every light blazed. The air recycler hummed at maximum capacity. She cooked dinner without checking the battery levels. They continued sweeping and extracting right through the night under the blue glow of Earth.

The first visitor was a lean miner from just east of Tycho Crater. Li watched him on her external cameras, a single figure in a battered pressure suit, following their tire tracks across the crater floor. She suited up and walked out to meet him.

“Saw your lights,” he said over Local Band. “Not something you see much this time of night.”

“Name’s Sam,” he said. “My claim’s about ten miles east. I’ve heard of you before. Tan, right?”

Li winced at the name but let it pass. “What brings you out?”

He shrugged, the motion faint in the suit. “Mostly curious. I can’t work at night, so I was just looking around.”

“Most people can’t work at night,” Li said.

“Yeah, but you can. I was thinking maybe I haul some of my dirt here, you run it through whatever you’re doing. You take a cut, I can dig more.”

Li glanced at Darrin. They’d talked about this before, but this was the first time someone had walked in with the offer. “What kind of cut?”

“Ten percent. I do the digging, you do the extraction.”

She shook her head. “It’s more than just running it through. We pull almost double the gas, strip the junk, and bottle the ^3He pure in a cryocontainer. Less weight, better price.”

“Cryocontainer? You’ll lose a lot to boil-off. I use K-bottles. No boil-off.”

“I use a pulse tube cooler to boil off sacrificial ^4He . Using cryocontainers cuts my freight bill, and the mechanism prevents boil-off. I send back ten times as much ^3He per kilogram of freight. If I process your helium, you won’t see a freight bill. I’ll cover that.”

Sam paused, thinking about the numbers. “Freight’s killing me. Fifteen percent gone before I see a credit. At even half that yield boost, it’s worth it.”

“Run a test batch,” Li said. “If we don’t beat your numbers by fifty percent, it’s free. If you stick with us, I want twenty percent.”

He grinned. “If you can hit those numbers, twenties are cheap. I’ll go get ten tons of my regolith.”

“Bring twenty,” Darrin called out.

When Rook came back, they dropped his bags into the oven. Darrin eased the heat up, slow at first. At 700°C , the helium started to flow, along with everything else. The mass spec matched Sam’s best RF extractor runs. Rook still looked like he was doing the math in his head.

Then Darrin pushed it higher, up to $1,000^\circ\text{C}$. Pressure climbed. The display ticked past anything Rook had ever seen, one hundred eighty-two percent of his best yield.

He stayed through the rest of the run, watching the ^3He get distilled, weighed, and bottled in the cryocooler. Li sent the credit before the cooler was even sealed.

Sam stared at the number in his account. “I’m used to waiting weeks for a Tug to pick it up, take it to Earth, then wait for Helion Dynamics to pay out.”

“I don’t mind paying fast,” Li said. “It makes accounting easier. So, do we have a deal?”

Sam looked up. “You’ve got yourself a customer.”

A pair of miners Rook knew from east of Tycho asked him why he wasn’t selling helium anymore.

“You know, since I hooked up with Tan, I don’t need to sell helium. She pays me for it right away, and she ships it.”

“How much does she charge for that?”

“Twenty percent. That’s not much more than I was paying Lunar Tug.”

“Yeah, we are generally between ten and fifteen percent on shipping. But I wouldn’t give up that five percent.”

“That’s what I thought until I tried it. She nearly doubles the yield.”

“Fills the bottle with trash?”

“She does her own fractional distillation. You’ve seen her huge cryocontainers. Ultrapure ^3He .”

“Nice.”

“Better than nice. I shut down my RF extractor. Without that energy hog, I can sweep a lot more. I am pulling four times as much as I used to.”

“Sam, you have to introduce us to Tan.”

Soon Li was processing their regolith, too. She got two more customers at the next rendezvous, then ten more at the rendezvous after that.

The next rendezvous, Ocean State Resources came knocking. She charged them twenty-five percent, a small surcharge for their rocky start.

Darrin ran the intake. With several clients bringing bags of identical-looking regolith, he needed a way to keep the bags straight. He printed QR codes on the bags, associating each bag with a miner and an account. Every bag got scanned for

its QR code, weighed in, logged, and sent to the ovens. The return bags went out with a readout of ^3He recovered, and the money was deposited in their accounts.

The miners called it Tan's Refinery. They'd butchered her name. She liked it, in a way; it gave her just enough distance to keep people from thinking they really knew her, and the money kept flowing.

They made use of the other gases, too. The water they were recovering was small, but there was never enough water on the Moon. They started selling it at rendezvous.

Darrin ran a test on extractions. After the first pass through a pile of regolith, he pumped hydrogen back into the oven. When he brought the heat up again, the sweep pushed out another eight percent helium. After that, he made the hydrogen gas sweep part of every extraction.

At rendezvous, Rook got a large shipment of heavy aluminum tubes. Darrin was intrigued and asked about them.

"I hate hauling my dirt over that crater rim. I plan to make a cableway from the base of the wall on my side to the floor on your side. I have a heavy Dyneema cable and a low-power drum to keep it moving."

"That sounds fascinating. Can I help?"

They spent the entire lunar day getting the pylons anchored and the cable run pulled over. Darrin kept the drive drum on his side, so it was easier to power.

When it became lunar night, Rook held his breath and clipped a ten-ton bag onto the line. It sagged, but it didn't touch the ground, and the cable carried it away. Rook started loading more and more bags, and they all just ended up in a pile where Darrin wanted them. Rook came over to see how it was going.

When he arrived, Darrin was looking at the pile of bags. "I love this! I'm going to build this inside the crater, running in a loop. I won't need to carry bags on the crawler anymore; they'll carry themselves. Think you can extend yours? I'd like more people to use it."

Sam hesitated, one hand on the cable. "Most folks are happy enough not to have to climb the crater rim, but sure. I have time during the lunar night. I can probably extend it another twenty miles."

Other miners, who hadn't been using Tan's Refinery, saw the cableway and came to investigate. Soon, they were using it, too, and Tan had more customers.

A month later, Li saw a new series of towers stretching west. Ocean State Resources had copied Rook's cableway.

Darrin worked with an engineer from Ocean State Resources, and together they built a transfer station. The station would read a bag's QR code and automatically route it correctly. After that, Darrin could put empty bags back on the line, and they'd go to the right place. Ocean State Resources also agreed to extend their line another fifty miles west, drawing still more customers for Tan's Refinery.

Inside the crater, the cableway loop dumped processed regolith tailings against the northeast wall. The pale, inert waste piled up slowly, like industrial snow. Li had a depth probe with her to estimate the volume. She saw another miner in Ocean State gear poking at the same pile. His name tag said Tseng.

"I think I've seen you around here before," she said. "What are you doing?"

Tseng smiled and pulled a compact mass spec from his belt. He dropped a sample in and vaporized it with his survey laser. "Just seeing what it's made of," Tseng said. "I am a geochemist by training. Mining helium is more profitable, but less interesting. I sometimes wonder what metals are in this waste pile."

"Waste?" she said. "These tailings are my way of scoring how much ^3He we have extracted." Li made a note on her slate. Tseng loaded another sample.

Darrin still worked his pit when he could, but most of his time went to managing the reactor, keeping its heat curve steady and regulating intake. Regolith arrived constantly, bags flowing in from every miner within reach of the cableway. Li had stopped mining altogether. She was too busy.

New QR codes began appearing on bags from farther out. Some miners built their own cableways, connecting to the network that Rook and Ocean State Resources had started. Darrin built a third one going north towards Ball Crater. Li's days blurred together under the weight of constant movement.

One evening, she and Darrin were staring out across the crater. "I loved the peace we had out here," she said softly. "What happened to that?"

"Progress," he replied.

"Darrin, the regolith in this crater has maybe a fifth of the helium as most places on the Moon. We are making as much from our cut of Rook's helium as we are from processing our own."

“I expected that,” Darrin replied. “This is a young crater. It has only been accumulating solar wind for a hundred million years, not the billions of years it has accumulated on most of the Moon.”

“Well, we’re making more money than ever, but I feel like I’m always running. When I was mining, I had a place I needed to be. It was... relaxing, in its own way. I wish I had somewhere to slow down.”

“You want to buy a larger space? Shipping will be enormous, but you can probably afford it.”

“I’ll think about it. Right now, I need to sleep.”

Li was running intake when a miner came by the refinery the next day, dropping off five bags of regolith. “This is my last load. I’ve been here long enough. Time to get back to Earth and spend some of that helium money.”

Li loaded the first bag into an oven. “What are you doing with your gear?”

“I just left it there on my claim. I can’t bring it back to Earth.”

“Yeah, that would be crazy. But I’ll give you five grams for your hut. Another five for the rest of your gear, if you put it all on a cableway.”

“That’s ten grams for nothing. I’ll go get it now, while you are still processing my load.”

The hut arrived sealed and holding pressure. It was followed by his sweeper, batteries, panels, and an RF extractor. Li moved most of the gear to the eastern edge of the crater floor, but kept the hut near the prominence.

Within a week, a dozen more huts came in, clipped to the cableway. They looked odd among the ten-ton bags. Some were clean. Most were scuffed, dented, and patched. It didn’t matter. Darrin hauled them into place with the crawler, welding modules together and cutting through walls between them. Quonset huts were not the easiest shape to use, but over the next few weeks, the cluster began to take shape as a single structure, thirty by thirty, fifteen feet high.

Li and Darrin set up a large kitchen. A crew pulling out from Shackleton sold her a convection oven and some leftover food, including a dozen #10 cans of tomato paste. Darrin mounted speakers in the ceiling panels and hooked them into an old media server loaded with rock and Chinese pop. The floor was still bare metal, and the air always carried the scent of lithium grease. But it felt different. It was a place to sit. A place where you could let your mind wander.

Li watched the cable. A miner was attached. He had come in out of Lexell Crater. His claim lay in shadow now, the sun gone for two weeks, and his crawler was useless without solar power. He clipped his last load to the line, then clipped himself on behind it and followed it in. The QR code on his chest matched the code on his bags, so transfers were automatic.

The ride was silent except for the faint vibration of the cable through his harness. Below him, the Moon stretched away in black and silver, illuminated only by blue earthshine.

When he got to Tycho Crater, he was dazzled by all the lights. This was his first time seeing what was at the end of the cableway. He noticed the large structure they had built. He stood outside for a while, wondering what it was. One section was a mirrored glass window. He looked at himself, the QR code on his chest identifying his claim, dust packed into the seams of his gloves. The lights inside were warm in a way lunar light never was. Li saw him through the window and waved him in.

“There’s stew,” she said when he came in. “Real carrots. Whiskey too, if you’re paying.”

He sat down like he’d forgotten what a chair felt like. The warmth hit first, then the smell of broth and roast beef. He sat, ate everything in the bowl, and drank more than he planned to. He wept as he ate. When he was done eating, he looked out the Earth window, lost in his own thoughts.

The next day, three more miners showed up. They all came from claims where the sun had just dipped below the horizon. Each had dropped off a final bag and decided to follow it down the line, like the miner from Lexell crater.

She didn’t run it at a loss. Everything cost. A hundred milligrams ^3He for a hamburger, twice that for a beer. A shot of whiskey could be as much as a gram. The prices were high, but ^3He mining paid well. Once the sun set and the work stopped, the food drew them in.

She stayed open through the lunar night. The air recycler couldn’t quite erase the smell of lithium grease. Miners came in, lingered longer than they meant to, and left full.

Li liked serving the guests, but every hour in the cantina was an hour she wasn’t mining. That was the real problem. She put up a sign, “Cook wanted,” with no expectation anyone would answer. Mining paid too well.

One night, a man asked about it. He was making decent money on his claim, he said, but the solitude was grinding him down. Li offered five grams a day. He took it immediately.

With a full-time cook, the quality of the food improved, and Li got even busier serving. Another miner came in asking if she was still looking for a cook. His sweeper had broken. Regolith seeped into everything, and he was done with the constant maintenance. Li hired him as a bartender.

Li still filled in on occasion, but she had time to herself again. People loved the simple meals, warm drinks, and old music. Nobody rushed you out. That was enough. Miners called it Tan's. Li liked that. She hung a sign outside the airlock. "Tan's Cantina".

While they were getting ready for bed, Li talked about the hires. "The cantina gives me a place to think. That's great. But what really made it work was hiring some staff. Miners get better food and better service. I think we can hire more people to improve operations."

"We can post some job openings in the cantina. See if anyone is interested," Darrin said.

The first new hire was Pavel Kerensky. He was a naval reactor tech before he came to the Moon. He'd spent five years in nuclear subs and knew the Soviet reactor's internals better than anyone else on the Moon.

He didn't talk much, just circled the reactor pit with a dosimeter and a notebook, checking welds, testing the coolant flow, running diagnostics on the control systems. On the third day, he came back with a list of modifications.

"The control rods are sluggish. The Soviet spec called for replacement every eighteen months of operation. Belov put in fresh ones, but I want spares on hand."

"How many spares?" Li asked.

"Full set. Plus redundant coolant pumps." He looked up from the notes. "You want this thing running for years, yes?"

"Yes."

"Then we do it right."

Pavel continued, checking the hot loop with a Geiger counter. "Darrin, where is your polonium trap?"

Darrin was confused. “There is no polonium, Pavel. This is just lead and bismuth. Lunar background is higher than on Earth.”

“It was lead and bismuth. The reactor makes polonium.” Pavel ran the counter slowly over the pipe, noting the spikes. “We need an extractor.”

Pavel walked Darrin through the sparger design. “We inject helium here, where the metal comes out of the reactor.”

“How will we force the helium in without hot metal pouring out?”

“Sparge on the lee side of a wedge and the metal won’t come back into it. We only need to overcome the Laplace pressure. A standard air pump will have adequate pressure. Not much volume.”

He drew a diagram on his slate. “Inject here, at the bottom of the hot pipe, right where it comes out of the reactor. Three feet later, we collect the bubbles. Run them through a Graham condenser.”

“Like in chemistry lab?”

“Same thing, but made of tungsten to block radiation and cooled with ammonia. We make two cartridges, so we can swap them out. We will need someplace to dump the polonium, though. Far away and easy to get to.”

“How about the ridge near the solar panels? The cableway goes by it, so we can get there easily, and it is miles from anyone.”

“*Da*. That will work well. I’ll bury a heavy steel pipe there. We bring the condenser up, heat it electrically, and the polonium will drip into the pipe. Then we cap it.”

“Won’t it fill up?”

Pavel shrugged. “When it does, we can make another. The half-life is short. In five years, it is quite safe.”

Li noticed that Pavel was asking the questions they had never considered. She gave Pavel the budget and let him shut down the reactor while he tinkered. He ordered the parts through Lunar Tug, spent a month installing them, and another week running the reactor through every failure mode he could simulate. When he was done, he pronounced it safe.

“Safe enough for what?” Darrin had asked.

Pavel shrugged. "For us to sleep at night."

After that, the reactor became his. He ran it steadily, kept the temperature curves smooth, and made sure it didn't kill them.

Luke Miller was hanging around Tan's Cantina until Lunar Tug had space to bring him back to Earth. He saw Darrin eating and sat beside him. "I see you are landing regolith faster than you are processing it. What's your plan?"

"Eat fast, get back to processing."

Luke chuckled. "That's a good plan. I was noticing you winch the bags up before dumping them in the oven. I'm stuck here for a couple of days. Mind if I make a suggestion?"

"What do you mean?"

"I've been watching the cableway drop bags on the crater floor, and you climb down, attach them to a cable, and winch them up. I was an industrial process engineer. I'd set it up so the cableway drops the bags on a chute conveyor. If you have a wide impact bed, you can easily handle surges, and your operations will be smoother and faster."

"I don't have time to build that."

"You don't have time to do what you're doing now, either. Can I ask how hot the waste dump is?"

Darrin's chest puffed a little. "We're running them at over a thousand degrees."

"I thought so. You can alter the ovens for heat recuperation. Getting four times the bake with the same amount of thermal input would be easy."

"You want to do all that?"

"I thought you'd never ask."

With that crew in place, Li and Darrin had time again.

In the morning, the automatic lights in their hut hadn't come up yet. Earthshine spilled pale blue across the bed. The Earth hung unmoving outside the viewport. She lay still, listening to the hum of the air circulator and the slow, even breathing of Darrin beside her.

“You awake?” she murmured.

“Mmhmm,” came the answer, muffled by the pillow.

She stretched beneath the blanket, her muscles sore in a different way now. “We need more huts.”

Darrin rolled onto his back. “For workers?”

“Sure,” she said. “But that’s not what I meant. Miners come in during lunar night. They have a meal, maybe a drink. But when they are tired, they have to go back to their own hut in the cold, in the dark, to sleep. They spend their batteries. Then they wake up with nothing to do, so they come here again.”

“So you aren’t talking about a few huts.”

“No. Dozens of rooms.”

“A hotel?”

She swung her legs over the edge of the bunk, the floor cold under her feet. From the drawer, she pulled her undersuit, sliding it on with practiced ease. “You can call it that. Air, heat, water, and they would be close to the cantina. If that existed, miners might even work a little into the dark, because they wouldn’t be afraid of darksiding. I’d keep rent low, like half a gram per night.”

Darrin laughed. “Low.” Putting on his own undersuit, he continued. “You’re not wrong. Just sounds like a lot of work.”

“They’re coming here anyway. We should give them a bed and a door that locks.”

“You’re turning this into a town.”

“Not yet. But we’re past the camp stage.”

“Alright. I’ll lay it out. Water, power, oxygen. Just don’t ask me to handle linens.”

“You’re the best city planner I’ve got.”

Two days later, Darrin had a plan: a five-by-ten grid of used huts, with utilities, thermally anchored under regolith.

People started staying through the lunar night. Some stayed into the day. It wasn’t enough. By the next month, he was laying out fifty more.

The music system had burned out again, leaving only the low hum of the recycler and the occasional clink of a glass. Darrin sat in the corner, tracing a cableway route on his slate. A lot of miners had built their own lines from their claim to the main route, and he was trying to make sense of the tangle. The new links made people more social. The regolith still all went to Tan's Refinery, but the miners also visited their neighbors. Without plan or permission, the cableways had become the Moon's primary transit system.

Li sat with her boots off, legs stretched, staring through the viewport at the growing pile of processed regolith. Mounds of lifeless gray ash lay half-lit by Earthshine beyond the huts.

Tseng had taken to spending evenings here, claiming the background noise helped him think. He was relaxing in the cantina at the far table when Li noticed him.

"Remind me again," Li said without turning, "what you said was in the tailings piles."

"You have different minerals coming in," he said. "Ilmenite is particularly valuable, and you can pull it out magnetically."

"Why would we do that?"

He shrugged. "It's an iron-titanium oxide. We could, in theory, extract the titanium from it. That ore is over thirty percent titanium by weight, and valuable by metal standards. ^3He is measured in parts per billion. The titanium is actually worth more than the ^3He per ton of regolith."

Li took a sip of water and leaned back in her chair. "What would it take?"

Tseng set the slate down and leaned forward. "You serious?"

"Serious enough to listen. You said titanium. Helion Dynamics is paying very well for titanium right now."

He rubbed his jaw. "Alright. The first problem is heat. You'd want the FFC Cambridge process. It strips oxygen from ilmenite and leaves you with titanium and iron without using a lot of other chemicals or air."

"If you need heat, we've got a reactor."

"Sure, but it's already busy. You'd need to slow down other processing. Better yet, get a second core just for the furnace."

Li gave him a look. “The reactor’s barely running.”

He frowned. “You’re pulling at least two megawatts.”

“Electric,” she said. “Yeah. The generator gives us ten if we want it, but we’re pulling around two. Add another megawatt thermal for the ovens. But the reactor’s rated for a hundred fifty megawatts thermal. We’re barely tickling it. With Pavel on board, we can pull the full one fifty.”

That made him sit up. “A hundred fifty megawatts?”

“If we feed it right. You give me a reason, and Pavel will give you the power.”

He leaned back, already calculating. “That changes things.”

“How do we separate the titanium from the iron?”

“That’s trickier. You’d need carbon monoxide. The Moon is carbon-poor, but the gas gets recycled, so it’s not impossible. You can buy some from Earth, and we will just keep reusing that.”

“And then we’re sitting on titanium bars?”

“More like sponges. You’d still have to melt them. That’s phase two. Clean it, melt it, cast it, ship it.”

Li let the silence stretch. Outside the viewport, the slag pile lay pale in the Earthlight.

“We can send bars back on Lunar Tug,” she said.

Tseng shook his head. “No. That will never work. Helium works that way because it’s light and pricey. Metals, even at Raines’ rates, don’t fly well.” He paused, eyes narrowing. “But if you sling them with a mass driver, that changes things. Aim for the Pacific, and let them drop.”

She laughed. “You already picked a splash zone?”

“Not yet,” he said, lifting the slate. “But if this project is a go, I will work it out.”

“How big?”

He tapped a few numbers. “Thousand-ton slabs. Big enough to recover, small enough not to vaporize the ocean. Slow them with surface area and an ablative heat shield, keep them under Mach two at impact. Most of the heat shield burns off, and

the rest shatters on impact with the ocean. Make them bowl-shaped for stability and cast an eye into the back so a salvage crew can winch it out. It will land flat, and it won't bury itself in the mud."

"You've thought this through."

"It works," he said. "Raines will pay for titanium, even if it's salty. He'll fish it out without complaint."

Li brought up her slate and began calculating. "Tell me more about the mass driver."

"We'll need a precision instrument from Earth," Tseng said. "We can mount it on the ridge. Hitting Earth is easy. Getting it to slow down right, break into the atmosphere correctly, and land where it's supposed to? That takes real calibration. No shortcuts."

Li went to the window. The tailings piles caught the faint light, their long shadows stretching across the basin floor, marking her life, devoted to power. Tseng's idea would erase that pile.

"It's not just about helium anymore," she said.

Tseng looked up.

When she saw her reflection in the window, she thought about her family photo. She realized her parents were smiling. This was what goodbye felt like.

"Neither am I."

She turned toward the kitchen. The stew would still be warm, and someone would be hungry soon.

Chapter 7: Salty Metal

Price of ^3He	\$7,800,000/kg
Current Production	32 tons/month
Current Consumption	40 tons/year
Time Since Start	28 months

José Torres was glassing the sky at night when he saw the fireball. He cranked *Peces de la Luna* up to forty knots, watching it disappear beyond the horizon. Seven minutes later, a second fireball appeared.

He glanced at a picture of his kids above the wheel, riding their bicycles.

Peces de la Luna was designed for a specific task: ten rapid deep-sea recovery operations at a time. The hull design was borrowed from a high-speed Italian ferry, the MDV 3000 Jupiter, allowing it to reach forty knots when fully laden. The ship was painted bright yellow, with a dark blue racing stripe, the fastest ship in Lázaro Cárdenas. Tonight, it was prepped for eight recoveries.

A second fireball showed as the first one hit the water. The sound of the splashdown took eight minutes to arrive.

Lunar librations gave a one-hour launch window each month. Tseng had launched eight targets during the window, three days ago. As they arrived, the Earth's rotation was stretching those impacts into a thousand-mile line.

When the target struck the seabed, the thud played on the ships' loudspeakers. The acoustic analyst triangulated the signals, calculating precise coordinates. Precision didn't always mean accuracy. Five miles out, José called for the crew to shut off the engines. The sea around *Peces de la Luna* was suddenly still. He released a UUV, which swam a search spiral around the calculated impact site, finding the target less than a thousand yards away.

José ordered a recovery glider to be prepared. An air compressor kicked on, inflating six support pylons attached to the fantail. When they were filled, the stern crane lowered the glider down onto them. The ropes, hawsers, and the lift bag had all been packed in the tail with the recovery kit.

Once the glider was secured, a chute laid three hundred tons of construction rubble evenly across the ballast box. Liquid oxygen and liquid hydrogen, each in corrugated thin-walled cans, insulated by syntactic foam, were strapped in place, completing the recovery kit.

José held off on launch. It took ten minutes to get a clear image from the UUV over the acoustic modem. When he was satisfied that the target's shackle was undamaged, he signalled for the

drop. The pontoons deflated, and the glider slipped off sideways. In a few seconds, the glider disappeared from view. He dropped an acoustic transponder buoy at the site so he could continue to communicate with the UUV, and *Peces de la Luna* sprinted to the next destination, two and a half hours away, to do it all over again.

The glider dove fast. Every minute spent in the water meant more heat transfer into the cryoliquids and more boiling loss. As the pressure rose, the pressure relief bubbles became more of a shimmer than individual gas pockets. In thirty minutes, the glider flared and settled beside the target, seventeen thousand feet below the waves.

The hawser extending from the lift bag was fourteen inches in diameter. The UUV guided the latching end to the shackle and locked it in place.

Secured, the lifting bag was spread over the hydrogen tank, and the valve was cracked open, letting gas vent slowly into the bag. At this pressure, it was not so much a gas as a supercritical fluid, but it still rose in seawater. The seawater that came into contact with the hydrogen flash-froze and sank to the bottom of the bag. As the bag took shape, it centered itself over the bowl. The feed rate was increased, dumping tons of hydrogen into the bag. The UUV sprayed seawater into the bag at high pressure, warming the hydrogen. Snow fell softly through the supercritical hydrogen, collecting at the base of the bag. This continued until the hydrogen was warmed to ambient, around 3°C. The bag was forty percent full, pulling five hundred tons.

The UUV fed oxygen in through a catalytic combustor, heating the hydrogen from abyssal cold to 400°C. The heat expanded the hydrogen until the bag was filled, reaching eleven hundred tons of buoyancy. The thousand-ton bowl weighed less than eight hundred tons in seawater, but the lift wasn't enough. Nothing moved.

José relayed another command to the UUV, ineffectually trying to rock the bowl. He then had it circle the bowl, using its thrusters to blow the silt away.

As he sat watching the UUV work, he looked at the size of the bowl. "A billion pesos for just that one bowl, enough to keep the bank from seizing my company."

Forty minutes passed before the bowl shifted. The lift of the bag overcame the suction of the silt. A snap sounded over the ship's loudspeakers as the suction let go and the bowl lifted free. The crew on *Peces de la Luna* cheered.

The bowl rose quickly, spilling gas from the base of the lift bag as pressure decreased. The bowl was fluttering, jerking the hawser violently. José had the UUV pull the vent line, reefing the base of the Kapton gas bag. At five hundred feet, the vent was reefed further, slowing ascent to six inches per second.

When the bag breached, the bowl's momentum kept it rising, slackening the hawser. With no lifting force, gravity won. The bowl descended, snapping the hawser taut, pulling the bag back under. It continued bobbing dangerously for a minute before the bowl hung dead in the water. José made a note: "Next time, slow to four inches per second."

In two hours, *Pleamar* arrived, well deck already awash. Divers pulled a rubber donut under the bowl and inflated it until the rim of the bowl rose above the waterline. The gear was hauled up onto the dry half-deck, where it was washed and fogged with deionized water. The ropes, hawsers, and lift bag were all transferred into a freshwater soak tank with Salt-Away.

Pleamar's heavy three-hundred-ton crane could not lift the bowl, but it had enough authority to precisely position it over a pedestal. Once in place, the donut was vented until the bowl made solid contact with the wooden shoes on the pedestal.

Pleamar purged its tanks. When the cargo deck was dry, they retrieved the donut, washed it, and stowed it for the next recovery. The bowl was rinsed with soap and deionized water. Recovery had taken four hours from arrival to having the bowl strapped in place. *Pleamar* was again underway, headed to the second target.

After the final drop, José called Felipe. "Every bowl was intact. A couple of shackles were bent, but no real problems. How is the mill?"

"*Mano*, good to hear from you. I have the crew scrubbing the walls and ceilings now. This will be a surgical cleanroom when you get back. How far out are you?"

"We don't have enough fuel to sprint back. We dropped down to thirty knots. I should be back in two days."

"Should I deploy the grillage?"

"We can do that after I get back. The cargo won't arrive for another six days."

"We will scrub the dock down, anyway. Now, *carnal*, I have someone here who wants to talk to you."

Felipe handed the phone to Maria. "José, the bank called again. They are starting foreclosure proceedings."

"It's going to be OK, *vieja*. Once I sell this load, we'll have enough. They won't take our villa. Where are the kids?"

"I brought them here with me to help Felipe clean the factory. Mateo is scrubbing the lunchroom floor. He works very hard for a ten-year-old man."

"And Sofia?"

"Sofia is why Mateo needs to scrub the lunchroom floor."

Pleamar picked up the five nearest bowls over the next two days, and *Bajamar* got the three farthest ones. José stayed on the phone with the captains for every recovery. There was no time for more than short naps on the trip, and when he got back, there were more preparations.

Bajamar made it back to port first. José and Felipe had laid out the plywood and two-inch steel plate grillage before it arrived. When *Bajamar* backed into its berthing slot, they extended the grillage to cover the ramp and the ship's deck.

Then SPMTs came like an army of ants, lifting the bowls from their pedestals, carrying them slowly off the ship to the fabrication room.

The alpha case was milled off under argon. The clean bowls were melted under vacuum and poured into ten-ton molds. The outer three millimeters of each ingot were milled away, remelted, and cast into industrial titanium. Each bowl took thirty hours. When they were done, the aerospace-grade titanium was loaded onto different boxcars than the industrial titanium. Felipe swept up the spatter and alpha case swarf and sold it separately to ArcelorMittal.

José signed the shipping documents, and it was the CPKC Railroad's responsibility to deliver the titanium. One week until the next operation.

Maria put away the bills when she saw him coming in the door. "José, we haven't seen you in so long."

José opened his arms, hugging her and his kids all together. "Maria, I'll be home more this week. What's for dinner?"

He fell asleep during dinner. Maria and Mateo moved him to his bed, where he slept sixteen hours straight. When he woke, the kids were already at school, and his wife was getting groceries. He sat in his chair and drank a Coke, staring at the clouds.

The next day, *Pleamar* and *Bajamar* were preparing to leave. He helped scrub the decks. They each took four hundred tons of MGO. When he paid the fourteen-million-peso bill, he wished they ran on LNG. Or at least were painted more brightly.

Over the next few days, the gear was all inspected and prepped for the next operation. One of the hawsers showed internal fraying under X-ray and was discarded. Two lifting bags needed patching, and one of the gas bags tore and needed to be replaced. All the eye thimbles were hammered and had to be replaced. The marine supply store behind his mill had woven all his bags and ropes. They were happy to do the repairs and replace the parts.

The crew was loading up *Peces de la Luna* when José got a call from Elin. "You sent me seven thousand tons of aerospace-grade titanium and seven hundred tons of industrial-grade. I got a report back from metallurgy. The quality is outstanding. I can pay seventy dollars per kilogram for the aerospace-grade titanium, and thirty per kilogram for the rest."

José calculated his cut. Twenty per cent would give him \$1.8 billion pesos. Suddenly, his mill's two hundred million peso debt didn't seem so bad. "Yes, ma'am. Thank you, ma'am."

"Do you plan on sending us any more?"

"I'd like to. I am about to pick up some more."

"Look, I am behind my targets, and your titanium is really helping. You run a steel mill? Do you know how to fabricate high-purity titanium?"

"Sure," he lied.

"If you could send me twelve-gauge titanium sheet metal, I'd pay double. I can send you the specs. You would really be helping me out."

"Of course, *Senora*."

"Thank you, José."

"*De nada*."

José called Felipe. "I just got off the phone with Dr. Elin Sarai."

"From Biloxi? Did she buy it?"

"She loved it. The money is already in the company's account. She wants more."

"Well, she'll get more. You are shipping out tonight, right?"

"Yes, but she wants us to make sheet metal. She will pay double for twelve gauge. Can we do that?"

"I'm on it, *carnal*. You go get the next delivery."

José put to sea. He had twelve hours to the first deployment. He glassed the sky, but there was nothing yet.

He was in the mess hall when Li called. "I saw the deposit in my account. Four hundred million dollars."

"Yes. How is that?"

"That means my cut is fifty million per bowl. We just launched ten more."

"Tan, she said if I can roll it into sheets, she will pay double."

"Whatever you want to do is fine with me, but if you screw up the sheets, and Elin won't buy them, I still expect my five hundred million dollars."

"Yes, ma'am. I have been wondering, why me? I mean, there are much bigger names in metal and shipping and whatever you want to do. Acero del Pacífico is small fry."

"Your location is convenient for me. I could have asked ArcelorMittal to help instead, and I am sure they would have agreed. This would be one of their most important product lines. I would sit in the front row. That's nice. But with you, I don't sit in the front row. I sit alone. I told you I wanted you to have ships, a new dock, and cogeneration ready in a year. ArcelorMittal would agree, but there would be analyses and meetings, and two years later, they still wouldn't be done. I lent you the money; you did the impossible. I told you I wanted to get paid, regardless of any other problems. You said, 'Yes, ma'am.' I trust you. Not because I know you or you are trustworthy, but because you have no choice. That takes the stress out of it, for me."

"I understand. But Tan, you have lent me ten billion pesos to buy the ships and build out our dock. It is so much."

"The loan is real, and I expect to get paid back, but don't hurt your business to pay me back. When you are comfortable, let Elin give me part of your share. The first months are hard, but it will not be hard forever."

The run had ten targets. A couple of the shackles were bent, as before, but this time, one shackle sheared off completely, making it unrecoverable. José called on the way back. "Felipe, I am coming in. The third bowl was unrecoverable. The shackle was shorn clean off."

"Tan is going to take that out of our cut."

"Not if I recover it. I am not sure how. I need you to set me up with two packs and a bunch of extra hawsers. I don't have a real plan yet."

"If you are shipping out as soon as you get back, does that mean I will be handling the offloading and processing myself this time?"

"Are you OK with that, Mano? You only need to process *Bajamar's* load. *Pleamar* is keeping station now for the rescue."

"I will take care of it. I am sending the truck to Liquide Aire to get the oxygen and will have two hydrogen containers and all the gear ready when you arrive."

"Gracias, carnal. I am two days out."

Pleamar had held station for three days when José got back to the site of the unsalvaged bowl. The captain had already sent down two UUVs to search for the shackle and found nothing. He dropped a fresh UUV and the first glider.

When they got down, he had them wrap a long hawser around the bowl, then used a mechanical cinch to tighten it. When the lift bag started pulling, he saw the problem. The bowl had been settling since it hit the bottom. If he had taken another week, it would probably be covered in silt. The sticktion force was beyond anything he had anticipated. He kept the lift bag hot at eleven hundred tons of buoyancy while he sent the second glider.

This time, he used a net made of one-inch rope. Two UUVs pulled it under the bowl, moving through the silt. Progress was imperceptible. José had them saw it back and forth under the bowl. Progress was still slow, but he could see it, every stroke a few inches. Four hours later, the net was completely secured. José connected it to the second lift bag and put in a little hydrogen to give it tension. The net could only hold about two hundred tons safely.

The problem now was the first bag. It was losing lift fast. The first oxygen tank was empty, the hydrogen was cooling off, and slowly permeating the Kapton inner bag. The second glider still had cryogens, but the syntactic insulation was meant for a fifteen-minute dive, not four hours of sitting.

José put another five tons of hydrogen in the second bag, but didn't heat it. He moved what was left of the second hydrogen tank to the first bag and used the oxygen to heat that.

While the first bag warmed and expanded, the UUVs roped the two gliders and the cryotanks to the net so they wouldn't get lost. When the temperature rose to three hundred degrees, less than a thousand tons of lift, José thought he saw the bowl lifting. There was no pop. He asked the crew if they were seeing what he was seeing. As they tried to decide if it was moving at all, it sped up. There was no pop, but the bowl was rising.

The net had broken the sticktion. Pulling that through had given water a hundred channels to flood in behind the bowl as it was lifted from the mud. Everyone cheered. The Pleamar captain blew the foghorn. José loved that and blew *Peces de la Luna's* dual-tone foghorn.

Because of the noise, José didn't notice it at first. There was a metallic crashing sound as the bowl rose. The UUV showed the culprit; two gliders were smashing into each other and the cryotanks, now filled only with water, were beating each other like drums.

He reefed the hem of the first bag, but by now, the hydrogen in the second bag had warmed to ambient, and as the pressure dropped, it expanded. He heard a loud pop. A UUV flew under, and he could see that a link in the net had broken. He was past the safety margins. He gave the command for the UUV to reef the second bag, but as it started going up, a couple more segments popped, then it was like tearing fabric. The second bag was now lifting a torn and empty net and the gear. It shot up like a rocket, hitting the first lifting bag, forcing it to burp its lifting gas. One of the gliders hooked on the first bag, dragging it up a ways before letting go. The equipment on the second net was going far too fast to be stable. The tanks and gliders were kiting back and forth, smashing into each other.

It hurt to watch his equipment smash itself to pieces, but it could all be replaced for a million pesos. The bowl was worth nearly a billion pesos. He tried to keep it in perspective.

The screen had the rate of rise. The bowl had good momentum, and the lift bag was still pulling, but the numbers were slowing down. Twenty meters per second dropped to fifteen, dropped to ten. If it stopped rising, there was nothing he could do for it. As long as it continued to rise, the gas was expanding, increasing the lift. He was helpless and didn't know what speed would show failure.

The second bag breached. His crew pulled it and the smashed gear up on *Peces de la Luna*'s deck. José kept watching the rise rate, which had dropped to less than a meter per second. He remembered he would slow them down more than that when they breached. If the issue was buoyancy, he was fine. If it was momentum...

At seven thousand feet, the number went back up to a meter per second. José let his breath out. He realized he hadn't breathed for over a minute.

The rise continued and accelerated, but José kept it reefed tight. No extra strain on the gear. *Peces de la Luna* and *Pleamar* both waited, giving a half-mile clearance for the assembly to surface. This time, *Peces de la Luna* took the lift bag and hawsers. When the bowl was secured on *Pleamar*, the teams got together for a moment of celebration before going home.

José called Felipe on the way back. "We got the bowl."

"Fantastic. How did it go?"

"Smooth and easy. Sort of."

"José, you need to talk to Tseng. He can fix this problem."

"It wasn't a problem. I just had to go out again."

"Carnal, this was a problem. Talk to them."

José called Li, Darrin, and Tseng. "On the first drop, there were bent shackles, but I didn't worry."

"How badly?" Tseng asked.

"I didn't see it as a problem. We were able to recover them. But on this run, one of the shackles was shorn off completely, probably on a sideways splashdown. We never found the shackle."

"You lost one of my bowls?" Li asked.

"No, ma'am. We recovered it. It took an extra week, but it is on a pedestal now, heading back to port."

"Then what's the problem?"

"Well, Felipe has suggested it before, and I agree, we need something sturdier."

Darrin was drawing on his slate. "I've seen similar problems in the North Sea. A shackle works well in port, but at sea, a crossed shackle is much stronger."

Tseng agreed. "Of course. Next shipment, cross shackles, and heavier bars."

“*Gracias*. Tseng, if you can send Felipe the geometry, he is going to make specialized castings for the connector hardware for the crossed shackle. My current hardware is too weak, anyway.”

When José returned, Felipe had already processed the five bowls from Bajamar. They had a moment to talk.

“José, the sheet metal is not so simple. I have ordered the equipment we will need, but it isn’t only the equipment. We need to hire more staff.”

“Of course.”

“Yes, well, the men we need are specialists. Harder to place. And it is not just the staff, either. This has to be done in a separate building. That is the only way to deliver high-purity titanium sheets. I made an offer on the marine supply shop behind us. I think they will take it. Then I will need to build a new building and extend a connection to it.”

“I like the maritime supply shop. They make our gear.”

“They still will, but from a new storefront.”

“I wanted to make sheet metal this month, or worst case, next month. Now it sounds like it may take six months.”

“No, carnal. I will not leave you waiting for six months. But you need to know the constraints.”

“*Gracias*.”

“How is Maria?”

“She has bills to pay, and nothing to pay them with. And I am gone more than ever.”

“It’s not easy.”

“How is Lupita?”

“She is doing well. Speaking of Lupita, I need you to come with me for dinner at her father’s house.”

“Of course. Should I bring Maria?”

“Yes, and your children.”

“And my children? Is there a special occasion?”

“*La Petición de Mano*.”

“*¡Felicidades, carnal! Ya era hora, ¿no?*”

“Sí, I should have done this a year ago. But, *mano*, I need to show you something.”

They walked back to the main office.

“When we processed the titanium last month, we got a pile of alpha case millings and copper-contaminated swarf.”

“We were pretty efficient. How much swarf was there?”

“Less than half a percent, but that was still thirty tons.”

“What did you do?”

“I talked to a buyer at ArcelorMittal. They wanted titanium since Elin has been buying it all up. They gave me a finder’s fee off the books.”

“How much are we talking about?”

“See those two duffel bags?”

José opened one, finding mostly five-hundred and two-hundred peso notes. “Well done, Carnal.”

“Bring one to Maria and see if it brightens her day. For me, I think mine is enough to start a family.”

José ate dinner at home that night. “Maria, work is not going to get easier for me. But last month’s load sold, and Acero del Pacífico is now solvent.”

Maria looked up at him in tears. “José, they set a date. We have to pay in a week, or they will force us out.”

José dragged out the duffel bag and set it beside his wife. “This is for you. It is enough to keep the villa.”

Maria opened the bag of cash. She cried and hugged him. “We still miss you.”

José fell asleep watching TV.

When he came into the office the next day, his secretary updated him. “José, you have a visitor.”

“It seems I always do. Can I meet him after lunch?”

“No, not this visitor. I sent him into your office. He has been there half an hour already. You’ll want to see him.”

“Marcia, we’ve discussed this. Do not let visitors into my office.”

“I think this was the right time to make an exception.”

José walked into his office. A tall, blond man was staring out the window overlooking the Acero del Pacífico docks. José cleared his throat. "*Buenos días, Señor.*"

The man turned to face him. Tobias Raines. José dropped his coffee.

"I was just looking at your dock here. You import steel from Asia?"

"Yes, that is my business. And some small fabrication jobs."

"I see. I talked to Dr. Sarai last night. She mentioned you had just sent her over fifteen thousand tons of titanium. Nice job for a steel importer."

"Yes. Thank you, sir. Elin has been very generous. I hope to do more for her."

"I hope you do, but I am curious. Where is the titanium coming from? I am looking at your fleet. You are doing marine salvage?"

"Yes. Salvage. I find the titanium on the ocean floor."

Raines looked away. "I notice most of the money goes to an offshore account, and the rest goes to Acero del Pacífico's Banorte account."

"Yes, I only get a portion."

"That offshore account, I know it. I buy ^3He from that account. The most prolific miner on the Moon. But I don't know his name."

"It's not a man, it's a woman. Tan. She said she would drop high-purity titanium on the ocean floor. I pick it up. She gets eighty percent of the ingot fair value, I cover all expenses and losses, and keep the rest."

José went over the details, Tseng's bowl designs, and the salvage process. He took Raines on a tour of *Peces de la Luna*.

"Why did you use LNG?"

"It is six hundred tons lighter than MGO, which helps. It cuts maintenance costs on my engines. But, mostly, the fuel costs half as much. This is a steel mill. I already have gas mains. That is also why I have a cogen plant and a hydrogen reformer."

"Rubble ballast?"

"I need three hundred tons of ballast per recovery. I wanted something inexpensive. I have a barge collecting construction rubble. People pay me to take it."

"Doesn't it wreck your hull?"

"I deadbed with silt from channel dredges. It's worked out well. I rake it when I get back to port."

"You've built a very nice system. I have a question for you. What would you think of expanding your recovery operation?"

"Well, sir, *Peces de la Luna* is built for ten recoveries. I could go out more often, but Tseng only launches once per month."

"What if it were someone other than Tseng launching, and the metals were something other than titanium?"

"I am completely ready for any metal, if they follow Tseng's design."

Raines knew he wasn't.

"I'm not ready to ship now. I won't be for a couple of years, at least. But when I am, we can do good business, if you don't mind expanding."

"Yes, sir. If I need to expand my fleet, can you give me notice with a contract I can show a bank?"

"We can do that. We can do much better than that."

They shook hands.

Chapter 8: Ceres

Price of ^3He	\$6,000,000/kg
Current Production	95 tons/month
Current Consumption	91 ton/year
Time Since Start	37 months to 60 months

Raines sat alone in the narrow conference room, eyes fixed on the empty chair across from him. The air was too cold. His tea cooled, untouched, beside a decision he did not want to make. He disliked what he was about to do, but the list of palatable options was gone.

He faced a wall of demand that was both exponential and geopolitical. Every nation wanted a terawatt-class tokaplex, and they wanted it now. A terawatt meant energy sovereignty, freedom from oil and coal, and independence in every negotiation.

Between that ambition and reality stood refractory metals. Ten-gigawatt plants were feasible, but costly; a terawatt-class core would consume more strategic metals than Earth could produce. The fusion age demanded mass.

Doubling the spot price had increased mining. Production had risen an incredible twenty percent. But a twenty percent increase was never going to build terawatt tokaplexes. Raines had not taken the meeting with Chen Zhengrong lightly. He needed a tenfold increase, and he was done asking.

The Asteroid Belt held the minerals, but it was untested, hazardous, and expensive. Raines saw no alternative. Helion Dynamics would build its own supply chain of ore pulled from the asteroids, refined, and sent Earthward. A heavy-metal pipeline from the Asteroid Belt to the Pacific.

Hundreds might die before the first shipment reached Earth. That did not change his decision. Success would depend on the director.

Catherine Dahl had run hazardous mining projects: Antarctic mining under miles of ice, deep-sea mining at enormous pressures, and magmatic brine extraction from the roots of active volcanoes. She had always delivered the ore, even when the loss of life was catastrophic.

Ceres would be worse. The odds of the first crew returning whole were slim to none. They might all die. He needed someone who would do it again.

There were many people whom he might ask to run this project. Many would refuse, squeamish at the thought of blood on their hands. Others would paper over the problems with endless procedures. Then, when something inevitably did go wrong, there would be investigations, recriminations, and resignations. A stalled project.

Dahl was different; she understood projects where miners die. She acknowledged the risk to investors and to the miners upfront, then delivered anyway. Not from indifference, but from clarity.

Raines didn't need sentimentality; he needed metal. Dahl could deliver.

The door opened without a knock.

Catherine Dahl stepped in, unhurried. No slate, no assistant. Just a canvas jacket worn smooth at the elbows and a look that said this was already taking too long. She scanned the room, gray eyes fixed on Raines.

"Catherine Dahl! It's a pleasure to meet you. I trust your flight was pleasant."

"What do you want?"

"Helion Dynamics is in a bind. We need refractory metals in enormous quantities. The Earth can't supply us. We are going to the Asteroid Belt."

Dahl pulled out a chair and sat.

"Ceres," Raines continued. "Not a pilot project. Our supply chain would depend on steady, massive output from this operation. Mine the Asteroid Belt, smelt it on Ceres, ship it to Earth."

She studied him, measuring the gap between words and meaning, then leaned back. "I've never mined in space," she said dismissively.

"No one has," Raines responded. "Who do you think would do a better job running this operation?"

"It's not just technical," she said. "Ceres will put crews under pressure, long cycles in low light, and almost no gravity. The kind of work where people make mistakes. Keeping them focused and productive will be the real job."

"I have to have this metal."

She held his gaze, then gave a single nod. Not acceptance, but willing to talk.

“Finding people for technical jobs on Earth will be straightforward. Hiring people to go willingly to Ceres is different. Were I to be involved, I would be honest about the risk upfront. The hires would need to be men who won’t fold when things go sideways. You cannot send men who can’t do the job. The training must be as brutal as the work, or they will break when they get on site. I’ve seen it happen.”

“I am not looking for a butcher. I expect the miners to be treated with respect and integrity. While that won’t make this operation safe, we can work out hazard pay and insurance policies honestly.”

Dahl watched the Jet d’Eau through his window. “I can get you your metal,” she said. “I expect to be fairly compensated.”

“Of course. Helion Dynamics will compensate you more than fairly. Bonuses for meeting early delivery goals will ensure our motivations align.”

“Yes, but there will come a time when I will take the fall.”

“Helion Dynamics would never do that, Catherine.”

“I appreciate what you are saying, Raines, but this is not about you.”

“What is it, then?”

“You are willing to accept harsh conditions and a high death toll. As we become successful, society will accept neither, and you will no longer be in control of the story. This is not about you, it is about Ceres.”

“What do you need?”

“A substantial stake in Helion Dynamics. Five percent.”

“Elin’s share is five percent.”

“Elin won’t be scapegoated.”

“You are asking too much. So am I. I will bring it to the board, but they will not approve this lightly.”

“Then make them understand the importance of this project. At that point, they will have no choice. When do we get started?”

“Here, now. This is not some far-off goal. I need metal yesterday. I have already lined up two General Atomics ships for this project. You can do whatever you want,

but cargo launches in two months. Crew follows fifteen months later on a faster ship.”

Dahl did the math. “Power, life support, base construction, full mining and refining. All in orbit in two months?”

“You figure out what you need. But I expect metal flowing six months after the crew lands.”

She gave a low whistle. “That’s aggressive, Tobias. It will take everything lining up.”

“That is the requirement.”

Dahl looked in his eyes to see if this was theatrics. “OK, we will do it. The first interplanetary operation.” She was in.

“Who’s on the ground on Ceres?”

“No one yet.”

“You’re not sending me?”

“No. You stay Earthside. But you choose who goes. Your team, your rules.”

She closed her eyes for a moment, letting the scope of the job take shape in her mind. She pictured the crews, the equipment, the operation in motion. Then she said, “Dr. Silas Knox.”

Raines leaned back, skeptical. “Knox? He’s radioactive. Half the ports on Earth have his face on their banned miners list.”

“I know. That’s why he’s still sharp. No safety net. He can’t afford to be mediocre. And his brilliance makes him worth the risk.”

She slid a slate across the desk. Knox’s record was there in black and white. Brilliant work, often illegal.

“Knox doesn’t get second chances. He can’t waste a shot, can’t afford to screw up and wait for someone to catch him.” She leaned forward slightly. “He’s exactly as good as he has to be, which is better than anyone else.”

Raines studied her, then the file. “You sure he won’t bolt when things turn bad?”

Dahl’s mouth tilted toward a smile. “Out there, there’s nowhere to run.”

“It’s your team, Catherine. Your call.”

Knox had been Scripps Mining’s golden boy, a rising star in the cutthroat world of deep-sea mining. He didn’t just survive it; he thrived. He pulled off impossible recoveries, turned marginal claims into record-breaking yields, and built fortunes for his employers and himself.

For a time, he lived like a king: a mansion in San Diego, cars bought on impulse, parties where champagne flowed from taps. But it was never about the money. It was the sense that he could beat the ocean itself.

The ocean doesn’t play fair.

One bad payout, a mine collapse, furious investors, and panicked insurers left him over-leveraged. The mansion and the lifestyle became a noose.

He panicked. Took a quiet side job. Helped a shady outfit steal ore from a competitor’s reserve, replacing it with sea rock convincing enough to pass. No one saw it happen. It was perfect.

Until it wasn’t. A man he’d worked with was arrested on an unrelated drug charge and, in a plea deal, he named Knox.

Knox did time. Lost his license. Lost everything except his ability.

After that, he worked where no one asked for papers, the gray zones where contracts were enforced with guns, and betrayal was just another line item. The jobs were brilliant. They were also illegal. Eventually, he stopped caring.

Two days later, Dahl found him in a half-lit cargo bay, elbow-deep in a crate of stolen spectrometry modules. The ship belonged to black market ocean miners, rough, half-legal, trawling the Atlantic floor for minerals no one could prove they had stolen.

“Dr. Silas Knox?”

He didn’t look up. “Depends who’s asking.”

“Catherine Dahl. I’m building a mining operation. Legitimate work. You would run site ops.”

That earned a flicker of interest. His eyes flicked to the shadows, still wary. “What kind of mining?”

She stepped closer, boots ringing on steel. “Ceres.”

Knox snorted, dry as cracked concrete. He straightened, wiped his hands on a rag black with grease.

“You serious?”

“Dead.”

He studied her, eyes sharp and cold, like a man measuring exits with nowhere left to go.

“You think I’ll live long enough to deliver metal?”

“You might.” She didn’t flinch when she said it.

The silence stretched. Dahl didn’t press.

Knox sealed the last crate, slow and methodical. When he straightened, his eyes were on the floor, as if the decision had been made before she even arrived.

“I got into ocean mining for the freedom,” he said. “You dropped off the map, ran your own operation, answered to no one but the weather.”

He glanced at the rust-streaked walls, the flickering lights, the smell of salt, oil, and rot.

“That’s gone. Now it’s just jurisdictions and contracts and lawyers with maps. You try to cut a trench, and five governments file injunctions before your rig hits mud.”

He met her gaze, and for a moment, the cynicism cracked.

“I used to build things that weren’t supposed to work. Projects that should’ve failed in weeks, yet I made them hold.” He swallowed. “I don’t do that anymore.”

Dahl said nothing.

“I thought maybe I’d crawl my way back. Do enough gray work, stay off the grid, wait for someone to forget.” He shook his head. “Earth doesn’t forget. Every mistake’s on record, every port scans your name, and the only thing they want is you gone.”

He looked toward the open hatch. Outside, the hull creaked with the tide. A cold wind cut across the dock.

“Ceres is a rock. No borders, no laws, no one telling you what can’t be done. It’s not clean or safe, but it’s empty. And empty’s the best deal I’ve had in a long time.”

He turned back to her. “I’m not doing this to run. I’m doing it because I remember who I was before I had to run.”

He stepped closer, boots ringing on the deck. “You’re offering me a hostile place with no rules and no margin for error.” His voice was steady. “That’s where I started. Maybe it’s where I belong.”

Dahl held his gaze. “When can you be ready?”

He stood up and walked out of the cargo bay without looking back. “I already am.”

Dahl built the home team. None of them would ever set foot on Ceres. Their job was to make sure everything that did go worked. She brought in Dr. Feld, a geologist with a long history in remote analysis. He would be searching the Asteroid Belt with a space telescope, cross-referencing spectral returns with mineral data. When samples came back from Ceres, he would read their structure, grade their potential, and flag new targets. He would be in charge of choosing which asteroids were mined.

Dahl’s system engineers went over the equipment lists. Make sure the team survives. Her chemical engineers focused on extraction. What to refine, how to do it, and what form the metal should take to survive reentry and recovery. If the chemistry didn’t work on Ceres, there would be no way to fix it, no way to refine that metal.

The General Atomics spine ship: a long, skeletal truss with a nuclear-electric drive at one end, was already parked in high Earth orbit, its frame slowly filling with extractor ships, tenders, folding habitats, smelter kits, environmental modules, anything bulky or heavy they would need. It looked less like a spacecraft than a junk cart piled high. Once the ship left, there was no means of correction. That was final.

Food was the last thing anyone wanted to talk about. It came from algae and bacteria, grown in sewage and nutrient broth. The tanks would convert it into a slurry of macros, micros, and amino profiles. Balanced. Efficient. Knox sampled a swig during prep. It hit the back of his throat like rot and copper. He gagged. Swallowed. Didn’t complain. Told Dahl it was an acquired taste. Then checked the manifest for a redundant water processor.

Knox didn't involve himself in the logistics. He skimmed manifests, asked a few questions, and turned back to what mattered, the crew.

While Dahl engineered the infrastructure from Earth, Knox focused on the people who would inhabit Ceres. His job was to hire the people who would live and die on Ceres. Miners, riggers, smelter operators. The crews who would drill into rock, refine metal, cast it into reentry shapes, and fire it toward Earth. They answered to Knox and Knox alone.

Dahl wanted to start with fifty men on Ceres. Men who would not flinch. Skill was optional. Desperation was not. She told him to recruit two hundred.

"Do you want two hundred or fifty?"

"Fifty, plus a few that may go later. But I expect most candidates to wash out. If they are going to be a liability, I don't want you coddling them. I want them out. If they get to Ceres and can't handle it, everyone's lives will be at risk."

"What happens to the washouts?"

"That's not our concern."

"It is mine. I will know these people."

"Be careful, Silas. Don't get too close."

He didn't like it, but couldn't fault her logic. He began recruiting. First, he talked to miners who had started as children and were now trapped in the most hazardous corners of the trade. They had lived with danger for so long that they treated it as routine. They had watched friends die and developed a practical instinct for when a job was drifting off script. They were quick on their feet. Ceres would be harsher than anything they had faced, but the pay exceeded what they were used to.

He next recruited from deep-sea mining crews. They worked in conditions that punished small mistakes. They understood pressure, machinery, and the routines that kept both from turning lethal. Knox had worked with some of these men, and they respected him enough to consider the offer seriously. Only a few were willing to sign on. They were not as desperate as the others and understood the risk. Knox was satisfied. The few he got would be enough.

He couldn't promise them safety. Only work, and a paycheck if they lasted long enough to collect it. Their estates would be paid if they didn't.

Dr. Knox finished selecting two hundred candidates three months after the cargo ship departed. In San Diego, they boarded two C-130J-30s. One held all their equipment, gear, luggage, and Catherine Dahl with a dozen of her technical staff. The other plane, the sardine can, was packed with two hundred candidates, facing the rear cargo ramp. Dr. Knox acted as their flight attendant.

“Buckle up, this is going to be a long flight,” Knox said over the rising whine of the four turbines. “If anyone needs to use the bathroom or get nauseous, raise your hand and a plastic cup with a lid will be passed to you. When you are done, pass that back. I will empty it and rinse it in the washroom. There will be no food or beverages on this flight. You’ll thank me for that.”

They stopped at Dutch Harbor to refuel. As they descended, Knox noticed one of the fishing warehouses seemed to have a Helion Dynamics logo. He could only wonder why.

The stop was worse than the flight. The pilot parked on an incline, with the front of the plane being two feet higher than the back. A firehose was brought to the front door, and the rear cargo bay was opened. The cold air hit like a blizzard, turning the interior to fog.

“Everyone, feet up!” Knox barked.

A few seconds later, seawater came out of the hose, washing the floor and pouring out the cargo ramp. When Knox was satisfied that the water was running clean, the hose was pulled, and the hold hissed shut. The temperature started to rise, as did the humidity and the marine stench.

Knox reappeared with a long pizza peel. “One cup per person. Drink it down, throw the cups forward.” He set the water on the pizza peel, eight-ounce cups, twenty at a time, reaching them out to the candidates.

When everyone was done, they took off for Dahl’s training camp in the Gobi Desert. The plane only had enough fuel to get there. An hour past the Kuril Islands, Knox spoke up. “Brace yourselves. We are going into a toboggan maneuver.”

Their stomachs felt it as the plane pitched down, losing five hundred feet per minute. The plane sped up as it descended, pacing the KC-130 tanker. Sixty thousand pounds of fuel were transferred in about twenty minutes. They disengaged, and the pilot struggled to level the flight, then began a slow ascent.

Six hours later, they saw the airport, a hardened, dirt track two miles long. They arrived on April first, during the Karaburan. The winds were over forty miles per

hour, and the air was dark with dust. The sun was a pale disk. To the candidates, it looked like heaven.

The other plane was already in a hangar and had been unloaded. They towed the sardine can into the hangar beside that, and everyone offboarded and scrambled to get their gear. They picked up their duffels, put on their coats, and headed to the barracks, a low, circular building made of rammed earth.

The training hall had once been a warehouse. Fluorescent tubes buzzed overhead. Rows of folding chairs faced a cargo net.

Day one, Knox called them in for orientation. “I am looking for fifty people to go to Ceres, with another twenty in reserve. A couple of you may become trainers. For the rest of you, thank you for coming out. While you are here, you will become an expert with every piece of gear we will use on Ceres. If you cannot master it, a truck will take you back to town, and Helion Dynamics will fly you back to San Diego. Four of your classmates did not enjoy the flight out here and have already departed.”

He made a slow circuit while the trainees waited. Each had a survey laser in front of them, a foot long and an inch and a half in diameter, oversized for gloved use. Beside it sat a rock, a dummy mass spectrometer, and stainless steel cams they’d use to stay anchored.

When he had finished the circuit, Knox stood on a half-buried 55-gallon drum. He spoke calmly. “The rocks in front of you are hematite iron ore,” he said. “Dense, but the rocks on the Asteroid Belt will be even denser.”

“The extractor ship lands first on three legs. It drills in slowly. Anchors screw in, locking it to the asteroid. That is your home. You are always connected to the extractor.”

He gestured to the gear on the tables.

“Once the extractor’s anchored and your tethers are clipped, the hatch opens. Gravity’s point-zero-two. You’re not walking; you’re crawling. Every motion’s a maneuver. Every ten meters, you hit the rock with the survey laser, open a crack, and jam in a cam. Ours are stainless steel for durability. The extra weight won’t matter at the G forces you’ll face.”

He stepped down from the drum and moved between the rows.

You don't trust your boots. You don't trust handholds. You trust the line. If it's not clipped, you don't move."

He paused.

"You fall off, gravity won't bring you back. You'll reel yourself in slowly. Stay calm. Pull gently. If you come off the tether, the extractor's bolted in. It's not coming to get you. Maybe another miner can, but don't bet on it. Their hold is as tenuous as yours."

A few chuckles. Knox didn't smile.

Off to the side, Dr. Feld watched the projection. Then he looked down at the rocks.

"Go ahead," Knox said, waving to the trainees. "Use the survey laser on your rock. Break off a chip." He demonstrated, steady and simple.

The class leaned in. Gloves on. Fingers careful. A soft snap-crack echoed through the hall as laser pulses sheared off thumb-sized fragments from the training rocks.

"Drop the chip in the spectrometer," Knox continued. "Lid closed. Ready."

There were a few fumbles. One trainee dropped his sample. Another held the laser backward. Knox ignored it.

Feld stepped up. "A mass spectroscope requires hard vacuum. Out there, you will have it. That's why these little bottle-sized devices work. Simple time-of-flight measurements. When you put your sample on the sample plate, hit it with the laser. Full strength, but millisecond duration. You will vaporize a point. The spectrometer reads the cloud and tells us what minerals it sees. The mass specs you have here are dummies, since we have no vacuum."

Knox was walking between rows, observing.

Feld continued. "You'll get a green or red ping. Green means it read. The ship relays that data to Ceres. Someone there will make the call or send the data to Earth. "Sometimes it'll take longer. Especially early on. Engineering will model the sample against the recovery cost. That's when we decide whether to move the rock. As we gain familiarity, both on Ceres and on Earth, the response times will improve."

Silence. The class looked down at their equipment. The task was simple.

Feld continued. "Once the ore gets back to Ceres, the samples go through another round of tests. The planetside mass spectrometer is much more informative. We also run other assays, chemical, density, hardness, magnetic, the works."

Feld folded his arms.

Knox let them fumble for the first ten minutes, answering questions, demonstrating technique. But then he started tapping them on the shoulder, directing them to go talk to Catherine. At the end of the exercise, there were eight fewer candidates.

Feld had worked out the doctrine of asteroid mining. Lunar mining taught them how to deal with low gravity and fine regolith, but asteroids were different beasts, tumbling, porous, occasionally hollow. Tools had to change. Cutters needed adaptive bracing systems. Drills required multi-point stabilization. Every rig needed to lock itself to rock or drift uselessly into space. Explosives were out. Precision pulsed cutters and hydraulic crackers were in. The crew practiced with each machine. "99% of this is simple, routine operations," he stressed. "But if you can't do the other 1%, you won't last."

"When you're working the dig, you'll load ore into these," he pointed at a ten-ton polymer bag. "Cinch it closed. Haul it back to the extractor ship. On Earth, it's too heavy to move, but on an asteroid, it will be light. At the ship, start loading up one of the tumblers," he gestured to truncated steel cones that looked like paper cups scaled up to handle hundreds of tons. "Two tumblers screw together at the wide end, containing the load. Then the extractor's mass driver launches the pair back to Ceres. You're not firing it at Ceres, you're giving it a shove. From there it coasts, maybe for weeks." The projector behind him showed a simulation of the tumbler pair arcing away from an extractor in slow motion, curling toward a pale gray speck labeled CERES.

A hand went up. Young guy. Tattoo on his neck. "What if it misses?"

Knox didn't blink. "They will always miss, but not by much. Ceres will send up tenders to make the catch. But if you are mining, don't worry. The extractor will do all the calculations and launch the load itself. You just fill bags and pack them in the tumblers."

Dahl watched from the side, clipboard in hand. She watched how Knox was doing the presentation. He had a way of making bad news sound like instructions. Every night, she got the staff together to go over the day, making sure that the next time, when Knox was gone, they would be able to train new recruits.

The candidates drilled hard nonstop. They got in and out of mock-up extractor ships in suits. They trained with survey lasers, drills, cutters, loaders, and everything they'd use in the Belt. Every move was deliberate. Just like it would be in space.

Two months in, the program switched up. Knox brought them to a mockup of the Ceres receiving station. There, they found tumbler pairs filled with ore in a designated impact zone, transport sleds, and ore processing, smelting, and casting equipment.

"Ceres has low gravity. That helps. Tumblers land softly. We haul them to the plant for comminution. First pass is a jaw crusher, then a burr grinder. It turns rock into coarse sand."

Someone in the back asked, "Why not just skip that and smelt it whole?"

"You burn energy doing that," Knox said. "Smelt garbage, waste heat. That reduces throughput. Beneficiation saves time and energy. Basic gravity separation with vibration tables. Light stuff floats up: Those are the tails. We keep the heads, the dense fraction. That's the value."

"What happens to the light stuff?"

"We use some of it to make the ablative shields for the metal we are sending to Earth. Most of it will go to tailings piles."

He turned back to the group.

"Today's smelting practice isn't a game," Knox said. "You'll haul tumblers on sleds, crush ore, sort by density, then smelt. Each step can kill you if you're careless."

His tone shifted.

"Molten metal runs at thousands of degrees. One slip, and you're cooked. Respect the process. No shortcuts."

The crusher was loud. The team was focused. Two hours in, someone screamed.

A trainee had leaned into the burr grinder to clear a jam. Just a bit of rock caught on the lip. He reached under the guard plate. The machine caught the fabric of his shirt, then the arm, then everything else. It was fast, just seconds. No one could have stopped it.

Knox stood beside the blood-slicked housing. He didn't look away.

“It’s not fair. It’s not just. Any one of us could have made that mistake. He should’ve gotten a second chance. None of you will,” he said. “Every machine out there is stronger than you, and it doesn’t care. Respect it, or it kills you.”

Feld threw up. He stayed in his room the rest of the day.

Catherine got a few people on her crew to clean up the grinder. The drills resumed an hour later in silence. Seven more men were heading to town.

At the end of the day, Knox lifted a 100-pound casting from the mold.

“Small batch here,” he said. “On Ceres, it’ll be thousand-ton ingots. Mistakes don’t just kill people. They stop production. Remember, production is why we are there. This isn’t safe work. It’s not meant to be. But done right, it’s productive. And that’s the difference between a graveyard and an operation.”

That night, Knox found Feld in the cafeteria. The fish stew was tasteless, which was a blessing. Knox sat across from him. “Bloody day.”

“Yes.”

“Mining is like that. It will be worse on Ceres. What I need to know is if you can train the next crew.”

”Surely you would train them.”

“There is a good chance, according to Catherine, that none of us will survive. But you will survive, and you understand the operation. Can you do the job?”

“I will make sure they are safe.”

“Then they will surely die on Ceres. Make sure they are alert.”

“Of course I will do both.”

“That death today probably saved ten lives on Ceres.”

“He didn’t do anything wrong.”

“No one does. There is a bus leaving tonight. Are you going to be on it?”

Training continued. They were copying Tseng’s system for delivering titanium from the Moon: thousand-ton, bowl-shaped ingots, a retrieval eye on the back, wrapped in an ablative glass mesh.

Feld brought up a video and explained the last step.

“Crew uses the crane to load the ingot on the launcher. Aiming, timing, and firing are automated. A small, solar-powered, ionic-thrust rocket will make fine corrections until the package is about half an astronomical unit from Earth. At that point, the guidance package separates, goes around the sun, and heads back to Ceres. The ingot continues to Earth. The flight is about eight months.”

They trained hard with Knox beside them the whole way, demonstrating, guiding, and helping them become a skilled team. After six months in Dahl’s desert camp, the crew was down to a hundred-twenty.

Catherine and Knox talked. “Fantastic job, Silas. Training went really well.”

“Thanks. Are we ready to send these guys to Tycho Crater?”

“I am ready to send the hundred you select.”

Knox didn’t like it. In the morning, a bus brought twenty men back to town, while the hundred loaded back on the sardine can. It was a lot easier with half as many people.

Their trip went from the Gobi to Florida, where they were lifted to low Earth orbit and then ferried to Tycho.

Darrin met the incoming crew as they disembarked. “Welcome to metallurgy in a vacuum.” They all moved to their huts. Four to a hut, narrow racks against the walls.

Darrin gave them an overview. “Earth sees this as a titanium operation. That is what we sell them. But we also make iron for Lunar infrastructure, and our most valuable product, oxygen,” he said. “We make enough for breathing, and tons more to power the Lunar Tug ships. They load up with Hydrogen in LEO. Hydrogen is light and relatively cheap to launch from Earth, but oxygen weighs much more. We dramatically reduced the cost of deliveries.”

“On Ceres. You’ll get enough oxygen to survive. Recyclers will split the carbon out of CO_2 , so you can keep on breathing, but every time an airlock opens, you lose a little. Mistakes will happen. You’ll need to replenish. Ore buys you backup oxygen.”

The crew performed well. Drilling, transport, sorting, smelting, they followed protocol, worked efficiently, and made no excuses. Darrin and Knox were both impressed.

Darrin's mass driver differed from the Ceres design, but the operation was the same. The crew loaded the mass drivers and waited to watch as the thousand-ton payloads were launched towards Earth, delivering metal on target in the central Pacific.

They trained on the crater walls too, rock climbing in pressure suits, cutting ore from vertical faces. The suits were uncomfortable, but they adapted.

Early in the third session, a trainee cut his leg on a fractured edge just below the knee. The calf depressurized, but no one panicked. He felt the pressure drop instantly, sealed the tear with a wrap of suit tape, and made his way back to operations for repairs. His fast action had minimized the trauma to his calf.

Darrin was watching. "These pressure suits look new. Different seals?"

Knox beamed. "Multiple seals at every joint. A cut glove doesn't vent the suit. Just the hand. Same for knees, elbows, and shoulders. You lose the segment, not your life. Built for field failure."

He brushed a hand over the reinforced joint. "Dahl had them designed after talking with lunar crews. Ripped suits have killed a lot of people up here."

Darrin gave a low whistle. "I want some for my team."

Two weeks later, they were working a steep section of crater wall. The man ahead of Lennox reached for a handhold. The rock crumbled. His cam popped free.

He dropped. Lunar gravity was weak, so it looked like slow motion, but the fall was real and could be deadly. Lennox moved without thinking. He lunged, hooking the other man with a carabiner to his own suit. They both fell off the face, but Lennox's cam was secure. Both men slammed against the crater face, but held.

The man gasped into the comm. "Jesus. Thanks."

Lennox didn't answer immediately. He was breathing hard, checking the anchor points, making sure everything was secure. Finally: "Check your clips before you weight them."

That night in Tan's Cantina, they bought Lennox a drink. Knox watched from across the room and made a note on his slate. Lennox was solid. Quick thinking. Good instincts. He'll make a good team leader.

After five months, Dahl called from Earth. It was time. Lunar Tug took fifty, including Knox, down to high Earth orbit, where their transport ship was waiting to take them to Ceres.

Their ride was a General Atomics nuclear thermal rocket, stripped down with no space and no concessions to comfort. Fifty seats packed into a hull that looked more like a shipping container than a spacecraft. The mass budget was tight.

“You guys remember the sardine can?” Knox asked. “This will be tighter and will last a full year. But no toboggan maneuvers, so relatively pleasant.”

It was still six months before the cargo would land when they started. The nuclear thermal rocket was much faster, so the trip that took the cargo twenty-one months would only take them twelve. They passed the time quietly, reading and playing cards.

Six months in, the monitors came up. The cargo ship reached orbit around Ceres. With rapt attention, the crew watched as the cargo ship began deploying descent pods down to the rocky south pole.

From the cargo ship, cameras showed the first wave of machines being activated. Their programming unfolded without voice or oversight, executing commands written years ago.

First, the surface was insulated. Self-expanding foam burst from canisters, boiling outward in slow waves. It spread over the South Pole, curing in vacuum to a fragile, velvet skin.

“That’s too fragile. When we step on it, won’t it be crushed?” Lennox asked.

Knox smirked. “Keep watching.”

Next came the hard foam. It was black, dense, and load-bearing. It formed the structural surface: walkable, buildable, and thermally resistant. Low albedo, high emissivity. Ceres had to be shielded from heat. The south pole was rocky, but most of the dwarf planet was not. Too much warmth, and it would erupt with steam, destabilizing everything.

Smelting and forging would bring immense heat. Every watt from the reactors, every ton of molten metal poured in the foundries, would soak downward. The foam had to stop that.

Only once the insulating layers had cured to spec did the rest of the deployment begin.

The hauler finally landed, its massive spike pointed skywards. It was built for only that one trip. Its nuclear reactor, still humming from the long burn out of Earth, was now repurposed to power the camp. It would power the foundries, the workshops, and life support.

The barracks modules unspooled themselves, delicate and empty as abandoned eggshells. A thin layer of black foam spread over the building. The crew watched, imagining life in their new home.

“Those look too weak to live in,” Jameson said.

Massive, blocky foundries followed. They landed hard and locked themselves onto the cooling slab. These were set on an extra meter of black surface foam, with coils of ammonia for active heat removal.

Storage barns. Fuel depots. There were five extractor ships, all parked in a row. Water processors, food supplies.

They still had six months before they arrived. That was time enough to get the systems running and confirm stability. The future was coming. It would not pause for tungsten, or for tantalum, or for anyone.

Chapter 9: Ceres Operations

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The flight had been long, crowded, and without reprieve. No one slept well. The air thinned over time in moisture, warmth, and patience. They expected space to be an empty void, but the cramped ship, poor food, and lack of diversion became its own form of pressure. For nearly a year, they floated in the slow, shared erosion of appetite and privacy, caught in a loop of chemical meals and scrubbed breath.

When Ceres finally came into the viewport, no one spoke.

It expanded from a dot to a looming face of dust and fracture. Gray ridges and black cavities etched a surface older than Earth's oceans. There were no soft curves, just a slow-turning mass, pocked and grooved, where sunlight caught on crater ridges.

The ship descended into low orbit. The station appeared on schedule, a square kilometer of hard foam, angles, and scaffolds. It looked temporary and skeletal, a foothold blown into place by machines.

Most eyes went to the barracks, a dull rectangle lit in amber.

Knox stood at the forward viewport, arms crossed, gaze fixed. The station looked like a thin exoskeleton clinging to rock. But he could already see its future; the rhythms were already in his head: power draw, tailings piles, feedstock flow. He had revised the protocols, rewritten the drills, and practiced until decision became reflex.

"We go in thirty," he said.

They suited up. Pressure gear sealed. Packs checked. When the hatch opened, ten of them spilled out, crowding onto a rover built for two. They jostled for footing,

half-laughing, half-shouting in Local Band. The wheels groaned under the load as they lurched toward the barracks.

From the outside, the module looked intact. The lights were on. The air lock didn't pressurize and open, though. They cycled through by hand and stepped into cold metal. Emergency lights pulsed, casting the room in a slow, amber rhythm. There was no air. No warmth. No weight.

For a few seconds, no one moved. The silence wasn't just in the room. It was on the channel too.

Knox scanned the room, jaw tight. He had run this scenario in his head a dozen times. And here it was. Real.

"Start tracing," he said. The voice was flat, and his order steadied them.

They moved fast. The leak was easy to find. Frost marked a seam on the outer paneling where a misaligned duct junction was improperly fused. They clamped it, sealed the breach with epoxy, and reinforced the joint with a crossbar from the maintenance rack.

Once the repair cured, they restarted the air feed, slowly. Full pressure would take hours. They watched the numbers rise as the module stabilized.

The gauge inched past the safe threshold. The station was breathing again. The crew cheered and started moving into their rooms.

Knox made his rounds through the other modules. Life support: green. Power: steady. Food systems: operational. Water reserves: full.

He checked the water processor status panels.

Three showed green. The fourth had a warning indicator.

He pulled up the diagnostic log. The warning had been active for five months. Processor four: catastrophic failure.

He suited back up and went to inspect it physically.

The housing was intact from the outside, but when he opened the maintenance panel, he saw it immediately. A puncture the width of a pencil was punched clean through the primary heat exchanger. The metal around it was deformed, petaled outward. Micrometeoroid strike. The damage to the tubing and electronics was

beyond their ability to fix. The processor was destroyed. Completely unsalvageable on Ceres.

He stood there for a moment, staring at the hole. One piece of rock, smaller than his fingertip, traveling faster than a bullet. It had hit during transit or early setup, and they'd been running on three processors ever since.

He checked the production logs. Three processors had kept the reserves full during automated operations. Three was enough.

He made his way to his office and stopped in the doorway. The L desk. The storage locker. The soft chair. He'd walked this room a thousand times in his head, checking sight lines, testing layouts, and memorizing every detail. Now it was real. The door had his name stenciled on it. Dr. Silas Knox, Captain.

He sat down, pulled off his helmet, and queued a voice packet to Earth. "Hi, Catherine. This is Silas. We arrived. Air leak in habitat unit fixed, habitat pressurized. One water processor is shot. Water reserves are full."

She wouldn't receive it for half an hour. He sat for a few moments before going to dinner. He could smell lithium grease in the air.

The people pulling kitchen duty had found unexpected items in the freezer: cake and ice cream. The cake said, "Welcome to Ceres." Knox had sent that with the cargo ship years ago. It had come with the supplies and sat frozen for six months in the mess hall freezer. The men cheered when they brought it out. There was plenty for seconds.

After dinner, Knox stood on a table. "We are here to get metal. The first people to set foot on an asteroid." He pumped his fist in the air.

His voice shifted register. "Don't get carried away with the excitement of it all. Asteroids are more unpredictable than any place you have mined before. Keep your heart rate low."

Knox paused in silence. "I designed our system. I will lead the first crew. I need three volunteers."

Three others slowly stepped forward.

Knox logged the names and sent the list to Dahl, his own name at the top.

They prepped in silence. Gear was checked, packs filled, timelines drawn. As the crew suited up, his terminal blinked. A private message had come in from Paris.

Knox read it alone.

Silas, you're going to have to be the adult now. That whole first crew may be killed. If they die, your job is to make sure a second crew goes out and does it better. If you go and you're lost, then everyone on Ceres dies. That's the math. Stay put.

He stared at the screen, jaw locked. Verified the timestamp.

Radio delay meant he didn't need to respond right away. He could think first, or pretend he didn't get the message until after he left.

Knox put on his suit and the leader armband. He read the nametag. Dr. Silas Knox, Captain. He strode into the mess hall and watched the three volunteers. They were calmly waiting for his direction.

Then he looked at the rest of the crew. Forty-six other men. The cleanup crew was hauling the plates back after dinner. Their faces looked the same.

If he died, what would happen to them? He looked around the room.

He removed the leader armband and returned his attention to the crew. "I need a fourth," he said.

Lennox stepped forward. Knox gave him the armband.

Knox pulled up the regional map. A target was marked in red on the display.

Knox explained the plan. "Feld believes this rock has tungsten ore. We are going to find out. You will be on that rock for one month. You will be bringing a thousand bags and a hundred cups, stacked just like a pack of Dixie cups on top of the extractor. Fill the bags, tie them off, connect them to the tether, and let the extractor pull them back to itself. The bags hold ten tons. Gravity is weak, so they feel light. You can lift them. Don't be fooled. They retain their momentum, so the key is to let it move slowly. The cups hold ten bags each, twenty when you screw them together. You load that tumbler pack onto the extractor's mass driver."

"How are we going to lift two hundred tons onto the top of the extractor?"

"Gravity is weak. The tumbler will weigh about fifteen pounds. Get Jameson to help if it is too much." Joe looked at Jameson.

A soft chuckle went through the room.

Knox continued. “Right now, our target is approaching. It will reach periapsis in a month. At that time, you come home. Get the job done. That asteroid will drift farther and farther away, so we won’t be able to access it again for decades.’

The first crew boarded. The target was loaded into the extractor’s autopilot, and the ship calculated its own approach. They came in slowly and smoothly. No burns inside two kilometers. Hall effect thrusters hissed in short bursts, guiding them toward the sunlit ridge. The surface emerged slowly, revealing a rugged texture of silicate rock interspersed with metallic veins. The asteroid bore the scars of countless impacts. Dusty hues of gray and brown mingled, highlighting the mineral richness in its ancient crust.

They coasted forward until a flat segment emerged, broad enough for gear deployment. Landing gear touched with a whisper. They were on the asteroid.

The crew stayed strapped in, eyes on their readouts. With sub-milligee gravity, even a shift in posture could break contact. The extractor balanced on three legs, each tipped with a fast-spinning, low-bite drill. The bits skated, then caught. The frame hummed as the drills bored downward, slow and steady. No weight to help them, only friction and time. Once the holes were cut, the anchors extended and began threading into place. One turn at a time, the ship moored itself. When the procedure was complete, a green light came on. They were fixed to the rock.

The sigh of relief was audible as they unstrapped. They checked their maps again, running overlays, marking terrain features, and blind zones. Then they clipped in, one by one, and, hearts racing, fanned out over the surface.

Moving slowly enough was difficult. A few bounced off the surface, rising into long, slow arcs. Tethers snapped taut. They reeled themselves back slowly and tried again.

The footing was unreliable, with loose regolith in some places, hard plate in others. They moved carefully, staying low, using gloved hands for balance and traction.

At a ridge near his waypoint, Lennox steadied himself and unholstered his survey laser. A short pulse traced the edge of a visible seam, heating the surface until a fracture opened along an old fault. He waited, tapped it with a glove. The split held.

He reached back, retrieved a cam, and worked it into place. Locked. He tested it, then moved on, slow and low, anchored with each step.

At the next site, he used the laser again to chip a sample, dropping it into the mass spectrometer. Seconds later, data reached the extractor and was relayed to Ceres.

Knox read it: scheelite.

He forwarded it to Earth but didn't wait for confirmation. He keyed back: "Tungsten ore. Begin harvest."

Lennox returned to the extractor and retrieved a ten-ton ore bag, drill, expanders, and saw. Moving one handhold at a time, he made his way back, anchored the saw, verified the clamps, and powered it up.

The blade bit deep. He followed, inserting hydraulic expanders, breaking rock, and packing it into the bag. At the end of the first cut, he tied the first bag shut and sent a signal to the extractor to pull it in. Slowly, the bag inched away from him.

He shifted the saw a meter, and started it again. He went back to cutting and inserting expanders. From the corner of his eye, he noticed something drifting. The saw had cut through his tether smoothly. He stood reflexively to grab it, but he stood too fast. He continued up, his boots losing contact with the asteroid.

"I'm drifting," he said over the radio.

That was when the others looked up.

One miner froze. Another dropped to all fours and began crawling toward the ridge, pulling hand over hand. A third unhooked a tether line and threw it. The shot was wide, but, under tension, the working end moved towards Lennox.

"I'm close," Lennox said, reaching for the line. It spun inches out of reach.

A second tether fell short.

"Still got eight hours of air," he said. "Plenty of time to think."

Silence followed.

The miner who'd been crawling reached Lennox's cut site. The saw was set firmly, as solid as the extractor. He clipped himself to the saw housing and pulled a tether from his pack. He jumped toward Lennox, paying out line as he went.

As he got close to the end of his own tether, he threw another line to Lennox. The throw was accurate. The working end was going straight towards Lennox. The reaction from the throw sent the rescuer tumbling backward. Suddenly, the rescue line snapped taut. The working end of the tether pulled back when it was about ten feet from Lennox. Close enough to see the clip spinning.

The rescuer reeled himself back in, repositioned, and tried again. This time, the line was thirty feet short. Lennox continued his slow drift, now over a hundred meters out.

Eventually, the crew turned back.

Bags were filled, sealed, and labeled. One by one, they were brought to the extractor's rail. The extractor calculated trajectories and launched each in a low delta-v path toward Ceres.

Four hours later, drifting cold and clear, Lennox understood the arc wouldn't return him.

He unlatched the survey laser from his belt.

"The site's solid," he said. "Keep harvesting."

Then he angled the beam and fired once through his own faceplate.

The transmission stopped.

The miner anchored to Lennox's saw watched the tether drift. "Alright. Back to work."

Back on Ceres, the crew had begun preheating one of the smelters. They were expecting the first tumbler to arrive the next day. The crew moved slowly in thermal suits, checking and rechecking seals and insulation. Every motion was deliberate. Hot mirrors redirected much of the radiated energy back into the furnace, keeping the crew just outside the danger line.

Water use was constant. Cooling circuits, steam traps, and intake purges. Knox tracked it all by hand, logging fluctuations. The system was closed-loop, but not perfect. Recovery was never one hundred percent.

By mid-shift, the reserves weren't rebounding, despite continuous cycling.

He ran a full diagnostic. Another water processor was nonfunctional. Corrosion had eaten through a weld where the titanium plating had flaked away. The remaining processors would also have a limited lifetime before the corrosive water ate through their welds, too.

He stared at the readouts, checked them again. The station could function with three processors. Two would force rationing and a slower mining cadence.

Hydrolox production, drinking water, food systems, and, as they saw now, thermal regulation for smelter operations were all affected.

He drafted a notice to Dahl.

Second water processor dead. Two functional but strained. Corrosion failure - titanium plating compromised. Remaining units likely on same timeline. Water more corrosive than anticipated. We're in trouble. Have not shared this with the team.

The response came three hours later.

I saw the telemetry. Keep mining.

Dahl's message confirmed Knox's decision. The station existed to extract metal; nothing else was guaranteed.

A second crew was sent while the first crew was still working. Another target, another haul. Tantalum this time, mixed with niobium on a stony asteroid.

In Paris, Dahl stared at the water system logs for hours. The text blurred. Her window overlooked the Seine, but sound, color, and distance all felt muted. She thought about Lennox. She put away her slate and left her apartment.

She walked down the street, turning where it was unfamiliar. A few blocks later, after a couple more turns, she found a bar she didn't recognize. She walked in, looking to see if she recognized anyone there.

She ordered a carafe of the house red, downing her first glass in a single gulp. The second went down more slowly. Her shoulders loosened. She drank again.

Her eyes fixed on an old poster on the wall, with a redheaded woman wearing a white hat, then returned to her wine glass.

"Of course, Ceres is dangerous. They knew. It's not my fault." She sipped her fourth.

"Why Lennox?"

The couple next to her moved to a farther table.

"The reserve tank is full. I hate corrosion. The manufacturer promised me it wouldn't corrode."

“I can’t send anything. I don’t have a rocket, and the transfer window won’t open for seventeen months. You think I wouldn’t? Why is it corroding?”

The bartender asked her to keep it down. She ordered another carafe.

“They can’t just stop. Do nothing, wait to die? They can keep pulling metal. I’m not a monster. I am giving them purpose. I’d send them the Seine.”

“They’re all going to die up there. I just don’t know how fast.”

Another patron moved closer to her. “Is it water?” he asked.

She turned. He was sitting at the bar, finishing a beer. She hadn’t noticed him before.

“That’s part of it,” she said. “It’s all of it.”

“What would it be worth, if I could get you water?”

She blinked. Laughed, briefly. “What, like magic?”

“Let’s just imagine it’s possible,” he said. “What would the number be?”

She studied his face. Calm, curious. She pulled a napkin from the bar and wrote on it.

One hundred tons. Pure. FOB destination: Ceres. Weekly.

She paused, then added the numbers and signed it.

*Two hundred dollars per kilogram. Paid on arrival.
Catherine Dahl, Helion Dynamics.*

He put the napkin in his pocket, and the two of them drank and talked through the night.

Chapter 10: Magazine Beach

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Catherine woke to the ringing of her phone and a splitting headache.

“Hello?”

“Hello. That contract you signed last night, do you still mean it?”

She blinked at the ceiling, trying to place the voice. “What? Who is this?”

“My name is Jakob Raskin. We met last night. You said you would pay two hundred dollars per kilogram for water delivered to Ceres. You asked for one hundred tons weekly.”

She remembered the stranger she met last night. She even remembered the contract. “Sure,” she replied. “There’s no way to do it, but if you could do it, I’d be more than willing to pay that much.”

“You may be right, Catherine, maybe there is no way. But I can tell you this: I will do everything I can to get you that water.”

“If you make any progress on that, let me know, and I will get an official contract from Helion Dynamics, signed by Tobias Raines.”

After the call, Jakob opened his contacts. He didn’t need many. Just the right ones.

That evening, he boarded a flight to Boston. The next afternoon, they all met in Magazine Beach Park in Cambridge. They sat in the shade at a picnic table, the Charles River flowing slowly by.

Mira Davis was the youngest, eighteen, still in her freshman year at MIT. She’d shown up in work boots and a tank top with grease under her fingernails.

Tom Wilson had come straight from his shift at Eversource, still wearing the utility vest with the company logo. He had his slate out, already working on something.

Ellie Thompson sat with her arms crossed, radiating impatience. Her degree was in Ocean Engineering. She had worked as an undersea medic and moved into habitat design. Sharp edges, the kind of person who'd seen too much to be impressed by mystery.

Pachakuti Quispe was the only one who looked relaxed, leaning back against the table, watching the river.

"Thanks for coming," Jakob said.

"You didn't say what this was about," Ellie said. "Just that it was a huge opportunity."

Jakob pulled a napkin from his jacket and spread it on the table. Wine-stained, wrinkled, covered in hasty handwriting.

*One hundred tons. Pure. FOB destination: Ceres. Weekly.
Two hundred dollars per kilogram. Paid on arrival.
Catherine Dahl, Helion Dynamics.*

No one spoke for a moment.

"That's..." Mira leaned forward, squinting. "That's twenty million dollars. Per week."

"It's real," Jakob said. "I met her in Paris two nights ago. She's desperate. Ceres is dying. Two of their water processors failed. They need water, or everyone up there dies."

Tom looked up from his slate. "Where do you get water?"

"Mars," Jakob said.

Ellie laughed. Short, sharp. "You're joking."

"Mars has ice," Jakob said. "Polar caps, subsurface deposits. We mine it, purify it, and launch it to Ceres. They pay on delivery."

"We," Pacha said quietly. "Meaning us."

"Meaning the five of us here at Magazine Beach. Whoever's willing to go, anyway."

Silence. The Charles River flowed past, indifferent.

“No one’s on Mars,” Ellie injected.

“Not yet.”

“There’s no infrastructure. No base. No support.”

“Can we build it?” Jakob asked.

Ellie burst out, “This is insane! There is no way we can do this.”

“It’s twenty million dollars a week,” Mira said softly.

Tom was staring at his slate, fingers moving across it. He’d gone quiet.

“Even if we could survive,” Pacha said, “how do we get the water there? Mars to Ceres isn’t simple.”

“No,” Jakob conceded. “It’s anything but simple. If it were simple, they wouldn’t be offering eighty million dollars a month. I don’t have answers. I think, though, that you four might. Not right away, but if you think about it, discuss it, I think you can work something out.”

“We’d need some equipment,” Tom stated. “We would need a mass driver to launch the ice blocks to Ceres. Similar to the system the Moon is using to deliver titanium to Earth, but much stronger. It will take time to spec that out. Getting out of the Martian gravity well all the way up to Ceres won’t be easy.”

“What will we eat?” Pacha asked.

“I was hoping you would tell me. Do you think we will starve?” Jakob asked.

Pacha hesitated. “Amaranth, oca, cavies. High altitude crops, low pressure tolerance. It’s possible. But I’d need a closed system. Water recovery, waste recycling, full nutrient loop.”

“We’d have sewage,” Mira said.

“That’s the point,” Pacha said. “Everything recycles. No waste. Nothing is replaceable. Everything has to be reused indefinitely.”

Ellie’s mind was reeling with the dangers. “If the habitat breaches, we die; if the power fails, we die; if the crops fail, we die. This is a long list of easy ways to die.”

“What would you need?” Jakob asked. “For the habitat.”

Ellie exhaled slowly. “3D printer. Regolith processor. Plenty of polymer. Get life support components from lunar suppliers. Those are proven. Air recyclers, heaters, air locks. It’s doable. I’ve done pieces of it before.”

“Power?” Jakob asked.

Tom was quiet, still looking at his slate.

“Tom?” Jakob said.

“Mars gets about forty percent of Earth’s sunlight,” Tom said distantly. “We’d need a lot of area. And dust is a major problem. Wiring, batteries, redundancy.” He paused, looking at his slate. “Kicker is, launch window opens in six weeks.”

Everyone turned to look at him.

“What?” Mira said.

“Earth to Mars transfer window,” Tom said. He looked up, and his expression was strange. Not excited. Not afraid. Just... certain. He turned his slate to show them the chart. “Favorable alignment. If we don’t launch in six weeks, the next good window isn’t for two years.”

The weight of that settled over the table.

“Six weeks,” Ellie said. “To build a Mars base.”

Pacha was watching the Charles. “If we go, we’re not coming back. Not for years. Maybe never.”

“I know,” Jakob said.

Mira leaned forward. “How much do we get? If we do this.”

Jakob smiled slightly. “That depends on how long we survive. But if we run for six months, that is twenty-six deliveries.” He let them do the math.

“A hundred million dollars,” Mira said quietly. “Each.”

No one spoke.

Tom was still staring at his slate, but his hands had stopped moving.

Pacha broke the silence. “What do you need from us?”

“A plan,” Jakob said. “Enough that I can pitch an investor. Can we actually do this? Can we survive? Can we eat? Can we save Ceres? I need a loose plan by tomorrow morning. “

Ellie was skeptical. “We’d need to test everything,” she said. “Habitat, life support, food production. We can’t just guess.”

“We have six weeks,” Jakob said.

“That’s not enough time.”

“No, but it’s the time we have.”

Ellie looked at the others. Mira was leaning forward, eyes bright. Tom was calculating something, lost in numbers. Pacha was unreadable.

“I need to know you’re serious,” Jakob said. “If I get the money, if I get the launch, are you in?”

Ellie pressed Jakob. “How much money are you asking for?”

“Money for equipment. I am hoping you guys can figure that part out. Plus the launch.”

“How much is the launch?”

“Half a billion. Maybe more.”

Ellie whistled. “Who’s paying that?”

“I’m working on it. I have a lead.”

“You don’t have the money yet.”

“Not yet.”

She stared at him. Then at the napkin. Then back at him.

“If you get it,” she said slowly, “I’ll go.”

Mira nodded immediately. “I’m in.”

Pacha studied Jakob for a long moment. “You’ll need a better plan than this.”

“I know.”

“But yes. If you can make it real, I’ll go.”

They all looked at Tom. He was still staring at his slate, silent.

“Tom?” Jakob said.

Tom looked up. His face was calm, but his hands were shaking slightly.

“Six weeks,” he said. “We have six weeks.”

“Is that a yes?”

Tom closed his slate. “Get the money. I’ll make the plan work.”

Jakob looked around the table at this group. Four people who barely knew each other. Four people who were about to save Ceres.

“Alright,” he said. “Let’s figure out what we’re actually building. I need to leave this in your hands. I have a flight to catch.”

The group continued their discussions without Jakob. He headed back to Logan International to catch a flight to Lagos, Nigeria. He had an appointment with a man he had only ever heard of, never met. A man who might listen.

Max Juno woke with gravity in his bones and smog in his lungs. It was his seventh month back on Earth. It still felt wrong.

The Moon had been sterile, hostile, and dangerous. He missed it every day.

He’d been a ^3He miner who got a polar claim in perpetual darkness. He had set up his solar panels on a ridge, giving him nearly perpetual light. It was incredibly productive. After two years of mining, he leased out the claim, returned to Earth, and continued to collect royalties. Now he paced through his penthouse suite in Lagos, irritated by the weight of the air.

The meeting was at a half-finished skyscraper downtown. He could smell wet concrete when he got out of the cab. He didn’t need a palace. Just a clean room and a screen. This room would be adequate, and the carpet was plush.

Jakob was already waiting when Max entered.

“You’re the lunar exit, right?” Jakob asked.

“And you’re the Mars guy,” Max said. “Which is funny, because there’s no one on Mars.”

“Yet.” Jakob gestured, and a screen flickered on. No graphs, no pitch deck. Just a satellite photo: a pale streak of ice, buried in red Martian dust.

“Ice?” Max asked.

“Mars has plenty. Poles, craters, and pockets under the regolith. The volatiles are clean. All I need is a shovel and a mass driver.”

Max narrowed his eyes. “You plan to sell it?”

“Two of the Ceres water processors failed. They’re begging Earth to fill the gap, but Earth can’t help. Launch window’s years out, and even then, it’s a long ride. They’ll be dry before anything gets close. I can send water for a fraction of the cost. And Mars has a launch window early next month.” Jakob pulled a folded napkin from his jacket. “This is the offer.”

Max skimmed it. The money was incredible, but so are the expenses in space. He looked at the signature, scrawled crudely at the bottom.

“Is that the Catherine Dahl from Helion Dynamics?”

“The same. I met her in a bar in Paris.”

Max rocked back. He started to believe the opportunity was real. He also doubted this scrawl would count as a legitimate contract when it came time to collect.

“Obviously, I will need to talk to her about this.”

“Of course. Of course you will. I have her number; you can call her right now.”

“For now, I will listen. If you have something to say, maybe I will call her.”

Jakob was nervous. “I have a crew from MIT. I met them all at school and alumni events. Our plan is to go to Mars, set up a stable living environment, and send large blocks of ice by mass driver to Ceres.”

“How are you going to hit Ceres at that distance?”

“No way. That isn’t our goal. Ceres has a small gravity well, and there are lots of other asteroids to mess with it. But they are already collecting ore. They have tenders, small ships that work in and around Ceres, catching payloads and guiding them in. We just have to get within tender range.”

Max had lived on the Moon. He knew how hostile Mars would be without Lunar Tug to bring them supplies. "What about food?"

"We'll grow it. Algae's simple. But one member of our team is planning to grow high-elevation crops, since they can grow at lower pressures. He has a plan to make regolith into dirt."

"You're serious."

"I just have to feed five people. Nothing fancy."

Max studied him. Gaunt. Sharp. A little unhinged. Definitely broke.

"What's your name?"

"Jakob Raskin."

"What do you want?"

"Maybe thirty million for equipment. Plus a ride to Mars. With that, I pay the team and build what we need. You arrange the ride. In return, you get forty percent of the company. First Martian exporter. First to deliver offworld water."

Max looked at the photo again. Ice, red dust.

"And if it doesn't work?"

Jakob shrugged. "Then we die on Mars."

Max frowned. "You say that like it's obvious."

"It is," Jakob said. "This isn't the Moon. Resupply is nearly impossible, and there's no lift capacity for a return stage. No launch window for two years, even if you could afford to send a rescue ship. We go, we live off what we brought and what we make. If that fails, we die. The crew knows that, and they are working right now to ensure we don't die."

Max sat back. He knew the math. The Moon had always offered at least the illusion of rescue. Mars didn't. If they failed, they wouldn't be coming back.

Jakob watched him. "We have to make it work."

Silence. The kind of silence you only get when someone actually means it.

Max thought, remembering his days on the Moon, knowing even if he did everything right, a stray micrometeoroid could end him at any time. Risky, but exhilarating. This project filled him with the same exhilaration. So many ways to fail, but there was a way to succeed, too.

“Jakob, let’s see what we can do. I’ll fund your shopping spree. You guys get started now. I have some other phone calls to make, starting with Catherine Dahl. We will see about getting you your ride. Maybe Lunar Tug has something. In the meantime, what can I call your company?”

Jakob looked at Max, caught completely off guard. He had no answer. Then it came to him. “Magazine Beach.”

Chapter 11: Martian Ice

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Five weeks after his meeting with Max Juno, Jakob stood alone in the cargo bay as the last of his team trickled in.

The ship was built for function, not comfort. The five of them were packed in with their equipment and a few covies in cages that reeked of hay and ammonia. The seed stock and tubers were sealed in insulated crates. Three air recyclers stood lined up against the wall. One was all they needed, but if anything went wrong, help was years away.

They gathered in the main bay, where Jakob had been staging equipment for a week. The warehouse still felt temporary. Crates lined the walls. Cables coiled like snakes. The air smelled of steel, plastic wrap, and lubricant.

Jakob let them settle. No introductions. No speeches. Just the team and the tools.

“We have one shot at this. Don’t screw it up.”

His voice was neutral, tired, stripped of anything unnecessary.

“Max has paid for this cargo hull and all our equipment. He bought a rocket from Lunar Tug that they were not using. Helion Dynamics graciously fronted us the money to get this all into LEO, where we will connect to that rocket and head to Mars.”

“This isn’t a dry run. There’s no backup. Mars doesn’t care if we make it. If something breaks, Max might be able to send a replacement, but launch windows only open every twenty-six months.”

He gave them time, then continued. “If you don’t think we can do this, you shouldn’t be here. If you do, then you already know what’s required: working without a net. Each of us is doing more than we were trained for. Living with

whatever comes. But if we pull it off, if we send clean blocks of ice, hit delivery windows, and hold the contract, then we become the first Martian exporters. More than that, we have a chance to build something real up there. And maybe, down the line, find a way back. On our terms.”

He looked at each of them in turn. “Got it?”

They nodded, one by one.

Mira cracked a grin. “You won’t have to worry about me.”

It broke the tension just enough. Jakob didn’t return the smile, but he moved forward.

There was no time to waste. Each person checked their own manifest, then verified someone else’s. No second chances.

They rode a heavy-lift to low Earth orbit in their cargo hull. Mira loved the heavy acceleration, then weightlessness. Tom worked through the transfer sequence, moving them in their cargo hull from the heavy lift rocket to a rocket Lunar Tug no longer wanted. This one would take them to Mars.

Mira was excited. “I love the feeling of acceleration. I can’t wait for our burn to Mars. It just totally crushed me down into the gravity couch last time!”

Jakob humored her, counting down to launch. “Five, four, three, two, one, go.”

Tom threw the switch, and the hydrolox rockets lit with a roar. The gentle push barely registered. The look of disappointment on Mira’s face brought laughter from the rest.

“Not what you expected, Mira?” Jakob asked. “This is a shuttle rocket, not heavy lift. It will burn for an hour, but at 1/10th G, enough to insert us into a Hohmann transfer orbit to Mars. After this, it will be six months of true weightlessness.”

Mira pouted.

The time was spent in tight, unchanging air. The ship was the size of a bus, but none of the space belonged to them. Every surface was spoken for: cable runs, storage racks, sealed containers, and equipment. Every tool, part, and system they would need was packed inside with them. There was no separate cargo, no second flight, no margin for error. The entire mission was compressed into this one hull.

They slept belted into stacked bunks and moved through narrow corridors. It was warm and humid, so they dressed in the bare minimum. Most days, just underwear.

There were no showers. Just sealed hand wipes.

The first time Ellie had to clean herself, she wedged into the corner by the water recycler, back to everyone, and worked fast. Her shoulders were tight. No one said anything, but everyone looked away. Studied their screens. Checked equipment they'd already checked.

She pulled her shirt back on immediately after, still damp.

"It's fine," she muttered to no one. "It's fine."

By week three, she stopped retreating to the corner. Just wiped down at her bunk, quick and mechanical. Still wore the underwear. Still kept her back turned.

By month two, the underwear started to smell. Everyone's did. The fabric held moisture and body oil with nowhere for it to go. The water purifier was not built for laundry. Pacha stopped wearing his first. He seemed completely natural that way. Mira followed a week later.

Ellie held out longest.

Month three, Jakob woke to find her sitting on the edge of her bunk, underwear balled in her fist, staring at nothing.

"There's no point," she said quietly.

She tossed it into the waste bag and didn't put another pair on.

After that, no one wore anything unless they were cold. Hair matted. Skin dried and cracked. The air smelled like polymer, machine grease, and breath. No one mentioned it, but everyone noticed.

Ellie stopped turning away when she cleaned herself. Just did it, same as eating, same as checking the boards.

One morning, Jakob caught her looking out the tiny porthole at Mars, growing slowly brighter in the distance.

"Can't wait to have a door again," she said.

Jakob nodded. “Yeah.”

“Dinner time!” Pacha called out cheerfully.

No one was happy. Food came in pouches, nutrient-balanced and soft. No flavor, no texture. No one wanted to eat. You ate because the others did. They were all losing weight.

Tom didn’t like that. He and Mira set up a stand to hold a cargo bicycle in place and connected it to some paddles, like an assault bike. Everyone got thirty minutes a day. Jakob hated it, but he knew Tom was right. As their leader, he showed enthusiasm he didn’t feel.

They also used resistance bands. Not the same as real weights, but it slowed how fast they were wasting. Ellie kept their schedules for resistance bands. Push, pull, legs, rest.

Time passed in repetition with seemingly no progress. Same lights. Same air. Same bodies. Nothing changed. No distance closed. They slept and woke in the same space. Thought the same thoughts. Waited for the gravity to shift.

The only way to be alone was to close your eyes and stop reacting. Then Jakob made an announcement. “We are about to do our slowdown. Watch out the window.”

Mars was big. No longer a red dot, the landscape was clearly visible. And, as they watched, the craft tumbled. A controlled and slow reversal, the craft now pointed away from Mars, and the engines came on. They ramped up slowly, giving the crew the feeling of gravity. Softly, for about ten minutes at a tenth of a G as they performed a Mars orbital insertion. Tom put them in a highly eccentric orbit. They stayed in that orbit, heating at periapsis as they entered the atmosphere, cooling during apoapsis. The heating cycles grew longer each time, and the cooling cycles shorter.

Then the real entry. The heat shield was ablating as they came in. The whole frame began rumbling. Jakob kept his hand on the parachute deployment lever, while Tom watched the gauges. At Mach 6.0, Tom called out, and Jakob pulled the lever to deploy the high-speed parachute. When it unfurled, it felt like an impact as the craft was suddenly decelerating at two G. Mira didn’t like it this time. The G force decreased as the square of the speed. Tom kept a steady eye on the gauges. At Mach 2.0, the force had decreased to about Martian gravity. Tom called out, and Jakob pulled the second lever. The high-speed chute detached and drifted away, and the much larger medium chute deployed, putting them back to two G. Pacha threw up,

but no one could clean anything. When they got to 1G, Tom called out for the final chute. It was enormous, but they had already slowed significantly. They only reached 3G. As they neared the landing site, Tom started the engines. This burn was harder than they had used for the Hohmann transfer orbit burns. The soft jolt of touchdown came like an afterthought, as if the ship had changed its mind at the last second and landed anyway. They sat at an angle, but the legs slowly adjusted until the deck was level. Everyone cheered when it stopped. Through the portholes, they saw a new world. Red, with a rosy sunrise. 75°S, 30°W. Smooth, near the edge of the ice cap.

Pacha began cleaning himself up. Mira came over and lent a hand.

Jakob felt the impact in his ribs before he felt it in his boots. The noise was low and wide as the ship settled. He stood too fast, while the legs were still leveling the ship. He cut his head on the cabin bulkhead and swore.

“First blood on Mars,” Ellie muttered as she cleaned him up and added a bandage. “Good omen.”

Jakob didn’t smile. He wasn’t here for moments. He was here to keep the mission from falling apart one piece at a time. He called them together. “OK, we have eight months of summer remaining. The ice cap is retreating, so each day we will go a little farther to collect. Be greedy. Pull as much as you can as fast as you can, because it is all going away anyway. We have thirteen days before we launch on the shortest transfer orbit to Ceres, and we have a lot to get done. The clock has already started.”

He posted the basic tasks, taping the list above the airlock.

- Solar Panel Deployment

- Dig foundation for Agriculture dome

- Collect Ice in ten-ton blocks.

- Deploy mass driver

- Start growing crops

- Build ag dome

- Make fiberglass ablative boxes to carry the ice to Ceres

Sol 13: launch package 1, one hundred tons of pure ice, shortest transfer time to Ceres.

The solar arrays went up first. Nothing moved without power.

Tom and Pacha were out of the lock within minutes, boots crunching through red dust. The panels were flat and wide, packed awkwardly for transit. One caught halfway through its extension. Tom swore, dropped to one knee, and cleared the rail with a multitool.

“Dude. Dust already?” Pacha laughed as he drove a stake into the regolith a few meters away.

“Fine as talc,” Tom muttered. “Static pulls it into everything.”

“I bet the first storm is going to knock this all down,” Pacha smirked, watching a distant dust devil.

“Low atmospheric pressure means low force from wind,” Tom replied. “But when the first storm hits, I’ll be watching.”

He finished the bolts and moved to the field generator: carbon frame, three-phase, clipped into the central rack. When he flipped the switch, the panels hummed faintly, and a shimmer passed across the glass.

“The alternating electric field pushes dust off,” Tom said. “Indium tin oxide electrodes on the surface. A slow three-phase charge walks the grit downhill.”

Pacha watched the effect, the dust slipping downward in slow motion, as if peeled by invisible hands.

“Reduces the buildup,” Tom said. “We will still need to rinse the panels weekly to remove what this doesn’t.”

Pacha nodded. Then turned back to his line of panels and kept working.

Mira passed them without slowing. She’d pulled out the saw sled and had assembled the disk. Six pie-shaped wedges were screwed together to create an enormous wheel fifteen feet in diameter. She got it all assembled and went straight to the edge of the shelf.

“This ice better be worth it,” she said over the radio.

Jakob followed. The regolith cracked faintly under his boots. It didn’t feel like dirt. More like baked ash. He stopped before joining Mira. He found a shovel and went about a hundred yards off to dig. After ten minutes, he stepped back and took a picture for Max. The words “Magazine Beach” were dug into the regolith.

The shelf wasn't flat. That was the first problem. The models had shown clean, horizontal strata. What they found was a tilted slab with wind-carved seams and dust-packed hollows.

Mira stood at the edge, eyeing the slope. "The grade is too steep," she said. I can't cut straight on that."

Jakob reached for one of the adzes. "I'll fix that."

The adzes were carbon steel heads on collapsible shafts. Mira joined Jakob, knocking down the surface in ragged swings. Tom joined in. The crust came off unevenly in slabs, shards, and soft sections that crumbled like stale bread. Red dust had seeped into everything. Some layers were dark, almost black from meteorite soot. It wasn't easy work or fast, but the surface had to be flat before the saw could track true.

Inside, Ellie was running diagnostics. The water recycler was green. Air mix: oxygen levels were good. CO₂ was low enough, hovering near 1600 ppm. The organics were too high, though. She tapped at the console like it had insulted her, lips tight, not speaking. Numbers came in slowly, sometimes delayed by dust interference. She looked at the catalytic cleaner, a U-shaped pipe packed with nickel dust. Countercurrent flow minimized heat loss. Normally run at 300°C, she set it at 500°C and turned up the fan. "This whole vessel needs to be scrubbed with bleach," she thought.

Meanwhile, Pacha moved gear off the ship. Cargo that could be outside was stacked and covered with tarps staked against the wind. With space cleared in the aft bay, he began assembling the grow tent. Ellie watched as he snapped the framework together, section by section. She said nothing, but her expression softened.

Jakob keyed the team channel. "Tom, Pacha, thank you for getting the arrays live. This is summer here, so the sun won't set for long, and twilight will persist much longer than we are used to. Too easy to work past the edge of exhaustion. From here on, we follow a schedule. Dinner, chat, then sleep. Your body won't tell you when it's tired, so we'll use the clock instead. He let that settle, then continued. "Two weeks. We cut, refreeze, package, and launch ice for Ceres. If we miss it, we don't get paid. And Ceres runs dry. This isn't theoretical, it's existential."

The team line stayed quiet. They already knew the plan, but now in the light, they understood it.

Pacha called them to dinner. “Hey, guys, are you getting hungry? We are working now, so we need more calories. Eat up!”

There was some grumbling about the continued lack of meat. After a few minutes, Pacha spoke up, “We have limited feed pellets, so we won’t breed the covies until their feed is growing. And not just algae.” He gestured toward the tent in the bay. “It is germinating now. Once we have an agriculture dome, I will plant. Give me twenty-five sols, and I’ll get us roots and leaves. Then we breed one sow. No more. Crops come in, we expand. Not before.”

Tom grumbled, but everyone understood. Rations would hold.

The sled creaked its way toward base, trailing a fine mist of powdered frost in its wake. Mira stood behind it, boots planted wide for stability, pushing hard. The traction assist wheels pulled, sometimes slipping. On top, the sled held a triangular prism of ice, seven feet high, twelve feet long, a mass of ten tons, faceted and glinting faintly in the sunlight. It looked like an enormous gemstone from a fairy tale.

She parked the sled with practiced slowness, locking the wheels into place beside the melt chamber, which Pacha had already hauled out. It was a pressure-sealed tank with water filters. Jakob stepped out to meet her, brushing ice dust off his gloves.

“Not bad for a first haul,” he said, voice crackling over the suit comms. “Smooth cuts.”

Mira smiled. “There was a seam full of rust-dust. I had to trim it on the fly with the adze.”

They both stared at the block for a moment. Jakob keyed the panel, and the chamber hissed open. Using the sled’s crane, they lifted the block up and into the chamber.

Inside, air pumps brought the pressure up to 1 bar, coils of heating wire glowed faintly red. The block started warming up from about a hundred degrees below zero up to where it could start melting. That operation was slow. As the water came off, it was filtered and poured into the massive freeze chamber, making an enormous lemon shape that would weigh a hundred tons when complete.

At night, their beds were uncomfortable. Gravity felt weird now. The ship creaked in ways no one liked, and the silence outside pressed in like a weight. With the equipment unloaded, they had a lot more space, though, which helped.

Jakob stayed up late. Max had pinged him from Lagos just before comms cutoff:

“Nice picture. I put it on my wall. Ice ships in twelve sols or you lose the money.”

No pressure. Outside, the ice shelf waited. Tomorrow, they would cut nine more blocks.

The second sol on Mars started with shovels. First, the foundation needed to be dug. They moved the top meter of regolith without encountering any large rocks. Next, they sorted out the finest dust. They filled the hopper of the 3D printer with that powder and set it to make foam. They laid out six inches of foam, about half fine powder and half polymer. The reddish-orange color looked beautiful to Ellie.

They let it cool for about an hour, then changed the settings. Now they were making the solid walls. Ellie jammed her shovel blade into the regolith, scooping up load after load, and dumped it into the intake hopper of their 3D printer. The dust didn't puff like it would on Earth. It fell sluggishly, clinging to itself in dry clumps, static-charged and oddly heavy for something so fine. The printer's auger rumbled beneath the housing, pulling the dirt toward the heating chamber.

Pacha stood at the controls, eyes on the temperature readouts. The polymer reservoir stayed just above the melt point, viscous and glassy, ready to bond with the crushed stone and dust. It made the regolith printable. Stronger than concrete once it cured, or so the specs claimed.

No dramatic hum of motors. No hiss of steam or whine of servos. Just the low mechanical growl of the extruder and the broad scrape of oversized shovels against broken stone.

The print pattern was simple: a low hemisphere, six meters wide, cut cleanly on one side for an airlock. Shelter from radiation and micrometeoroids. No windows. A transfer box with a pipe for air, water, and sewage. This was the agriculture dome, but they would be living in it for a while. After this came the equipment storage barn. Eventually, the habitat.

Ellie paused, leaning on her shovel with its wide blade waist-high, more like a snow scoop than a spade. In this gravity, it worked. The regolith was loose, fractured, and dry. She watched the curve rise slowly with each pass of the nozzle. It looked layered, almost clumsy, but she could already picture it finished. Not a home, exactly. But the first place on this planet built by hand and meant for life.

“Load's holding,” Pacha said in his headset, voice clipped and dry. “Poly's stable. We keep pace, we finish the shell by bedtime.”

Ellie nodded, though Pacha couldn't see it. She just kept shoveling.

It was hard work. No gravity to help the shovel bite. Just pressure, leverage, and grit. But neither of them stopped. There was nowhere else to be, and nothing more important to do.

The sun hung high. No shadows moved. No breeze stirred the dust. Just the dome, growing layer by layer, like a promise they meant to keep.

By midmorning, they'd settled into a rhythm, trading roles when the one on the shovel was too tired to continue.

Pacha took the shovel, breathing slow and steady inside his suit as he fed the hopper with one scoop after another. Ellie stood at the printer's interface, fingers flicking through the displays, keeping the temperature in the right band, watching the extrusion for skips or warps. The dome was waist-high now, a rough curve taking shape against the orange dirt.

From the east, a figure approached, slow and broad-shouldered in his suit. Tom. He dragged a sled behind him with a rectangular shape strapped tight beneath a dust blanket.

"Hey," his voice came through their headsets, distorted a little by the local mesh relay. "Mass driver's online. She hums like a dream. Got time for a door?"

Ellie glanced over and nodded. "We're at the cutout. You're just in time."

Pacha stabbed his shovel into the ground and leaned on it, panting. "How many tons to Ceres?"

"None yet. But she's aligned, powered, and ready. I'm gonna do a test launch tomorrow. Could hit a boulder on Deimos from here."

Ellie raised an eyebrow. "Please don't."

Tom just chuckled and stepped around the printer. He pulled back the blanket and revealed the airlock, composite, pre-fabbed, pressure-ready. Not big. Just wide enough to crawl through in a suit. Airtight seals, no windows. Practical. Ugly but adequate.

He propped it next to the rising wall. "You want me to anchor it now, or wait?"

Ellie waved him over. "Do it before we print too high. We'll match the layers around it."

Tom got to work without another word, pulling out anchor rods and the little bolt gun they'd brought for this kind of thing. Ellie watched for a minute, then turned back to the readouts.

Polymer holding at 145°C. Flow rate steady. No signs of clog. She gave Pacha a nod. "Back at it."

He groaned, picked up the shovel again, and dug.

"Your hole," Tom said to Ellie, with a flourish after he finished bolting it in place. "I'm going to grab lunch and yell at Jakob about calibrating the mass driver scope. You two shovel heroes can finish the shell."

Ellie checked the interface again. Still green. "We've got three more layers to seal the door. Then we start capping."

Pacha kept shoveling. "Better get moving, then. Before Tom eats all the rations."

"I didn't think about that," Ellie exclaimed and turned the machine to standby. They finished filling the hopper together, then started walking back to the ship.

Five more blocks had been cut by lunch. They were lined up by the processor, going in one at a time. Everyone was in high spirits as they ate. Mira talked about the ice. "That saw is a monster. Huge, and when it cuts, it is like a blizzard, but it cuts smoothly and fast. When we get the hang of it, I bet we can cut two hundred tons a day. I cut fifty so far."

Tom jumped in. "The mass driver is ready to go. I tossed a rock with it slowly. It flew about ten miles. I will calibrate it tomorrow with a bigger push, but it is at least functional."

After lunch, Pacha and Ellie returned to the printer, and Tom joined Jakob and Mira on the ice field. Jakob was starting on block seven. The sled saw ran smoothly. His movements were measured, calm, and smooth. Eyes on the cut line. Depth consistent. No rush. The shelf was cooperating.

Then the whine started. Jakob hit the kill switch immediately. Too late. The saw shuddered once, then died. A coil of white smoke unspooled from the motor housing.

"Mira," he said into the comm. "We've got a problem."

She was there fast. No wasted movement. She didn't ask what happened. Just knelt, glove hovering above the housing. She could feel the heat radiating from it.

“Something’s cooked,” she said.

Jakob bent down. Even through his gloves, the heat was obvious.

“Motor’s burned,” Mira added, flat.

Jakob didn’t answer. He stood, slowly, swearing low and hard. He yanked the saw blade off and put the saw and cutting sled on a cargo sled. He and Tom started pushing it toward the ship, boots grinding over the ice shelf. Mira got up with the saw and began opening up the motor.

“All hands,” Jakob said over the team channel. His voice cracked. “Emergency. Sled saw motor’s blown. Meet at the ship.”

Chapter 12: Ice Saw

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The line went silent.

Pacha dropped his shovel and ran. The printer was still running when Ellie flipped the switch to standby and sprinted. Dust kicked up behind her boots.

When they arrived, Mira had the casing split, revealing the motor's internals. The coils were scorched black. Red dust clung to the interior walls, packed into seams and corners where it should never have reached. The insulation had frayed and burned, peeled back like scorched fabric. Mira stood beside the motor, arms crossed, eyes locked on the damage.

Jakob didn't look up. He pointed at the motor guts. "No backups," he said. "I brought blades, belts, bearings. Not the one thing that matters."

Tom stared at the open casing. "You didn't pack a spare motor?"

Jakob didn't answer.

"I should have," Jakob said. He didn't look up. "I knew better. Two is one, one is none. I say it. I believe it. I missed this one."

Ellie leaned in, frowning. "We can maybe rewind the motor. Depends on how deep the damage goes."

Mira stayed quiet. Her jaw was tight.

Pacha finally said what everyone else had been circling. "Can we repair it?"

Jakob didn't answer right away. He exhaled, long and steady. "Not here. Not with what we've got. Not in time. We have eight days to cut and process three more

blocks. Then the launch window opens. That's the contract. That's the water Ceres needs. If we miss it..." He stopped, not finishing the sentence.

Ellie knelt beside the housing, touched it with one gloved hand. "We don't have another way to cut blocks?"

"Not the way we need to," Jakob replied. "The sled gave us depth, speed, and control."

He looked at each of them. He had no answers.

Mira finally spoke. "So what do we do?"

Jakob looked down at the wrecked saw. Then out the airlock window, where the sky bled into dust.

"We figure it out," he said. "There is no backup here. We figure it out."

No one spoke.

Dinner was hot stew, thick with lentils and dried vegetables. No one said much. Forks scraped quietly.

Mira sat stiff, back straight, eyes forward. She hadn't touched her food. Tom pushed things around on his plate with slow, absent motions. Ellie sat, not talking, not eating much either.

Jakob didn't join them. He sat at the end of the main corridor, near the airlock. Helmet off, still in his dusty suit, elbows on knees, staring at the saw motor on the floor in front of him. The casing was cracked open, the blackened copper glinting under the overhead lights. He hadn't spoken since dinner started.

They finished processing the last two ice blocks quietly. The filtration system hummed in the background, water melting in the chamber, trickling through screens and mesh filters into storage. Steam rose in thin threads, caught the lights.

When they finished, Ellie stood up. "I'm going to take care of the printer." She then wordlessly put on her gear. Her boots thudded in the corridor. The printer had to be cleaned before they went to bed, before the polymer inside started to break down. If it did, it was a loss they couldn't afford.

Back in the corridor, Jakob hadn't moved.

He sat alone with the motor at his feet, the weight of the entire mission heavy on his shoulders. They had planned for everything. Redundancies, backups, contingencies. Two is one, one is none. That had been the rule. And now here they were. One saw. None left. No launch. No contract. No way forward. Jakob didn't say it out loud.

The hum of the water processors. The clink of utensils left in bowls. Jakob composed the message with mechanical fingers. No grand speech, no dramatics. Just the facts, numbers, part codes. Three sled saws. Ten motors. Redundant systems for a world that didn't forgive mistakes.

To. Max Juno

Subject. Equipment requisition – Priority

Three sled saws. Ten compatible drive motors. Full specs attached.

Ship ASAP. No substitutions.

-Jakob

He stared at the screen for a while after it sent. He didn't expect a reply. He had no idea what time it was in Lagos. Max would get the message when he got it. But no matter what Max did, no help would come soon enough. Years, if everything aligned perfectly. He needed it in days.

He closed the message. Then stood. There was still a job to do.

Back in Nigeria, Max Juno was already halfway through his second cup of coffee when the message arrived. Max sat for a long moment, staring at his own reflection in the glass. He opened the procurement system, out of habit more than hope. Three sled saws. Ten motors. Full specs. He filled in the form and stepped through the menus.

Estimated delivery: thirty-four months. He sat back. By then, it wouldn't matter. The cost was hundreds of millions.

Max stared at the screen, then reached for the cancel key. He didn't place the order.

There was no point pretending he could help.

Back on Mars, Ellie dragged a power conduit across the red dirt. It snaked from the ship's auxiliary port to the half-finished dome. The doorway stood open, rough-edged, unsealed.

She moved deliberately. Not quickly, but without allowing hesitation where despair might settle.

Pacha joined her at the printer a few minutes later, ready to help with the shutdown. She looked at him oddly.

Then she spoke. “Pacha, I can’t do anything for that motor. I wish I could. But there is something I can do, if you’ll help me. Let’s finish this dome.”

Pacha nodded. They had worked hard all day and deserved rest, but the whole team needed someplace better to live than the ship.

Ellie watched the controls while Pacha shoveled regolith into the hopper. Then she joined him. Shovel after shovel. The printer extruded each layer in a slow, molten spiral. The curve of the dome was closing now. More overhang. They had to print slower, wait between passes for the material to cool and hold shape. Ellie adjusted the print speed manually.

Four hours in, they had said nothing, but looked at each other, complicit in their transgression.

Pacha pumped his fist when the printer stopped. Ellie didn’t smile, but she looked up at the finished dome and exhaled. Dust already clung to the surface. The polymer looked dulled, aged, like it had weathered years instead of a sol.

She ran the electrical hookup and initiated the systems check. Ellie watched the green light blink on the control panel, steady and soft. The pressure test held. Airflow was stable. The dome was sealed. It would hold. The compressor started pumping in the Martian atmosphere, and the recycler struggled to keep up, converting the CO₂ into oxygen and soot.

She stood for a moment, one gloved hand resting on the junction box. Then she turned and walked back to the ship.

Inside, she passed Tom and Jakob without a word, ducked into storage, grabbed her sleeping roll, and cycled through the airlock again.

She didn’t say anything. She didn’t need to.

Mira guessed what was going on. She came next, pack slung over one shoulder. Then Pacha, carrying the covies in a pressure box. “The floor will be warmer than my bunk,” he said.

That was as close to an announcement as there was.

They were going to sleep in the ag dome tonight. It wasn't fully ready. It had air and heat, though, and space. And they built it. It was theirs.

Tom and Jakob brought their gear last.

Even if it all failed tomorrow, even if they missed the launch window, they would spend this one night in the world they were trying to build. A night in a home they built. A claim on a distant planet staked in sleep.

They put their cots near each other, on the side opposite the cavy pen. As they all prepared to sleep, Jakob took a moment with Ellie.

"The setting here is much calmer. I still don't know what we will do tomorrow, but we will sleep better tonight. Thank you."

In the morning, the dome was dimly lit. Pacha programmed the lights to start low, build toward midday, then fade again at the end of the sol. The air inside still held a hush, like a cathedral at dawn, the natural reddish hues of the regolith lending a regal feel.

They stirred from their sleep slowly. It had been the best night since launch. The sleeping pads were soft, the heat even, the air dry but warm. There was space to stretch out. No clanging bulkheads, no humming walls a meter from their heads. Just quiet.

When Mira sat up and rubbed her eyes, she didn't bother to put a shirt on. She looked around the peaceful dome. She could smell the polymer faintly, but mostly she could smell the soil and the covies and the people. It was calming.

Pacha was the first one moving with purpose. He checked the cavy pen. They all looked warm, nestled together like furry little coals. He topped up their water and scattered feed, murmuring to them in Spanish like they were his children. If the mission had any hope of surviving long-term, it ran on covies and calories.

After the animals, he turned to the stove. It was just a hotplate and a pot, but it was enough. Oatcakes, protein mash, dried fruit. He called them to the table, and they all gathered. Some of them had put on light cotton thermals as they got out of bed. No one wore shoes. This was home.

They ate together, quietly at first. A few soft laughs. It felt like something new had started. The motor was still broken, but the world hadn't ended last night.

When Jakob came by the table, he had shaved. Just a rough pass with the clippers, but it was the first time since Earth. It made him look less ragged. He watched the others as they ate, not with concern, but with clarity.

When the dishes were cleared, he stood.

“Six blocks,” he said. “Mira?”

She nodded. “Done. All filtered. All refrozen. Sixty tons are ready.”

He looked around the dome. “We lost the saw. We lost time. We can’t afford either. But we’re not finished. Ceres still needs water. That’s why we’re here.”

He met their eyes, one by one. “Here’s the plan.”

He stepped forward into the light. “We are not in danger of dying today. Let’s be clear. We have power. We have heat. We have clean water. The dome holds pressure. The cavies are fed. Pacha’s crops are germinating.”

A pause. “But no one’s coming to get us. Not because they don’t care, but because they can’t. Max said he can get us the parts in three years. The only ship that brings us back to Earth is the one we pay for. We pay with water.”

He gestured to the ceiling above them.

“This dome, it’s better than the ship. It’s warm. It smells like soil. We made it. We can live here.”

Ellie, quietly, pulled on a thin undershirt. Tom put on boxers.

“We’ve got sixty tons of clean water ready to go. If we ship that now, Ceres gets a week’s supply. Longer, if they ration, and they will.”

He exhaled once, steady.

“Last night I thought we were done. But now I’ve got a plan. It’s brutal, maybe stupid. But it will work.”

He let the silence stretch.

“We brought three adzes. They’re meant for smoothing the surface, not cutting. We’re going to hand-cut the next forty tons. Muscle, breath, repetition. I don’t know if we can cut even a ton at a time, but whatever size we can cut, we stack it up on the sled.”

He looked to Mira. Then Tom. Then Pacha. Then Ellie.

“You swing until your grip gives out. Then you rotate. Fix a filter. Feed the cavies. Make a shipping container for the ice. Then back to the adze. We keep going.”

No one moved.

“We’ll burn a lot of calories. So we eat more. I’m not rationing. If we run short, we wait on Pacha’s harvest. That’s the gamble. We move forward. If we succeed, we launch on time. If we fail, we lose the window. We launch a partial load, or we launch late. But we don’t stop.”

He met their eyes again.

“I’m not asking for volunteers. There’s no committee. This is what we will do.”

Silence. He looked around at a team, people who knew what they were doing.

When Jakob got to the ice field, he looked around.

Pacha had used a pole to chip out the final cut on the seventh block. His arms hung heavy, shoulders low, but the job was done. Tom and Mira were guiding the ice onto the sled with ropes and winches, careful, methodical. Jakob watched them work.

Ellie was starting an adze cut. Lots of chips, but chips are also ice. She was cutting a long line. They would break off one-ton chunks later.

They weren’t moving fast. But they were moving. And that was enough for now.

Tom’s voice came over Jakob’s radio, a direct call. “Jakob, we still need to calibrate the launcher. I need your help.”

Jakob started walking out to the Mass driver. He caught up with Tom, who was carrying a few boulders on a sled.

Tom began speaking. “I need to walk you through something.” He pulled up a slate layered with elliptical arcs, transfer windows, and annotated times. “How do you see this project right now?”

Jakob frowned. “Twelve sols until Day Zero. First launch. Then one shipment per week for ten weeks. Then we wait for the next window.”

Tom shook his head. "That's the wrong perspective." He turned the slate. "Ceres doesn't care about launch dates. They care about delivery dates. If we want a shipment to land there every week, we have to hit a moving target."

Lines flickered across the screen: curved orbits from Mars to Ceres. Tom tapped the first. "Day Zero is the fastest transfer. But to get them a shipment one week later, we can't launch one week later. Ceres has moved. The second shipment takes two days longer to arrive, so it launches five days after the first. Then three days later. The curve compresses."

Jakob leaned closer. "We're overlapping launches?"

"It inverts," Tom said. "Week eight ships before week seven. Week nine before week six. The curve bends backwards." He changed windows. Four early orbits glowed green. "These launches happen before Day Zero but arrive at Ceres in weeks eleven through fourteen. Four extra shipments. Eighty million we didn't count on."

Jakob stared. "Send ice early; it gets there late."

"Exactly." Tom swiped again. Longer curves swept the far side of the solar system. "And here - if we launch the other direction before blackout, we hit weeks sixty-three and sixty-four. Nine extra deliveries. That cuts the blackout from sixty-one weeks to forty-eight."

They got to the mass driver, and Tom loaded up a boulder. He programmed the machine and asked Jakob to reload every time there was a launch. Tom hiked down range a couple of miles and set up a monitor. The first boulder flew, landing ten miles downrange.

"Hook, two minutes off," Tom called out, adjusting the parameters.

Jakob loaded another. He reflected on Tom's plan. "Some of these blocks will go halfway around the solar system."

"That's how you hit a moving target."

The second boulder launched fast. It also landed about ten miles downrange. "Perfect. Dialed in."

Jakob loaded the third boulder. He looked down at the slate again. This was an opportunity. A narrow one. Exciting, if they had a saw. But now it meant nothing. They would be lucky to make their first shipment. Hitting extra shipments was not possible.

Tom spoke, “Look, Jakob, I know we have a lot of stress right now, and I am not helping with this. I wanted to explain this to you before, back when we were making ice. Now it seems pointless. But I think you should know. It changes all our scheduling.”

The third boulder launched faster. It flew over a hundred miles down range. “Still perfect. We are dialed in,” Tom shouted. He walked back to Jakob, and together they returned to the dome.

“We’re burning out,” Jakob said. “Everyone’s trying, but the adzes are killing us. Shoulders, wrists, lower backs. Another week like this and someone’s going to tear something.”

Tom didn’t argue. He’d seen Mira wrap her wrist at lunch. Ellie’s gait had shifted. Everyone was compensating.

Jakob kicked at a rock. “What if we reforge the adze heads? Hammer them out into axe blades. Something that swings straighter.”

Tom gave it a second. “I can do the work. Might help with bite depth. But it won’t solve the strain. Still heavy tools, still tough ice.”

Jakob nodded. “I know. I just don’t see a good option.”

Tom didn’t answer right away. He kept hauling the empty sled back over uneven rock. Then he said, “We’ve still got the cargo bikes. Both of them. The ones we used for cardio.”

Jakob frowned. “You want us biking across the ice?”

“No. Mount a bike frame on a rig, use the drive wheel to turn a belt. I’d make a smaller saw blade. No tungsten carbide. Not eighteen feet in diameter. Maybe six feet in diameter. The bike spins that. It won’t be fast, but faster than hand-chipping. And we’d shift the strain to legs instead of arms.”

Jakob turned that over. “You think the bikes can handle that kind of load?”

“They’re built for cargo. I’ll weld a stand to a sled to hold the frame steady, and fix a belt to a pulley. Add a guide system for the blade. Maybe a screw mechanism to control cut depth.”

“We’d still wear ourselves out.”

Tom shrugged. “We’d be cutting ice, not cartilage. It’s an upgrade. Overall, it is much less effort, smoother, and uses stronger muscles than cutting ice with the adzes.”

Jakob exhaled through his nose. “Make two. We’ll rotate crews through them.”

“I’ll start tonight.”

Jakob looked at Tom seriously. “Let’s do it.”

Tom looked quizzically at Jakob. “Yeah, I just said I would.”

Jakob shook his head. “No, the other thing. Let’s launch tomorrow. Let’s hit every extra launch date you suggested. We are depending on your bike saws, you realize, but let’s do it.”

Tom’s pace picked up. “Yes. I mean, uh. Yes, I’ll get on it. Mira is good at this stuff. She and I will deliver two bikes by tomorrow.”

Jakob didn’t know if he was a fool or a genius. But right now he felt excited.

When they got to camp, Jakob went in the ship. He pulled on a headset and took a seat at the comms board. The air felt thinner here, despite the pressure seals. He stared through the window, scanning the empty horizon like Knox might somehow materialize from it.

Calling Ceres was always strange, like shouting into a canyon and waiting half an hour for the echo.

“Dr. Knox,” he began, voice steady, measured. “I’ll be direct. We lost our saw. We’re cutting with adzes now. I don’t know how much ice we’ll be able to harvest.”

He paused, then continued.

“But here’s the thing. We’re prepping the shipment for week 14, launching tomorrow. We believe we can still reach weeks 11 through 13 if we push. I’ve attached the full launch schedule and estimated arrivals. It’s aggressive. Maybe too aggressive. But we’re going to try.”

He ended the message and queued the transmission.

The red sky hadn’t changed. The sun burned low on the horizon, indifferent. Dr. Knox didn’t yet know the message had been sent.

When Jakob got down to the icefield, Tom was hard at work. Jakob took Pacha's adze and started cutting. Pacha exhaled, exhausted. He gave Jakob a status update. "Jakob, I gotta tell you, the ag dome's working. I'm washing and breaking down regolith, building soil. I added some carbon powder from the air recycler and some liquid fertilizer I brought up. So sweet. Sprouts are growing. It's like a dream. The covies aren't cramped anymore, and they're actually happy."

Then he paused.

Jakob heard it before it came. "But?"

"I need a greenhouse," Pacha said. "Not just grow lights. Real surface exposure. I'm doing what I can under LEDs, but if you want real food, we need sun-grown plants."

Jakob closed his eyes. "You need space. Light. Water. Air."

"Water's not the hold-up right now," Pacha said, gently. "It's space."

Jakob nodded, but didn't answer. The truth was starting to tighten around him. Water for Ceres. Food for Mars. There was no room for both to be first.

Then his radio crackled. A long delay, followed by the calm, even voice of Dr. Knox, the kind of tone you develop when every decision has consequences.

"Jakob, good to hear from you. I want you to know we see what you're doing. It's dangerous work, and we're grateful. What you're giving us, we can't get any other way."

There was a pause. Knox continued.

"We'd been planning around ten shipments. That's what the window looked like. But if you can really hit fourteen, we'll make it worth your while. I'm not offering charity. We're rationing already. Your ice lets us keep the mines open longer, keep the factories warm, keep people working. It buys us time."

Jakob stood for a moment longer, processing.

Then he started swinging his adze again.

In a few minutes, the comm pinged again.

"Jakob, one more thing," Knox said, almost sheepish. "We've got iron. Thousands of tons of it. Comes out with the molybdenum. But we've got no carbon. We can't

make steel. If you ever stumble across graphite out there, we'd be happy to pay for that, too."

Jakob let out a short laugh, low and sharp.

Iron with no carbon. Ice saws with no motors. Plants with no space.

Everywhere, a piece missing.

Jakob looked over the ice. The others were already swinging. Mira was steady and rhythmic, while Ellie leaned into each strike with the quiet focus of someone burning calories just to stay warm and functional. The blocks glinted under the low sun, their edges still sharp from the adze work. They would pile up about five tons on the sled, now, a mixture of smaller blocks and chips from the adzes. They had processed eighty tons, and the next load was almost ready.

Tom spelled Ellie, and she came over to talk to Jakob.

Jakob was direct. "Pacha needs a big greenhouse."

That stopped her. She exhaled sharply, not quite a laugh. She tilted her head. "You want that printed now?"

Jakob shook his head. "I want it, yeah. But not yet. We can't live on cavies and algae powder. Pacha's right. We need real space for things with roots. He says five acres. That was always the plan. But that plan went out with the motor. For the next ten weeks, we are all ice cutters first. Everything else waits."

Ellie understood. There was nothing to debate. She stepped back, caught her breath, and watched him swing. "I'll start planning it, though. We get through these ten weeks, I will build it fast."

They hauled another sled up to the processor. Only five tons at a time, to keep it from falling off. Three more loads to go. Jakob hoped to get another one loaded before dinner time, but he was going to be strict about no cutting after dinner. They could get two loads tomorrow.

Pacha came back from processing the ice. He called them all to dinner. "It's seven O'Clock according to the ship. Time to eat."

Jakob looked at the load, only half full. Whatever they loaded would be enough.

After dinner, Tom caught Mira. The two of them suited up and went to the ship. Jakob knew why. No one else even thought about it.

Morning came quietly. The warmth inside was a minor miracle, and no one wanted to leave it. Everyone ached from the work the day before.

Pacha got up first. He moved silently past the others, pulling on a light shirt as he made his way to the covies. They chattered softly at him from their enclosure, little noses twitching. He fed them from a bin of pellets, checked their water, then turned to set the table. No one had said it was his job, but it had become one of those things that simply happened because he did it.

Ellie woke slowly, stretching in the warmth of the dome. After six months of cramped ship quarters, the cot and gravity felt luxurious. She stood, rolled her shoulders, felt her body move properly, then went to find her base layer, letting it slide on slowly. Mars was supposed to be harsh, but right now? Right now, it felt like freedom.

Pacha glanced over, then back to his work.

“Morning,” she said, grinning.

“Morning,” he replied, not looking up. “There’s hot water if you want it.”

She sat at the table and sipped. As the others woke,

The others drifted in, rubbing at sore shoulders, flexing their hands. The work left its mark, even if no one complained. Tom came in from outside. The first rig was standing, a cargo bike locked into a welded frame, a belt looped down from the rim to a saw spindle, rough depth screw bolted in. It wasn’t elegant, but it spun.

At breakfast, Mira couldn’t stop. “Guys, we built a pedal-powered saw,” she announced, arms wide. “Not a toy, either. I mean, it’s smaller than the old one. We had to make blades for it. Six foot, not eighteen foot. But that’s still real. You can feel the bite. Smooth cut, no chatter. Way faster than the adze.”

Ellie looked up. “Seriously?”

“Seriously, seriously. No shoulder pain, either. You use your legs. I could ride that thing all day.”

Pacha raised an eyebrow. “That’s a promise?”

Mira grinned. “You want a turn? We’re making a second one today.”

Jakob leaned in. “Is it stable? Will it hold alignment?”

Tom nodded. “Good enough. And if it isn’t, we will fix it.”

The mood shifted. Not giddy, but lighter. Bodies still ached, the math hadn’t changed, but the path forward suddenly seemed a lot better.

Jakob looked around the room at tired faces, calloused hands, too-thin bodies made lean by too much work and not enough food. No one had given up, but this couldn’t last.

They all left together to try it out. Tom popped it up on a sled to get it down there. The sled didn’t roll. Tom just put it on another sled, and they all went down except Mira and Jakob. Mira crouched beside the sled that didn’t roll, her gloved hands steady. Jakob brought her a tarp. Soon, it was dotted with tools. Mallet, wrenches, a grease gun, and a box of spare parts. She didn’t look up. Her focus was on the left front wheel housing, already disassembled and resting on its side like a tired limb.

The sleds were simple machines, flat aluminum frames mounted on twelve aluminum wheels, six to a side, with minimal suspension and no steering. They were built low to the ground, squat and stable, each one rated to carry a hundred tons. Their only concession to technology was the electric assist built into the hubs, small motors connected to an onboard battery, just enough to take the edge off pushing.

“This wheel’s been dragging,” Mira explained to Jakob. “Bearings are shot.”

She tapped the old bearing out with a soft mallet. It dropped onto the tarp with a dry metallic click. The outer ring was scored, and the ball cage looked warped, with too much grit in the housing, too much wear.

“Dust gets in,” she said, mostly to herself. “Regolith fines. Starts grinding everything down. Same as the motor.”

From a foil-lined pouch, she took a fresh bearing. She expected this before they left. She had maybe a hundred in the supply locker. She pressed it into place with slow care, aligning the motor housing before bolting it shut. The work was quiet and methodical, practiced. When she was done, she spun the wheel, then cleaned up.

“These motors aren’t for pulling,” she added, almost offhand. “They’re too weak. They are just there to keep you from wrecking your spine. Light motors mean I could pack more replacements. Looks like we are going to need them.”

By lunch, both rigs were running. The bikes weren't comfortable, and no one rode them for fun, but the saws cut faster. They were cutting one-ton triangular blocks, but these stacked up well. They would load up eight tons at a time. Soon they had finished a hundred tons, and Tom and Jakob loaded it into a fiberglass carrier. They loaded more ice into the processor.

Tom grabbed some food at lunch, then He and Jakob left to load up the first hundred-ton block of ice in the mass driver. Together, they slid the block into place. The box it rode in was shaped like a giant lemon, made out of glass fiber spun from molten regolith. Ellie had made four before the saw disaster. Tom pulled up his slate, checked the programming, and triggered the sequence. The launcher responded with a hum, its alignment shifting minutely. Jakob watched the arc plot stabilize. It didn't launch. The driver would wait for the right microsecond and launch then.

Without another word, they turned back toward camp. Inside the dome, the others were wrapping up lunch. Mira looked up as they stepped through the airlock.

"Launch is set?"

Tom nodded. "Automated. The clock handles the microsecond window. We'll know in an hour if the launch worked."

"Like hell," Ellie grinned, putting on her gear. "I'm going to watch the launch."

Pacha started putting on his gear. "Dudes, no way I am missing this."

The rest followed. Tom and Jakob looked at each other. Tom spoke up. "Jakob, there is no reason to watch. If it doesn't launch correctly, there is nothing we can do. I will program in a new launch tomorrow."

"No, Tom," Jakob answered. "We are not going to fix it if it doesn't work. We are going because we want to watch the inaugural flight of our business. Everything may look kind of hard right now, but we are still launching a hundred tons of ice to Ceres on a launch window we didn't know existed. Let's go."

The team got to the mass driver about ten minutes before the launch. Tom propped his slate up with a rock, showing a countdown timer. They chattered excitedly, the pains from the day's work receding, replaced with excitement and a sense of accomplishment.

They all counted the last ten seconds together. "Ten. Nine. Eight. Seven. Six. Five. Four. Three. Two. O..."

The railgun clicked, and the package started moving before they finished “one.” It was faster than they could believe. Even Tom, who designed it and knew exactly how it worked, was amazed by the launch. They could see a tail of red dust coming up behind where the ice had already passed. The Martian horizon is only two miles away, so they couldn’t watch it go down the entire path, but they did see it when it started across the sky. A glowing red ball, faster than they could believe. Everyone cheered.

The team was excited on the way home. Mira spoke up. “We did it. We actually did it. And the ice saws Tom and I built. Are they great or what?”

Pacha laughed. “They are great, Mira. I still finish exhausted, but not all beaten up as the adzes left me. And they are so much faster. So, does that mean Ellie might have some free time to build for me?”

Ellie thought about it. “Well, we only have two saws. If we are keeping up with the schedule, I will start your greenhouse. If we fall behind, I stop. Deal?”

“¡Trato!” Pacha shouted. “Let’s do it.”

The math had changed. They could load multiple shipments a day. They returned to the ice fields. No time to celebrate. They kept cutting. But if they kept working, they would be able to keep Ceres well stocked.

Their third morning in the dome was quiet but not slow. People moved with purpose. The sleds were loaded. The ice was pure. The mass driver hummed and launched again, without an audience this time. They were actually cutting faster than they needed to. Tom had found a way to send more water than they had ever planned.

For now, it was working.

Chapter 13: Success Breeds Hard Work

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The shutters of *Le Puits Perdu* were drawn against the light, though the Parisian street outside was already beginning to hum with delivery bikes, mopeds, and the low murmur of pedestrians threading through narrow corridors of the sixth arrondissement. Inside, the café was quiet. The tables were unset except one, where two white plates rested on clean linen, and the smell of baking butter and yeast softened the cold stone walls.

Emmanuel, the owner, was somewhere in the back. He had opened early for Catherine Dahl and no one else.

Catherine sat alone, her coat folded neatly beside her. The waiter brought a basket of pain d'épi, still warm, its golden crust flaking under the cloth, and a terrine of whipped salted butter. Beside it, a small plate of Saint-Marcellin, a wedge of Ossau-Iraty, and two slices of a dense Alpine tomme. The second plate held a small galette filled with caramelized onion and mushroom, topped with soft goat cheese and a single runny egg. Her espresso was untouched.

When Max Juno arrived, the door opened without ceremony. He stepped inside, blinking once at the dim light, then smiled as he spotted her.

"You look rested," he said.

"I fake it well." She gestured to the empty seat. "Come in. It's good to finally meet you."

Max slid into the chair opposite her. The waiter appeared without prompting, placing a fresh plate for him, along with his own galette, tomato, melted comté, and thin slices of dried lamb shoulder.

"I told Emmanuel you eat what's in season. If you'd like something else, tell them."

The waiter returned again, placing a small dish of fig preserves, a bowl of clementine segments in thyme syrup, and a second espresso.

Catherine reached into her folio and placed an envelope on the table. She slid it forward without commentary.

Max glanced at it but didn't pick it up. "That for me?"

"Four hundred twenty million," she said. "That's more than the agreed price. Payment in full for the first fourteen shipments, with a fifty percent bonus for delivering full loads all fourteen weeks."

He reached for the envelope now, weighed it briefly, and tucked it inside his coat. He never dreamed of this much money, even when Jakob had explained it.

"You didn't think we'd pull it off."

"I didn't think you'd go to Mars," she said. "When I made the offer, I thought Jakob would try to source from Earth. I knew there was no way he could move that much water from here. He never mentioned Mars. He's full of surprises."

Max smiled faintly. "He is."

"Still," she said, slicing a piece of bread, "he did it. That water's not just for drinking. It's in our smelters, our reactors, our cooling cycles. It's Hydrolox for our ships. The algae tanks are running full tilt, and the team is taking showers. What you delivered is the reason my project made it through its first phase."

Max spread a bit of fig preserve on a corner of bread, took a bite, and nodded. "Then we're in business."

Catherine wiped her fingers on a napkin and pulled out a second envelope. Thinner than the first. She placed it beside her plate.

"This one isn't a check. It's a proposal. New contract. Triple the water. Same quality, same schedule. I want to add one hundred tons of refined carbon per week. And food."

Max raised an eyebrow.

"Meals," she clarified. "For my crew. Prepared. They are to include meat, vegetables, and bread. What they are eating is horrid. I tried it. I want the miners to remember what a meal feels like."

He looked at the envelope but didn't open it.

"You've got production running. Why upscale so fast?"

"Because Raines wants more. Because demand from Earth isn't slowing. I am sending another sixty miners, ten replacements, and fifty new headcount, bringing the total up to a hundred. What we've built on Ceres works, but only just. We sent people up there to dig. Jakob's team is living. Mine is surviving."

Max leaned back slightly. "You know what this would cost you from Earth."

"Which is why I'm not sourcing from Earth. You have a working supply chain. The margin is yours."

He took a long sip of coffee, savoring it. "I'll talk to Jakob. No guarantees."

"I wouldn't trust you if you offered one. See what Jakob can do."

They ate quietly for a few minutes, the soft scrape of cutlery and faint clatter from the kitchen the only sounds. The bread was still warm. The clementines were perfectly balanced, acidic, sweet, and herb-laced.

After a while, Catherine added, almost as an afterthought, "Tell him thank you."

Max looked up.

"Not for the water," she said. "For listening that night. For letting me feel like someone out there actually cares and wants to help."

He smiled. "Jakob is remarkable. When he first presented his plan, I almost dismissed him, but I saw his determination and unique perspective. He is inventive. He never ceases to impress me."

When Catherine was done, they said their goodbyes, and Max stayed. Not a lot of people came down to Lagos, and Max liked the idea of investing in other ventures. He had made some calls and set up a few meetings while he was up in Paris.

Some of the people he met, he didn't trust. Some other ideas were good, but didn't speak to him, per se. Max was focused on space. His time on the Moon had left that hunger in him.

By midmorning, the café had opened to the public. Max lingered at a side table near the rear window. Emmanuel, out of sight but never far, kept the space open for him between bookings.

One fellow impressed him. An Australian arrived without fanfare. Broad across the shoulders, wind-chapped, and moving with the careful assurance of someone accustomed to uncertain footing, he stepped inside and paused just long enough to scan the room. He spotted Max, gave a brief nod, and crossed over.

“Callum Haldane,” he said. His voice carried a dry, gravelly cadence, shaped more by red dust and engine noise than formal instruction. He set down a narrow duffel and offered a calloused hand.

Max shook it. “Glad to meet you.”

The waiter arrived without being summoned and poured coffee for them both. Callum watched in silence, waited until the cups were settled and the waiter had retreated, then leaned back slightly and took stock of the room.

“I’ve been watching what Raines is buying,” he said. “He’s driving up the price of beryllium, trying to stockpile for those tokaplex builds. Ceres gives him tungsten, tantalum, molybdenum, vanadium, all the big refractory metals. But beryllium’s not there in ore form. Too diffuse. No concentration.”

He paused for a sip of coffee. No reaction to the taste.

“That’s what got me thinking about Mars. Volcanic history, past water flow. You get both, you might get proper ore bodies. So I pulled old Martian data. Started combing through rover sweeps. Found a crater about sixty klicks east of Olympus Mons. One of the polar satellites passed over on a low-angle sweep. I found a density anomaly. Off-pattern. I ran the spectrum against Earth-side samples. Closest match was beryllium.”

Max said nothing. He took a slow sip, watching Callum over the rim of his cup.

Callum continued. “I’ve got three others ready to sign on. Crew from Western Australia. We’ve worked lithium fields, deep copper. We know how to run a mining site. What we don’t have is infrastructure.”

Max nodded slightly. “What are you missing?”

“Transport. Shelter. Food, water, air, heat, sewage, the full stack. We’ve got rigs, partial automation, but we’re not building from bare regolith. You get us to the site and keep us alive long enough to dig, we’ll pull ore.”

He glanced at Max, then added, “And we both know it won’t sit on the shelf. Raines has a standing buy order for every ounce of beryllium he can find. Those tokaplexes aren’t made of dreams and kind words.”

Max rested a hand on the table. “I think we may be able to help you with that.”

Callum nodded once, almost to himself. “You drop Mars-pure beryllium in Raines’ lap, and he’ll crawl over his own funding board to get more.”

Max leaned back slightly. “I can fund that. I can put you where you need to be. I want forty percent. What do you think?”

Callum’s expression didn’t change, but he took a moment before answering.

“Forty’s heavy,” he said. “But you’re taking the risk up front. If you’re covering transport, infrastructure, and coordination, I can live with that. But once we’re operational, the terms stay fixed. You don’t touch the technical, and we don’t chase investors.”

Max gave a slow nod. “Not exactly. I’ll buy equipment and pay your early expenses, but you’ll pay those expenses as soon as you start making money.”

They sat for a moment, the soft sounds of the café rising around them. Max stirred his coffee once, then looked out toward the street.

Eventually, Max nodded. “There is a team on Mars. Magazine Beach. They can get you housing. They are on the South Pole now, but they need to move to a temperate location. They might be happy to move near Olympus Mons. They’ll want to be paid. They’ll also need mineral mines for fertilizer. Could your people help locate that?”

“If we get dug in, yes,” Callum said. “We’re not agronomists, but we know how to find what’s buried.”

“You’ll need to handle smelting and casting yourselves.”

Callum gave a small nod. “We’ve got someone lined up. Not fully industrial, but enough to get clean metal. We’re not planning to ship raw ore.”

“The South Pole team can take care of getting it to Earth.”

Callum rose, adjusted the strap, and met Max’s eye.

“If you’re in,” he said, “we’ll be ready.”

Then he turned and left, without looking back.

Max stayed seated a few minutes longer after Callum left. The maps were still in the duffel. Nothing on the table but a cooling cup of coffee and a few grains of sugar near the saucer's edge.

Forty percent. Housing, logistics, launch. If the site held, it would be worth ten times that. If it didn't, well, they'd burned fuel on worse guesses. And they could help Magazine Beach move.

Outside, the sky was quiet. Dust curled just above the regolith, lifted, then dropped by cold crosswinds.

Max sat with a new cup of coffee and opened the second envelope Catherine had left that morning.

It was a full draft contract. The figures were typed cleanly in the margin.

Max set the paper down, took a slow breath, and looked out the window. The city moved with its usual rhythms. Nothing was scarce here.

But out there, between three other worlds, scarcity defined everything. And now, just barely, a few people had begun to price survival like a market.

It was time to speak with Jakob. He keyed in the transmission and let the recorder settle. The slate light blinked once. Then he spoke.

"Jakob. Pacha. I'm sending this from Paris. Just a short update to begin with..."

On Mars, the dome lighting was fixed at midday levels, but Jakob hadn't adjusted. He and Pacha sat at the main table in the fabrication wing, both with slates open, half-finished reports abandoned in the corner of the screen. A tone sounded, low and deliberate. Message in queue.

Jakob pulled it up and scanned the header.

"It's Max. He's in Paris today," he said.

Pacha pushed his tablet aside. Jakob tapped play.

Max's image appeared, seated alone at a narrow café table. He looked rested, measured, maybe even a little too calm. The message played.

"Jakob. Pacha. I'm sending this from Paris. Just a short update to begin with. Dahl was pleased. Ceres got more than they expected, and the ice was cleaner than they hoped. She gave a fifty percent bonus. She wants more. Three times more ice, but

also carbon and food this time too. Processed, sealed, frozen. She is impressed with your agriculture and wants to feed her team better. I attached the contract.”

Another pause.

“I also talked to Tan. She wants ice and food as well. She is willing to pay what Dahl is paying. She has less need for water, but more need for food. She runs a cantina there and is feeding a lot of Lunans. Right now, she is feeding them Earth food, but believes Martian food would be a real hit.”

Pacha sat forward when he mentioned food. Jakob said nothing, but his eyes narrowed slightly at Tan’s name.

Before it ended, Jakob was already flagging Tom, Ellie, and Mira. They needed to hear it. Jakob replayed the message when they were all there.

Max continued, his voice lower and more deliberate. “What I’m about to propose wasn’t on the table when we last spoke. But the opportunity came fast, and I think it’s real.”

He glanced down, as if reviewing something. Then back to the camera.

“A miner named Callum Haldane came to see me this morning. Australian. Solid. His team’s been analyzing Raines’ buy orders and looking for new sources of beryllium. According to some old rover data and old orbital telemetry, they found a crater about sixty kilometers east of Olympus Mons with a high-density anomaly. Spectrum scan lines up with beryllium signatures.”

Another pause.

“They’re miners. Not settlers. They’ve got the tools, partial automation, and one smelter tech. But they have no life support, no base. They need housing, power, food, and water. They need someone to keep them alive long enough to get the ore out.”

Max leaned slightly toward the camera.

“I told them I could fund it and coordinate logistics. But that only works if you’re willing to establish the site. You’d build it for them, not just for you. If it goes ahead, they’ll arrive with a nuclear thermal ship. Once they land, the engine becomes a reactor. You’d have more base power than you’ve had since the landing day. Enough to expand food, keep lights on, and maybe even grow.”

He let the weight of that settle.

“They’ll refine the metal on site. If you can get it to Earth, we can sell it. Raines needs it.”

He looked straight into the lens.

“I know this complicates your move. But Olympus might be the right place to build for real. They can help you move if you can take care of them. Think about what you would need.”

When the message ended, there was a brief silence. Then Tom gave a short, dry cheer. “Bonus! No one said anything about a bonus. And now three times the water? Sweet!”

Mira tore up some paper to make confetti and threw it in the air. “Four hundred twenty million. We’re all rich. And next year. Another three billion. Yes!. She began doing cartwheels, alarming the covies.

Ellie just fell on the floor, half crying.

Tom and Pacha sat thinking. This was a lot of work for them. The money was on Earth, not here. They both started making lists of equipment they would need. They hoped the Aussies could bring it.

Jakob was all business, maintaining his calm composure. But his mouth didn’t get the message and was grinning of its own accord.

Jakob let them have their moment. Then, when Ellie sat up and wiped her eyes, when Mira stopped bouncing, and Pacha looked up from his notes, he leaned forward. “We need to talk about what this actually means. Four months more here, then we relocate north. If we don’t move above the Antarctic Circle before winter, we’ll die. No sunlight means no solar power means no oxygen.”

The team sobered. Pacha nodded slowly. “And the food scaling. Feeding the four miners is just more of what we’re doing. But a hundred people on Ceres? That’s industrial agriculture. Different greenhouse designs, different water chemistry, and completely different waste management. We’d need to figure out how to even ship it without it spoiling.”

“One step at a time,” Jakob said. “First, we tell Max we’re in. We can handle the water and carbon. The miners bring the reactor, and we build them a base at Olympus Mons. Food production, we’ll scale up as we figure it out.”

Tom was already sketching on his tablet. “If the miners arrive first, that actually helps us. We can use their ship’s reactor to power the move, build the new base with real power instead of scraping by on solar.”

“Exactly.” Jakob pulled up the recording interface. “Let’s tell Max what equipment we will need. By the time the Aussies arrive, we’ll be ready.”

Back in Paris, Max was packing up. He thought about the other groups he had talked to that day: a chemical engineer ready to build a chemical industry on Mars, a hotelier ready to provide hospitality for tourists and Cereans on vacation, and a lunar helium miner who thought Mars would offer easier living than the Moon and still be profitable. All good pitches, but not tonight.

He thought about what made Mars special.

Earth had power. The Moon had money. Ceres had metals.
Mars had a business plan.

Chapter 14: Moving Day

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Mira and Pacha had been out cutting ice all morning.

Jakob had pulled up insolation charts. “Tom, looking at our current solar power output, do you believe we can still make the trek north on our own?”

“It’s a race against decreasing sunlit hours and how far we can travel in a day. By delaying until now, the decision to wait has been made.”

Jakob knew that when he had not pushed back during the call from Paris, there was no other possible outcome.

Ellie sighed. “Yes, we need to go north, but not to Olympus Mons. The ice is at Planum Boreum. We need the water, and, to be frank, so does Callum. He just doesn’t know it.”

“When we first arrived, we already knew we needed a way to get to Planum Boreum for this season,” Jakob reminded them. “Callum is that way. We stop a month or so where he is, get his team operational, then he will take us north to Planum Boreum, and we will be back in business sending ice to Ceres. It’s a shorter flight for Callum, too, so he can supply his own camp.”

“As long as no one gets attached to the miner’s site,” Ellie cautioned. “We have to have water. We have to move on.”

Jakob looked back at Ellie. “OK. Now, what I need you to do is get ready for the move. And be excited about it. Agreed?”

As Ellie nodded, the airlock began cycling, and Mira and Pacha came in.

Mira was excited. “Tom, we dropped ten blocks in the cycler. Another launch is ready.”

“Good. I’ll slow launch that load tomorrow. That will give us one more week before we need to resume shipments.”

Callum hadn't arrived in ten hours or twenty. The hope of a daylight landing was gone. When it got dark, Jakob climbed atop the ag dome and began slowly waving a beacon light northward, scanning the horizon. After a few hours, he saw a new light near the horizon, too bright to be a star. Jakob froze, afraid to move in case the light was some trick of dust or fatigue.

The radio crackled a moment later. "G'day, mates. Callum's my name."

Jakob's shoulders relaxed. The calm message erased a mountain of worries. As the gyroplane aligned with the makeshift runway, its huge coaxial rotors set it down softly. Callum unlatched his suit from the ship's oxygen and electrical feeds and stepped out. Tom stepped forward, wordlessly hooking up the hydrolox lines to refuel.

"Welcome to Magazine Beach!" Mira called out, her voice warm and inviting. "Come on in."

They all went through the airlock into the ag dome. When Callum removed his helmet, he was hit by the sudden, thick scent of damp soil, growing things, and the musk of the covies. It felt earthy and alive.

"My crew's livin' in our ship for the last four months. Rank doesn't even cover it. Your place smells sweet and full of life. Is this what you're gonna put together for us?" Callum asked.

Ellie explained her priorities. "More or less. When I get up there, building an ag dome is my first priority. That lets us raise livestock and gives Pacha a place to start crops and mess with the dirt. We will all sleep there until food production is stable."

Pacha gave him a tour, showing him the plant beds and trays under light. Callum leaned closer, smiling when he saw the covies in their pen. After months of algae and supplements, the sight of anything fresh made his mouth water.

"So you're Pacha," Callum whispered. "Max sent me these, said to hand 'em over quietly." Callum pulled out two baggies of soil, one dark, one more multicolored. "One is Andean, one is Himalayan, but they both just look like dirt to me. What's the go with 'em?"

Pacha moved between Callum and the rest of the crew. "Thanks, man. Keep it down. The rest of the team doesn't need to know. My legal mycorrhiza cultures are failing, hurting crop yields. I need a healthy soil biome. The robust, wild fungi and

bacteria in the dark Andean soil and the more colorful Tibetan soil are promising for our low-pressure environment. But, you know, we aren't supposed to do it this way. I've just run out of options."

Callum gave him a wink, and no more was said. Pacha put the soil away in a drawer, and they rejoined the rest.

Jakob was sitting at the table. "Hey Callum, you must be famished after that flight. Pacha is a first-rate cook. Let's have some dinner."

Callum took a seat. "Nothing could sound better. I'm starved."

Mira set the plates and cups of water. Pacha first served up a simple salad of microgreens with freshly baked dinner rolls.

Callum devoured it. "It's been too long since I ate real food. These rolls are amazing. They have an earthy flavor I love, and they are actually chewy. After months of paste, it is great to have something to sink my teeth into. What are they?"

"Amaranth is our staple here. It grows well at low pressure, even without mycorrhiza, and is more nutritious than other grains. I'm glad you like it, because you'll be eating a lot of it."

Pacha then served the main course, roast cavy with sliced oca. The aroma of roast meat enveloped Callum. It was mouthwatering. He picked it up and began eating, letting the fats roll down his chin. Then he looked around at the rest of them, using knives and forks. "Sorry, mates. Reckon I got a bit carried away. That meat is delicious."

Pacha was amused and threw Callum a napkin. "I'm glad you like it. It's the only sort of meat we have here, but we have it with every meal. How's the oca?"

"Is that what these little potato things are? Yeah, like lemon zest. Their crunch goes great with the meat. My boys are going to love this food."

After dinner, Callum excused himself to get cleaned up. Mira showed him the restroom.

Pacha smiled. The Magazine Beach crew had always appreciated the meals, but never so viscerally as Callum.

When Callum returned, Jakob brought up the plan. "We'll send Tom and Ellie with you first to get started on a new ag dome."

“That’ll work,” Callum said. “We’d love a dome like this.”

Jakob nodded. “It’s Pacha’s dome, really. Agriculture trumps everything else. We can stay as long as we are not in his way. We need to get everyone under cover before the move from here is final.”

Tom and Ellie began loading the sleds. Ellie also needed a glassmaker for plumbing. Most people looked at a dome and thought the walls were what mattered. Ellie knew the real work was waste management, lighting, power, air, laundry, and running water. Even in the ag dome, that was what counted.

Callum checked the straps himself. “That’s all you need to get started?”

Ellie pulled her visor down. “It’ll do.”

The gyroplane had been designed for this trip. It needed a full tank to make it back. Tom checked the fuel level before disconnecting the hoses. “All filled. Let’s go.”

Ellie stopped to hug the people staying behind before leaving. “Mira, you remember how to clear it if the sewer backs up, right?”

Mira’s eyes were wet with tears. She didn’t answer.

Jakob cleared his throat. “We will keep things running here.”

With that, the three of them got in the gyroplane. They plugged their pressure suits into the gyroplane’s oxygen and electricity. Callum checked everyone’s connections, then started off.

Jakob was fascinated watching the gyroplane ascend. The six-bladed tractor fan was large by Earth standards, twelve feet in diameter, but in Mars’ thin air, it didn’t have enough bite to get the machine rolling. The electric landing gear provided most of the acceleration, with the tractor fan maintaining speed as it lifted off. Then, at fifty feet, the enormous pusher prop extended and unfolded, its thirty-foot radius delivering efficient thrust that carried them steadily north. Jakob watched the bird slip past the horizon, then turned back to the dome.

The gyroplane kept a steady pace. Callum had set the autopilot almost as soon as they leveled out. “We should arrive around three. I need some shuteye.”

After Callum nodded off, Tom followed suit. Ellie stayed up. The surface of Mars didn’t even look black; it just looked like she was travelling in space, and there were no stars below. The stars above were spectacular. Earth was visible, and Jupiter. Ellie felt at peace among the wandering planets. The gyroplane hummed evenly, a

mix of the electric drive whine of the tractor motor and the deeper pulse of the huge pusher prop.

It was past 3 a.m. when Olympus Mons came up on the horizon. She couldn't see it, but it began blotting out the stars. The gyroplane descended in a long glide. Callum woke and retracted the pusher prop. The basal scarp soon blotted out everything forward except a small light on the landing strip. They gently autorotated down with the tractor fan still pulling, landing on the pediment, a mile south of the scarp. Ellie imagined how this would look when it was built. Larger, stronger, better domes. Enormous greenhouses. Mines. It needed a name that fit both the miners and the Magazine Beach crew.

When they got out, she grabbed a shovel and started digging. Twenty minutes later, she stood back to admire the word carved deep into the red soil. "Bounty."

Callum's transit ship stood a hundred yards away, legs planted deep into the regolith. The hull showed streaks of re-entry heating and red dust accumulation. Around the base, the three solid rocket boosters looked like battered iron pillars. Their bottom ten feet were crushed and splayed outward where they had absorbed the final impact.

"That ship is enormous," Tom commented. "It looks beat up."

"Yeah, nah. You blokes ordered some pretty massive gear, and my crew's got extensive mining kit. They assembled her in high Earth orbit. She never felt a lick of Earth's pull, so size wasn't as much of a problem. Landing her was dicey. None of us thought we would live. It was all autopilot. The mass injection rate increased first. We could really feel the thrust increase after months of barely any gravity. Then we started the skip, bringing us down to subsonic. The parachutes came out, getting our speed low. We dropped just below the shield that way, then the candles lit. Those solid rockets have the raw thrust we needed to avoid making a crater. They stuck out, so when they hit, they crumpled, giving us a soft stop."

Callum, Tom, and Ellie were all in the airlock. As soon as the pressure normalized, Callum made a sweeping gesture. "Welcome aboard."

The smell hit them as soon as the inner door opened: lithium grease, warm metal, and the stale tang of long-cycled air. The interior, for its size, was still close-packed, crates strapped into the passageways, personal kits lashed to bulkheads. Tom and Ellie remembered their own trip. The Mars transit is rough. They took off their pressure suits and slept on the floor until morning.

In the morning, Ellie got up first. She nudged Tom, and they began cooking pancakes with some amaranth flour from Magazine Beach. The miners appreciated the break from standard rations. As they finished eating, Ellie spoke. "First task is to build the ag dome. Can you guys dig the foundation for me?"

"We'll get that dug out in no time," Callum replied. "One thing us boys can do is dig."

They suited up after breakfast and began unloading. Tom was already at the hatch, helping unstrap the sleds. The miners hauled down the excavator, a tracked backhoe with a frontloader. They had a slew of other equipment for mining, from jackhammers to sizing screens.

Tom and Ellie hauled out the Sabatier unit for polyethylene production. It was small, capable of a ton an hour, matched neatly to the output of the Bosch carbon powder line. The water inlet was underneath, and had an H_2 pipe coming up from it. Ellie studied it to find the hydrogen bypass. She found the water intake and looked to where the electrolysis section broke off into hydrogen and oxygen. There was the inlet where she could inject hydrogen directly, as long as she kept a couple of gallons of water in the cracker to keep the gas contained.

Ellie started talking to Callum. "This ship was a nuclear thermal rocket," she said. "Hydrogen propellant?"

"That's right. Fastest way out here."

"Making polymer can eat water," Ellie said. "Back at Magazine Beach, we didn't care, because we had plenty of water. Here you are parched. The plan is that I am going to use your hydrogen. Is that OK?"

Callum thought for a moment. "We used most of it. Maybe thirty tons are still in the tanks. Take all you want. It will boil off soon, anyway."

Ellie did the math, then nodded. "That's two hundred tons of polymer. We can make a larger dome and a few greenhouses."

Callum smiled. "That sounds great. There is a hydrogen takeoff in the cabin and another one near the door. Now, see that place over there where Petey is standing? Is that where you want the dome?"

Ellie nodded. "Roughly, yeah. Let me mark it out. I am going to need graded regolith, so please move it over by the sizing screens."

Ellie drew a circle on the ground showing where she wanted the dome, and the excavator came alive, putting a low, electric vibration in the ground. Ellie watched the first cut into the regolith, the loader bucket peeling away the surface in broad, light scoops. Dust slipped through the teeth like dry powder, curling into the thin air before falling back. They fed the first load through the shaker box, separating the dense minerals that block radiation from the light silicates used for glassmaking.

Callum gave a rowdy wave as he headed back to the gyroplane. “Righto, listen up, you lot! Be good to Ellie and don’t go actin’ like a bunch of drongos. We’re in dire need of a decent home. I’m flyin’ south now to grab Pacha and bring him up.”

The miners waved as Callum left, then got to work helping Ellie. They understood processing dirt. Sorting and grading went fast. On the pediment, the regolith was shallow. They pulled up five feet of regolith, then started on the bedrock with drills and jackhammers, digging three more feet to finish the basement.

While they chipped away, Ellie started making polymer with the hydrogen takeoff near the door. That ran by itself while she took the silicate tailings and started up the glassworks. By the time excavation was complete, she was ready to line the basement with glass wool and set up rebar to support the domes.

The miners were fascinated when she pulled out one of the 3D printers and began laying in polymer/dust foam. The foam insulated the basement floor and walls, locking the rebar in place. They were running out of daylight when Ellie switched over to using ninety percent regolith to make the stronger surface walls. She barely got the double wall structure started, sealing in the airlock and the utilities block before dinner time. The rest of the dome would have to wait.

While Ellie was working on the ag dome, Tom was working on getting power. The ship had come in on a small nuclear thermal rocket. The reactor made two hundred megawatts of thermal energy. But for electricity, they used a thermocouple that only managed two hundred kilowatts. That was plenty for running the ship in transit, but not enough for operations on Mars. Tom and Petey had stayed in the ship, rooting through the gear to find the generator.

“Have a squiz at this, mate,” Petey said, grinning as he hauled out a large crate. “They reckon this little beauty can pump out a hundred megawatts of spark if you give it some welly. Got a fair few pipes and bits and bobs with it. Reckon we can piece it together, or are we dreamin’?”

Tom's eyes widened. He opened the crate and started inspecting it. Simple design, built to interface with the General Atomics reactor. He just needed to send supercritical CO₂ through the pipes instead of hydrogen.

The system needed forty tons of liquid CO₂ for the working fluid. Tom pulled out his slate and did the math. Callum's big twin screw compressor could make the CO₂ in two days. He set the pressure relief valve to eighty bar to drain off the nitrogen and other unwanted gases. Tom started the motor, then checked with Petey. "You think we can get this together in two days?"

"Rockets aren't exactly built to be pulled apart, are they? But once we've yanked that nozzle out, I reckon we'll be right as rain. I'll get the jacks and blocks sorted and get stuck in straight away."

Tom helped him set up the blocks and jacks on a couple of sleds. As soon as the nozzle was supported, Petey worked his way up into the nozzle connection and rows of nuts. His pneumatic wrench strained with each one.

Tom was impressed with how quickly he got to work. "Need a hand?"

"Bit of a squeeze in here for the both of us, mate, but if you can worm your way over to the hydrogen tank connection, you can turn the thrust shutoff valve and disconnect the reactor intake pipe. Once that's done, we can get this show on the road."

Tom looked at where he needed to get to and thought about how much better Mira would be at getting there. He squared his shoulders, grabbed a nut driver, and started winding his way in.

By evening, Tom had the tank shut off, and the reactor intake pipe was connected to the turbine return pipe. Petey called out as he got the last bolt off the nozzle. Together, they lowered the bell until it cleared the hot pipe and trucked it out from under the ship. They slid the turbine into the space. With no nozzle, this was no longer a rocket; it was a power plant.

Tom and Ellie ate with the miners on the ship that night.

At dinner, Ellie began updating people on construction. "We have finished the subgrade work. Tomorrow, the dome should go together pretty easily. We have one more night with the smell of lithium grease and recycled air. Tomorrow we'll move into the dome."

Tom's fork stopped mid-motion. "You'll finish tomorrow?"

“Just the structure, and air. Pipes, wires, and furnishing have not begun.”

“Well done. Me, I don’t have good news. Just work. I need a one-acre section of bedrock scraped clean. All the regolith piled up near the ship. And I need a dozen six-inch holes about four feet deep. Does anyone here know how to do that?”

Luca gave a bit of a sheepish grin. “I’m your man for that, then. The regolith’s only about five feet deep in these parts. No worries, mate. I’ll have it knocked over in a couple of hours.”

“Hours? Perfect. That space will be our primary heat sink. The reactor makes the CO₂ hot, but I need something to make it cold.”

The next day went very smoothly. Ellie and Angus worked on the inner dome first, while Luca cleared away regolith and Petey drilled the holes. Tom followed Petey, using a three-foot wrench to twist each heat sink, expanding the diameter and locking them into the bedrock. He set up the spider connections to and from the heat sinks, and piped the return back to the impeller. Tom and Petey worked together to attach the intake of the turbine where the nozzle had been. When everything was in place, Petey slowly opened the valve to feed CO₂ into the turbine until the liquid reached the fill line on the sight glass, about halfway up. Tom opened the intake valve coming from the reactor and hit the starter. The system bogged, trying to move the liquid up into the reactor, but after a minute, there was no liquid, just a hot, supercritical fluid, putting out real power for Bounty.

When Luca came back with the excavator, dome construction sped up. He could keep the hoppers of both printers full as well as dump regolith in the space between the walls. By lunch, the dome was sealed. Ellie started compressing the air inside it and set the recycler to keep the CO₂ under a thousand ppm.

Callum and Pacha arrived just as they were all sitting down to lunch. Pacha brought all the covies into the ship and opened the air valves on their flats.

“Those guys hate the flats. I’ll let them breathe regular air again.”

Ellie looked around the ship. “I am not sure this is any improvement, Pacha, but you do you.”

Angus looked at Callum. “Hey, we finished the dome. Want to check it out?”

Callum looked back at Ellie. “Is that true, lass?”

“We have to wait for air. The recycler is running hard, but the CO₂ level is still too high. When we have enough oxygen in there, a green light will come up on the

utilities block. Give it another hour. Just so you understand, the dome walls are done, but it is not yet plumbed or fully wired.”

Callum smiled. “When it’s ready, let’s unload all Pacha’s stuff to the dome, and you can give me a quick tour before I go.”

After lunch, they brought a couple of sleds over to the gyroplane and started unloading his terra preta, plants, seeds, and two baggies of dirt. When Callum saw those, he looked over at Pacha, but Pacha said nothing.

They brought it all back to the dome, but the light wasn’t green yet.

Ellie took the opportunity to remind them of the rules. “Now, listen. Since it is not plumbed yet, for the next few days, if you need to go to the bathroom, you will need to use the head on the ship. I’ll get the plumbing sorted soon, and then we can use the restrooms in the dome.”

They nodded. Petey looked baffled. “So, if I need to hit the loo in the night, you want me to chuck on my pressure suit, head through the airlock, stroll over to the ship, go through its airlock, pull off my pressure suit, hit the loo, then put the pressure suit back on, head through the ship’s airlock again, come back to the dome, go through the dome’s airlock, take off my pressure suit again, and then finally head to bed? I’d totally miss brekkie doing all that!”

“Yeah, mates, no worries. I’ve got a better idea,” Angus whispered.

When the green light finally came on, they all helped move Pacha’s gear inside. Pacha set his covies loose in the dome and began setting up soil regions for the plants. Everyone moved their own gear, too. They brought cots, blankets, and a few buckets with clothes and personal items.

This time, he was planting a sunflower relative called goldeneye. “It’s good for the soil biome,” he said, looking meaningfully at Callum.

Callum couldn’t stay. After a brief look at the dome, he was back in the air, heading down to pick up Jakob and Mira.

Ellie continued, bringing Pacha downstairs. He was glad to have someone to talk to. “Ellie, I missed you guys. Magazine Beach feels empty, now. Callum landed while I was asleep. I hope I got everything. It was so dark the whole ride up here, so I couldn’t see anything. It was like our transit to Mars. Just stars and blackness. Callum just got in the air and went straight to sleep. I didn’t even have anyone to

talk to. But it was dawn before we got close to Olympus Mons. It is so huge. The scarp is just everything when you come in. Sorry if I am talking too much.”

Ellie laughed. “Well, I’m glad you’re here. What do you think of the vertical garden space? The basement gives us more flexibility.”

“Hey, Ellie, this dome is totally beautiful. Exactly what I need. Thanks,” he gushed. “This wall will be so much easier on my back. Love it.”

Ellie showed him his quarters. “I also set up a private bedroom down here for you, when the upstairs is overrun with plants and livestock.”

“Perf, Ellie. You always were looking out for me. But I gotta know, what about the greenhouses? For five of us, we just needed a few acres, but if I am feeding hundreds, which sounds like the plan, I am going to need a lot more space.”

“Yes. I am aware. I’ve laid out plans for a greenhouse district east of here, a ten-by-ten grid of ten-acre structures. We aren’t planning to build them all right away. One at a time. But we’ll get you all the space you need.”

Pacha looked at the plans on Ellie’s slate. “That’s cool. That is space for crops, pastures for covies, and land where I can grow moss and lichen just to build soil. Fantastic.”

That night, dinner included cavy meat. This was the first meat the miners had had since they left Earth. “Cheers, Magazine Beach! Meat, vegetables, and bread. I haven’t eaten this well since we left Earth!” Petey called out, to a round of cheers from the other miners.

Pacha did a fake bow. The teams were all exhausted. After dinner, they talked a little before turning in.

In the middle of the night, Ellie woke to the sound of water spraying onto metal. It made no sense. She hadn’t attached a water supply to the dome yet; it was completely dry. She sat up, alarmed. Then the pungent odor hit her. “Miners are going to be interesting to live with,” she thought, and went back to sleep.

A few hours later, Ellie heard the sound again. “Miners,” Ellie thought and rolled her eyes. As she listened, the footsteps came back over by where she was sleeping. “Pacha. What are you doing?” she whispered. “I expect that from the miners, but not you!”

“Ellie, it’s so convenient,” Pacha replied, sheepishly. “I was about to suit up when I realized they’re right.”

Ellie didn't fight it, but when she woke the next morning, she pulled him aside. "I am not sleeping in a room that smells like an outhouse. You and I are completing the sewage system now."

Pacha nodded without argument. They gathered their tools and began the work. Ellie showed him how to heat the glass stock and draw it over the mandrels to make the tubes they needed. Once he understood the technique, she let him handle most of the forming while she assembled the anaerobic digester, the photobioreactor where they grew algae, and the distiller that gave them back pure water. Pacha used the dry minerals left by distillation as soil amendments. The system they built was four times larger than what Ellie had built at Magazine Beach, with capacity for twenty people. With Pacha's help, the ag dome bathroom was operational by noon.

That same morning, Angus got on their cargo trike and rode out to the southwest slope and then up to the shield. He set an anchor ten yards back from the edge and tied himself off, then set another one a yard away and laid a polymer trough over the edge. Angus looked out, almost three miles above Bounty.

Petey was watching him through binoculars. Local Band couldn't reach that far, so they were on FM. "Angus, Petey here. We see you on the edge. Are you tied off? Over."

Angus pulled the jacketed spectra line from the cargo bike. "Tied off and solid, Petey. The view's a ripper. Volcanic chaos to flat sand. I'm ready to send down the cable. You lot ready? Over."

Angus started feeding the cable down slowly, then realized he could never hold the weight. After a couple of hundred yards, it pulled out of his hands. He called out, "Cable released. It's running! Take cover! Over."

The last two hundred yards hit the talus apron hard. Petey and Luca anchored the base. Angus attached the electric seat and took the first ride down the cable. After that, they could move easily up and down the scarp. They spent most of the day at the mine site, coming back down before dinner with rock chips and core samples from the mining site. The miners were talking to each other.

"Looks like that vent shrivelled while it was still pulling out beryllium. We have both Bertrandite and Phenakite."

"Two minerals? That'll complicate the processing. Did ya give the gangue an assay, then?"

“Bang-on, mate. I checked that first thing. A few iron-rich bits and some feldspar. The feldspar will need to be knocked out chemically.”

“No wazza. Callum was dead set on that during the trip, wasn’t he? Reckon he’ll be wrapped.”

During dinner, Petey was talking to Tom. “Look, mate, you’ve got all this power now. I need to start making steel bits. You mind if I wind up the power? I have an FFC Cambridge refiner that can drink the whole hundred megawatts if I let her, and we need the steel.”

“Sure, knock yourself out. We aren’t doing that much right now. Just the glassworks and the Sabatier machine.”

To make a cableway, they needed metal, both for the towers and the cable. They had planned for that and set up their FFC Cambridge refiner, using a magnet to pull iron ore from the regolith. After an hour, they had a large pot of sponge iron. They alloyed in carbon from the Bosch recyclers and made simple steel.

Ellie’s next concern was equipment maintenance. Cleaning was impossible in the wind, on the regolith that was causing trouble in the first place. Electrostatic attraction meant there was always more dust coming in. She knew they needed a spacious building to use as a garage. She planned to add more later, but the first one was built toward the cableway. No airlocks, just oversized doors in and out. It had thin walls supported by glass rebar. The garage gave them a place to work, sheltered from the sun and dust, well lit, with a clean floor and tools handy. The digger and rovers could be kept there when not in use, out of the dust and sun. She moved three sleds in, too.

With the garage in place, Ellie started defining Bounty. All traffic would go in and out through the garage. She began marking the borders of the settlement. She used a pole to draw a line around five acres. The ag dome was the center. There was also space for apartment domes, the sewer system, a metal shop, and glassworks. The ship and generator were a few hundred yards outside the border, and the Sabatier machine was kept still farther away because Ellie deemed it too dangerous. She then built a small wall along the boundary, a foot high and a couple of inches wide. Not straight, it looked like corrugated tin on edge. This made it stiffer and allowed her to save material. When the entire wall had been built, she got Tom and Pacha to help her pull out rolls of *LDPE* sheeting. They used tape from the ship to quickly assemble a bubble cover while she anchored it on the wall using fitted caps.

When it was ready, she called everyone over to watch. “Are you guys ready to see Bounty become your home?” she asked, as she hit the air pump switch.

Slowly, the bubble inflated. No real pressure, less than a millibar, but enough to give it shape. They would still wear pressure suits inside, but there was now a space with no dust. Pacha cheered, and Tom gave a thumbs up. The miners loved their home.

Polymer accumulated slowly. Pacha understood that, but he didn't like it. When they had enough for the base, he and Ellie went down to where the greenhouses would go. They were not as fast moving dirt as the miners, but they excavated the first greenhouse site down to bedrock. Ellie laid down an inch of foam and set the glass rebar. Pacha started the hard work of converting regolith into soil. Using the miners' ten-ton ball mill, he added five tons of regolith and one ton of precious ice. He pumped up the pressure to three psi and began tumbling. As the ice tumbled, it melted and washed out the perchlorates and rounded the grains of regolith into something more like sand and clay. As it processed, he would regularly draw out half the water and vacuum distill it to leave the perchlorate dust behind, recycling the clean water back into the mill. When the water ran clean, he pulled out the sand and laid it in the middle of the base. The Herculean task was going to take several days, but he didn't stop until Ellie came by and told him it was dinner time.

He and Ellie set to work preparing dinner. Jakob and Mira should have arrived by now. Radio would only give them a five or ten-minute notice of their arrival. Outside, the temperature in the covered yard was dropping to the night's cold, but inside, the smells of the night's meal were intoxicating. The miners had never eaten oca. Pacha was roasting it with herbs and algal oil, and it smelled intoxicating. Just before dinner began, Callum arrived with Jakob and Mira, bringing one ton of frozen food, another ton of ice, and a small pressure box. Before anyone even said hi, Mira blurted out in a singsong, mocking tone, "Hey, Pacha, you forgot one of your cavies."

Pacha was beside himself with guilt and quickly took charge of the animal, soothing it and giving it some amaranth leaves to nibble, then returning it to the herd.

Mira and Jakob were pulling off their gear, hanging up their outer suits and pressure suits. Mira sighed when she got her thermal regulator off. As she began removing the base layer, Ellie jumped up and nudged her. All the miners were staring. "Uh, Mira, we have guests. Just step behind that screen." Mira and Jakob went behind a changing screen to change into soft clothes and slippers. It was the first privacy they had had since the transit to Mars. It felt awkward.

Callum entered. He had gone back to the ship to get a bottle. It was nothing special, but it loosened the dust in their throats.

Jakob began. "It was bittersweet closing up Magazine Beach. Mira and I stacked four hundred tons of ice in the ice shed. We couldn't fit another snowball in there. When we were packing, Mira heard a squeak, and that's how she found the missing cavy. After that, we turned the place upside down to make sure there weren't any more."

Mira cut in. "Pacha, we were thorough. Really, it was just the one."

Pacha smiled, though he was still upset he had left one.

Jakob continued, "We pulled out all the frozen meat and vegetables. I could not believe it added up to a full ton. We also got a ton of ice and waited for Callum. I left the hydrolox generator running for future trips. The days are getting short, so it might take two or three days to generate a full load, but it will let us go down again to get ice or pick up anything we left behind."

Callum nodded. "Yeah, we need more ice. No question. The thing is, it takes all day. I hope to get in another three trips, though. We really need the water."

Jakob nodded. "Thanks, Callum. I'd like to pull a lot more than that, but the trek is too long for that." He continued with his story. "It felt bittersweet turning off the recycler and the heater. Once we lifted off, I looked down, watching the words 'Magazine Beach' fade in the distance as we headed north."

Mira jumped in, excited. "You guys flew at night. Jakob and I flew during the day. Mars isn't as interesting or varied as Earth, but it was still fascinating to watch the features go by. And coming up to Olympus Mons was incredible. It reaches out forever and up to space."

Jakob finished his story. "When we came in, I saw the bubble and their enormous ship, so I knew where we were. But, as we landed, I saw the sign 'Bounty.' That touched me. We walked in through the garage, where everything was so tidy and organized, then came into the bubble and saw your huge ag dome. This is really something, Ellie. It's going to be hard to go to Planum Boreum."

Ellie nodded. She had built exactly what was needed. Leaving would be hard.

It had been a long time since the Magazine Beach crew all sat together. Ellie brought everyone up to speed. "We have a lot to build. We have food to last for maybe three months. We can extend that with algae, but, as always, we need to get Pacha what he needs."

Jakob nodded. "Of course. Food is always number one."

Ellie paused a moment, then continued, “I would say maybe, because we have another shortage. Water. The plan has always been that we need to move up and set up a second ice camp on Planum Boreum. I am not sure how long we can stay before we move.”

Tom sat with a slate propped against his tray. “We slow-shipped six months’ supply on the old contract. If we don’t have new ice established in those six months, there will be a crisis on Ceres. Planum Boreum has a huge ice cap and is closer than the Magazine Beach, but it is still a five-hour flight. We have time to work this out, but we need to use that time to establish a new ice supply.”

Ellie’s face clouded. “I need to build them an ice shed before we leave.” She turned away before they could see her eyes wet.

“Hey, mates,” Callum said. “You lot looking for water? Surface ice is nice and all, but as an old miner, I can tell you Mars has loads of buried ice. The Medusae Fossae Formation’s got as much as you could ever want. Problem is, it’s a fair bit south of here.”

Jakob sat up. He knew about the ice caps. Not this.

“Yeah, it can be miles thick. I have shallow radar. We can fly around with it and see if there are any good deposits.”

“That sounds like a lot of ice. I didn’t see any coming in.”

“Naw, it’s not on the surface. You need to dig down a ways. It’s like permafrost. I bet we can find some rich deposits. You game?”

Ellie started. “Jakob, this sounds promising. I told you guys not to get attached, but from the time I named this place Bounty, I have not wanted to leave.”

Jakob responded, “Look, we have time. Tom said we have six months. We can try some ideas. This ice sounds like a totally different beast, though. I understand saws. I don’t understand underground permafrost ice. And our sleds are meant to go a mile or two in a day. I don’t know if this can work, but I’ll check it out. Callum, can you help me scout out these deposits?”

“Nothing easier. Let’s get in the gyro at dawn and take a look, see if we find some.”

Jakob was satisfied they had a direction.

Callum tore a piece of bread and dipped it into the stew. He was eating real food again. Life was grand. Max had set him up with this crew that kept him fed and housed. But now he understood he would also need to work to help them.

For a moment, no one spoke. They could hear the soft hiss of the recycler in the corner, the low murmur of water moving through the new plumbing, and quiet chewing.

Chapter 15: Martian Industry

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Jakob was up before dawn. “You ready, Callum?”

“I reckon we can take a swing down and find you some ice,” Callum said.

Jakob did not need convincing. They went to the runway and climbed into the gyroplane, ready to check every ridge and hollow on their route south. Callum showed Jakob how to use the shallow radar.

About twenty miles out, Jakob leaned toward the screen. “I found ice,” he said, unable to hide his excitement.

Callum studied the return. “You did, but it is barely anything. We should keep going.”

They crossed a series of shallow basins. Jakob picked up several more signals, each one thin and scattered. With Callum’s guidance, he learned to distinguish lava tubes, dense ore, and ice. He found a few more deposits, but Callum waved them off as insignificant.

Then, a hundred twenty miles out, the display lost its structure. The basin filled edge to edge, every depth reporting the same return. Jakob leaned closer to the screen. “Callum, take a look. It’s saturated. It’s ice all the way down.”

“That is the good stuff, mate. Looks miles thick. Plenty for what you want.”

“Can we set down?” Jakob asked.

“If you like,” Callum said, setting the gyrocopter down on the plain.

Jakob grabbed a shovel from the cargo space and started digging. “I want to see how clean it is.”

After several shovelfuls of dry regolith, Jakob grew frustrated. “Hey, this is dry as dust. There’s no water here. The reading showed solid ice.”

Callum laughed. “When the radar gets saturated like that, you know there’s a lot of ice, but you can’t tell how deep it starts. Could be thousands of feet down. You can’t reach it like that. Petey and I will come back later and do a proper reading.”

Jakob’s shoulders slumped slightly as he realized his shovel would never reach that ice. He put the shovel away, and they started the flight back.

When they returned, Ellie was bent over a block of carbon, checking measurements on her slate. The generator’s hum filled the air. She did not look up until their boots crunched across the dirt.

“Did you boys find anything?”

“We found water. Callum says it’s an enormous field, beyond the horizon, more than we could ever use.”

Mira had come over, wiping regolith dust from her gloves. “How far?”

“Hundred twenty miles south,” Jakob said. “The radar lit up solid blue.”

Ellie straightened, interest flickering across her tired face. “Fantastic! If we can tap it, we don’t need the polar runs.”

“Right,” Jakob said, already thinking through logistics. “We’ll need to set up an extraction site. Same process as Magazine Beach. We haul it all back here, use the settlement’s power to process it.”

Callum set his helmet down on the workbench with a heavy thunk. “Nah, mate. Won’t work like that.”

Jakob turned. “What do you mean?”

“Magazine Beach was surface ice, yeah? Frozen ice sitting right there. Easy as. This is different. This could be thousands of feet down. Permafrost under pressure. You can’t just saw it.”

“So how do we get it?” Mira asked.

“No worries. My team’s done plenty of in situ salt extraction in Southern Australia. Same process. We can bore that for you once we get our own mine sorted. Petey’s making coring bits for us, and he can make your drilling gear, too. We should be ready to take a look in a couple of weeks.

Tom recognized this was out of their league. “So, Callum, what can our team do?”

“The main thing is you are going to need hot water. You build a way to make hot water, you inject that, it does the melty thing down below, you pump out cold water.”

Jakob felt he misunderstood. “We inject water to get water?”

“Too right. You’re not mining blocks of ice, you’re pumping liquid water. ”

Ellie started thinking about power. “How hot?”

“Hot as you can. Ninety degrees going down. Maybe more. Four degrees coming back up.”

Ellie and Tom sidebarred on heating. “Tom, do you think we should go pull the solar panels from Magazine Beach?”

“I was thinking the same thing, Ellie,” Tom confided. “But I have a better idea. You can make glass sheets. We mirror those with a tiny bit of aluminum. Solar thermal. More efficient and much easier to make.”

“OK, I can make the reflectors. Are we going to bubble the extractor side?”

Tom nodded. “The bubble here on Bounty works great. Everything is so much easier without the dust. It will keep our collectors clean, too. What do we do at night?”

“Stop injecting, drain the collector. Start it up again in the morning. But we will still need a small electric panel to run the pumps.”

“That should be no problem. Callum brought a bunch with him. Mostly deployed up by their mining site, but if they have spares, we can use those. Otherwise, it is a long ride down to Magazine Beach for Mira and me.”

When they rejoined the group, Callum was talking about gas relief. “You get deep, the trapped gas is under huge pressure. CO₂, mostly. Trapped in the ice for millennia. It’ll fight you if you don’t vent it controlled-like. Just a reed valve on top. You want some pressure.”

Jakob felt a sinking realization settle in his gut. This wasn't something his team could improvise. "Look, my team hasn't got these skills. We don't know how to do this."

"I do," Callum said simply. "My boys drill and mine for a living. We'll handle the extraction site. You lot handle the heat and getting the water back."

Mira's years on oilfields kicked in. "If it is coming out liquid, the only realistic plan is to pipe it back."

Everyone knew she was right. Production meant there had to be a pipeline.

"Hundred twenty miles of pipe. That's a big scale for us," Pacha said quietly.

Ellie nodded slowly. "I can make that much glass, but it will be slow to lay and fragile."

Callum grinned. "Glass pipe. You lot do everything the hard way. You need something flexible. How about polymer pipe? An inch may do. Go with two. One ton per mile. You can probably move a hundred tons of water daily with that if you push it. More than you need, right? Petey can make you some dies for that."

Jakob looked at Callum. "How long to drill?"

"Two weeks if the ground cooperates. Maybe three."

Jakob nodded. "That would be appreciated."

Callum grinned. "She'll be right, mate. I'm getting back to my crew. Let us know when you need us."

The next morning, Callum's team suited up and got on the cableway. Callum was pleased with their success in building that. It carried crew and equipment right up to the mine and right back down to Bounty. Jakob watched them secure the rig components, each piece strapped down with the casual efficiency of people who had done this a thousand times.

"You'll want to come see it once we're set up," Callum said. "Right at the base of Tom's new mass driver."

"We'd love to see it," Jakob replied.

Jakob turned back to the settlement, where Ellie and Tom were already laying out the pipeline route on a terrain map.

“Straight shot, one hundred twenty-five miles. Let’s do it.” Tom said.

“Nah,” Mira said. “Straight pipes always break. Pipes need to zigzag so they can handle the heat and cold. Expect to add about thirty percent, turns every ten miles. That’s what we did in my dad’s oil fields.”

“One hundred sixty-five miles of pipe,” Ellie calculated. She looked up at Jakob. “That’s a lot, but we can do it.”

Mira was already calculating on her slate. “If we lay five miles a day, it will take less than two months, but that means we won’t be making enough polymer for Pacha. If he doesn’t get his greenhouse, we are all going to stop eating soon.”

Jakob had faced this problem before. They have to deliver, and they have to eat. Last time, they had to deliver. That worked out. This time, there were more people involved, and their margins were thinner. They had to eat. “Alright. The first greenhouse is the priority. After that, half and half. Callum’s crew has their own stuff to do. We let them make steel when they need to. Need to make them carbon for alloying with the iron, too. Can’t stop that, but it won’t be a lot, and we should be able to make a lot of polymer at night. Meantime, Tom and I can start surveying and digging in that trench. Mira, can you make the solar heating troughs? I think we have enough to do for the next month while Ellie and Pacha get that first greenhouse built. Pacha, will that work?”

“Yeah, boss. We can swing it. No worries.”

Mira didn’t complain. “I’ll start fabricating the trough mirrors today. We’ll need at least two hundred square meters of collector surface. “

Tom seemed to be calculating something on his slate. “I think the trenching will be pretty straightforward. With Callum’s excavator, we can probably trench ten miles in a day.”

Jakob watched them work through the details, felt the weight of the project settling onto all their shoulders. This wasn’t as simple as their first operation. The infrastructure had more moving parts and affected more people. But the upside was huge. Permanent, load-bearing infrastructure that would provide water for the entire inner solar system.

“Let’s get started,” he said.

Petey was satisfied with only five tons of carbon. Ellie got that done the first day, so she didn’t mind. What she hated was when he made iron. That stopped polymer

production for up to twelve hours. He was extracting iron, then alloying it with carbon to make mild steel, then forming tubes. He never ran at night, though, and he spent at least two days a week sharpening and grinding, which didn't use much power.

Pacha just focused on dirt and plants. He trimmed the cavy herd back a little. He had to conserve their feed. That meant extra meat for a couple of days, then down to half as much as they were used to. More algae soup, so less algae biomass for his terra preta.

After two weeks, Tom and Jakob had trenched about ten miles using the excavator. They were eager to start getting pipe in there, but Ellie was focused on making polymer for the greenhouse.

When she finally had enough for the greenhouse, everyone but Mira went with Ellie to start laying down the first large greenhouse. They brought three sleds, the excavator, and a 3D printer.

Ellie, Jakob, and Pacha each had a sled. Tom got on the excavator and started pulling out regolith. Dumping it on their sleds. They hauled it off, just twenty feet north of the greenhouse. As they got closer, Tom asked for direction. "You want it two feet deep? Any slope?"

"Pacha was out of breath from pushing the sleds. "No slope. Two feet straight down. That's my soil level."

As soon as the excavation was done, Ellie jumped up on the 3D printer and set in the instructions for putting in the foam floor. There was plenty of fine powder to strengthen it. As soon as the base was done, the corrugated wall started. This one used dust, sand, and stone. It was just like the perimeter wall at the settlement, but this one was thicker and went up four feet, straightening at the top. Tom and Pacha bolted the rails on the inside for Pacha's equipment to ride. It was a long print, and even with the excavator to keep the hopper loaded, it was tough work. By evening, they had stretched the polyethylene sheet over the structure and secured it with Petey's steel mesh. Tom brought the power cable over before dinner time, and they turned on the compressor, aiming for one-half psi. The air recycler was set for 0.03 psi partial pressure CO₂. Nice and rich for the plants.

They went back and started dinner.

Callum and Petey were talking as they ate.

Callum was eating the small ration of roast cavy, telling Petey the news. “Yeah, tomorrow is the first blast. We drilled a ring of twenty holes. Electric cardox charges. That why wThat’sed the big twin screw.”

“Callum, I don’t mind sitting down here making steel all day long. I like making sure the pipes and bits are made right. But you can’t cut me out of the exciting parts. I have to be up there to see the seam blow.”

“Oh, yeah, mate. Of course. We should invite the Magazine Beach mob, too, don’t you think?”

Jakob leaned back. “We would love to see what you guys have up there. We will take the day off tomorrow to watch your team work.”

The next morning, the team was excited. They needed the rest after building the greenhouse. Mira loved the cable ride. She sat by Pacha. They started out low and easy, but when they got to the scarp, it kept the same speed while going straight up the face. A couple of horizontal towers kept them off the face. Then they were up on the shield. Mira was wiggling around, looking out over Mars and over the shield. “So different up here,” she whispered to Pacha.

He was doing the same thing. “Amazing,” he replied. “I feel like I can see back to our old Magazine Beach site.” They both laughed.

When they got to the mine, it didn’t look like they expected. Crushers and grinders, with the sun glinting off rows of solar panels. Drill rigs and hardware. The actual mine itself didn’t look like anything.

Callum’s team handled the steel cartridges with the careful precision of miners who’d spent years around explosives. The cardox charges were steel cylinders filled with liquid CO_2 , an electric heater, and a rupture disc. They were not going to detonate from a bump, but old habits ran deep. Every charge was treated like it could kill you, because they had handled other charges that could. The habit was what kept them alive. Twenty charges were lowered into holes in a ring about forty feet from the center. They covered the top with stones and regolith and tamped it down.

Then Callum got them all back behind a small metal wall two hundred feet away. He carefully inserted each wire in the trigger box. Picking up his slate, he started counting. “One!” and there was a huge thump. The ground shook, and the rock broke. Debris shot straight up. But there was no pause. He just said “Two!” and the event repeated. He counted out each hole until all twenty were blown.

Mira looked at Pacha. “I wasn’t expecting that.”

“Yeah, I totally assumed he would count down or something. That first one was a shock. But the second one, I think I actually pissed myself. They both laughed.

“I know! I could feel the thud run right up my legs. That was amazing.”

When the dust cloud settled, Jakob looked at the broken rock piled in the center. He could see streaks of green.

Callum came up. “How was the show?”

“That was amazing. We were so close. I can see the green streaks you are after in the rock.”

“Yeah, Cardox lets you get close. Really feel it. But those green streaks, those are just copper. The stuff we’re after is gray. I don’t know if you can even see it.”

The miners went in and retrieved the Cardox cylinders and began putting them on their racks.

“Beautiful blast, that. We’ll be mucking this out for weeks.” Callum gestured at the pile. “Which reminds me, Jakob, we’re going to need to keep the excavator up here now. We’ll be shifting overburden daily.”

Jakob felt his stomach drop. “How long?”

“Permanently, mate. This is just the start. Once we clear this lot, we’ll be opening the next section.”

“But we still need the excavator. You can’t just take it; we’re using it,” Ellie said.

Jakob interrupted. “Callum, that excavator has been a fantastic help to us. Thanks for letting us use it for so long. We’ve got some refiguring to do, but we’ll be fine.”

Ellie interjected again. “How about you guys use it during the day, and we work at night, just shuttling it back and forth?”

Jakob tried to quiet her, but Callum spoke first. “Moving the excavator isn’t just a ride up the cableway. My whole team will work for a full day to get that thing safely up the scarp. Multiple ropes. Then, when it’s up here, it is a ten-hour drive to the mine. Back and forth won’t work. But don’t worry, mates; it’ll work out.”

They rode down in silence. Jakob and Mira had reached the ten-mile mark. Not done or even halfway, but something.

Ellie was thinking about building more greenhouses without the excavator. She could do that. A one-day job would take a week or so, but it could happen.

Pacha knew this was trouble, but his mind was on the first greenhouse and getting that going. He remembered using shovels to keep the 3D printer running, and wasn't concerned.

Tom knew this was a disaster. There was no way they could keep trenching with hand shovels. They would be lucky to lay one segment a day. Forty thousand days to finish this. He would pretend, but this was not real anymore.

As they came down the scarp, the pipeline stretched out like a fragile thread connecting the settlement to a distant promise to nowhere.

The next morning, Jakob and Tom suited up before dawn. They loaded two sleds with thirty bundles of ten pipes each, a pipe welder, and shovels. The settlement was quiet with only the soft hiss of recyclers breaking the silence.

Jakob was welding the pipe, and Tom was laying it in place. After ten segments, they went back and started pushing the berm back into the trench. They put a small metal plate on the front of one, which made pushing easier, but not easy. They were coming home exhausted, but after a week, they had finished ten miles that had already been trenched. It was slow, but it was progress.

The next day, they set to digging. Tom was looking at a manual. "The trench needs to be three feet deep. With the angle of repose of this regolith, the top is almost five feet from the center."

Jakob looked at the line stretching ahead and did the math in his head. For every twenty-foot segment, they'd need to move nearly three hundred cubic feet of regolith. And they couldn't just toss it aside because it would slide right back in. They'd have to carry it at least nine feet away from the center.

He picked up his shovel and started digging.

The regolith came up easily at first, light and powdery in the reduced gravity. But after the first few inches, it got denser. Compacted. He had to put his weight into it, feeling the resistance through the shovel handle, even with the dulled feedback from his gloves.

Tom worked beside him, establishing the width. Five feet from the centerline on both sides. They marked the edges with stones, then began working their way down.

Scoop. Carry. Drop. Return. Scoop. Carry. Drop.

After an hour, Jakob's shoulders burned. The suit's cooling system was running hard, trying to keep up with his body heat. His breathing echoed loud inside his helmet.

They kept going. The trench slowly deepened. Six inches. A foot. Eighteen inches.

Then Tom's shovel hit something solid with a dull clang. He stopped, crouched down, and brushed away regolith with his gloved hand. A rock the size of a melon sat embedded in the compacted soil.

"Great," he muttered.

He wedged the shovel underneath and pried. The rock shifted but didn't come free. Jakob came over, and together they worked it loose, rocking it back and forth until it finally popped out. They carried it twenty feet away and dropped it.

Back to digging.

Two feet down. Two and a half. The walls wanted to slump inward. They had to widen the top edge to maintain the angle, which meant more material to move.

They stopped for lunch. In pressure suits, the only way to eat was paste. They ate paste lunch in silence, sitting on the sled, staring at the endless plain stretching south. They had cleared thirty feet of trench to full depth. Jakob stood at one end, looking at the long, neat cut in the Martian surface. It had taken six hours. Two hours later, they had dug enough for two segments. Installing the segments was easy. The welder only took a few minutes, and the pipe was light.

Tom looked at the rock sitting twenty feet from the trench. "That rock we moved? Dumb. We could easily have just gone around it."

Jakob nodded, exhausted and humiliated. They had laid two segments today. He brought up the slate, but didn't log it, yet. He didn't want to see the number. Two segments. Tens of thousands to go.

That night at dinner, no one talked much. Ellie was running numbers on polymer production. Tom was sketching something on his slate but kept erasing it. Pacha was happy, sorting seeds for the greenhouse.

Jakob's entire body ached. His hands were stiff, his back tight, his legs trembling slightly from the day's work. He flexed his fingers, trying to work feeling back into them.

"How far did you get?" Ellie asked.

"Two segments," Jakob said.

Ellie nodded slowly. She opened her mouth, closed it, said nothing. Everyone knew the math.

The second day was worse. The novelty was gone. Now it was just hard, repetitive, grinding labor.

They dug. They carried. They dumped. They dug again.

That evening, Jakob logged their progress.

Jakob logged it on his slate.

Day: 2

Segments done: 4

Segments remaining: 40,916

"This is impossible," Tom said finally.

Jakob didn't answer. They both knew that from the day they visited the mine. The numbers didn't lie. Two per day. Fifty-six years. Half that if two crews worked. Jakob smiled grimly to himself.

That night at dinner, Magazine Beach got together. Jakob spoke first.

"Guys, this is not working. We are laying a section or two every day. At this rate, we won't get paid, and the Cereans won't get the water they need to live. Bounty is nice, but I think we need to face reality. We cannot meet our obligations here. I don't know what else to do. We need to set up a site on Planum Boreum and start cutting ice. Ice delivery is why we are on Mars. Ceres needs it. Bounty needs it."

Tom looked up. "Yes, I looked at the numbers, too. I don't know what to do, but I agree with your proposal. We have to move on. Pacha, can you fly up sometimes and bring us food? "

"Dudes, I don't want to hear this. You think I can grow food here without water. This is a mess. The first large greenhouse is growing well. I am washing and

tumbling regolith to get nice dirt, adding carbon and organic matter, but the real thing is, I have put in plants to nourish the mycorrhiza in the soil. But to keep it going I need water, too. If you move, what am I going to do. Fly in water?"

"I hate it as much as anyone," Jakob explained, "But we cannot lay that pipe fast enough to make a difference. Our first obligation here is to ship ice. The time we bought ourselves with the slow shipments is coming close to an end. We have three months left to start shipping or it will be a problem for Ceres."

Petey was eating by himself at the next table. He overheard them, but didn't say anything.

That night in the ag dome, the team was down. Mira cried as she packed up.

Jakob kept it inside for the sake of the team, but it hurt. "Guys, pack very light. We need to bring hydrolox for power. It is dark where we were."

Ellie just stared at him. "What are we doing? Are we going to build a new dome?"

"I am open to suggestions, Ellie, but yeah. We will move our solar panels up, and build a new dome on Planum Boreum. Probably smaller."

Tom was stoic, working on his slate. He wasn't sure hydrolox would provide enough power for the mass driver.

Mira wasn't taking it well, but didn't want to talk about it.

At dinner that night, Callum and his team came down from the mine. They looked tired but satisfied, their suits streaked with dust and rock powder. Callum dropped into a chair with a satisfied grunt.

"Four days on, three days off," he said to no one in particular. "That's the rhythm. You work yourself ragged, then you rest proper. Keeps you sharp. Keeps you safe."

The miners talked to each other a few minutes, then Pacha brought over plates of roasted oca, algae stew with a little cavy meat in it, and flatbread.

Callum took his gratefully. "How's the pipeline coming?" he asked Jakob.

"Slow," Jakob said.

"Yeah, Petey said as much. He wants to help."

Jakob looked down, depressed. “Callum, we have looked at our options, and have reached a sad conclusion. My team is going to have to make the move north. We have an obligation to get water to Ceres and to your crew, and this pipeline is not working. We are going to need you to fly us up, and bring us food regularly.”

Callum laughed, a genuine, relaxed sound. “Well, before you build a new camp, I want to show you something Petey made for you this morning.” Callum left for a few minutes and came back with a big piece of steel, flat with a curved tube on it. “This here is a plow share for what we call plow laying. You feed the pipe in the top tube and pull the plow along. It deposits the pipe three feet down, easy as. Well, not exactly easy, but you can move with this. No digging.”

Jakob stared at him. “What?”

Callum set down his fork. “Right. Sometimes I forget you lot aren’t miners.” He pulled out his slate and sketched quickly. “You’re digging trenches by hand, yeah? That’s brutal. What you need is a plow. You pull it through the regolith, it lays the pipe. And It’s easy to plow this regolith. Just keep moving, welding the sections of tubing together. You can drive it straight through. Not zero work, but easier than digging. And much faster.”

Mira leaned forward, staring at the sketch. “That works in regolith?”

“Better than anywhere. Loose soil, almost no rocks, piece of cake. Petey heard you were having trouble, thought this would help. It doesn’t strain the pipe. Much smoother.”

Jakob felt something unclench in his chest. “How fast does it work?”

“Depends on the ground, but in this stuff, an experienced crew could lay five miles a day. You lot aren’t experienced. We’ll see.”

Jakob looked at Tom, saw the relief flooding across his face. Then at Ellie, who was smiling faintly.

“Callum,” Jakob said, “can you show us how it works?”

“Sure, mate. First thing tomorrow.” Callum picked up his fork again. “So you know, we all are hoping you folks don’t have to move back south. We like having you around.”

Tom and Jakob talked that night, privately. “Look, Jakob, this project is beyond us. I love the plow laying concept, but we need to be real. This may be another way to fail, and we don’t have much time before we need to start launching ice.”

Jakob looked at him with eyes that could swallow Mars. “I’ve been thinking the same thing. We’ve bitten off more than we can chew with this project. To move, to make a Magazine Beach North, that’s going to take time. But we are going to see how it goes tomorrow.”

Tom and Jakob kept the conversation private and went to bed.

The next day, Callum gathered them around. “OK, first step is to break into tasks. Plow laying is a continuous process. Your pipe is in twenty-foot segments. How are you connecting the segments?”

Ellie spoke up. “I was going to use compression fittings because they are so easy to use, but making so many fittings was daunting. We use butt fusion. Mira helped make the machine, and it only takes a minute to make welds as strong as the pipe.”

“Good on you. I want you two to start welding the pipe. It will take time, but we want a whole, single, continuous pipe from one end to the other. Does that make sense?”

Mira spoke up brightly, “Yeah, we can do that. How close does it have to lie?”

“Oh, just wherever. Anyplace within about ten yards will be fine. The pipe will move itself into the plow. Just make sure the lengths match.”

Ellie and Mira smiled, eager to begin.

“OK, you lot,” Callum continued, looking at Jakob and Tom, “You boys are going to be doing the real work. You push that sled along the track. Add some weight to it, maybe so the traction assist can help. Make sure the pipe is moving in smoothly and follow your map. Sometimes it will hit rocks. Stop, get out the shovels and picks, and figure out the problem. You can back this up a little when you need to, and the pipe will come out. Clear the obstruction and move on. If you can see an obstruction coming up, you can navigate around it. Pipe doesn’t need to be straight. It is stronger when it has some bends.”

Tom considered the task. “A harness will let me pull. I think that will be stronger.”

“You do what you like. Drag that plow from one end to the other of this whole thing, and you will have laid the pipe. Got it?”

Tom and Jakob brought the sled for pulling the plowshare. They carried the plowshare, picks and axes, and ten bundles of pipe. Callum and Petey went with them.

Ellie and Mira brought another sled with them to carry the pipe from Bounty. The sled could carry thirty bundles of ten pipes. Mira bolted the butt welder to the sled, and she and Ellie got started. They could weld pretty fast.

Callum gave a lecture first, to make sure Jakob and Tom understood the plan. Then they demonstrated plow laying. They laid the first mile about as fast as walking. Then they turned it over to Jakob and Tom to see how much they learned. When Jakob and Tom started, everything went wrong. Just getting the pipe to feed properly was far harder than it looked. It was under tension and would twist up. The hard problem was preventing the pipe from getting torqued as it was laid. The sled was another problem. Without enough weight on top, it didn't have the traction to help pull the plow. If Tom pulled too hard, the front wheels lost all traction. If Jakob pushed too hard, the wheels went under the regolith. Coordination wasn't easy. The real problem, though, was simply moving the plow, straining their legs and backs for every foot of progress. Jakob and Tom got only half a mile done before dinner. Yet when Jakob glanced back, he saw the line of pipe running straight and true behind them, the plow holding steady despite their inexperience.

Jakob caught Tom's eye. "Look at what we've done. We can do this. This is working."

They walked back to dinner. Partway back, they met Ellie and Mira.

Ellie smiled. "The welding went well. We welded forty bundles of ten segments, went home, and got thirty more bundles. All told, we covered over two and a half miles today."

Jakob and Tom were in too much pain to talk. Just walking was pretty rough.

Mira laughed. "Get on the cart, you old men."

They happily got on, lying down, making sure not to move any muscle for the entire ride home.

When they got back, Pacha had dinner ready.

Tom spoke up. "You two made it look so easy. That is hard work right there. But the thing is, we did it."

Callum and the miners broke out in laughter. "She'll be right, mate. Don't wreck yourselves. I am scheduling time to drill your wells in a month. Reckon you can get it sorted by then."

The next day, Callum went with them, but Petey stayed behind working on tools. Tom and Jakob were getting smoother. Mira and Ellie were very proud of their innovation. Mira had welded tie-downs on the sled, and they could carry fifty bundles of ten pipes. “We’re gonna weld twice as much pipe as last time,” Mira said with a smirk.

“You do that,” Callum replied.

Tom and Jakob were noticeably better, even though their muscles were sore. They usually kept the sled flat, and they were getting good at putting just the right amount of twist in the cable before it went under, so the cable on the ground didn’t get all kinked up. They got two miles in.

But this time, despite their enthusiasm in the morning, Ellie and Mira were in trouble. They could haul more in a load, but their endpoint moved. Each time, they were farther from Bounty. Longer back to town, longer back to the site. They got three miles in, one and a half loads. They left the sled and twenty-five bundles of pipe there and walked back. They ran into Callum, Jakob, and Tom.

Callum clapped Jakob and Tom on the shoulders. “Great work. You went more smoothly and faster. You’ll get better. Hey, Mira, no ride back for these boys?”

Ellie answered. “We didn’t finish our second load today. Walking back and forth takes too much time. In a week, we are going to have to start walking beach before we start welding. I don’t know what to do.”

Callum smiled. “We can jabber on that over dinner tonight. I’m not worried.”

At dinner, Callum sat by Tom. “So I hear you were a pilot back on Earth.”

“Yeah, general aviation. I had a private plane. Not a career or anything.”

Callum nodded. “What would you think about flying the gyro?”

“I mean, I can see that would be really helpful to the team, but I obviously have work to do, and we are going to need to figure out how to bring more pipe with us tomorrow. I am not sure I have time.”

“Oh, you’ve got time. You just don’t know how to use it, yet.” Callum said with a wink.

In the morning, Callum woke Tom early. “Let’s start loading the pipe. We can try ten bundles at a time.”

Tom and Callum started loading up the gyro. When they loaded the third bundle, Callum explained the plan. “We are going to move this stuff into position. Doesn’t have to be right, has to be close. What I need you to do is watch me closely. See every part of this, from takeoff, climbing, cruising, and landing. Got it?”

Tom looked at the controls. “Wilco.”

With that, Callum read the wind and faced into it, and engaged the driven landing gear. Tom watched the indicators as Callum lifted off. As they flew along, they saw Ellie and Mira’s sled, with 25 bundles on it.

Callum brought up his slate and figured out where they would be at the end of those bundles. “OK, mate. Kick out the first bundle.”

Callum tilted the gyro, slowing it and letting it go lower. Tom kicked out the first bundle. They repeated that maneuver every two thousand feet, following the map on the slate. When the last one was out, they headed back.

Tom watched as Callum brought it in, flaring, bringing it almost to a stop, then settling on the ground smoothly.

They loaded up another ten bundles, but this time, Callum had Tom fly, guiding him the whole way. Tom engaged the landing gear and tractor prop. When he saw the lift rotors turning, he pulled back and hopped about twenty feet.

“Let’er run a little further. See the mark I etched on the airspeed indicator? Don’t try to fly until you hit that.”

Tom kept at it and was soon flying like a pro. When they got to the next drop-off, Callum asked Tom to land. “Don’t worry about slow descents yet. After you have some miles under your belt, you can work on that. For now, let’s just unload this stuff.”

Tom found a place to set it down. He approached pretty steeply, not noticing how fast he was going down. They hit pretty hard, but nothing was broken.

Callum checked himself first, then explained. “The gyro works a lot better with forward airspeed. You can work on short landings later. For now, bring it down smooth like your plane from home. Deal?”

Tom looked sheepish. “Deal. Sorry.”

They unloaded all ten bundles, and Tom flew them back to Bounty smoothly. Tom thanked Callum. "Callum, this has been great, but it's dawn, and I need to help Jakob."

"Not today. Petey'll work with Jakob today. Today you are doing prepositioning."

Tom wasn't sure what to say, so they went and grabbed another ten bundles. They worked like that all day, back and forth. Tom got much more comfortable with takeoffs and landings. Ellie and Mira set to welding, easily putting down twenty miles that day, and Jakob and Petey were fast. When it got late, Callum had Tom pick them up.

Ellie and Mira were first. Mira was excited. "Thanks, Tom! No trips back, we always had the stuff we needed on hand. I bet we welded fifteen miles of pipe today."

Next, they stopped for Jakob and Petey. "Thanks for the lift. Hey Tom, no offense, but with Petey we made eight miles today."

Tom laughed. Callum reminded them of his schedule. "OK, guys, we did a lot in three days. Tom, you keep flying people out. I wouldn't bring any bundles when you do it now, though. You are going to have to work the plow with Jakob from now on."

Tom nodded. That week went pretty well. They would bring a couple of bundles with them each morning when they flew to the starting point. In the evening, Tom and Mira would drop off another load. By the end of the week, they were laying four miles a day. By the end of the second week, they were up to six.

In two weeks, the pipe was fully welded. Ellie went back to helping build greenhouses, and Mira went back to making the solar thermal troughs. In a month, the pipeline and all two thousand square meters of troughs were completed.

As soon as the pipeline was done, Ellie wanted a test. She hooked up a large compressor to the Bounty side, and Tom flew Mira out to the extraction site.

Mira held a strip of Mylar next to the outlet. It didn't move at all. "Tom, we might have torn that pipe anywhere along the way. We wouldn't know; it was buried without us ever inspecting it."

Tom nodded. "I guess you're right. We will fly back over that area. If we can see a geyser of dust rising up, that will be our leak."

Mira continued watching. She thought she saw some dust stirring. "OK, clearly the pipe is damaged, but something is still getting through. It must not be too bad."

The wind grew steadily, lifting a small swirl into a rising column. Mira blinked, taking it in, and Tom's mouth twitched into a grin. Mira's chest tightened, then exploded. "It works! It actually works!" She spun toward Tom, laughing and crying at the same time.

Ellie had time to build a glass distiller to process the extracted water. She even started making the glass filament packages to carry the ice blocks. The rest of the Magazine Beach crew was helping Pacha build up more greenhouses.

After lunch, Jakob saw Tom and Petey loading up some twenty-foot casing joints on the gyroplane.

"Hey, Tom, can I join you? I want to see the drilling."

Tom nodded quietly. "Sure, Jakob. We're ready to go."

Jakob always enjoyed the view as the gyroplane took off. Mars looked so different than Earth. The smaller radius meant the horizon was closer. It felt small. As they approached the drill site, the first thing Jakob noticed was the rows of heating troughs Mira had built, ready to heat injection water when this whole thing started. Soon, he saw the short derrick where Callum and his crew were working. The short derrick worked with the twenty-foot joints. Tom set the gyroplane down twenty yards from the structure.

Petey and Tom went in, carrying a single joint between them. Jakob followed.

"Hey, Callum, how is it going?"

"We're at twelve hundred feet. Hit the first ice layer yesterday. Pressure's higher than expected, but manageable. We are running out of joints, though."

Petey and Tom set the joint down. "We got nine more in the gyro, if your crew can go get them. What is the pressure?"

"Nothing bad. Yet." Callum tapped the gauge. "We're at nineteen hundred psi downhole. That's higher than I'd like, but it's holding and well within our safety margins. The pressure manifold is venting CO₂ faster than I expected, is all."

"Will it blow?"

"Nah, we'll be careful. But when we start injecting heat, the pressure's going to spike. We'll need to balance heat injection with venting rate, or the whole formation fractures."

Jakob looked at the manifold, the maze of valves and sensors. “How do you balance it?”

“Gently.” Callum grinned, but there was no humor in it. “We’ll start slow. I’m running CO₂ for a while until the temperature climbs to forty below. Probably half an hour. At that point, most of the frozen CO₂ will have sublimated. Then we will send down warmer air until we get near zero. When we are ready for the water, we trickle it in, five liters a minute. Watch the pressure. If it climbs past twenty-two hundred, we shut down and sit while the gas vents.”

“When do you plan to start?”

“Tomorrow morning. Soon as the sun heats the troughs.”

Jakob didn’t sleep well that night. He lay in his bunk in the ag dome, listening to the soft hiss of the recycler. Outside, the Martian night was silent and cold, the stars sharp and unblinking.

At dawn, the team flew down and gathered around the manifold. Mira had the solar troughs aligned, and the holding tank was reading ninety-two degrees Celsius.

“Ready?” Callum asked.

His team nodded. One of the miners opened the injection valve slowly, and heated CO₂ began flowing down into the frozen ground.

Jakob was surprised they didn’t inject water. They had done all this to make hot water to inject, after all, but he had learned that Callum talked in straight lines, but operated in a complex manner that was second nature to an experienced miner.

Callum saw Jakob’s confusion. “We are not sending water down yet. It’s too cold down there. We would just end up with a solid ice block at the end of our borehole. We’ll pump warm air until the temperature gets to about ten below.”

Jakob trusted Callum, even if he didn’t say anything. Still, he appreciated the occasional explanation.

For the first minute, nothing happened. The pressure gauge held steady at nineteen hundred psi. Then it ticked up. Nineteen ten. Nineteen twenty.

“Easy,” Callum murmured.

The pressure climbed. Nineteen fifty. Two thousand. Two thousand fifty.

“Gas vent is maxed,” one of the miners reported.

“Hold steady,” Callum said. “Let it equalize.”

The gauge crept higher. Twenty-one hundred. Twenty-one fifty.

Jakob felt his pulse hammering in his ears. The numbers kept climbing. Twenty-two hundred. Twenty-two twenty.

“Shut it down,” Callum ordered.

The injection valve snapped closed. For a long, terrible moment, the pressure kept rising. Twenty-two forty. Twenty-two fifty.

Then it stopped. Held. Slowly began to drop.

“Relief valve’s doing its job,” Callum said quietly. “We’re venting faster than we’re building now.”

The pressure fell back to nineteen hundred and stabilized.

“We’re going to inject slower,” Callum said. “We can’t risk the formation.”

“Slower? That’ll take forever.”

“She’ll take as long as she takes.” Callum looked at him. “We don’t rush this, mate. We rush, and the formation cracks. It cracks, we lose the whole site.”

Jakob nodded, throat tight. “Of course. I don’t mean to rush you.”

Instead of a half hour, it took four days before they were ready to start injecting water. Callum’s team adjusted the flow until the pressure held steady at twenty-one hundred psi. On the fifth day, cold water began flowing from the recovery well.

Well, not really water. Sludge. Jakob was there when the first liter came up. He was devastated. Expecting clear water, the sight of this heavy mud was too much.

Callum saw the pain in his eyes. “She’s got some color to her. We’ll need a basket centrifuge here to clean it up before sending it back to Bounty. I think it looks pretty good. Petey will take care of us.”

Ellie was listening. “I can work with Petey to make fiberglass filters that will fit his baskets.”

“That’s the spirit, Ellie. But I need to warn you, the filters will wear out. I don’t know how fast, but probably a new one every couple weeks. How’s that sound?”

Ellie thought about it. “I make a lot of fiberglass anyway. Pulling a little aside for filters will be easy.”

Petey didn’t worry about making a great centrifuge. He used woven steel mesh and only designed it to spin up to a thousand rpm. But it had volume, over a ton capacity, and he made two, so they could switch when one got blinded. That would catch all the sand and rocks.

Next, he built a bowl centrifuge. This would fill up, too, but slowly. All the remaining fine sediment would end up on the walls. The water would come out transparent, good enough to pump to Bounty.

In a week, they had it operating. Two of each centrifuge. When they filled, they would switch to the other one. Ellie had used a sixty-mesh in the basket centrifuges. Pulled out all the big stuff. The water came up like sludge into the basket centrifuge, and exited like chocolate milk. Then it went into the bowl centrifuge, which spun faster, four thousand RPM. The liquid that came out looked like Lipton tea.

Tom connected the recovery line to the pipeline.

“Send it,” he said.

The pump engaged, and water began flowing north toward the settlement. It would take six hours to reach the distillation plant Ellie had built. Tom and Mira flew back, slowly, going over the track of the pipeline, looking for any leaks. They went slow and low, but saw no problems.

Jakob was at Bounty when the water started flowing. The still was connected to the waste heat from the generator, which boiled it fast. The steam went through a cooling coil filled with cold ammonia, which dumped its heat into the Martian atmosphere with a radiator. The water that condensed was clear and pure. Jakob watched as it flowed into the hundred-ton freezing tray. He pulled out his slate and sent a message to Dr. Knox. “The system is working. Your next shipment is freezing now.”

Jakob didn’t notice the concentrate. That coffee colored mother liquor was freeze-dried. The dry chemicals were stored, and the water was condensed, providing enough water for Bounty.

Jakob stood beside the pipeline, watching the pressure gauge hold steady. He felt pride. Extraction, cleaning, pumping, purification. So much more complex than Magazine Beach, but actually easier to operate. The two teams had really pulled together.

Ellie came over and joined him. She checked the water output from the freeze dryer. She loaded up a bottle of five gallons, and went to find Pacha.

Pacha was working on greenhouse five. Inside, the air was warm and humid, the soil beds ready for planting. Goldeneye was spread around, and the buds were showing. In the middle of the greenhouse was a lined area for a large pond, gently sloping to about a foot deep in the center. This structure was running at three psi to sustain cavies. While he used the same seeds, he was not planting for crops; he was planting for pasture. He scattered duckweed on the pond and planted water spinach. Fast-growing and nutritious, these plants would keep the CO₂ levels low, and the cavies could eat them. He walked the length of the greenhouse, checking the irrigation lines, the grow lights, the ventilation ducts. Everything worked. Everything was ready. He planted amaranth first, because it made great fodder for cavies, but he planted buckwheat and Tibetan lentils as well for nutritional and textural variety. Oca would follow in a few weeks. None of it was in neat rows; it was scattered around. Here, the herd could thrive.

Pacha was still in the greenhouse planting when Ellie found him. "This looks great, Pacha."

"Right now, only the goldeneyes are growing. They are helping build the soil. Don't say anything, because it isn't exactly legal, but Callum brought me some inculums from mountainous areas. They seem to be thriving. That's why I planted the goldeneyes. Let's see what it looks like when it starts greening up. The soil here will be wonderfully healthy. That'll make expansion easier. Growing soil is the hardest part. We need twenty greenhouses just to handle all the orders we already have."

"I think the flowers are wonderful. Are you going to put the cavies in now?"

"If I put the cavies in here now, they'll eat the seeds. I need to wait another six weeks."

Ellie listened, but really she was bursting to share what they had done at the Medusae Fossae Formation. She calmly walked over to the pond. Pacha's jaw dropped when she nonchalantly dumped in the five gallons of water she brought. "Well, I have good news, too. No one is moving. Water is flowing to Bounty."

"I knew you guys would get that done. Fantastic. So we're back in business."

Ellie nodded. "It's all working."

At dinner, Mira sat by Jakob. "We did it."

"We did," Jakob agreed.

"So what's next?"

Jakob thought about the ice field, thousands of cubic miles of frozen water waiting beneath the Martian surface. He thought about the beryllium mine, the greenhouses, the settlement that was no longer a camp but a colony.

"Honestly," he said, "I have no idea. But whatever it is, our team can do it."

"Yes, we can."

Outside, the Martian sun was setting, casting long shadows across the plain. The solar troughs were folding for the night, the mass driver was idle, and the settlement lights were coming on one by one.

Jakob watched it all and thought: This is real. This is permanent.

And for the first time since landing on Mars, he believed it.

Chapter 16: Quinceañero

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Maria stood on the roof of her villa, watching *Peces de la Luna* come in. At forty knots, it was riding high on its plane.

Mateo Torres was at the helm. Tonight, they were coming back from a fifty-drop run. When he reached the first buoy, he throttled back. The wake caught up to the boat as it lost its plane and settled in for maneuvering.

When it docked, Mateo jumped off the ship and got on a scooter to ride home. He ran up the stairs to José's office.

"¡Papá! I'm home. Make the call."

Felipe and José were waiting in the office.

José made the call to Bounty. "Callum, I hear you are ready to send us metal, too. Each metal from Ceres has posed special changes. I hope you are sending something easy."

It would be at least fifteen minutes before there was a response.

Mateo told them about the run. "Mixed metals. We got thirty bowls of manganese, fifteen of tungsten, and five of tantalum. They overshot the target by almost a hundred miles."

"They can't overshoot the target zone. It's a hazard," Felipe said. "Don't we have a penalty for that in the contract?"

"I called Catherine Dahl directly about it. She accepted the penalty, waiving their cut on the value of the bowls that missed. That was two bowls of molybdenum," Mateo said. "It's much more profitable for us when they miss, but from what she said, I don't think they will be missing much in the future."

"Frankly, I'm afraid to talk to her," José said.

"She's sweet, but she doesn't play around. You need to get used to her bluntness to see her kindness behind it."

José's slate crackled. "G'day, mates. Raines tells me you lads can grab up our metal. He's been hollering for some beryllium."

José was excited. “Absolutely. No problem. You know the bowl dimensions?”

As soon as the message was sent, Felipe spoke up. “*Carnal*, we cannot handle beryllium. It’s toxic. Are you going to let Mateo haul it in?”

“I’m not a child. I’m fourteen. I can handle it safely.”

“I don’t doubt you can, but the ocean can’t.”

Mateo turned to his slate.

Fifteen minutes later, Callum’s voice came through. “Thing is, we don’t have all that much. Two ingots.”

Mateo sent Callum a diagram from his slate. “Two ingots are OK. Can you send us bowls clad in an inch of CuBe1? Use the CuBe for the crossed shackles.”

After the voice message was sent, Mateo described his plan to José and Felipe. “The copper-beryllium alloy is strong and won’t corrode in the seawater. We can inspect the cladding for damage. If there is none, we can bring it back and melt off the cladding, then maybe scalp a couple millimeters to leave the beryllium pure.”

Felipe grimaced. “It sounds like a lot of work for only a couple of bowls. Maybe we pass on this.”

Mateo was shaking his head. “No, *tío*, we do not pass on this.” Mateo brought up his slate again, showing what Elin would pay. “Our cut for two bowls would be forty billion pesos. For that money, we can do it.”

José’s eyes were smiling as he listened to Mateo.

Felipe looked at the numbers.

Mateo continued, switching to another window. “We can process the swarf, too. Even a single bowl will give us twenty million pesos. Half that if it is cash from ArcelorMittal.”

José finally spoke. “*Carnal*, you are not wrong. This is a risk. But I will vote with Mateo. He can set up the new facility. And Callum won’t stop with two bowls. He’ll find another vein.”

Felipe reflected for a bit before responding. “I remember when we first talked about tungsten. You had already shot your mouth off to Raines, saying it was easy. We were lucky he gave us eighteen months. Nothing about tungsten was easy. But it was valuable. Now we routinely handle eight different, difficult metals from Ceres. If Mateo here thinks he is ready, I offer my full support.”

“Thank you, both, *familia*. You won’t be disappointed.”

Callum came back on José's slate. There was a pop before he started talking. "Mates, I talked to Petey. It'll be easy to make that. Just like making a candied apple. How soon will you be ready for us?"

José looked to Mateo.

"Callum, sir, we can be ready in eight months."

Felipe facepalmed. "Why did you tell him eight months? Do you have any idea what can go wrong on a project like this?" Felipe asked.

"Many things can go wrong, but delay won't be one of them."

"Felipe, this isn't the old days. Mateo won't be showing contracts to a bank. He will be making a small withdrawal."

Felipe knew he was right. "OK, I will go along with you. But charge an extra five percent for hazardous materials."

Mateo looked at his father, who nodded. José and Felipe left the room. José left his slate with Mateo.

Two hours later, Mateo emerged with a signed, printed contract: twenty percent plus five percent hazmat fee. "Tom is going to send us a bowl of iron first, to ensure they have the launch calibrated right. Once that is ready, and we are ready, they will send the beryllium. I backtracked on eight months. Thank you, tío."

"Fantastic," José said. "Let's go to the square for *paletas*."

When they got out of the taxis, Lupita walked slowly, with Felipe's help. He found a large stone table. Mateo found a *paletero* and brought him to the table. Lupita asked for coconut with spice, but the *paletero* didn't have that, so she got a regular coconut *paleta*. Felipe got a creamy spicy mango *paleta* and traded with her when she didn't like the coconut.

Mateo checked the ship locations on his slate. "I was expecting *Corriente* to be back by now."

"Which ships did the far pickup?" Felipe asked.

"It took all three recovery ships today. *Estele* took the far pickup. It won't be back for days. *Ráfaga* took twenty in the middle. But *Corriente* is only pulling ten this time, so it should be back soon."

"*Bajamar* and *Pleamar* could only ever carry five."

"Papá, that was the old way. Today we have too much work for that system."

Maria looked at Lupita. "How are you doing?"

"I am queasy sometimes, and hungry."

"You look the same, but you walk a little differently," Sofia said.

"I don't feel the same," Lupita replied.

Maria was looking towards the *kiosko*. "I remember my *quinceañera* was here. It was so bright and festive back then."

Mateo laughed. "They still have *quinceañeras* here, Mamá. Our neighbor had one here last week. And they are as bright and festive as ever. Just not today."

Maria turned to Sofia. "What do you think? Will you have yours here?"

"I don't know, Mamá. That's a long time from now."

Maria turned to Mateo. "We don't do big things for boys on their *quinceañeros*. It's unfair."

"It's not unfair. I wouldn't want all that fuss."

They continued eating. Felipe let Lupita eat the rest of his coconut *paleta*, too. She whispered in his ear, and Felipe addressed the family. "The titanium sheet metal business has been very profitable for us. It paid for the new recovery boats. I believe we could do more like that."

"Sheet metal from other metals?" Mateo asked.

"Sure, but not just sheet metal. Mill works. Pipe, beams, plates. And alloys. The reason Elin mostly takes ingots is that she needs to alloy them before forming the products."

"We can afford to build that, but I don't know where we could put it."

Felipe looked at Lupita, who nodded imperceptibly. "Mobile, Alabama."

José stopped breathing for a moment. "What are you thinking, Felipe?"

"Here is our plan. Elin is running a huge operation, and just getting the right metals is a constant strain. We build a new site. Mobile is connected to the CPKC railroad we use, but also other tier one railroads, including the CSX that Elin is on. And not all metals come by rail. Mobile has a great seaport. That will allow me to ship to Elin by sea when needed. And it is a very nice place to raise a family," he said, placing a hand on Lupita's belly.

"Uncle, what about Acero del Pacifico? We still need you."

"There are many competent people in this town, and we know most of them."

“Felipe, Mateo is right. I still need a partner to keep this operational. This is your dream, and this town should not limit you. I will support you in any way you need, but I need you to help me find the right person,” José said.

The next day in school, Mateo was distracted, sketching out ideas for the beryllium processing mill while pretending to pay attention in Mexican history class. He was playing with the idea of how much could be done on a single pedestal.

After school, he went down to the mill and found Felipe. “I was looking at how we can build the beryllium mill. The SPMTs will deposit the ingot on this pedestal.”

“A wash pedestal?”

“Of course, the first step is to wash it. The crew can use high-pressure hoses and detergent to wash it off.”

“How will they get to the bottom?”

“The edges are hydraulic lifters that can turn it over. It has a center drain.”

“Very nice wash station, Mateo.”

“It’s not simply a wash station. Once it is washed and dried, the people leave, and we flush with argon. The same pedestal has induction coils and a cap with more induction coils. We heat it in place to melt off the CuBe, which drains down the center.”

“This is no longer just a pedestal.”

“Under the pedestal, the liquid CuBe can pour into molds easily. While it is still hot, a mill moves into place and scalps the bowl. The lifters turn the bowl over until the milling is complete. We wash all the swarf together.”

“How do you wash it? It will react with water.”

“But we used water to wash it before, and you thought it was fine.”

“It was fine; you were washing CuBe. Beryllium reacts with water. You need to find something nonreactive.”

“OK, then the swarf is swept up and washed with something else. I will keep working on it.”

Felipe laughed. “That is a nice design. You kept it very compact. With only one pedestal and no crane, it will fit easily between the other buildings. What is the pedestal made of?”

“The exterior is beryllium oxide, a nice ceramic that won’t contaminate the pure beryllium. Once the pedestal and bowl are clean, we replace the casting system to cast pure ingots.”

“The idea is very good. I will help you find someone to build the pedestal.”

“Felipe, I need to make another change. We need one more place to process tungsten.”

“Why? We are doing OK.”

“No. We were backed up three weeks on tungsten last time. I left twenty bowls just sitting in the *Ráfaga* until there was space to unload them. It delayed the recovery of lunar titanium.”

“Have you talked to José about it?”

“Not yet. To build the expanded tungsten facility and the beryllium facility, I am going to need to completely tear down the original steel works you two built. How do you think he will take it?”

“Explain it to him. He’ll listen.”

At dinner that evening, Mateo waited for a lull in the conversation. “Papá, I showed my plan for the beryllium facility to Felipe. He likes it. It is very compact. It will be slow, but that’s fine.”

“Wonderful. When will you build it?”

“I contacted a construction company. We are looking at companies to make the machinery. But, also, I need to make some other changes.”

“You often do. Why do you bring it up at dinner? This is time for family.”

“Because, Papá, it means tearing down the old steel works. I don’t want to, but Ceres is just too productive, and we need more space.”

José lifted his wine glass and looked Mateo in the eye. “Well, either you tear it down or the rust will. Felipe and I should thank you.”

That evening, Mateo had another call with Mars. José relaxed with Maria and Sofia.

“José, Callum here. Tom and I want to send a test drop. He never tried to hit Earth before. So we were hoping to send a bowl of just iron.”

“Sorry, José isn’t here now. It is seven pm here, and he is with his wife and daughter. This is Mateo. We are ready for a bowl of iron. Let me send you the coordinates and times so you know where to hit. Once you launch, I will coordinate with shipping.”

As the months passed, Lupita got out less and less. Maria missed her and gave her a call. “Lupita, you need to bring Felipe over here for dinner tonight. I heard people in Mobile eat oysters chargrilled, so I am cooking that tonight. Come tell me what you think of them.”

They arrived at six o’clock and were greeted by Sofia. “Aunt Lupita, you look enormous! I mean, come in.”

Sofia led Lupita to the table, and the family gathered there, while Maria finished preparing dinner.

Sofia put out pickled green tomatoes and okra early.

"I heard the first one is hardest. I hope so," Lupita said, snacking on pickled okra.

Maria called out from the kitchen. "How many are you planning to have?"

"Six may be enough. We'll decide when we get there."

Dinner was simple, with only the pickles, oysters, and toast. They all continued sitting around the table, chatting, but when it came time to go, Lupita asked to stay.

"Of course," Maria said. "You are always welcome here. We will see how you feel tomorrow."

Maria washed her face, then went to bed. At four, she woke Felipe. "Manuel is coming! Get me to the doctor."

Felipe jumped up and dressed, but Lupita couldn't get out of bed. Soon, the whole family was awake. Jose called a doctor, while Maria and Sofia took care of Lupita.

Mateo, Felipe, and Jose waited in the living room. Mateo brought the ladies extra towels. Sofia took them.

"Bring me a bowl and pitcher of warm water," Maria requested. Mateo ran to get that, but Felipe had overheard and was already filling the pitcher when Mateo got to the kitchen.

The doorbell rang, and Mateo went to answer it, letting Felipe bring the bowl and pitcher to the ladies.

As the doctor stepped inside, Manuel's first cries rang out.

"I guess I know where the mother is," the doctor said, bringing her things and running to the guest room.

When she arrived, Lupita looked a wreck, and Felipe was holding his son, wet and sticky everywhere.

Dr. Ramos quickly took the baby's vitals, then the mother's. She busied herself with paperwork. Sofia was washing Manuel gently.

There was no hurry to go home, so Lupita, Manuel, and Felipe stayed for a week. Sofia was their constant nurse.

But time came when Lupita wanted to go back to her home. With hugs and kisses from everyone, they said goodbye and went back home.

Manuel had arrived, and the iron bowl was now a month out.

Maria cooked for Lupita and family and sent the food with Sofia, who became Lupita's assistant. Lupita was grateful for her help, especially at night when the baby cried.

Mateo came home from the dock and found his father in the office. "Papá, Martian iron is coming. Let's go pick it up."

"Why? It's just iron. Record the impact site and pick it up next time we are out there."

Mateo went to his room and started analyzing the costs and the value. He knew his father was right. Iron wasn't valuable enough to pay for recovery, but it was out there, and so should they be. He called ArcelorMittal. The best they could give was five million pesos. Looking at the fuel, crew, and lift reagents, he calculated they might come in under three million pesos, as long as nothing went wrong.

That night, Felipe, Lupita, and Manuel came over for dinner. Maria and Felipe helped Lupita up the stairs. José opened a bottle of Nebbiolo, pouring for Felipe, Maria, and himself. Maria brought water for Lupita.

While they ate, José brought up the subject. "Felipe, my son proposed we go hunting for the iron test ingot. It will be here in a few days. What do you say?"

"I think we would have a great time, but you know I need to take care of Lupita and Manuel now."

"Do you? Are you saying you don't trust Maria and Sofia? Lupita can live here and be waited on hand and foot. We three men go fishing," Jose answered.

Felipe looked at José and then at Mateo. "One bowl of base metal. I doubt it will bring a profit."

"I talked to ArcelorMittal. It could bring a small profit. I would like to see that the shape is correct."

"Lupita?"

"Go, Viejo. Maria and Sofia take better care of me than you do, anyway. Go catch that Martian fish."

José was at the helm of *Peces de la Luna*. He insisted on packing three recovery kits. Lupita and Maria took turns on the telescope watching the bright yellow ship with a blue racing stripe pull slowly out of the harbor, then lift up on its plane as José hit the throttle. With only three lifting kits, it was light and moved fast. José enjoyed the heel as he turned first a few degrees south, then a few degrees north, just to feel the ship handling.

Maria turned to Lupita. "That man is still twelve. I think he loves that boat more than his own family."

“No, Maria. He loves that boat, but he loves you and Sofia much more.”

It took them a day to get to the holding point, 18°N, 120°30'W. Felipe went below deck for a moment and came back with three huge fishing rods. “We’re here a day early. Let’s see who catches something first.”

The pilot took a wide circle at five knots. They all went down to the fantail. Every three or four minutes, José would throw out a live sardine. The three of them let their lines run. The sun kept moving, but no one caught anything.

Mateo saw the first one, blue and yellow, jumping high enough to be seen. “Dorado!”

Each of them started adjusting, pulling the line in, releasing it a little, hoping to get a bite.

Felipe called out, “I got one!” as his line pulled hard. Immediately, the fish started greyhounding over the water, flashing blue and yellow.

“He’s as beautiful as my boat,” José said. He and Mateo pulled their lines in and stowed their rods.

Felipe controlled the line as the fish started cutting first to one side, then the other. Slowly, he pulled the bull in closer. Mateo got the gaff when suddenly, the bull dived in a spiral.

“Pull the head up!” Mateo called out.

Felipe was way ahead of him, but the bull was strong. He got it up to the surface, and Mateo was again ready, when it dove again. José had the pilot slow down a little. Soon, the school was all around them. José kept dropping sardines, and when a cow got too close to the fantail, Mateo grabbed her with the gaff and slid her into the scupper. Felipe continued fighting the bull. When Felipe’s bull was exhausted, Mateo hooked it, too, and dropped it in the scupper.

Two fish gave fifty pounds of meat. They cleaned them right on the fantail and had fish tacos for dinner.

José was at the helm at four in the morning, watching for the fireball. Nothing came. Mateo checked the schedule. It should have hit the atmosphere. A space telescope gave him telemetry, and he recharted the course.

“Papá, it’s late. It’s going to hit outside the safe zone.”

José gunned the engine. Mateo plotted the course. It was going to miss the target by a thousand miles and the safe zone by five hundred miles.

Mateo called the US Coast Guard. “*Sécurité, sécurité, sécurité.* This is *Peces de la Luna*. Information update regarding kinetic event. The impact zone has shifted west. The new target coordinates are eighteen degrees north, one thirty-five degrees west. All vessels are advised to maintain a wide berth. The event is expected in one hour. Out.”

It took a day to reach the impact site. *Estele* was another day out. Since all they had was a splash location, José dropped three UUVs to scan the area. They found it three miles from the calculated impact site.

Mateo ran the glider deployment. The gliders were really just rails around the bags. They were filled on the second deck overlooking the fantail. With the aluminum/sodium hydroxide bag ready to go and preattached to the glider, the ammonia was poured into the rear bag, and the glider went off the launch ramp, diving fast.

“It looks much smaller than the old tanks.”

“The best part is no cryogenics, so it is stable. Easier to carry, less concern when it’s on the bottom if there is a lift problem. We can even resupply from a freighter if we need to.

Two UUVs surfaced while the other one guided the recovery glider to the bowl. The heavy hawser was snapped on, and the water vents on the aluminum bag were opened. As the temperature rose in the ammonia tube, it began pulling ammonia faster and faster through the nickel wool reactor. The lifting bag caught enough hydrogen to lift up, pulling the glider and reactants with it.

The reaction continued until gas started to spill out from the bottom. At fourteen hundred tons of lift, the iron only looked stationary. It was climbing, a millimeter at a time.

José suggested they blow the silt away. Mateo told him to wait. The motion had been imperceptible, but not zero. As they watched, suddenly it became obvious, and they heard the snap of the sticktion release.

As they headed home, they hailed *Estelle*, still coming in. Mateo suggested they transfer and let the crew take *Peces de la Luna* home.

José had never been on the new recovery ships. The twelve-hundred-ton crane was amazing.

Mateo was carefully monitoring ascent speed, running about one foot per second, slowing slightly in the last 50 feet.

José saw the numbers. “Mateo, slow down. I always go four inches per second. Bobbing can destroy the lift hawser.”

“Papá, don’t worry. The coil spring connector takes the shock out of it.” Mateo slowed the ascent, anyway.

Estelle moved slowly, and the UUV dragged the gear under the crane. The crane lowered the connector, and the UUV switched the bag lift for the crane lift. At sea state four, the active heave compensation made the lift smooth and easy. One attached, the crane simply winched it up, letting the ACH deal with the waves. The UUV hauled the lift bag and gear forward in the tunnel,

where a light fifteen-ton crane pulled it all up and transferred it to the second deck, where it was washed, fogged, and secured.

The entire recovery had only taken forty-five minutes. The three men called home. Felipe talked first.

“How are you feeling, Lupita?”

“Maria is very kind to me, but your son misses you.”

“Does he? You put him on the line.” She held Manuel on her lap and put the phone to his ear. “Manuel, you need to help your mother.”

Manuel cooed.

Mateo spoke next. “We will be back in four days, Mamá.”

“I’ll watch for the fast yellow ship with a blue stripe,” Maria said.

“No, Mamá. That ship will be back before us. We transferred to *Estelle*, a catamaran.”

José took the phone call into his cabin to talk to Maria.

By the time they got the iron ingot back to Lázaro Cárdenas, Mateo had worked out a deal with ArcelorMittal for the iron, if they could break it down into pieces no larger than one hundred tons.

Mateo put up temporary walls around the bowl and filled it with dry ice. When it was cold, a single drop from a wrecking ball shattered the bowl into manageable pieces. José kept one piece, and they sold the rest for five million pesos.

As soon as they returned, Mateo called Mars. “Callum, what happened? You guys missed by a thousand miles. It was an emergency shipping hazard.”

There was no response for thirty minutes. Finally, “Sorry, mates. I brought Tom. Let’s work out what happened.”

Tom had sent his targeting data already. “When we talked before, you said you were in Mountain time. You calculated the impact site and the time of day for that impact, and sent it to me. I input that on the targeting software on the mass driver and on the navigation pack. By my calculation, it landed as scheduled.”

“I am not in Mountain Time, I am in *Tiempo del Centro*. Why is this scheduled for four AM, mountain standard time?”

Mateo sat quietly for the entire fifteen-minute wait.

"When we talked, you told me the time. It was the same as Denver, which is Mountain Time. That was Daylight Savings Time, but now it is standard time, isn't it?"

"Mexico doesn't have daylight savings time, and we are not in that time zone."

Mateo waited again.

"I am so sorry, Mateo. I see what I did wrong. We will use Mexican Central Time from now on," Tom said.

Callum was as upbeat as ever. "Well, we got that sorted. But, given the timezone issue, we actually hit our target spot on. Are you lot ready for the beryllium?"

"It is still under construction, but will be ready in a month or so. You can send whenever you're ready," Mateo replied.

A few weeks later, Mateo was at the docks when they finally tore down the first building. When it came down, Mateo found an old sign, a little smaller than the one he was used to. He brought it home.

"Papá, I thought the company name was Acero del Pacífico. This says, 'Hermanos Acero del Pacífico.'"

"That's the full name when Felipe and I started this. He was twelve at the time. Grew up in it. That is still the legal name, but Acero del Pacífico is snappier, so we made a new sign."

The next few days, José and Felipe spent more time together, up in the villa and down at the mill. They were talking very seriously, and whenever Mateo came by, they would stop, then talk about lighter subjects.

Four days later, Mateo was walking down by the dock when José found him. They bought some burritos from a truck.

"Happy birthday, Mateo."

"Thank you."

"What do you think about Felipe and Lupita moving to Alabama?"

"Papá, we still need him."

"We will miss him, but we will survive without him. When I bought him out, he also let me buy twenty percent of Planetary Metals."

"How will we operate?"

"We will continue to recover interplanetary metal and sell to Felipe, but he will be buying metal not only from us, but from around the world, and doing millwork and even fabrication there in Mobile. It's easier for the railroad, and easier for us."

"OK, Papá. Do you know when they are going?"

"Felipe is going to scout out locations next week. He plans to move in a month. You know that Martian Iron?"

"Sure. You kept a piece."

"Yes, for a new sign. I pounded it out in my blacksmith's shop in the villa. You can see the workers are putting it up."

"I can't wait to see it."

"You know, Mateo, with Felipe gone, I will need a partner. Since you were a boy, I always dreamed of this becoming 'José and Son's Acero del Pacífico.'"

"I've been watching you, though, this last year, and I don't think that really makes sense."

"What do you mean, Papá? I've worked very hard. The beryllium project is going well."

José held up his hand. "Don't try to change my mind. I discussed it with Felipe while we were doing the pickup, and more recently, when I was buying him out. We had to refile the paperwork, anyway. Felipe recommended one man he has worked with, who has been in metals all his life and knows the deep-sea recovery business. I have talked to that man, and agree with Felipe."

"Sí, Papá. I trust you and Felipe."

They had been walking as they talked and were now standing in front of the sign. The workers were bolting it in place. When José looked at the workman, he pulled the paper wrap off the sign. It read "Mateo's Acero del Pacífico."

"Happy Quinceañero, son."

Chapter 17: Lunar Control

Price of ^3He	\$8,000,000/ton
Current Production	2 tons per month
Current Consumption	8 tons/year
Time Since Start	4 years

Abundant water from Mars changed operations at Tan's Cantina. No more gritty, chemically clingy frost scraped out of lunar shadow pockets. Mars had overcapacity, and the Moon had thirst. The pipeline was open.

Steel production, using Lunar iron and Martian carbon, expanded the settlement. The cableway grew, and new habitat shells could be assembled on site. Miners arriving from Earth brought just the core with them, and bought the steel structure in Tycho.

Li and Darrin kept their original hut.

Martian food was the staple now. The cantina smelled like roasted garlic, with cavy and amaranth chicha the most popular menu items. Earth food seemed foreign.

People could come in at any time to get a hot meal, but they did more; people lingered. They talked and laughed.

And Li made money. From the cantina, of course, but also because taking care of miners increased ^3He production, and she got her twenty percent.

Helion Dynamics was now serious about building the terawatt tokaplexes. Biloxi had built a large, crescent-shaped artificial island to support their construction. Raines was stockpiling every gram of ^3He he could get. When the terawatt tokaplexes came online, each one would consume five tons of helium per month. The Moon was producing two.

Li was in the cantina when a Lunar Tug pilot radioed in. "Tan, we've got four cheechakos. Certified claim about a hundred miles from Shackleton. They wanted a direct drop, but I told them the nearest Odd Point Rendezvous near there is in two weeks. They don't want to wait, and I've got nowhere to park them. I need to refuel at Tan's Depot to make the hop, and that's going to cost them. Can you work some of your magic on them?"

Li smiled. “Tell them that while you are refueling, they can come over to Tan’s Cantina, meet some fellow miners, and get a free chicha.”

Six hours later, the cheechakos arrived. Li didn’t say anything, but the bartender slid them some chichas, and they began talking. One of the miners took them to see the big map on the back wall of the cantina showing active sites, mined sites, and cableways. Li was strategically seated nearby. They were shocked to see their site had already been mined. Li stepped up. One of the miners introduced the cheechakos to her. “Tan, these guys just came up. Still drinking their free chicha. Looks like their claim is down south, and has already been mined.”

Li shook their hands. “You know, Darrin just put in a spur on the southern line, if you like that latitude. Makes dropoff easier, if you want us to process your regolith.”

“What do you mean? We have an RF extractor. We can process it ourselves.”

“You can. That’s up to you. Do you want to show me your gear?”

They all put their pressure suits back on and left the cantina. Li showed them how to clip their suits onto the cableway, and they rode back to the depot.

She made comments as she looked over their gear. “That RF extractor is pretty small. I have some larger units for sale. Of course, you’d want more solar power, too. How are you fixed for extra pressure suits? Do you all plan to live in that one hut? I have larger ones, if you’d like.”

Li was thorough, and they appreciated her attention to detail. They replaced some of their equipment and left some with Li. After a tour of the ^3He extraction facilities, they agreed to let her process their regolith. Her twenty percent cut seemed cheap compared to the cost of learning how to operate an RF extractor efficiently. Finally, they agreed to work the location she asked them to mine, as long as Darrin would show them where it was.

They called back to the pilot. “You’ll never believe what happened while you were getting ready for the hop.”

The pilot suppressed a laugh. “What happened?”

“We were in the bar, and we met a character. She helped us out and set us up with a better location. So I guess we won’t be needing the hop.”

“Was her name Tan, by any chance?”

“It was. Have you heard of her?”

“She does alright by the people nearby. Well, since you don’t need the hop, I am going to get some ^3He . Good mining.”

Li was using a hundred-kilogram cryo-cylinders filled with 99.9% pure ^3He . Helion Dynamics paid a premium for those shipments. They were reliably pure, and the large containers made for easier logistics. The pilot loaded three and headed back to LEO.

Tan was back in the cantina, adding the cheechako’s new claim to the map when Andrei called. “Tan, I just got a request. A man is looking to go to visit the Moon. He needs a hotel. I told him I would ask around. Do you have space?”

“Sure, we have space, if the price is right. Did you get a sense of how much I can charge this guy?”

“I don’t know. He sounds like a mid-level bureaucrat, but he works for the UN. Name is Tadeusz Vassiliev. Didn’t say why he was going, but if it is official, you can probably charge more. I should have thought of that. I just blurted out my rate.”

“You charge dollars. I charge the miners a gram per night, but this guy doesn’t have ^3He . Float \$20,000 per night and see what he says. Mention it comes with free oxygen and water.”

“What will he eat?”

Tan had to think. She was charging so much for meals here, no Earther could afford it, but if she cut him a deal, others would expect it. “Suggest he bring his own meals.”

That evening, Li talked it over with Darrin. He was curious. “Why would the UN be sending someone up here?”

Li looked around. The big map. The Martian food. Employees. None of this was actually authorized. It just all kind of came about after they got the reactor. Li froze. “Darrin, he’s an inspector. That is the only reason the UN would send someone to the Moon.”

“What would he inspect up here?”

“The UN still thinks they have jurisdiction up here. And they damn sure won’t like the reactor. That reactor is why we have success here.”

“What’s the play, Li?”

“We have a week before he arrives. When he gets here, Tycho will be a ghost town. Nothing interesting to report.”

Darrin looked at the cableway. “I don’t know how we can hide that.”

Li called a meeting in the cantina. She and Darrin explained the problem to Pavel, Luke, and Tseng.

Pavel spoke first. “Do we know that he is coming to look for a nuclear reactor?”

“No. We know a man is coming to the Moon and works for the UN. But what else would he be doing here?”

Pavel nodded. Li was right. “OK, if he is looking for nuclear power, he is going to bring radiation monitors: scintillators, neutron detectors, Geiger counters. I will shut down the reactor, but that won’t be enough.”

Luke jumped in. “Tan, when you shipped the reactor, you logged it as a double RF, right?”

“Yeah. That was Belov’s idea.”

“Well, if they are looking for a double RF, we give them one. A large sheetmetal structure with two doors and two compartments. I’ll bury a bottle of ^4He under the device. Put in tailings, send a puff of helium out into a collector.”

Darrin was looking at the equipment connected to the buried hot loop. “He will see everything else. I don’t see how this can help.”

Tseng nodded. “He won’t see anything. We will bury it all, set up Luke’s show closer to Tan’s Cantina. I will smooth it so it looks like none of us have ever been near the north wall. The tailing pile has more than enough regolith to do that.

Pavel liked the plan. “I am going to need to scrub for polonium. Increase the bubble rate of helium and slow the reaction. When I shut it down, we will switch out the cold trap. It will still be detectable if he knows where it is, but at a distance, it will be invisible.”

“I like the plan. If there is anything you want from Earth, you need to order it in the next two days, or it won’t get here before our visitor,” Li said. “Darrin has the accounts, so let him make the actual orders.”

With that, the meeting ended.

Tseng started bagging up tailings and moving them to a minor, secondary crater twenty miles east. Darrin reprogrammed the cableway to drop real bags in a crater just west of Tycho.

Luke set up the fake RF near a cableway pylon north of Li and Darrin's hut. It measured two meters by two meters by five meters, built from heavy eighth-inch steel. Hidden in the frame was a small tube, connecting a buried bottle of ^4He to an outlet nozzle. A tiny battery powered a few status lights. It looked convincing to anyone completely unfamiliar with gas extraction.

Tseng inspected the door. "Luke, this doesn't close properly."

"It latches. What's the problem?"

"It doesn't seal."

"It's a prop. No need to seal. Just look good."

"Doesn't look good to me."

"Let's hope Vassiliev is less discerning than you."

Luke practiced loading, collecting, and unloading until it looked natural. He kept it in for a minute, then hit the helium button.

Li watched. "Ten seconds is enough. If it's longer, he'll start thinking. We don't want that. Unload before the tension is drained. This is theater, not extraction."

Luke adjusted his performance.

Two days before Vassiliev arrived, Li sent out the news that Tan's Cantina would hold a one-day-only sale: half-priced meals and free chicha. Every miner dropped whatever he was doing to attend the blowout.

While they ate, Li briefed the miners. "If you see the inspector, talk slowly and move slowly. Stop by the cantina, but bring your own food. I will dispense water and oxygen. While he's here, nothing else will be served."

When everyone had gone home, Li vented the Cantina and removed the airlock. Looking around, she noticed the map. She pulled it off the wall and rolled it tightly, bringing it back to her hut.

Pavel scrubbed and shut down the reactor. Darrin and Tseng disconnected the ovens and the turbine and moved them out of the crater. They then buried the

entire site in a meter of regolith. As a last measure, Darrin turned down the speed on the cableway to about a mile per hour.

Li greeted Envoy Tadeusz Vassiliev in Tan's Depot while the pilot was unloading a pallet of meal kits.

"You're Tan?" he asked.

"Yes," she said. That was her first lie. "I can show you to your room. I will have your luggage brought over. Do you need a guide?"

"No, thank you. I'd just like to look around, get a feel for the place."

Vassiliev opened a case. Inside, Li noticed a Geiger counter, a scintillator, and a thermal scanner. She didn't let on that the last one surprised her. "Very well. If you'd like to meet up later, your room has a phone."

They put on their suits, and Li brought him to the cableway. "You can clip in and go wherever you want. Don't go too far or you can run out of oxygen." Not exactly a lie. She just hadn't told him other places he could get oxygen, or how long his suit would last.

As soon as Vassiliev was out of sight, Li called up Pavel. "Pavel, good and bad. No neutron detector, but he has a thermal scanner. Are we in trouble?"

"A thermal scanner? Yeah, those pipes were still hot when we buried them. He won't see anything under the sunlight, but he might during lunar night. Do you know how long he is staying?"

"He didn't say. I asked for a week's pay upfront, and he didn't complain."

That evening, Li returned to the depressurized cantina. Vassiliev came by while she was there.

"I saw your large RF extractor," he said. "It's just a double?"

"Oversized double core. For regolith processing. Much faster than processing out in the fields. Miners send their stuff here to save electricity." That last part was totally true.

"I noticed some regolith falling out while Luke was running it. I thought it was leaking. But, in the end, Luke collected two grams of helium. It must have been craft from loading. Where does the power come from?"

She looked up on the North rim of the crater. “Have you seen the solar panels on the rim? They get light even when the crater floor is dark.” True, she mused, for about ten hours a month.

“I’d like to see outside the crater.”

“Of course. Would you like someone to guide you?”

“I’d like to move around on my own.”

“The cableway is designed for just that. How much oxygen will you be bringing?”

“I’m not sure.” He turned without a word and left.

In the morning, she saw him with a K-bottle of oxygen, heading north to the rim. She watched him get off the cableway and check the solar panels. She was glad they never disconnected them.

Li nudged Darrin. They watched him with binoculars.

“Li, what is he doing up there?”

“He wanted to see where our power came from. I pointed out the panels.”

Darrin called Pavel, who was in his hut. “How far is the polonium waste tube from the panels?”

“About ten yards. Why?”

“Look at the rim.”

Pavel suited up and came out on the crater floor, watching with binoculars. Vassiliev wasn’t just checking the panels. He had his Geiger counter out. Then he stopped and knelt down.

“Oh, fuck! He found something!” Pavel shouted into the team channel.

“That’s in the middle of the panels. Nothing there, right?” Li asked.

“That’s not where the waste pipe is, but that isn’t how a man acts when he finds nothing. Maybe a drop of polonium fell out there?”

Vassiliev pulled out a trowel and began digging around. They watched him for twenty minutes. Finally, he put a sample in a bag and got back on the cable to continue north.

Darrin spoke first. “He doesn’t know what it is. There are other sources of radioactivity on the Moon.”

Li shook her head. “He doesn’t know. As far as he is concerned, you’re right. But he bagged it. It is going back to Earth, and they will find out.”

Pavel closed his eyes.

Li came back on the team channel. “Forget it. It is out of our hands now. Let’s make sure nothing else happens.”

She thought of other ways this could go wrong. A gabby miner might say too much, but anything contradicting Vassiliev’s own observations would sound like the ravings of a madman. Whatever was said, she might be able to talk her way out of.

If, however, he died out there, that would bring enormous attention. Not another inspector, but a battalion with the authority to dig, seize, and incarcerate. That could not happen. She had to keep him safe.

It was time for Rook. Li sent Darrin over the east rim to find him. Rook was on his claim, bagging regolith, when Darrin arrived. For a hundred grams, he dropped what he was doing and followed Darrin straight back. They met Li in the cantina.

Li was pacing. “Sam, I know you don’t like to talk. That’s fine. Lean into that. But I need you to meet this guy. He is carrying around a K-bottle.”

“A K-bottle? Why?”

“I might have suggested it would be safer to have oxygen on hand.”

Sam slapped his thigh. “Cheechakos are the best. No extra battery, but a bottle of oxygen.”

“Yes, well, I need you to accidentally bump into him. Ask if he wants help carrying the bottle. Never act like it is weird. Maybe even make a show of refilling it.”

“Sure. What should I say to him?”

“As little as possible. Move slow, talk slow, don’t know.”

Sam grinned. “That’s me.”

“One more thing. Can you bring this oxygen K-bottle with you? Make it look normal to carry a bottle around.”

Sam laughed. “Anything special I should do?”

“Just keep him alive.”

Sam agreed. Outfitted with the K-bottle, he headed north.

After an hour, Rook was ready to backtrack. He started mapping out different routes Vassiliev could have taken. Before he got off, he saw an odd sight: a miner was sitting on a rover, chatting with a man with a K-bottle.

“This is the talky part,” Rook thought to himself with a grimace. He got off the cable and started walking over. When he got in Local Band range, he stopped and listened.

Vassiliev was trying to figure out who was in charge. “How much regolith do you process in a day?”

The man blinked slowly, checked his bag weight. “Whatever we collect.”

“And your quota?”

“We don’t run quotas. I process my claim at my own rate. I take my own time to get my ton.”

“Your ton?”

“Whatever the claim has. Usually around a ton.”

Vassiliev had seen others, moving slowly, like they had all the time in the universe.

Sam decided to come over. “Hi, guys. I think I went the wrong way on the cableway. Can you help me out? I need to refill my K-bottle.”

The miner switched to FM, leaving Vassiliev out of the conversation. “What is with these K-bottles?”

“Makes our guy easy to spot.”

“Dumbest thing I ever saw.”

They switched back to Local Band. “Two strangers in one day. I can refill both your bottles.”

Vassiliev's eyes perked. "Thank you. I don't know how much I have left."

When the miner took their bottles inside, Rook plugged into the rover and signalled Vassiliev. "Hey, buddy, recharge. You don't want to freeze out here." He took a cable from the rover and plugged it into Vassiliev's hip pack. "Your charge is low. How long have you lived up here?"

Vassiliev wasn't sure what to say. "I am on vacation from Earth, and was curious to see the Moon."

Sam shut off his mike while he laughed. Vassiliev continued, unaware. "Look, if you have some time, what would you say to helping me? The Moon has dangers I am not prepared for. Tan offered a guide, but I don't trust her. I want someone she doesn't know about. I can pay."

"What's it worth?" Rook asked, barely looking up.

"How does five thousand dollars sound?"

Sam smirked. It sounds like a quarter of a chicha. "That would be fine. My name is Rook. Where we going?"

The miner brought back the full K-bottles.

Vassiliev grabbed his like it was a life preserver. "Thank you for the conversation and for the oxygen. I've taken enough of your time; we'll be on our way."

The miner looked surprised. "You two together? I didn't realize that."

"I just hired him," Vassiliev explained.

Sam kept track of where they were and made sure Vassiliev kept his battery charged. They moved around under the cableway and even past it.

Vassiliev watched miners from behind his visor. Watched a man bag maybe ten kilos in half an hour. No music. No jokes. No camaraderie.

They looked like ghosts.

"I expected... more life," Vassiliev said.

Sam looked at him. "They're mining."

Sam pointed at the sun. "If you don't want to get caught out here in the cold, I suggest we head back to Tycho Crater by sunset."

“Sure. Nothing to see out here, anyway.”

The ride back to Tycho Crater was long and quiet. No conversation, no confrontation, no evidence. When night fell, Vassiliev took out his thermal imager and started looking around.

When they came over the ridge, he imaged the entire crater. Li’s hut door and the tube for her Earth window stood out, and the area near the RF extractor. Tan’s Cantina glowed faintly. Some of the regolith in the waste pile was hot.

He dropped by the Cantina to talk to Li. “I took a thermal image of this crater as we came back. The only signature I saw was the things you showed me and some of the regolith on the tailings pile.”

“Probably something fresh out of the RF extractor.”

“Probably. Well, I’m checking out. Lunar Tug will be picking me up here tomorrow.”

“Luke and Darrin are in Luke’s hut, if you want to talk. I heard Luke has some beer, if you’re interested.”

“You know, if you can keep it quiet, a beer might be very nice right now.”

Li escorted him down to Luke’s hut. They came through the airlock and hung up their pressure suits on the wall. As promised, Luke had a six-pack of Earth beer. Pavel was there, too. They chatted for about half an hour. Pavel remembered he had some frozen Earth bread and excused himself to go to his hut. He was back in about ten minutes, and they microwaved it and enjoyed the food. When Luke mentioned he was tired, they all suited back up and left.

Li and Darrin helped Vassiliev get his gear back to the depot. While on Lunar Tug, he called his supervisor. “Tim, this place is like a graveyard. They use solar power, so they just sleep for two weeks at a time, hoping their batteries don’t die and they don’t run out of oxygen. We were looking for black markets? The people are all loners, barely able to talk to others. They couldn’t begin to start a black market. Honestly, it would be better if they did.”

“You didn’t find a reactor?”

“No, they are using solar panels. I verified the connections. Luke is very fast and efficient with the RF extractor. I came over the rim during lunar night and ran a thermal scan on the entire crater. If there was a reactor, even a buried one, it would have shown. There was nothing.”

“No radiation?”

“Yeah, well, yes, I did find some radiation. I searched the area and didn’t find any more, but I bagged what I found. I am curious to see what it is.”

“Get that to the lab. They’ll figure it out. Who is running the show up there?”

“Darrin. He lets Tan do all the legwork, but it is clear that Darrin is the one who is running things. Tan just sits in the cantina all day. She offered me algae soup in a squeeze bag so I could drink it suited. It was disgusting, but she loved it.”

He watched through the porthole as Tan’s Depot shrank in the distance.

“Look, no reactor. No illegal food service. No commerce that I can see. Honestly, Tim, I think we’re worried about nothing. These people are too dead inside to cause trouble. And I mean that literally. This place creeps me the hell out. Do not send me back here.”

Li watched the Lunar Tug shrink to a glint and vanish in the black sky.

Darrin came in and found her. “Li, the Lunar Tug pilot just sent me a recording of a call the inspector placed. He thinks I’m in charge.”

“Good. If they do come back, they won’t arrest me. Now, let’s get back to business. Turn it back on. All of it.”

Darrin didn’t hesitate. “With pleasure.”

Tseng and Luke dug out the reactor, and Pavel began the slow process of bringing it back online. Once there was power, the cableways sped back up to normal speeds, pulling in bags of regolith. Miners could see the motion from their huts and clipped themselves to the lines, gliding in.

Li reinstalled the airlock on the cantina and started repressurizing.

In about ten minutes, the pressure was enough that they could start the music. She played it loud and pumping. The staff flipped the tables upright and set up stools and chairs.

Li grinned. “All drinks are half price. Anyone who can’t get a room is welcome to sleep in the cantina tonight.”

Miners were coming in fast, leaving their pressure suits hanging on the wall. The bartender cracked open a jug of chicha. The earthy smell filled the cantina as he started pouring mug after mug of the cloudy amber.

Li and Darrin stood by the wall, watching the festivities. Luke, Tseng, and Rook showed up. The five of them pushed two tables together and started talking. About fifteen minutes later, they were joined by Pavel.

Sam was talking about the ride back. “It was night, and Vassiliev pulled out a thermal imager. Fine, nothing to see out there, but when we came over the rim, I totally knew the reactor was going to glow like midday. But nothing. Were the pipes really cold?”

Pavel laughed. “When Li told me he had a thermal imager, I knew exactly what our problem would be. I can’t let the pipes freeze. Electrical power was keeping the metal coolant molten. So, while he was gone, we scraped six inches of regolith and replaced it. We dumped all that hot regolith on the tailings pile.”

Sam slammed his chicha down. “Hah! That’s why the pile was hot. I wondered how a fake RF heated the regolith.”

Luke was next. “How was the Earth beer? I had to go with that, you know. Vassiliev isn’t ready for chicha. Pavel, you were smooth.”

“Who? Me? All I did was share some bread. And accidentally slipped a sample bag out of Vassiliev’s suit when I was putting mine on.”

“What did you do with it?” Darrin asked.

“I dumped it near the wall and refilled it with some normal regolith. Looks the same.”

The party continued for another six hours. Finally, Li and Darrin went to bed, leaving the bartender in charge.

Li looked up through the Earth window. “Productivity has made us friends and earned us enemies. Our allies will not always protect us.”

Part II

Jovian Resource Corporation

Chapter 18: The Charter

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Restrepo arrived early. The Kuwaiti delegation had called the meeting, but arriving first meant watching the others enter, reading their faces before the diplomacy locked into place. He chose a seat near the window where he could see both the room and the city beyond.

New York stretched outward in its familiar grid, financial towers pressed against older districts, everything held in place by quiet authority. For a century, oil had been so organized: predictable, distributed, controlled. Now ^3He threatened to redraw the entire map. And Venezuela, already bleeding, would be the first to die.

The others arrived in silence. Al-Rashid from Kuwait took the head of the table with the easy confidence of someone who'd orchestrated a hundred meetings like this. Morrison, the Canadian delegate, looked tired, his jaw set. Okonkwo from Nigeria carried himself with the alertness of someone determined not to be sidelined. Dr. Lais, representing Indonesia, moved with careful precision, her expression giving nothing away.

Al-Rashid opened his slate. "Thank you for coming. The numbers have worsened. Fusion uptake is accelerating faster than our models predicted."

Restrepo didn't wait for the polite circling. "We are running out of time. Venezuela cannot survive another decade of this. Oil revenue is the only thing between us and complete collapse." His accent felt thicker when emotion crept in. He forced himself to slow down. "Fusion is extinction for my country."

"Nigeria faces the same reality," Okonkwo said, leaning forward. "We are funding infrastructure, education, healthcare. All of it depends on petroleum revenue. We will not be pushed aside while others decide our future."

Morrison's voice carried a bitterness Restrepo recognized. "Alberta's been keeping Canada solvent for years. Now Ottawa vilifies us for it while spending every dollar we send. Fusion gives them an excuse to turn off the taps completely."

At least Morrison's resentment came from being unappreciated. Restrepo's came from watching his country starve.

Restrepo pulled up his own slate. "I've been looking at the output numbers from the Moon. They don't make sense. I know they're using unlicensed nuclear power up there. What did the inspector find?"

Al-Rashid winced. "Vassiliev found a radioactive sample, but nothing else. I've talked to him extensively. When we gave the sample to the labs, they found ordinary ilmenite. The lab says it must have been cosmic radiation. Based on what he found, I'm surprised the Moon is delivering as much ^3He as it is. We thought Tan had a nuclear plant, but he checked it personally. Solar arrays on the ridge of her crater and a very large dual RF processor and absolutely no heat signature anywhere else."

Morrison shook his head. "No way. They can't deliver those numbers without nuclear power. He didn't look hard enough."

"I share your frustration." Al-Rashid's tone carried the weight of someone who'd already fought this battle. "But Vassiliev is a good inspector. There may be a nuclear plant up there somewhere, but we need to accept we cannot find it. And even if we did, our options are limited. Besides, since we authorized General Atomics to build nuclear ships, Mars and Ceres just park the ships and use the reactors for power. All legal."

Morrison looked up. "Then we hit the supply chains. Can't touch the reactors themselves, but we can restrict what feeds them. Lunar operations depend on regular fissile shipments for auxiliary power. Mars and Ceres too. Slow those shipments, production tapers."

Okonkwo's arms crossed. "How?"

"Environmental safeguards," Morrison said. "Several delegations have already raised concerns about moving enriched materials through Earth's atmosphere. The language is vague enough to expand."

Restrepo saw it immediately. The logic was elegant, almost beautiful. They couldn't stop fusion, but they could starve the mechanism feeding it. "Frame it as international responsibility. A phased restriction tied to environmental review. No one will understand the impact. It'll pass overwhelmingly."

"Perfect," Al-Rashid said. "Most delegates won't consider the implications for lunar production. It looks prudent. Responsible."

Al-Rashid's expression darkened. "Next topic. There's an urgent development. Raines is pushing a Jovian Resource Corporation. He claims the Moon cannot supply enough ${}^3\text{He}$ to meet global demand. He wants the UN to charter a corporation to unlock unlimited fusion fuel from Jupiter."

Restrepo felt his stomach drop. "The vote?"

"Will pass easily," Al-Rashid said flatly. "Raines has built a coalition we cannot break. Too many nations see it as their path to cheap, unlimited energy. We cannot stop the charter."

Morrison shook his head slowly. "If Jupiter opens up, lunar production becomes irrelevant. Everything we're doing won't matter. We'll be finished."

Restrepo looked at the city beyond the glass and felt the old helplessness rising. Venezuela had been outmaneuvered for decades. OPEC had done it. The Americans had done it. Their own incompetence had done it. Now fusion was finishing what sanctions never quite managed. And Jupiter would be the final nail.

Lais looked up from her slate. "I think Jupiter may be an impossible problem. This corporation won't matter at all. Let the baby have his bottle."

Okonkwo's face darkened. "I was sold that promise before Ilha Fusão. Then they produced steady fusion. I was told they would never fuel it, but now they're mining the Moon. These people will find a way. They are very inventive, and we cannot tolerate that risk."

"Influence the board," Lais said, her earlier skepticism replaced by calculation. "Fill it with the right kind of people. Treaty specialists. Governance experts."

"People who move slowly," Morrison added.

Restrepo nodded. "People who look ideal for a new space institution. Impartial. Scrupulous. They conduct studies, hold consultations, write reports. Progress looks like it's happening, but the corporation never advances beyond planning."

Okonkwo uncrossed his arms. "And if someone notices?"

"They won't," Al-Rashid said. "It will be interpreted as best practice. No one objects to caution in space."

The room settled into understanding. Restrepo felt the weight of what they were doing. They were buying time for their economies. They were also sabotaging

humanity's expansion into space. He pushed the thought away. Venezuela was dying *now*. Whatever Jupiter might offer humanity could wait.

"We coordinate quietly this week," Al-Rashid said. "The fissile resolution needs the right sponsors. The board nominations need discreet support. Nothing overt."

They agreed without ceremony.

Restrepo stood, closing his slate. "My country is dying, and we are trying to buy enough time to find another way to survive. I hope you all understand what is at stake."

"We do," Al-Rashid said quietly.

A week later, the General Assembly chamber hummed with anticipation. Restrepo took his seat in the Venezuelan section, several rows behind the Helion delegation where Raines sat with his aides. The American looked confident, relaxed. He had no idea.

The fissile ban resolution appeared on the docket. Restrepo kept his expression neutral as the reading began, though his heart rate picked up. Everything depended on the sponsors holding, on the environmental language resonating, on dozens of delegations voting without thinking through the implications.

The electronic board flickered green and red.

The resolution passed. Overwhelmingly.

Coño. Relief and guilt hit simultaneously. The ban was binding. Lunar reactors, Martian grids, and Cerean foundries would all feel it within years. A decade at most. The installations would wither slowly, starved of fuel they couldn't acknowledge needing.

And hunger for energy would not go away. Venezuela would sell oil again.

Across the chamber, Raines rose to speak. Restrepo watched the man's confident cadence, the rehearsed gestures, every appeal to global prosperity. The Jovian Resource Corporation speech was polished, persuasive. Raines believed every word.

The vote was called. Overwhelming approval.

Raines sat to polite applause, relief visible in his shoulders. The man thought he'd won.

Then the board appointments were read aloud.

Dr. Emile Carson, the author of the seven-volume treatise on the Outer Space Treaty.

Ambassador Chidinma Okafor, the steward of treaty review committees.

Professor Liu Wei, a sustainable development theorist, cited in every UN resolution, but never put into practice.

Restrepo watched Raines' face. The smallest flicker of confusion. A tightening around the eyes. The man was beginning to understand that something had gone wrong.

Delegates swarmed Raines afterward, offering congratulations. Restrepo remained in his seat, gathering his materials slowly. He felt no triumph, only a hollow kind of satisfaction.

He stood and made his way toward the exit. As he passed the Helion section, he caught a glimpse of Raines accepting praise with a strained smile. The man understood now. The fissile ban would strangle the outposts. The Jovian Resource Corporation, his great victory, had been hollowed out before it began.

Outside, the New York evening pressed against the glass facade. Restrepo walked toward his hotel, phone already buzzing with messages from Caracas.

He had done what he could. The fissile materials ban would slowly strangle the ^3He supply, and Jupiter was decades away, if it ever happened. Venezuela was back. Maybe they would even find another path forward, diversify, prepare for a future that no longer needed its oil.

He told himself it was enough. This time, he believed it.

Chapter 19: Twinkle, Twinkle, Little Star

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Max Juno was the first to hear about it. Sitting in his Lagos office in the newly completed tower, he was on a routine call with a contact at Shibata Capital in Tokyo. They drifted through the usual topics: metals contracts, new alloys, logistics, until a small aside broke the rhythm.

“There’s a reactor program. Kagutsuchi Vortex. Early runs show stability at two, maybe three gigawatts. Machine about the size of a city bus. They claim it is very stable.”

Max noted the words, asked no follow-up, and let the talk slide back into routine. Only after the call ended did he allow the weight of it to settle. Small fusion, if real, could be very disruptive of the fusion industry Helion Dynamics was building.

He composed a message to Tobias Raines:

“From a source at Shibata Capital. New reactor: Kagutsuchi Vortex. Two to three gigawatts. 20m footprint. Early data promising. Sharing because you should know.”

He hesitated over the last sentence, aware it revealed more than he liked. After all these years he still did not know whether Raines’ driving aim was to remain the richest man alive or to see fusion carried into every corner of civilization.

In Geneva, Raines read the note at his desk overlooking Lake Geneva. He pressed a button and gave three instructions without pause.

“Get Max Juno on the line. Clear my schedule. Ready the jet for Tokyo.”

Max Juno was still in his office when Raines’ call came through. He excused himself smoothly and answered the call.

“Hello, this is Max speaking.”

“Max, how soon can you get to the airport? A jet is waiting for you there in Lagos. We’re meeting them in person.”

Max hesitated, just for a moment. “You’re sure?”

“I’m sure,” Raines said, and in that certainty there was something larger than business. something closer to faith. “You found this. You belong in the room.”

The line went dead, leaving Max staring at the polished surface of the conference table, his own reflection looking back at him, eyes sharpened by the weight of what was coming.

Max was already in a limo when he called his Shibata Capital liaison. “Secure a slot at your headquarters. Tonight. Raines is coming personally. He wants Kagutsuchi Vortex present; CEO and chief engineer, no substitutions.”

The liaison swallowed hard, bowed reflexively though it was over the phone, and set to work.

Raines next called Dr. Elin Sarai. “Elin, listen, I am going to need your technical expertise. Have you heard of Max Juno?”

“Max Juno. Maybe. Is he the guy who financed the Mars operation?”

“The same. Smart man. You’ll like him. He and I will meet you in Biloxi. We’re going to Japan.”

They converged in Biloxi, where Sarai’s manufacturing plant dominated the Back Bay shoreline. She was already waiting on the tarmac, helmet under her arm, coveralls marked with graphite dust from the shop floor.

Raines stepped down first. She studied his expression before speaking.

“Tobias, what is it?”

“Another fusion plant,” he said simply. “Smaller than ours. If the reports are true, it’s quite amazing. We are going to visit the company that made it.”

She had brought her crash bag, the things she needed if she was suddenly called away on a business trip. Together they boarded the jet bound for Tokyo. Max filled in the details he had.

In her excitement, Elin’s words spilled out fast. “They have a self sustaining stellarator? That’s amazing. We must see it.” As the flight continued, Elin gave an

impromptu lecture on the technology, bringing Max and Raines up to speed on what to look for when they arrived. “Stellarators are generally more stable than a tokamak, but none have been able to generate the confinement needed to sustain fusion.”

“We sell tokaplexes, not tokamaks. Why did you say ‘tokamak?’” Raines asked.

“The fusion core itself is a tokamak. It is also fairly small, though larger than the Kagutsuchi Vortex reactor. The process of using that and keeping it fed and stable and extracting electricity from it requires many more buildings and structures. That complex of buildings is our tokaplex. It is the turbines, cooling towers, emergency systems, and controls that take up so many square miles, not the reactor.

Max sat back, unsettled, staring at the darkening horizon. He had delivered the message because Raines deserved to know. Max had made his first fortune selling lunar helium to Helion Dynamics. Raines had backed Magazine Beach when no one else would. The loyalty ran deep. But even now, uncertainty gnawed at Max. Would Raines see the Kagutsuchi Vortex as . threat or an opportunity? Would he crush them, absorb them, or enable them?

At 35,000 feet, Max spoke briefly with Shibata Capital’s liaison, confirming the attendance of Kagutsuchi Vortex’s two senior figures: Kaito Ishikawa, the reserved CEO, and Dr. Naomi Fujimoto, whose early work on vortex confinement had become a foundation for the field.

“They are honored to meet with you,” the liaison said with practiced formality. “And eager.”

“Good,” Max replied, ending the call without ceremony.

The Gulfstream touched down at Haneda under a velvet blue sky. A black sedan waited, its driver and attendant already in position. Elin stepped out after Raines, a small carrycase in her hand, posture sharp but restrained. She glanced at Max, as though reading uncertain data, withholding judgment.

Raines turned to Max, understanding his uncertainty. “Max, you did the right thing. If this is real, it changes everything. If it isn’t, we lose a day in Tokyo.”

The car carried them through Tokyo’s arteries of neon and glass before arriving at Kagutsuchi Vortex headquarters, a low, modernist complex on the edge of the bay. Security was discreet but thorough. Waiting at the entrance was Nakamura, the director from Shibata Capital. His bow was precise, his English polished.

“Welcome. Ishikawa-san and Dr. Fujimoto are inside.”

Nakamura led them into a large hall where the reactor dominated the space, coils gleaming under hard white light. Around it, consoles and scaffolding suggested both research and ongoing construction. By the machine stood two figures waiting for them.

Kaito Ishikawa, tall and angular, carried a stillness that suggested calculation as much as patience. His suit was flawless, his hands folded behind him.

Beside Ishikawa stood Dr. Naomi Fujimoto, slight and sharp-eyed, a slate in her hand. Her posture was alert rather than ceremonial, the stance of someone who lived between theory and machine. Her eyes were already on Elin, measuring.

As Raines, Elin, and Max entered, Ishikawa bowed lightly.

“Welcome to Tokyo,” he said, voice calm and deliberate. “We have much to show you.”

Raines returned the bow, not as deeply, then gestured toward the machine dominating the hall.

“Let’s see what you have built.”

They moved closer. The stellarator loomed, coils curving in elaborate arcs, a twisted cage of superconducting fields surrounding the core chamber. Technicians stood back as Fujimoto touched a console.

“The central challenge, as you know, is maintaining density and temperature while keeping the plasma stable. We have achieved this with a field geometry of our own design.”

Lights across the console shifted. Nothing changed in the room. No noise, no light, no vibration. Inside the stellarator, the plasma kindled, a deep blue glow held in its magnetic cage. The confinement lines shimmered faintly against the chamber walls. Elin watched with fascination as the needles on the reaction gauges moved up and remained steady.

“It is stable,” Fujimoto said. “We have sustained uninterrupted operation for seventy hours.”

Max raised an eyebrow at the figure. Raines showed no outward change, but his attention fixed.

“There is some waste,” Fujimoto continued. “Deuterium and ^3He near the boundary must be vented to maintain stability. We replenish continuously. The venting also removes ^4He ash.”

Elin leaned in slightly, studying the graphs that appeared on the console. As it continued, the outer shell of the stellarator began to glow dull red from residual heat.

“Our present limitation is not fusion,” Fujimoto said, her tone matter-of-fact. “It is recovery. Our steam cycle cannot absorb the generated energy. Turbines saturate, and over ninety percent of output is discarded.”

Elin spoke without hesitation.

“Mechanical bottleneck. Not thermonuclear.”

“Correct,” Fujimoto answered, her eyes sharpening in acknowledgment.

While Ishikawa and Raines spoke quietly at the main table, Elin and Dr. Naomi Fujimoto stepped aside. The hum of the reactor hall softened behind them. A small cocktail table stood between them, forgotten by the others.

“Your machine is beautiful,” Elin said, her tone precise but sincere. “Can you make it larger?”

“There is a limitation in size,” Fujimoto explained. “At larger sizes, the field cannot contain the plasma once fusion begins. At smaller sizes, edge effects dominate, and ash sweeping removes too much reactant. We found this sweet spot at two gigawatts thermal.”

“How do you keep heat from destroying your magnets?” Elin asked.

Fujimoto gave a small smile. “This project is less about fusion and more about materials and heat exchange. Ishikawa sees the small footprint and calls it revolutionary. He compares it to your facilities without realizing most of your acreage is turbines, cooling, and power distribution.” She paused, then continued in a quieter voice. “The real advances are in the walls. We use silicon carbide for both the first wall and the containment shell. Then a spiral of magnesium oxide, so helium can enter cold, absorb heat as it winds through, and exit hot enough for generation. We’ve reached outlet temperatures over twenty-five hundred degrees Celsius. Different ceramics might allow us to boost that temperature, if needed.”

Elin's eyes sharpened at the detail. "That's impressive. You know, electricity isn't always the primary use-case. Offworld, they need steady process heat for smelting. Something like this could serve directly, without turbines at all."

For a moment both women stood silent, each considering how the other's work reframed her own. When they looked up again, it was with the recognition that they had learned something neither had walked in knowing.

Elin nodded. "What you have built here is phenomenal."

Elin talked to Raines briefly afterwards, explaining her thoughts. There was a pause, a silence where fortunes are decided. Raines nodded once to Elin, then broke the silence, his voice deliberate, cutting cleanly across the room.

"How scalable is production?"

Ishikawa lifted a hand. "I regret to say that development has exhausted our capital. We have no manufacturing facility. Even if we did, each reactor requires months of precision assembly. Hundreds would be impossible. We could perhaps produce a handful, each essentially handmade."

Raines did not look away from Ishikawa.

"You're telling me you can't produce a hundred. Fine. Can you produce ten, and can you commit to pushing output as far as you can take it?"

Ishikawa's reply came without hesitation. "Yes. We would love to. As much as we can, as fast as we can, but we have no manufacturing facilities."

"Then I have a proposal," Raines said. "I'll build you a facility in Gulfport, not far from Elin's base in Biloxi. Helion Dynamics has infrastructure there already. And when you start producing, I will buy everything you can make."

The room absorbed his words. For a moment, no one spoke. Then Nakamura exhaled slowly, the relief unmistakable. "He means to build with us," he murmured to himself. "Not against us."

Raines continued, "The Kagutsuchi Vortex will have many uses on Earth, particularly in remote areas, and we will help you find those markets. But right now the UN is blocking the export of fissile materials to space. The claim is safety, but the real goal is to stop fusion power. The reactors currently powering operations in space won't need refueling for years, maybe a decade if they're careful."

Ishikawa's eyes narrowed slightly, understanding.

“But your Kagutsuchi Vortex can replace those fission reactors and provide more power. Much more. That will expand lunar production, increase metal flow from Ceres. We need this technology, and we need it fast. These reactors aren’t banned, yet. If they’re in orbit before anyone moves to regulate them, we control space power. But I’d like this kept quiet until that happens. Agreed?”

Ishikawa’s composure cracked slightly, relief visible in his expression. “Of course. We would be honored to help, and we will keep this to ourselves.” He bowed, deeper than before. “We are very grateful for your faith in us.”

Raines then stood, ending the meeting with the same clarity with which he had begun it. “Good. Then we are aligned. Build. Scale. Deliver.”

The group rose. There were bows, brief handshakes, but no ceremony. The air carried the quiet charge of a compact sealed.

On the jet that night, Tokyo receded into a scatter of lights. Inside the cabin, the atmosphere had shifted. Raines sat with a drink untouched in his hand, silent. After a long pause, he turned to Max.

“This will affect you too. I want you scouting Martian ventures. Risk more. Even ventures that fail may help get the right expertise to Mars, and an entirely new venture may work. Mars is the natural core of operations off Earth.”

Max nodded once, already thinking.

The silence returned. Max glanced at Elin. “He just guaranteed the market before the machines even exist.” He paused, still processing. “I wasn’t sure what he’d do. He could have bought them out, buried the technology. Instead he’s building them a factory.”

“You thought he might crush them?” Elin asked.

“The thought crossed my mind,” Max admitted.

Elin’s tone was more contemplative than surprised. “He doesn’t see fusion as his. He sees it as humanity’s. He doesn’t want to own the light, just see it shine. The rest of us have to keep up.”

The engines hummed, steady and low, as the three of them sat with the weight of what had just begun.

Chapter 20: Baracoa

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The province of Guantánamo no longer belonged to Cuba. By vote of the United Nations, its territory had been transferred to the jurisdiction of the Jupiter Resource Corporation, an arrangement argued in back rooms then framed as visionary. Washington objected to surrendering its naval base, Havana to losing land for symbolism's sake. Amira Bastos, CEO, and Dr. Emile Carson, Chairman of the Board, were charged with turning this provincial corner into JRC's visible capital. Neither relished the assignment, but both understood its necessity. They were stewards of a prestigious enterprise, and appearances required that they be seen arriving here.

The seaplane banked wide, engines straining against the humid air. A Grumman Albatross, old and noisy, vibrated through the floorboards, its interior faintly scented of oil and damp canvas. Bastos and Carson exchanged glances. Neither liked it, though it was the only practical way here.

This was not the way they were accustomed to travel. Amira Bastos and Emile Carson had flown between New York, Geneva, and Vienna in wide seats with polished attendants. Neither spoke of it, but both regarded the flight as an indignity, the kind of compromise expected of lesser officials.

Through the windows, the coastline broke into ridges and narrow valleys. Rivers cut through the forest and spilled brown into the sea. Ahead lay Baracoa, pressed between the hills and the surf, its colored houses fading under salt and heat. To Bastos and Carson, it appeared not as a town but as a stage. What mattered was how JRC would be seen arriving here, what image they could project against the landscape.

The Albatross came down hard on the bay, its hull shuddering as spray burst across the windows. The fuselage groaned while the engines throttled back, leaving only the uneven slap of water against metal. Bastos tightened her grip on the armrest. Carson muttered something under his breath, then fell silent. For both of them, landing this way felt provincial, as though international executives had been reduced to tourists arriving at the edge of nowhere.

The seaplane drifted toward the spider dock, a wooden structure that swayed with the tide. Planks shifted underfoot when they stepped out, knocking against each other with every wave. Salt rose in the air, mixed with the odor of fuel. The dock was too narrow for ceremony, but that did not stop Bastos from imagining how it would look when foreign visitors first arrived. The sightlines mattered. The dock should lead straight toward JRC's offices, visible immediately, not tucked behind warehouses or houses.

Baracoa opened before them in narrow streets and painted walls, uneven rows pressed between forested slopes and the sea. Pastel facades were faded where the sun struck them and brighter in the shadows. Murals chipped but recognizable still carried scenes of boats and storms. Children ran through alleys, their voices carrying above the sound of waves. For Bastos, these details suggested opportunity: texture that could be framed as authenticity. Carson, watching in silence, saw only how small the town was.

They left the dock for the narrow street beyond, their shoes striking uneven stones polished by salt and years of rain. The air was heavy, almost sweet, carrying the scent of fruit and wood smoke. Vendors had laid out baskets on low tables: guava, plantain, mango. Children darted past without slowing, their games ignoring the two officials in pressed linen. Bastos slowed only long enough to place a hand against a sun-bleached wall, testing its rough plaster.

"This is where it should be," she said, her voice measured. "Headquarters at the water's edge. Not behind warehouses or on some hill, but here, facing the dock. A visitor should step off the plane and see us immediately. The message is that we belong."

Carson kept his gaze on the slope above the bay, terraced with houses and patches of green. “Belong,” he repeated, though his tone was unreadable. He tapped the stone with his heel, frowning at the uneven surface. “And where do we live in this arrangement?”

Bastos smiled faintly, as though she had anticipated the question. “The hillside. Villas already under construction. Terraces cut for privacy, each room with a view of the water. Gardens where you can walk after meetings. Not offices dressed up as houses, but actual homes. Mine near the beach, yours higher up where it is quieter.”

Carson’s expression softened, though his voice remained flat. “So, a retreat. That I can accept.”

Bastos nodded. “A retreat, but one that shows we came not to dominate, but to fit. Walls in color, courtyards open to the light, walkways shaded with vines. We will make the right impression. That is what matters.”

They walked a little farther, past a courtyard where laundry hung between two balconies and a mural of a fishing boat peeled against the plaster. The streets bent inland, opening onto a view of the fields before the ridges swallowed them. Bastos slowed, her hand resting on the edge of a gate.

“Headquarters, villas, docks, I can picture those clearly,” she said. “But what comes after, I do not know. We are chartered for Jupiter, yet Jupiter is distant. Something here should show we are moving forward, but what should it be?”

Carson studied the open land beyond the last houses. The fields were dotted with palms and cut by a narrow road, then rose into slopes streaked with forest. “An institute,” he said finally. “Not a factory, not a launch site. A place for research, debate, engineering trials. Just out of town, within reach, but separate.”

Bastos tilted her head, as though weighing the word. “An institute. Yes, that feels right. Not too large. A few hundred people. Laboratories, classrooms, housing along the cliffs. Visible enough to suggest progress, but not so much that we overextend.”

“It would buy time,” Carson added. “A structure people can point to. Something to fill the gap while we prepare for what comes later.”

She nodded slowly. “Yes. A language for Jupiter, without the risk of failure.”

They stood for a moment, both looking across the bay. The town continued around them; children were still shouting in the alleys, shutters half-open against the sun, a vendor calling out the price of oranges. To Bastos and Carson, the sounds were no more than background. What mattered was the picture that could be drawn from this place, and how well it would play elsewhere.

They turned back toward the dock as the afternoon light stretched across the bay. The ridges glowed green in the distance, and the sea flattened into silver. Bastos walked a step ahead, already outlining in her mind how the headquarters would look from the water, how the institute would rise against the cliffs. Carson followed, content with the thought of terraces and gardens above the town.

Neither spoke again until the dock came into sight. The Albatross waited there, rocking gently in the tide, its engines quiet for the moment. Bastos paused, watching the sun drop lower behind the ridge.

“It will work,” she said with certainty. “Headquarters, villas, institute. Everything in place.”

Carson gave a single nod, as though no more needed to be said. They had seen what they needed to see. The rest would follow.

Chapter 21: Lunar Power Up

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Darrin got off the call with Felipe. Planetary Metals had agreed to send him the tungsten pipe he needed, but they didn't have the tin. Catherine Dahl was the only person who could help him out. He had never called her before, but Li had given him her number. He went over what he was going to say in his mind. Tin was of no value on Ceres. Raines sent him the reactor. If she has any tin that she could send to him, he would appreciate it. However many times he went over it, he felt like he was grovelling. All he wanted was to ask. He waited until noon Paris time, put in the number, and closed his eyes, finally pressing the dial button.

Catherine came on. "Tan, what a pleasant surprise. I haven't heard from you in months."

The time delay meant she would wait two and a half seconds of silence before she heard his response. Not long, but awkward, and every moment he didn't speak only made it worse. "No, sorry, this is Darrin. She gave me your number. I am looking for Catherine Tin. Catherine Dahl. Tin. I need tin. Can you sell us tin from Ceres? Planetary Metals doesn't have much stock, and it is already committed."

The delay meant people spoke in longer segments. On Earth, the call would have gone back and forth several times with hellos and softer greetings. That didn't work with the delay.

Catherine was laughing too hard to speak, so she waited until she was composed. "Darrin. I've heard so much about you, but I don't think we've met before. Dr. Knox told me he often has leftover tin when he purifies nickel. He and Mateo have been talking about an iron-cased bowl, but so far, it has never been shipped. Do you want it processed on Earth or shipped directly to the Moon?"

"No, Ma'am. We don't want that at all. I mean, we want it shipped directly to the Moon. We will melt it here, so we don't need any fabrication. I can send Knox the location of the regolith-filled landing pit. He has sent us metal balls before. We need twelve thousand tons, total."

“I am talking like an idiot,” he thought during the delay.

Catherine sounded calm as ever. “I am sure that there is no problem. We can swap straight across for titanium, if that works.”

“Thank you so much. And a pleasure to meet you.”

The call ended. “Did I really just end with a greeting? Either way, the tin is coming.”

Planetary Metals delivered the pipe, and Darrin, Luke, and Pavel assembled it one mile west of the OK-550. Four loops like a cloverleaf, each two miles long. There was a takeout every five hundred feet, and the lines were insulated with a foot of aluminum oxide aerogel, fragile but able to withstand the heat.

Darrin and Luke set up the expanded cold ammonia radiators on the inside face of the north wall. Pavel was afraid of heights, so he acted as ground crew, keeping them supplied with pipes and reflectors, and using his survey laser to show them where to set each bolt.

Six months later, the tin started falling, in thirty-ton balls inside copper spheres. They weren't round after impact, but most of them stayed together. Darrin fished them out of the regolith landing pit and hooked them up to the cableway.

They pulled out the pure tin and ground it down to pebbles, which they poured into the pipes. There was a standpipe for adding more pebbles to keep it full when it all melted.

Finally, Lunar Tug brought the Kagutsuchi Vortexes.

The Kagutsuchi Vortex came in twelve crates, each containing an intricate structure of magnets and stainless steel. The enormous and complex magnetic fields made assembling the sections a difficult project.

It was lunar night, as it all came together. They were illuminated by Earthshine and worklamps.

“We are short one trim magnet, bottom-right quadrant,” Pavel Kerensky announced from the reactor floor. “And the manifest says ‘Molecular Turbine Vacuum Pump.’ There’s no pump. Just the large hole where it is supposed to be mounted.”

Darrin glanced at the hole, then at the gauge on his pressure suit showing the impossible vacuum of space. “We’ll be fine, Pavel. What pressure was it supposed to reach?”

“The manual says it needs to be pumped down to 10^{-10} torr, and I have no pump.”

“Check your gauge. My suit is showing 10^{-12} torr. The instruction manual is written for Earth engineers.”

“It is not about the vacuum,” Pavel muttered. “It is about the principle. The package is incomplete.”

Li Yun Tan descended from the mezzanine, a slate in her hand casting a pale glow on her face. “I’ll order another magnet, but that will take weeks. Let’s check again.”

The three of them spent the next two hours looking through every container and bit of packing material that had come with it.

Li found it. “The magnet was in the ‘Oven Mitts’ box. It’s huge. I thought a trim magnet would be a tiny thing. This is the size of a coffee mug.”

“It’s strong, too,” Pavel warned. “Keep it in the holder until we line it up.”

Pavel and Li worked together with the magnet, but the field strength made it impossible to line up well. Pavel set up the press, and Li began winding it slowly into place.

When the magnet was fully installed, he called out, “It’s ready. Let’s start melting tin.”

Darrin had been running a megawatt across the first loop for more than an hour already, but it wasn’t yet hot enough to melt the tin. He moved to the control panel on the OK-550 and increased the current until it was running two megawatts. “Pavel, you tell me when it reaches temperature.”

Pavel was measuring the temperature on takeout number eight. He called out the numbers for fifteen minutes, then, “Temperature’s not rising steadily. It must be melting.”

He moved around to the control panel and switched on the pump. It bogged, and he hit the off switch immediately. “Nyet. We wait thirty minutes.”

Luke got them to help install the new Ipsen tumbler ovens on takeouts three through nine. They went on easily enough, but it still took two hours. When they were ready, Pavel went back to the panel and tried again, and this time, a trickle of metal moved. With something moving over the walls, Pavel was ready to start.

“Say good-bye to this world,” he said as he started the Kagatsuchi Vortex. He kept the deuterium density low as he brought the temperature up. The walls were heating. Pavel increased the density of deuterium until he could see radiation from the D-D side reaction. Then he let in a puff of ^3He . There was a brief thermal spike, as expected, then all the gauges held steady.

“It’s working. I am feeding it again,” he called out, and started a longer, slow stream of ^3He . Gauges showed fusion increasing steadily. The tin hadn’t all melted yet, but the reactor wall was over five hundred degrees. As the hot tin mixed with the still solid beads of tin in the pipe, it melted grain by grain. Pavel increased the pump flow rate a little, but it quickly ran dry, and he pulled it back to the slow trickle.

While the electric current had been heating the entire loop evenly, the reactor was heating the beginning of the loop the most. That meant even though all the tin up to takeout three was liquid, there was only a little liquid on the bottom of the pipe by takeout twenty, where the generator was installed, and less by the pump inlet.

“No need to hurry it. The wall still needs to heat up.” Pavel increased the ^3He stream, letting the wall rise to a thousand degrees.

Luke was measuring the temperature on takeout nineteen. “Gauge is moving. Try ramping the pump again.”

Pavel didn’t increase the flow rate much, but it held. The tin had melted. It was under 300°C , but the pipe was flowing. Pavel brought the flow rate all the way up before increasing the density in the stellarator.

Once the tin was flowing, he increased the ^3He feed rate until it reached the Lawson criterion. All the electricity was still coming from the OK-550’s generator, but now that was just pumps and gauges. Pavel stayed calm. “We hit self-sufficiency. I am no longer heating the plasma.”

The temperature of the tin kept rising, and the pipe started glowing. Luke was watching the Pyrometer on the takeout when it glowed cherry red. He called out the temperature every hundred degrees until they hit white heat at 1300°C .

At that point, the takeouts were glowing too brightly, so Darrin draped them all with woven alumina blankets. “I didn’t know this came with free lighting.”

Pavel kept increasing the ^3He flow, driving the temperature to the theoretical maximum. “Look at that loop glow. We are going to have to cover it with regolith to save our eyes.”

Darrin throttled open the sCO₂ turbine. They held their breath as they watched electric generation move up to 200 MW. The temperature on the radiators began rising. “Running at forty percent capacity, output is steady, and the radiator bank is stable at 38°C. Pavel, you can shut down the fission plant’s electric generator. This one has us more than covered.”

While Pavel went over to the OK-550 to start the shutdown sequence, Luke started a tumbler oven with a hundred tons of regolith.

“Oh, this is smooth. With the bags automatically dumping on the conveyor, and a cableway to carry away the tailings, these will go a long way to helping me catch up with the increased demand,” Luke said.

“While I am shutting down the generator, can I shut down the entire OK-550 reactor?” Pavel asked.

“Not yet. I still need the old ovens running to help with the backlog. I’m going to be running both systems for a while. Darrin, I am going to need another twenty Ipsens, especially if operations continue to expand.”

When they were all satisfied, everyone but Luke headed back to the cantina. The blankets were glowing bright cherry red.

Pavel went to the bar and lined up four chichas.

Li and Darrin took Li’s favorite table, next to the claim map. “Darrin, we have the power, now. Can we distribute that power by the cableway? There was another darkside death last cycle. A solo miner. He didn’t push things, but his battery had been slowly degrading, and he didn’t notice. We can do something about this.”

“Tseng is making aluminum, and I use that for conductors. I can put HVDC cables on the cableway pylons. We can provide emergency power. What we can’t do is power everyone’s camps.”

“If we could power sweeping during the night, we would double our production.”

“The sweepers take too much energy. We can power lights and huts. We can provide emergency power. But we can’t power thousands of miners running heavy equipment. We can do is provide oxygen at the towers, though. Tseng is producing more than Lunar Tug needs. If anyone is in danger, they get in a crawler and drive to a tower, hook up, breathe easy and stay warm.”

“That’s what I want. If chechakos know this area is safer, then all the sites under the cableway will be filled fast,” Li said.

“I’ve been thinking about Tan’s Depot. Right now, we can refuel Lunar Tug ships, they have a pressurised warehouse for shirtsleeve loading and unloading, and the soft massdriver saves them fuel both ways. But what would you think if we were to install a new tower, coming over the rim to the northeast. It could be five miles long, and really cut Lunar Tugs expenses.”

“What would it do for us?”

“Tramp ferries will start charging less, and I am sure Lunar Tug would follow suit. And they could move more cargo. We have the power, now. I’ll get the materials from Tseng.”

“The depot has been very helpful. If you think it will be good for business, go ahead. I don’t mind.”

An hour later, Luke came in to the cantina, and Li and Darrin joined him and Pavel at the bar.

One of the miners, Rook, had sent a notification that he was running low on bags. Easy enough to send him more bags, but Darrin liked visiting the miners. He liked to see their camps and operations. The next morning, Darrin set up his pressure suit with extra batteries and set out.

The new cableway ran a circle inside the crater with a single north and south line connecting that ring to another ring outside the crater. The steel towers were fifty meters high, five times taller than the old system, fabricated from Tseng’s steel made from lunar iron and Earth carbon. In lunar gravity, their height allowed two-mile spans.

Six primary cable lines radiated outward from the external ring. The titanium 5 cables had a very low thermal expansion coefficient and were completely impervious to UV which had been a problem for the old dyneema lines.

Darrin watched the loaded bags coming in on the line across from him. The large bags held thirty tons of regolith each. Some miners were still using the old ten ton bags, but the thirty ton bags meant less handling and simpler operations. That drove most larger operations to thirty ton bags, and there were requests for larger sizes.

As he rode along, he watched a miner drop ten bags into the transfer station. The station would automatically load them on the line to Tycho Crater. The miner was still using the ten ton bags.

The visits helped Darrin understand the miner's needs and it let him connect to them.

Rook was the first miner Darrin met. He wasn't on the new line, a little south of the northeast line. Darrin dropped off and followed hauler tracks to Rook's camp. Just a squat hut burried in a pile of regolith and the long, smooth trails of his sweepers combing dust into low mounds. Darrin waited for Rook to finish what he was doing.

"You're early," Rook said. He wore an old pressure suit, logo faded, patched with silver foil. His voice was rough, like a man out of practice with conversation.

"You're on the schedule," Darrin replied. "You flagged the system. Said you were exceeding bag capacity."

Rook nodded and walked toward the docking rig where four bags hung from a low truss, each already sealed and tagged.

"New sweepers," he said. "Bought 'em from vendors at Tycho Crater. Tripled my coverage. Got more dust than bags now."

"You know we have new bags now, thirty tons each."

"My hopper is set to fill ten-ton bags. I just need more of them. I would like a cableway out here, though."

"If you can do the work, I can bring you some old towers and Dyneema cable. It only works for ten-ton bags, but that will be enough for you," Darrin said.

Rook thought about it. "That would be fantastic. I could stop wasting my time driving and focus on mining. Are they complicated to set up?"

"Mostly easy. Let us know when you are ready to go operational, and a technician will come make any needed adjustments and tension the lines for you."

Rook looked Darrin up and down. "Tan's already pulling twenty percent of my income. How much will that cost?"

"Right now, she would just be happy if you could send more regolith. No charge," Darrin explained. "The one thing is we expect that if you add a spur, you will extend it out to other miners beyond you. Does that work for you?"

Rook thought about it for a long moment, tapping his gloved fingers on the docking truss. When he spoke, the rough buzz of his voice was calculating.

“You’re offering me a faster turnaround on my dust, and all I have to do is run the line past my hut,” Rook summarized. He nodded slowly. “I’m good with that. But if I’m going to be the gateway for the whole sector out here, that’s real estate that costs me something, even if it’s just the hassle of dealing with technicians.”

Rook walked closer, his shadowed visor reflecting Darrin’s suit. “Here’s my offer. I allow the expansion, I place the towers where they don’t foul my sweepers. And I get a cut, maybe 10%, of the haul from any miner who hooks into the grid past my claim. That’s my premium for being the local node.”

Darrin shook his head before Rook had finished. His voice, clear and steady, cut through the silence.

“I appreciate the ambition, Rook, but that’s not how Tycho’s infrastructure works,” Darrin said. “The cableway is free to everyone. No fees, no cuts, no exclusivity. You profit from it. We pull out more helium and pay you faster. But you don’t get to charge the next guy your own cut. Darrin paused, letting Rook think about it.

“What I can offer you is a guarantee that we’ll maintain everything we set up here. We give you the tools for free, Rook, because we want you to succeed, but we can’t let you charge a tax on the rest of the network.”

Rook hesitated. “Appreciate that. With the system you built, I don’t need a crew. I drop the bag, and you guys do all the rest. I really like that. It lets me focus. OK, we have a deal.”

“Perfect,” Darrin said, retrieving his hook from the truss. “I’ll have the tower team prep the ground posts tomorrow. Expect a tech out here to set the anchors in two days. You just focus on filling more bags.”

Darrin and Rook spent a while walking the perimeter of Rook’s claim. Darrin loved seeing how each miner ran their site a little differently.

Finished, Darrin started his long walk back to the cableway. He gave Rook a final wave. He was pleased with the infill strategy, repurposing old equipment and extending the reach of the cableway. Darrin returned to the line and clipped onto the inbound cable, headed back home. He remembered when Li had flashed her laser across the crater.

Chapter 22: Sidestep

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	One year after the start of JRC

“I need to reserve the Oaks Boardroom again this evening. I will be having a guest from out of town. And he will need a room.”

“Wonderful, Dr. Sarai. We are happy to set that up. We trust your Cyprus Suite is still meeting your needs. I will set up his room next to yours.”

“Thank you.”

Raines arrived late. Elin met him at the airport. “Tobias, always a pleasure. I have your room key.”

“If I go to my room, I will sleep for a week. Show me the boardroom.”

After the taxi ride, she led him straight there. Raines keyed up a slide on his slate, projecting it on the wall. “Have you seen this!” Raines exclaimed, pounding his fist against the wall. “JRC is building palaces, not launch facilities. I read their entire quarterly report. Lovely pictures of their headquarters building and the streets of Baracoa. Shots of the JRI campus. Jovian Research Institute. A college? Who even authorized that morass? No mention of Punta de Maisi or launch facilities. The entire report reads like a travelogue. They have had a year, and they’ve done nothing.”

“They are diplomats. This is what they do.”

Raines didn’t stop. “We need action. Villas don’t power fusion. Right now, the one thing that is keeping us operational is rapid growth on the Moon, but it is not making enough to feed even one terawatt tokaplex. I have prepaid orders for ten.”

“The Moon is working hard. Tan called me recently with new designs for lunar sweepers. She thinks some sites can be twice as productive with them. I introduced her to Case.”

“Well done, Elin. Tan is an absolute engine. But even a factor of two won’t get us where we need to be. Do you know if Tan has a Kagutsuchi Vortex yet?”

“Yes, two were delivered to the Moon. Derrin powered one up. Mateo mentioned he is commissioning another fast boat to keep up with the increased titanium production. When I told him Mars and Ceres would be getting more power, too, he said he was going to order three. Manufacturing may not be the real constraint with them, though. With the fissile material ban, General Atomics told me they are retiring their one remaining fission-powered freighter after this next run.”

“So we are back to individually commissioned hydrolox ferries? Those are so slow.”

“It’s worse than that. Tan’s titanium production is tied to her ^3He production. Andrei will have a much harder time scaling than Mateo. His current fleet is built on repurposed used ships. I don’t know if he can handle the increased volume.”

Raines looked back at the projection on the wall of the JRC quarterly report, his eyes flashing in anger. “This cannot work. We need interplanetary transit. JRC is not going to step up. What are we going to do?”

“We are likely going to deliver another Kagutsuchi Vortex every two months. You might talk to Andrei. If he can afford to commission new fusion-powered ships, he could improve lunar service, and maybe open up mid-system service.”

“I don’t know what the lead time will be, but that gives me hope.”

Raines brought up a new chart. “Elin, I don’t know where you are with the terawatt Tokaplexes, but I have a lynch mob of impatient customers who want that power now. I have some ^3He in storage. Are you ready to start on China?”

“I only have one construction area for the terawatt tokaplexes, but I keep it busy. You want me to ship it now?”

Raines stepped away from the projection and looked out over the bay. He could see Elin’s artificial atoll in the distance, where the terawatt tokaplex was being built. “I believe Tan will deliver five tons per month soon. When she does, I want you to be ready to ship. And when lunar production hits ten tons per month, ship another.”

“If you are willing to dip into your reserves, I can deliver that, but these will drain your reserves fast.”

Still facing out the window, he began talking softly. “Elin, let me be frank. This business, with machines you built, is immensely profitable. Even so, we cannot afford to go to Jupiter. JRC has the money and the charter to do it, but it doesn’t seem primed to act. They have five projects in their charter. They need to make heavy lift cheaper, so they can get their equipment into space. They need to work

out interplanetary transport so they can get their equipment to Jupiter. They need to work out ^3He collection, and they need to figure out how to get that ^3He out of the Jovian gravity well. And they need to paint murals on their buildings. They are making excellent progress on that last project.”

He turned back to face her. “What I want us to do is to help them. If they have an idea, I want you to look it over and offer some suggestions. If they want to build something, we find a company to build it and, behind the scenes, cover part of the cost to make it cheaper for JRC. I want them to look like the luckiest team alive, because we will be backing them. We won’t get any credit. All our help will be disguised as third-party stuff. But we can sidestep their bureaucratic limitations, and, maybe, actually get somewhere.”

Elin was quiet for a moment. “Tobias, you know I will do whatever I can. But, irrespective of what I do, they may not succeed for decades.”

“I know, but I refuse to let that thought limit me. We need to make this work.”

“Right now, I am getting the tokaplexes built. The demand for the ten gigawatt units remains very high, and the terawatt units still require refinement. I am also helping Kagutsuchi build their vortexes. And, most importantly, I help Gonalo Silva keep the operational plants working. Time spent on JRC projects will necessarily slow other work.”

“I know, but I also know none of these systems will work without Jovian helium.”

Room service delivered some Gulf Shrimp Cocktails. They watched the ships move across the bay as they ate.

Chapter 23: Space Freighters

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Andrei Volkov and Max Juno climbed the narrow stairs into the jet. Once inside, the cabin closed around them, quiet except for the low vibration as the engines rose.

Volkov settled into his seat, eyes half-closed, thinking of the work ahead. “Max, what should I know about Nakamura? How should I speak to him?”

“Nakamura is a friend and the lead investor in Kagutsuchi Vortex,” Max said, seizing the point. “He is grateful for the way Raines made that investment so valuable. He will take care of you. He is well-connected with the Japanese industrial technology sector. He will be able to help us find the right companies to make the equipment you need. You just need to be open and honest. He won’t play any games.”

Hours passed in silence. Volkov’s hands were steady, but his pulse wasn’t. He was about to bet everything on ships that didn’t exist for markets that might never form. Just like the lunar helium rush. Buy rockets, hire crews, and pray the business materializes. It had almost bankrupted him then. Now the stakes were higher, and he was no longer young.

Mercifully, fatigue overtook him on the long flight. When Tokyo’s haze finally appeared on the horizon, Max leaned across the aisle. “We are here, Andrei.”

Volkov nodded as the jet sank toward Haneda’s runways, his mind spinning. He’d come to build his company from cis-lunar courier to the mid solar system’s transport operator. If Nakamura could help, fine. If not, he would find another way. He would not stop until Lunar Tug reached its next horizon.

They arrived at the Shibata Capital Tower in the early morning. Not quite empty, but very quiet. A receptionist was waiting for them. She led them through a wood-and-glass corridor to a small conference room with a low table, no screens, and insulated walls. Nakamura was seated inside.

Max gestured toward their host. “Nakamura, this is Andrei Volkov. He is the CEO of Lunar Tug. Raines has asked me to help him.”

Nakamura inclined his head, his tone measured but curious. “I know of Lunar Tug, though we haven’t crossed paths before. A friend of Max is a friend of mine.”

Volkov extended his hand, his expression direct. “A pleasure. Lunar Tug is at the point where expansion is critical. I need new infrastructure. I must have orbital hubs to reduce the number of intercepts. But that is not all. I wish to extend my range to Mars and Ceres. I will need a new class of interplanetary ship.”

Nakamura studied him for a moment, weighing the words. “Ambitious. Tell me more.”

Volkov projected an image from his slate. “The hub will be a large warehouse and an even larger pole warehouse. Refueling stations. Living quarters for up to a hundred people with light rotational gravity. Importantly, it will need a soft mass driver for both braking and launch.”

“How do you intend to power this, Andrei-san?”

“Solar panels and batteries will be adequate. The mass driver will only operate at one G, but it will be a ten-mile-long cable.”

Volkov switched to another image. “A major headache has been descent to Earth. I will need the hub to be able to handle descender craft so no one needs to send a rocket up just to bring cargo down. I think Mitsubishi Heavy Industries would be perfect. They have made modules for space stations before and I am sure they are capable of delivering. I won’t need many. One for Earth. Maybe two, eventually. Another for the Moon, Another for Mars. It’s a short run.”

Nakamura was silent for a moment. “One design, three orbits. That’s efficient. Once proven in LEO, replication becomes straightforward. Mitsubishi Heavy Industries works on space modules. They are exceptionally capable and would be an excellent fit for your hubs. However, I know of a smaller company in Oita, Kagayaku. They do exceptional work, and are hungry for a project like yours. As one of their primary investors, I can tell you they always deliver above expectations.”

Volkov leaned forward, skeptical. “An unknown sounds like a risk. Are you sure they can deliver?”

“They can,” Nakamura replied evenly. “The modules they build now are already designed for orbital assembly. That foundation is solid. While it is a small

company, they will prove to be exceptionally adaptable. Their success will be as important to them as it is to you. Mitsubishi can build a hub for you, but it will be their hub, their specs. I believe your requirements are quite specialized. Kagayaku will deliver precisely what you require.”

Volkov leaned back slightly, then straightened. “That sounds promising. I will speak to them. Now, let’s talk about ships. My fleet today is all short-hop craft, lunar surface to LEO and back. What I need are ships that can push farther, to Mars, Ceres, even Jupiter, and return safely. Can Kagayaku build them?”

Nakamura folded his hands, considering. “That scale of ship requires a larger yard. Ishikawajima-Harima Heavy Industries comes to mind. IHI can build hulls of that size and integrate advanced systems. The challenge, of course, is power. You know there is a ban on fissile material in space.”

Volkov gave a short nod. “I am not planning to rely on fission. I want the Kagutsuchi Vortex. It has the capability I need.”

Nakamura’s expression flickered with interest. “Fusion for transport. An excellent proposition. IHI could integrate the reactor. With that kind of drive, what exactly do you have in mind?”

Volkov sent the list of specifications for docking, fueling, and mass driver interface to Nakamura’s slate. “I need four categories. First, passengers. A fast bus, sixty seats, designed to minimize transit time. People must spend days, not months, between worlds. Second, a pole ship for oversized cargo: hubs, other ships, equipment to build a city, items that no other freighter can handle, but which can be safely attached to a pole. And third, general cargo. Enclosed, versatile, but not as heavy. The workhorse. It can carry passengers or cargo or both.”

Nakamura scanned the specs line by line. “And what is the fourth type?”

“A very fast ship. Multiple engines, less room onboard. Something that could reach the outer solar system, if we go there. We don’t need it yet, but I’d like them to work on it.”

Nakamura set the paper down, his expression unreadable, then allowed himself a slight smile. “It is a bold beginning. And with Kagutsuchi Vortex powering the fleet, IHI will see the prestige. But to produce so few is difficult for any firm. Designing a ship costs far more than building one. Perhaps we can consolidate the designs of the main three, so the differences are only superficial.”

Volkov thought a moment. With used equipment, development costs were someone else's problem. But this project would be engineering from scratch, and suddenly those costs were his. He wouldn't just be taking out a loan on his ships. His entire business would be in hock, and it still wouldn't be enough. He met Nakamura's eyes. The man wasn't selling; he was guiding. That guidance was what Volkov needed. "Yes. Seven total ships, based on a common frame. Three freighters, two passenger buses, and two pole ships."

Nakamura bowed slightly. "I will bring your request to IHI. Even with your detailed specs, they will not simply build this. You will need to be involved throughout the process. You understand that, right?"

Volkov smiled. "That sounds great. I look forward to hearing from them."

Nakamura grimaced slightly. "Much of what they will be talking about will be delays and cost overruns. We all hate that, but it is a part of the business."

Volkov cringed, but appreciated the honesty.

The three men rose together. They shook hands, and the room emptied as quietly as it had filled.

On the flight home, Andrei talked to Max. "Max. I spec'd what I needed, but the price I set is all I have and more. I have no way to deal with cost overruns. What have I signed myself up for?"

Max smiled warmly. "It's a massive bet. But you're building the infrastructure everyone will need. Raines knows that. So do I."

Andrei nodded but said nothing. Max poured two vodkas.

Andrei drained his. "Na zdorovye."

Chapter 24: The Seine

Price of ^3He	
Current Production	
Current Consumption	

Time Since Start	
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Max Juno stepped into Catherine Dahl's apartment and immediately moved to the tall windows overlooking the Seine. "God, Catherine, this view. How do you get any work done?"

"You get used to it," Catherine said, as she set down two cups of coffee on the small kitchen table. "Though I admit, some mornings I just stand there watching the boats."

The apartment had the comfortable, lived-in feel of someone who'd actually made Paris home rather than just renting space. Books in French and English lined the shelves. Her dishes appeared to be handcrafted. Max wondered where in her travels she had found them. The pictures on her wall were mostly of friends in amazing places. Max thought about all really incredible projects she had been a part of. One photo was of Dr. Knox on Ceres with a smelter in the background.

Max settled into the chair across from her, cradling his coffee. "So how are things on Ceres? Your last reports sounded optimistic."

Catherine's expression grew animated as she pulled her slate toward them. "More than optimistic. I want to show you what we've accomplished." She turned the screen so Max could see an aerial view of the ice mining operation. "Four massive steel piers screwed deep into the asteroid's surface, forming a square a mile across."

Max leaned forward, studying the image. He could see tiny figures of workers near the base of one pier for reference.

"We've got cables running between the piers," Catherine continued, swiping to show the system in operation. "We use them to pull scrapers across the area. Regular scrapers wouldn't have enough traction in the low gravity, but this cable system lets us load up enormous amounts of ice with each pass."

"That's ingenious. Whose idea was that?"

"Knox's team worked it out after we lost too many conventional vehicles to poor traction. Sometimes the worst constraints force the best solutions."

Catherine swiped again, showing processing facilities that looked like a cross between a chemical plant and a foundry. "Here's where it gets interesting. We take that ice and separate everything. carbonates, phosphates, ammonia, other salts and minerals. But Max, the volume of pure water we're extracting is enormous. Forty tons per day of double distilled water."

Max felt something cold settle in his stomach. “Forty tons a day?”

“Plus about six tons of organic compounds as byproducts. The process is so efficient we’re running out of storage space.” Catherine’s enthusiasm was evident, but Max was struggling to focus on her technical explanations. Forty tons of water per day meant...

“I was thinking,” Catherine continued, “we could send the phosphates and ammonia to Jakob. Those could boost the agricultural yields.”

Max managed a nod, though his throat felt tight. “I imagine so. Jakob’s team makes ammonia now, but that might let them expand.” But she was talking oddly. If their water processor was working so well, what was Mars doing for Ceres?

“That’s great. I’ll let Dr. Knox know!” she enthused. “Since we started this water mining operation, Ceres has become much more viable. I am planning to expand operations. We can support two hundred more crew.”

Catherine set down the slate and her expression grew more serious. “Max, I need to be direct with you about something. We’re going to end the water contracts. Also, the block carbon contracts.”

The words hit him like a physical blow. Max stared at her, his mind reeling. Those contracts were the foundation of everything they’d built on Mars. Without them...

“Catherine, those contracts are our entire business.”

“I know.” Catherine’s voice was gentle but firm. “And I can’t tell you how grateful we are for everything Jakob accomplished. If it weren’t for his team, the Ceres project would have failed. My crew would have died on Ceres and never delivered metal.”

Max felt like the room was tilting. He could already picture Jakob’s face when he heard. This would cut him deep, yet Catherine was leaving it to Max to deliver the news.

“We’ll work out fair compensation for the contract cancellations. We will pay for everything in transit and the value of one additional year to sever the contract,” Catherine continued. “We simply don’t need any more water or carbon shipments. Stop today. Ceres is completely self-sufficient now, as we had always planned to be.”

Max stared down at his coffee, watching the steam rise. His mind was racing through implications. Martian operations had been built around those enormous, steady, growing orders from Ceres.

“I’ll let Jakob know,” he said quietly.

Catherine nodded. “I know this is difficult. I can’t thank you enough for what you have done. We will continue to support you. Ceres can help Mars. We make plenty of steel now, but it is not worth shipping to Earth. And the ammonia and phosphates I mentioned. We can ship those to you free of charge, for old times sake.”

Max didn’t have anything to say. He was still thinking about how this would affect Jakob and his team.

Catherine paused, then slid a different document across the table. “There is something else I need to discuss.”

Max looked down at the contract. Food procurement. The numbers made him blink and look again.

“You want food for two hundred people? Three meals a day?”

“Two hundred and growing. Max, you can’t imagine what your food shipments have meant to Dr. Knox’s team. The bacteria and yeast slurry is horrible. I tried drinking that stuff once and gagged. That’s what they are eating three meals a day except when Jakob’s meals arrive. What Jakob’s been sending from Mars, real vegetables, actual meat, and that wonderful amaranth bread is not just nutrition. It’s sanity. It’s hope. It’s the difference between existing and living. That one meal per week is the world to them.”

Max studied the payment terms, his shock slowly giving way to amazement. The figures were extraordinary, more than he’d ever imagined food could command.

“The crew fights over portions,” Catherine continued. “They plan their days around meals. What you’ve built on Mars isn’t just farming, you’ve created something that makes life in deep space bearable.”

Max looked up from the contract, his mind spinning. “This is... Catherine, this is worth more than the water and carbon contracts combined.”

She smiled. “Good food is priceless when you’re stuck in the asteroid belt.”

Max sat at the table a moment, studying the numbers. He stood and he and Catherine shook hands. “Catherine, we are all excited by what you have been able to accomplish in the Asteroid Belt. And we appreciate your generous terms for closing the contract. I will discuss the food contracts with Jakob. I expect he will be as excited as I am.”

They stepped together for a brief *bise*. “Good-bye, Catherine. And thank you.” He then turned to walk down the stairs.

He crossed the street and stepped down onto the quai where he turned west. He replayed the conversation in his head as he walked. The original Mars business was dead but other businesses were growing. Pacha and his demands for greenhouses and livestock had always been a distraction. On the Moon and Ceres, people ate algae. That Martians had complex agriculture had always seemed like a distraction, a small tax he had to pay to keep them happy. That distraction was now becoming the main business. Mars was becoming the mid solar system’s grocery store, selling to the Moon and Ceres.

Callum’s mine had played out. He made billions from it, but it was done. Instead of coming home, Callum found a second beryllium vein. And copper. Copper was not rare enough to justify shipping to Earth, but, as CuBe, it worked. And Callum also found minerals that Pacha wanted to add to the soil in the greenhouses, improving the yield.

As Max walked along the Seine, he watched the water flow beside him. Upwellings would spread out and move downstream. He couldn’t find a pattern to it, but could see the dynamic.

Max was still mulling the implications when he stepped up off the quai and crossed the street to the *Le Puits Perdu*. He noticed they now featured “Astran Cuisine,” made with Martian ingredients. That food came from his team. How did they get it?

Max had never tried to be controlling, but he didn’t realize this was going on. Jakob never said anything. Jakob wasn’t there to ship ice. Jakob was a person.

Jakob had listened to Catherine in a bar. He would listen to others. They aren’t pieces in his puzzle. They are actors.

Emmanuel greeted Max when he stepped inside and showed him to the back where Tobias Raines was waiting.

Max took him aside for a moment before meeting Raines. “Emmanuel, I notice the Astran cuisine. Has Raines ordered yet?”

“No, he has not.”

“Then I want two Astran plates. What would you pair that with?”

“With amaranth chicha brewed on the Moon.”

“Perfect. Let me just pay now.”

“Of course, sir. That will be forty two thousand Euros.”

Max drew in his breath.

Then paid the bill.

As he made his way to the back room, Raines stood to greet him. “Max, good to see you. How was your meeting with Catherine?” Raines asked, sitting back into his seat. “How’s the Mars business?”

“A disaster,” Max explained. “The Cerean water business is the entire reason we went to Mars. That business is completely dead. So is the Cerean carbon business. But other businesses are growing, especially food. Financially, it’s the best day we have had so far.”

Raines studied the numbers. “The only reason your team ever grew food on Mars was for survival rations. When you came to me, I was impressed with how thoughtful your team was about food and shelter. You have fantastic housing compared to anyone else in space.”

“Yeah, I am constantly impressed by this team. They are making improvements in everything they do. Their new housing is much safer, agriculture is more automated, water collection is easier.”

“Why are you having them spend so much time on that instead of their business.”

“I’ve been asking myself that same thing. But I was looking at it all wrong.”

“You’ve been pretty successful for someone looking at it all wrong.”

“I thought Jakob had found a good opportunity. That’s not what he did. Jakob put a farmer next to a builder next to a mechanic next to an electrician, and set them loose on civilization. When we added a miner, it didn’t add a revenue stream. The entire team became more powerful.” Max leaned in. “Mars can do more. I am going to add two new groups.”

“What are you thinking of?”

“I’ve been talking to a chemical engineer, Dr. Marc Lafevre, an assistant professor at the Sorbonne. I met with him last time I was in Paris. He wants to set up polyethylene production on Mars. Ellie is making some, as she needs it, but he wants to have actual chemical factories. We’ve got abundant carbon and oxygen

from the atmospheric processors, some nitrogen in the air, and we are careful to recycle everything. With Callum's mining operation pulling rare earths as byproducts, we could build a real chemical industry."

Raines was quiet for a moment, processing. "That takes serious equipment."

Emmanuel brought their meals.

Raines looked up.

"This is the Astran plate, sir. Roast cavy, baked oca, amaranth salad with lunar microgreens, amaranth bread rolls, and chicha made from amaranth grown on Mars, fermented on the Moon," Emmanuel said. "Enjoy."

"I saw that on the menu, food raised by my team, and I had to have some. We should know what they are eating," Max said, picking up his cup. "Cheers."

They both took a drink. "Creamy," Raines said.

Max continued talking. "Yes, the equipment for the chemical industry will be exorbitant, and shipping will triple that, at least. I am planning on using Andrei's new ships."

Raines paused his fork with cavy meat. "Are the ships ready yet?"

"The first standard cargo ship will be rolling out in six months. I plan to hire it."

Raines nodded and chewed.

Max continued. "When you are in Paris, where do you stay?"

"Le Maurice."

"That is a very nice hotel. I was staying there recently, and met the director of operations, Bastien Laurent. He is fascinated by Mars, and wants to build a hotel there. Not rough crew barracks. He wanted to make a destination."

Raines finished a bite of salad. "A resort on Mars?"

"The first interplanetary destination. Hotel, casino, spa, maybe some kind of theme park. Think about it. Where do crews from Ceres go for leave? Where do their families live while they're working the asteroids? And when JRC finally tackles Jupiter..."

Raines was nodding slowly. "Mars becomes the staging ground."

Max continued. “And the leisure hub. Mining Ceres is brutal. Working Jupiter will be worse. But if crews know they can take leave on Mars, hit a casino, catch a show, see their families...” Max spread his hands. “Mars becomes indispensable.”

“The JRC is moving slowly,” Raines said, half to himself. “But if they ever are ready for serious Jovian operations, they’ll need infrastructure. Mars can provide that.”

Raines smiled. “You’ve been thinking about this for a while.”

“It’s been on my mind since I signed Callum,” Max said leaning forward. “But I didn’t understand it until after I talked to Catherine.”

“This is going to stretch you, Max. Can you cover it?”

“It will be tight. Andrei is stretched too. But we’re both excited.”

Raines called out. “Emmanuel, another round of chichas!”

After Emmanuel poured, Raines lifted his glass. “Vive Mars!”

Chapter 25: Jovian Research Institute

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The shuttle climbed inland from Baracoa, following the winding road past forest and coffee plantations until it reached the settlement of Jamal. Here, where the foothills of the Sierra Maestra began, the Jovian Resource Corporation had carved out something new from the Cuban countryside.

Amira Bastos stepped out into the humid morning air, squinting at the campus spread across the hillside ahead. It had been four months since JRC had established the Jovian Research Institute, the JRI, and this was her first visit to see what their money had bought.

The campus spread across a natural amphitheater in the foothills, its buildings arranged in terraced rows. Low-slung concrete structures that opened onto sunlit courtyards were connected by terra cotta walkways. Abstract murals of planetary forms caught the morning light, and the pale plaster buildings glimmered against the green hillside. Small clusters of students and postdocs moved between laboratories, carrying slates, tools, and cups of coffee.

Looking northeast, Bastos could glimpse the curve of the beach half-hidden, glistening under the sun, only a couple miles away.

Dr. Ji-Wu Kim, the Jovian Research Institute's director, appeared at her elbow with the practiced smile of someone accustomed to managing up. "Ms. Bastos, welcome to JRI. We're honored to have you here for our first major symposium."

"Beautiful layout," Bastos said, noting how the buildings seemed to flow naturally into the landscape. "Good turnout?"

"Excellent. Sixty-two researchers from fifteen institutions. Graduate students, postdocs, senior faculty are all here to tackle the Jupiter problem."

The Jupiter problem. Bastos liked how that sounded. Simple, focused, the kind of thing that looked good in board presentations. "And they're making progress?"

“Significant progress. Why don’t I show you?”

Kim led her along a shaded walkway toward the main complex. The central building was larger than the others, its conference rooms opening onto a courtyard where a fountain trickled over carved stone. Through the glass doors, Bastos could see the main hall was already packed.

“That’s Rithika Patel,” Kim whispered as they took seats in the back. “One of our senior systems engineers. She’s presenting failure mode analysis for atmospheric collection platforms.”

At the front, a young woman was explaining something involving a whiteboard covered in equations that might as well have been hieroglyphics to Bastos. The presentation involved charts and technical jargon, but she caught the key phrases: “yield optimization,” “operational parameters,” “viable collection rates.”

She turned to Kim. “Kim, I am going to need Rithika to write an abstract for me. I want the board to hear about the kind of work we are doing here.”

When questions began, an older man with precise gestures raised his hand. “What about material degradation in the radiation environment?”

“That’s Dr. Leland,” Kim murmured. “Our materials specialist.”

“Excellent question,” Rithika replied. “We’re modeling accelerated wear patterns, particularly for membrane systems and structural components. Current projections suggest...”

Bastos tuned out the technical details and focused on the energy in the room. These people believed in what they were doing. The details mattered to them.

After the presentation, Kim guided her through the hallways of the main building. Along the way, there were signs pointing to various interest group meeting areas: Heavy Lift - South Wing, Harvesting - Central Hall, Jovian Ejection - North Wing.

“The institute is broadly chartered with the impossible task of solving the ‘Jupiter problem.’ We have broken this problem into four subproblems.”

She ticked them off on her fingers. “First: heavy lift. Getting mass from Earth to orbit cheaply enough to build the enormous infrastructure needed for mining Jupiter. Second: interplanetary transit. That one’s handled. A private company, Lunar Tug, has announced plans to provide regular freight and passenger service throughout the mid solar system. Third: harvesting the ^3He from Jupiter’s

atmosphere. That is where most of our people focus. Fourth: ejection, getting that helium back out of Jupiter's gravity well economically."

Bastos copied the list of talks. This was the kind of clear problem statement she could work with and share with the board.

"The harvesting problem has the best characterized solutions," Kim continued, leading her to a different conference room in the central hall. "We've organized our harvesting research around three main paradigms, each with multiple variants."

Bastos pulled out her phone and discretely started taking notes.

Kim introduced Bastos to a graying woman. "Amira, this is Jackie Morrison, formerly with NASA orbital mechanics. She is our department chairman for Helium extraction."

Bastos smiled and shook her hand. "Pleased to meet you. What are you researching?"

Jackie launched into a catalog. "First approach: orbital intercept missions. The two main variants are elliptical orbits that dive into the atmosphere periodically, and low circular orbits that skim the exo atmosphere continuously. Both require significant propulsion to maintain their trajectories."

"Second, atmospheric aircraft. Fixed-wing designs optimized for the density and pressure environment. Some maintain level flight while others dive repeatedly. These craft experience much higher pressures, so collection is faster and easier than with the exoatmospheric collectors, but powering them is a huge problem."

"And third, buoyant platforms,. Various dirigible and balloon concepts for sustained atmospheric presence. The problem there is keeping them from being ripped apart."

Bastos looked at her perplexed. "Which one's most promising?"

"That's... complicated. Each approach solves different parts of the problem. The dirigibles are probably most feasible for sustained collection, but they are the worst when it comes to ejecting that payload from the gravity well."

By midday, the Caribbean sun had turned the courtyard into a furnace, and Kim suggested they take lunch in the cafeteria. The dining hall occupied the ground floor of the western building, with large windows overlooking a valley that stretched toward the distant sea.

Kim guided Bastos to a table with several of the researchers, picking at a plate of rice and beans while listening to animated discussions about membrane chemistry and orbital mechanics. At the next table, a group of graduate students were sketching diagrams on napkins, their conversation a mix of English, Spanish, and technical jargon.

“You know what strikes me?” Bastos said to Kim. “The energy here. These people aren’t just doing research; they’re on a mission.”

Kim took a sip of carbonated water. “That’s what happens when you give smart people an impossible problem and tell them to solve it.”

“Is it impossible?”

“Six months ago, I would have said yes. Now...” Kim gestured toward the windows, where they could see other buildings scattered across the hillside. “Maybe it’s not impossible. When you break the problem down, as they do, it is just many nearly impossible problems that need to be solved all at once.”

Through the windows, Bastos could see the landscape that had drawn JRC to this location. Jamal sat in a natural bowl surrounded by green hills, isolated enough for security but connected enough for logistics. To the north, the mountains rose toward the Sierra Maestra. To the south, the land dropped away toward the coast where Baracoa waited with its harbors and launch facilities.

After lunch, Kim walked her through the rest of the campus. The afternoon light slanted across the courtyards as students and researchers moved between buildings, animated conversations spilling out of doorways.

As they walked through the main hall, filled with people working on their slates, Bastos asked Kim, “The talks were fascinating, but surely they exchange papers, so they already knew all that. Why do you have the symposium?”

Kim gestured to a young man sitting in a chair staring at the ceiling. “Several reasons. I may poach someone we need on our team. I get people talking to each other, to see if they can engineer something. But, mostly, I do it to feed thoughts to the creative thinkers. Like Nikola Savic over there. He’s exploring alternative approaches. He’s not ready to present them yet, and maybe he never will be, but independent thinkers like him are our real hope.”

This was something the board would not appreciate, but Kim’s plan made sense.

“How long?” Bastos asked as they began walking back toward the shuttle pickup point.

“For what?”

“For answers. Real ones. Not just more questions.”

Kim considered this as they descended the terraced walkways. “If we’re talking about proof of concept? Demonstrable technology? Maybe ten years for collection systems. Twenty for integrated operations. I don’t know.”

Bastos nodded. She didn’t have a deadline, she simply needed a number. Twenty years was long enough to sound realistic, short enough to maintain political support.

As the shuttle pulled away from the JRI campus, Bastos settled into her seat and watched the buildings recede into the afternoon haze. The vehicle wound back down through the hills, past small farms where cattle grazed in fields bordered by royal palms. The landscape here felt ancient and settled, a stark contrast to the cutting-edge research happening in the labs they’d just left.

The road curved through a valley where a river ran clear over limestone rocks. On the far bank, she could see the remains of an old sugar mill, its chimney rising like a monument to an earlier age of Cuban industry far removed from space travel and helium.

As they descended toward the coast, the view opened up. First the red tile roofs of Baracoa appeared through the trees, then the blue expanse of the Caribbean beyond. The JRC headquarters building sat prominently on its hillside overlooking the harbor, where even now she could see ships moving between the piers and work lights beginning to flicker on as daylight faded.

The shuttle approached the JRC complex, with its security gates and administrative buildings and the gleaming headquarters. Bastos didn’t understand it all, but the people working on it seemed to.

And that was enough.

Chapter 26: L'Hôtel Bounty

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Bastien Laurent never sketched on his slate. He had a B3 hardboard portfolio case filled with cream sketch paper. Marc Lafevre was watching him work under the sterile lighting of Lunar Tug's Martian ferry. Bastien was drawing the interior of a dome. "Marc, look at Ellie's pictures. Bounty is beautiful. I love the domes, how they blend in with the land, like they are supposed to be there. My hotel must be like that."

"I'd have guessed those domes would be too plain for you, Bastien. Far from the classical French opulence of your *Le Maurice*."

"Not at all. Ellie's domes would look out of place in Paris, but Haussman architecture would look absurd in Bounty. I am working on a grand entrance to the Bounty Hotel. This would be travelers' first contact with *L'Hôtel Bounty*. It is here that the patron must move from feeling comfortable to feeling well loved. A spacious dome. Inside, there is no counter, no separation. A hostess will greet them with a simple sparkling water, and check them in with their gear. We require a large seating area maybe with a fountain. Patrons may wish to get their bearings. Some may wash up in the pool."

Marc looked at his drawing. It was a fifteen foot tall replica of Olympus Mons, with water running over the shield into a pool. Bastien had drawn two men in the pool *au naturel*. One was standing under a waterfall coming off the top of the shield. "That would be welcome. Using wipes to clean myself on this trip has been awful."

Bastien winked. "Agreed. That is why I imagined the fountain. I am imagining what I want to be there when we arrive."

"My goals when I arrive are much less beautiful. I need to increase ammonia production to support expanding agriculture. Did you try Astran Cuisine at *Le Puits Perdu*?"

"I just ate a roll. Earthy."

“I had the meal. Max paid for it. The absurdity of shipping food from Mars to Earth aside, the food is interesting and satisfying, and is in huge demand on the Moon and Ceres. They are expanding agriculture twenty fold. Max asked if I could expand *LDPE* production enough to meet that need. I can, but that is the wrong solution.”

“I thought *LDPE* was great. Admittedly, I heard that from Max, but what’s the problem?”

“*LDPE* is great. It is easy to manufacture, it mixes well with regolith, forming tough walls, and it is high in hydrogen, so it blocks radiation. I am planning to load it with boron for neutron shielding. It is easily foamed and it can be used to make sheeting, which is convenient. If you can only make one polymer, this is it.”

“You want to do more.”

“Right. Now they are buying ETFE sheeting from Earth for their greenhouse. We have some with us. It is stronger and clearer than *LDPE*, and is resistant to UV radiation. But shipping from Earth means they never have enough, and when it needs to be replaced, they have to sub in *LDPE* sheeting. I need to start making ETFE on Mars. I can reinforce it with aramid fibers and experiment with gauges until I find what we need. This means I need to find fluorine feedstock.”

“Where can you get that?”

“It’s not rare, but I need the right deposits. I need Callum to find them. Sulfur, too.”

“ETFE needs sulfur?”

“No, it’s a different issue. The primary use for *LDPE* is mixing with regolith to make buildings.”

“Yes, Ellie’s domes.”

“Exactly, but sulfur does that easier. *LDPE* is a radiation shield, but in places where radiation isn’t a concern, like foundations or greenhouse walls, sulfur-regolith concrete would be much faster, easier, and cheaper.”

“I am sure Max doesn’t mind the expense. Why are you worried about cheaper?”

“Cheaper means we can afford to build more, faster. Cutting costs means cutting total labor. If I can build the foundations for all those greenhouses with mined sulfur and loose regolith, I can put more effort into making ETFE sheeting.”

“Callum said he would point us to the right places, and I brought backhoes. Still, I don’t think Callum was completely happy with Max.”

“What do you mean?”

“Callum’s game is beryllium. Find a site, dig it up, make billions, find another vein, repeat. All these other minerals are a distraction.”

The ship fired its rockets to slow down and move into low Martian orbit. “Suit up,” the captain called over the loudspeaker. “We are about to depressurize.”

Bastien and Marc started getting their suits on, layer by layer. The captain had already begun moving the air into the long storage cylinders, bringing it from three psi travel pressure down to two psi minimum safe pressure. When everyone checked in that they were ready, the captain sped up the compressors, bringing the ship down to less than a tenth of a psi. Bastien watched a door close and heard nothing.

They all assembled in the cargo bay. The captain’s voice now came over Local Band radio. “I need you guys to start loading up the cups. I will bring them in, you fill them with cargo. It is all color coded so you will know what goes with what. I hope you guys are good at tetris and Jenga.” The captain opened the bay door and there was a short puff as the remaining molecules of air escaped.

He tethered and scrambled out the door and on top of the ship. He came back down with a dixie cup the size of a semitrailer. “Red.” The team started loading up the red pallets. Weight was no problem. Arranging was. When everything was inside, they twist-locked the lid on.

“What is this cup?” Bastien asked Marc.

“The moon is making more titanium than Earth, now. These are descenders used to drop from LEO to the surface. Lunar titanium cups and lunar heat shields, Earth made parachutes, electronics, and interiors. Did you think Lunar Tug was going to land?”

“Patrons would prefer that.”

The captain verified the seal, and put it on the launcher. “We launch twenty degrees before we get to Bounty. Tell me when you see it.”

Bounty was far too small to see, but Olympus Mons was very clear. The ship lurched as the mass driver slowed the cup enough that it would intersect the

Martian atmosphere. They watched as the lid started glowing white hot, then dulled, then a chute came out.

Green, Blue, Orange, and Yellow were all loaded and launched the same way. Then the captain came back with an unmarked cup with seats and life support gear. Everyone but the captain got in and buckled up. The captain moved them to the launcher. His voice came over the comm line. "Five Four Three Two One."

Instantly, Bastien couldn't breathe. The launch was at eight g. But less than a second later, it was over.

"Was that 'comfortable' or 'well-loved'?" Marc asked when they could breathe again.

"We are going to have some unhappy patrons," Bastien replied.

On impact with the ground, the cup flipped end over end twice, then rolled fifty feet before coming to rest against a boulder, leaving them upside down. No one said anything for a few seconds, until they were sure they stopped. One of Marc's engineers was the first to release herself from the seat. She held onto the chair and landed on her feet. One by one they all came down and sat on what they thought was the ceiling.

Forty minutes later, there was a banging on the hull. Over the descenders FM radio, they heard Callum's voice. "Goodday, mates. Anyone injured?"

Marc got on the radio. "Shaken, but not broken."

"OK. I need you guys to release the lid. I can only take two of you at a time."

Marc and Bastien watched the gyro take off with two of their staff. "Customer service may face some challenges," Bastien confided. Marc laughed.

It was almost dark when Bastien and Marc finally arrived at Bounty. Pacha had prepared dinner for everyone. The ag dome was the only place large enough.

Jakob sat at the head table. He raised a cup of chicha. "Today we welcome two new teams to Mars. Sorry about your gear. It is scattered over forty miles. Tomorrow we will help you bring it in."

Marc didn't say a thing. He just looked at Bastien, who rolled his eyes.

The food was good, better than what was served at *Le Puits Perdu*. They spent the night in the ag dome on cots. It was the best night they had in four months.

After breakfast, Bastien insisted on a tour. Pacha and Mira took him to see the greenhouses.

Pacha wanted to start with greenhouse five. When they got through the airlock, Pacha and Mira removed their helmets. Bastien followed suit.

“The air feels thin here. Are you sure it’s OK?”

“You see those?” Pacha asked, pointing to a cavy running under the leaves. “If they can run, we’ll be OK. Don’t exert yourself. We are at two PSI.”

“Why isn’t my blood boiling?”

“That’s not the real problem here. The real problem is the Armstrong limit. If the partial pressure of water vapor in your lungs exceeds the total pressure in the atmosphere, then you’re done. Here, the air is almost half water vapor, so you can breathe much easier. The other half is oxygen, with four percent CO_2 and the balance sulfur hexafluoride. Your voice sounds normal, we are fairly safe from fire, and we don’t put too much pressure on the membrane.” Pacha pointed up at the clear sheet covering the greenhouse.

“So you can just walk around here? That’s amazing. This place smells wonderful. Why can’t you do this under Bounty’s dome?”

“It’s too big. The sheeting would never hold. Bounty has ten pascals overpressure. The strength of the sheets limits the width of the greenhouse.”

“What are you growing in here?”

“Pretty much everything. The purpose of this space is to raise cavies, let them graze, and feed the soil. Growing soil is the hardest part of Martian agriculture.”

“How about trees?”

“The sauco can grow forty feet tall. Here I keep it trimmed to six.”

“Can I keep them in pots?”

“They may be hard to keep watered, and they won’t grow as tall, but you can do it.”

“Why are you asking so much about the sauco tree?” Mira asked.

“Forgive me. I don’t mean to interrogate. But this space you have is beautiful. I must have such a space for *L’Hôtel Bounty*.”

Pacha looked at him funny. “A pasture in a hotel?”

“A park. People walking slowly, looking up at the night sky, sitting under a tree during the day. Flowers blooming, little beasts running around. And your pond! I must have a pond. Do you have geese?”

Pacha laughed. “No way. Nothing bigger than a cavy. The only thing that flies are these butterflies,” he said, pointing to a few resting on the membrane.

“You have butterflies? You buried the lede, Pacha. Why do you have butterflies?”

“The air is too thin for bees. I have some beetles for pollinating, but butterflies do well for several plants. These are called Painted Ladies. No problem at this pressure. They seem to like the high humidity.”

“You are aware they are gorgeous, aren’t you? Are they poisonous or anything?”

“No, the cavies eat them sometimes with no problem. Not often, though. Cavies can’t fly.”

Pacha next took him to greenhouse twelve. When they came through the airlock, Bastien reached for the helmet. Mira noticed and jumped to slap his hand away. “No! Stop!”

She had knocked him over, so he stood back up.

“In the pasture, you can take your helmet off. Here you can’t breathe. Running less oxygen helps the plants grow faster, and we don’t keep it as warm, so there is much less moisture in the air. More nitrogen and argon. This is for plants only,” Mira explained.

Pacha showed him around. “The amaranth grow well here. We share this with the cavies. They actually eat more than we do. The leaves make a good salad or cooked greens, and the seeds are grain. That’s where we get our bread and our chicha. We grow many types of plants, but this is our staple.”

“These are all golden. In the pasture I saw some red ones, too.”

“I raise both, but more golden.”

“Pacha, have you seen a field of wheat or corn? Soothing, but not beautiful. These are beautiful.”

After the greenhouse tour, they walked back to Bounty. Pacha elbowed Mira and switched to an FM band. “Mira, this guy gets it.”

“Gets what?”

“He loved the greenhouses. Jakob thinks they make food. Bastien sees they make life. He thinks its beautiful.”

“Jakob had to eat only algae biscuit soup for months until you got it growing. He has a different perspective. When Bastien eats at a restaurant with two michelin stars he thinks he is slumming.”

“Whatevs, Mira. He gets me.”

That night, at dinner Marc and Bastien sat with their teams. One spoke up. “You know what they made us do today? They made us trek all over Mars and haul back our luggage. It’s not our fault the equipment was scattered all over.”

Bastien looked at him. “You are a great chef. We need you. But we need everyone else here, too. Whose fault was it? Jakob’s? The miners? Who should take time out of their day to take care of us?”

“The staff can get it.”

“That’s us. We are the staff. Tomorrow Marc and I will join you. How much is left?”

“We found all the containers, but only brought two back.”

“OK. Tomorrow we get the other three. And if we don’t finish tomorrow, you and I can finish the next day.”

“*Oui, Monsieur Laurent.*”

Over the next week, Bastien went everywhere in and around Bounty with his portfolio case, making hundreds of sketches, ignoring Lafevre and the rest of the camp entirely. Jakob kept asking him how it was going, and all he said was he was thinking.

Finally, Bastien agreed to describe his plans. “Beauty comes from order and contrast. The opposite of beauty is not ugly, it is disorder and monotony. Here we have the mountain, the scarp, and the plain. Beautiful. This ag dome. The colors and texture are of Mars. The shape mirrors the craters. Beautiful.”

Ellie adjusted herself in her chair, sitting straighter.

“*L’Hôtel Bounty* must fit in and stand out.” He took a drawing from his portfolio, the one he had been working on in transit. “Here is the grand entrance. A cableway brings patrons on a gondola. They emerge relaxed and calm, their baggage with them. They are immediately welcomed and cared for.”

Callum looked quizzically. “So, I pick them up from a descender and put them on a gondola?”

“That would never work. Do you know how you land on the Moon? Tan has built a small mass driver that helps slow incoming ferries and gives them a boost on launch. When it lands, the ship goes through an airlock. When you step out, you are in shirtsleeves. When you are ready to return, you are again in shirtsleeves. It is calm and relaxing.”

Jakob raised his hand. “Sure, but the Moon has much less gravity. Lunar Tug could never get back into orbit. It is tight as it is.”

“Tom, your mass driver is sixty miles long.”

“Sixty-three.”

“The Moon has a driver about a mile long. We can do this. The first stage of *L’Hôtel Bounty* cannot be a room. It is a depot. Lunar Tug comes in, refuels, and leaves. Before we disembarked from Lunar Tug, I looked at the captain’s face. I saw sorrow. Resignation. He was looking at six more months to get back, without even a break when he got here. Imagine instead, he came down, landed with us, ate with us, refueled, and went home much faster. That is the beginning of hospitality.”

“I’m pretty sure Max expects rooms and beds,” Jakob said.

“When you were telling me about Ellie and the domes, you said Ellie knows the difference between the structure and the building. Power. Air. Thermostatic control. Plumbing. Cooking spaces. Max expects patrons to enjoy their experience. We will get to structures, but that isn’t where we start. A depot, a gondola, and an entrance are more important than a thousand rooms and beds. If they sleep in the

ag dome, packed like sardines, they will have been welcomed and treated with dignity.”

Tom raised his hand. “Your design doesn’t have an airlock. Are those revolving doors?”

“Yes. An airlock is designed for efficiency. A revolving door can do the same thing, but it is designed for the user. Foot traffic enters and exits through a series of three revolving doors. Here, above the doors, is the air pump, constantly, silently moving air from the entrance side to the exit side. Patrons would be unaware of this. They only feel welcomed.”

Bastien pulled out another sheet, and stuck it with pushpins on the wall above him. “*L’Hôtel Bounty* is part of Bounty. You can see in this drawing how he must be designed. The entrance connects by a sealed walkway to the central building. It looks like one of Ellie’s domes, but it is larger and steeper. The ground floor has twenty suites. Not rooms. Spaces where a family can unpack their luggage and themselves. Above that would be *la grande salle*. Meetings, events, dining, dancing. The beautiful curved ceiling.”

Ellie said, “We don’t dance on Mars.”

“Pity. But perhaps that will change.”

Ellie blushed.

“Around this, we put in smaller domes. Each will have a restaurant and rooms. Smaller spaces. We will add more as we gain a measure of demand. We must satisfy Jakob’s need for rooms and beds.”

Everyone laughed. It was Jakob’s turn to blush.

“But here,” he said, pointing to the space next to the walkway between the entrance and the central building, “Nothing must ever be built. Connected by a revolving door, the pressure is lower here. This space is a park. Flowers. Trees. A pond. Meandering walkways and benches. Those little beasts running everywhere.” He pointed to the cavy pen in the ag dome. “No pen. Unfettered, inviting the patrons to enjoy the space as they do. Marc tells me he will be making a strong, clear ETFE membrane. We use this, allowing an unobstructed view of the stars, sometimes even of Earth. And these.” In a smooth gesture, he released five butterflies from his pocket, letting them flutter around the ag dome.

Chapter 27: Heavy Lift

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Amira Bastos stood at the floor-to-ceiling windows of her corner office, watching the afternoon light play across Baracoa's harbor. Behind her, Dr. Ji-Wu Kim settled into one of the leather chairs arranged around a small conference table. "Amira, you sounded upset when you called," Kim said carefully.

Bastos turned from the window. "Did you see Raines' interview in the Financial Times yesterday?"

Kim shook her head.

"I can tell you the board has. He was talking about the lunar ^3He project six years ago, how he solved it by offering loans and stable price guarantees, creating a rush of miners. Six months later, ^3He was flowing. Same with refractory metals. He funded asteroid mining operations, and within three years, asteroids were producing more tungsten than from all terrestrial mining combined."

"Those were logistical problems," Kim said quietly. "What we're working on is fundamental."

"Don't tell me that, Ji-Wu. I have a board meeting in two weeks and I need to show them progress, not excuses. It's no secret Raines wants new leadership at the JRC, and now he is making his case to the public."

Kim was quiet for a moment, studying Bastos's face. "Getting ^3He from Jupiter does not lend itself to quick solutions. I know just the group for you to talk to. The Heavy Lift group is tasked with improving the cost and cadence of launches from Earth to LEO. It is the most visible part of our charter."

The next day, Kim brought Bastos to JRI. They went to a two-story structure labeled "Heavy Lift".

Kim brought her to visit one of the chief researchers. "Amira, may I introduce you to Dr. Marcus Webb, a propulsion specialist who joined us from SpaceX. He has been looking at a set of ideas that I think will suit your needs very well."

Webb stood and shook Bastos's hand. "A pleasure to meet you. Ji-Wu tells me you want to build a new launch facility."

"The launch facility is Kim's idea. I need something that makes a statement about the value of JRC," said Bastos. "What is your plan?"

Webb turned to the whiteboard. "The fundamental problem with chemical rockets is that the fuel that accelerates the rocket also has to be accelerated. You need more fuel to accelerate that fuel, which adds to the weight, which, in turn, requires more fuel. This vicious cycle is called the rocket equation. Each increment in speed requires exponentially more fuel. While it looks grim, it has a bright side. Each decrement in speed requires exponentially less fuel. That is where we in the Heavy Lift group live."

"What do you propose?" Bastos asked.

Webb began drawing towers, tubes, and dotted lines in sweeping arcs. "The system my team has been working on involves a six mile pneumatic tube supported by a titanium lattice. The tube accelerates the rocket to Mach 2.3 at the top, at which point a ramjet stage takes over, driving the rocket to Mach five at ninety thousand feet. The result is we move from three percent payload to twenty percent payload, eight times as much."

Bastos looked blankly at Kim.

"Marcus, that is a very nice system, but we are on a deadline. Tell us about needed research, construction time tables, and costs," Kim said. "Your research is directed to finding the ideal system, but today what we need is expedience."

"There is another system we can use, designed to minimize dynamic load. It still uses a pneumatic tube, but rises a half a mile. In six seconds, the rocket exits at Mach 0.75. It stays subsonic up to the stratosphere, allowing for a lighter rocket body. If we combine that with air augmentation, the system can launch twenty percent more payload from the same fuel."

"Twenty percent doesn't sound so impressive."

"This system allows a high cadence. The rocket goes into the launch tube, gets fueled, and launches. Six seconds later, you are ready to load another rocket. A typical launch pad requires weeks to months of inspection and reconditioning after every launch."

Bastos made a note on her slate. Launch cadence had overwhelmed US and Chinese launch facilities. This could be sold to the board.

“Another advantage is reduced sonic loading. Takeoff typically has the engine noise reflecting off the ground into the payloads, which must be adapted to deal with that destructive force. With this, there is no launch noise. By the time the rockets are on, the ground is fading away fast. Tons of sound-absorbing blankets are no longer needed.”

“Coming back to your comments on cadence, did you say we could have a launch every six seconds?”

“No, that is just the launch itself. There are several other steps, but we could probably launch two or three times per hour.”

“What would it cost?” she asked.

“No matter what we build, it will cost around ten billion dollars. You always need a seaport, airport, and hangars.” Webb said. “The launch tower itself would cost ten billion beyond that.”

“Put together a full proposal,” she said finally. “Cost estimates, timeline, technical specifications. Everything. If I’m going to show this to the board, it needs to be bulletproof.”

“How long do we have?” Webb asked.

Bastos checked her watch. “The board meets in two weeks. You have three days to put it together, then I’ll practice my presentation with you until that meeting.”

“We’ll have something ready,” Kim promised.

Kim and Webb went to meet with other researchers. Bastos remained in Webb’s office, looking at the whiteboard covered in sketches of towers reaching toward space. She didn’t understand the thermodynamics or the structural engineering, but she understood opportunity.

The day before the meeting, Kim, Webb, and Bastos sat in her office. The images on the whiteboard had all been drawn by Bastos this time. “I feel I really got it this time.”

Webb walked up to the board and erased a four and wrote a three. “Try not to get too technical in the meeting. You got the key points.”

The board meeting started at eight A.M. Dr. Emile Carson spoke first. “Amira, JRC was chartered to do an enormous, maybe impossible job. That you have not accomplished your goal is no surprise. My concern, however, is that I don’t see any action.”

Bastos looked around the room. None of them would meet her gaze. Okafor had a large envelope in front of him, with her name on it.

She went directly into her rehearsed talk. “I have the top scientists in the field working on extracting helium from Jupiter. In the meantime, there is one element of that problem that has global importance, and that is heavy lift. I have provided documents on your slates.”

Despite feeling weak in the knees, she stood and moved to a whiteboard, as rehearsed. “Our primary limitation right now is heavy lift, from the surface to LEO. Canaveral is backed up for three months. Boca Chica is backed up six. Wenchang only takes reservations three months in advance and is completely booked. The lunar population is capped not by demand or technology, but by heavy lift capacity. The world needs more.” She drew a diagram of the kilometer launch tower and the cavernous warehouse. She graphed the rocket equation beside that, drawing a dotted line from the top of the tower to the graph. “While this tower doesn’t boost height or speed much, as you can see, it has an enormous impact right where the most fuel is being used. It can reduce fuel costs by twenty percent.”

Bastos stopped looking at the whiteboard and looked at the people. “Six months ago, I would have said fuel savings were paramount. But today, I see that as secondary. The goal is volume.” She continued drawing, showing a large arched roof connected to the tower. “The hangar has space to load payloads and stage several rockets at the same time. When ready, they are moved into a low-humidity pipeline.” She drew five rockets in a row. “The rocket spends twenty minutes in each station. First, we dehumidify the entire rocket body and all the internal bits to prevent frost formation. Bone-dry air is blown into every crevice of the rocket. After it is dry, we chill the entire rocket body to nearly freezing. In the third station, the body of the rocket stays close to freezing, the fuel tanks are cooled with refrigerated air to -60°C , then, in the fourth station, with nitrogen vapor, leaving the tanks colder than they need to be to prevent boil-off when the rocket is fuelled. The fuelling takes place in the launch tube. When ready, the tube is sealed. The launch itself only takes six seconds.”

Carson was calculating on his slate. “Do I understand you can launch three rockets per hour with pipelining, which would be seventy-two per day?”

“Theoretically, yes. However, we don’t expect to reach that cadence. The research team expects to launch forty rockets per day.”

Professor Liu Wei, the sustainable development theorist on the board, was recognized. “Fuel expenditure remains my primary concern. I feel that a twenty percent improvement is too small.”

“Professor Wei, there are many at JRI who agree with your perspective. They have suggested an alternative project. With the board’s indulgence, I will describe that process.” Bastos moved to another whiteboard. “Mostly, the project is the same. We still need seaports, airports, barracks, and hangars. But they propose an enormous tower six miles high. This will take several years to build and will cost twenty times as much, but the rocket will exit near the stratosphere at Mach 2.3. This resolves several rocket problems very cleanly. The rocket would not need to handle the transsonic domain; it would be through seventy percent of the atmosphere, and it would have saved...” She drew another dotted line from the tower top to the graph of the rocket equation, showing a third of the fuel consumed at that point. Bastos noticed the directors were engaged.

“At this altitude and velocity, JRI proposes a ramjet stage. While my first proposal uses a shroud for air augmentation, this ramjet is more complex and will require years of research before it can be viable. When it is viable, it will be able to accelerate the entire rocket to Mach 5.0 and release it at ninety thousand feet. At that point, the rocket lights. Much of the work has already been done. Even the nozzles that work at that altitude are ten percent more efficient. Net result, this system can launch around ten times as much payload from the same rocket. JRI exists to research academic ideas such as this. I don’t bring this to you at this time because I believe we need action now.”

The board dismissed her briefly and deliberated amongst themselves. When she was escorted back into the boardroom, she noticed the envelope in front of Okafor had been torn in half.

Carson spoke. “Amira, we have reviewed your documents. Twenty billion is steep, but we all knew a spaceport would need to be built. We are authorizing that operation.”

Bastos was about to speak when Carson held up his hand. “Professor Wei and Ambassador Okafor are both insistent that progress continue on the larger project, as well. While it will take time and money, the board believes that it is consistent with the chartered purpose of JRC. We have allocated an additional fifty billion per year for that project. Will you be able to fit both projects in Punta de Maisi?”

Bastos looked at each of them. “Yes, they will fit together well. Following your plan, we will be operational sooner, and eventually, we will be efficient, too. The shorter

tower will remain useful for non-standard launches. I will start immediately on the proposal I gave you, and get you a combined field proposal by next week.”

That night, Bastos brought Kim and Webb to Restaurante La Colina. The waiter asked what they would be having.

“Lobster in coconut milk and blue lagoons for everyone. And keep them coming.”

Kim asked first. “It looks like things went well. Spill.”

Bastos said nothing until she had taken a sip. “The board was prepared to let me go. It started off tense, but our prep worked. It didn’t even seem stressful. I just went through the motions we rehearsed. Carson understood cadence. But Wei hated it.”

“Of course she did. She hates combustion. And agriculture, too, if that came up,” Kim said.

“I felt that. But one thing she doesn’t hate is spending more money. The board approved the kilometer tower, but stipulated that we must also build the super tower.”

Webb dropped his drink. “Both! You’re a genius, Amira.”

The busboy came over to clean up the spill while the waitress brought him another drink.

“Speaking of genius, Marcus, you are going to have to help me build a new proposal.”

The board approved the updated proposal online, and she sent out an RFP. She got one of the younger new managers to oversee the process, Terence Halden. Not many companies could operate at this level, and preparing a bid took time. A month later, Bastos checked in on him.

“Seventeen responses so far,” Halden reported, scrolling through his slate in Bastos’s office. “Most of them are consortium bids. This is too big for any single company to handle alone.”

“Any standouts in the bidding?” she asked.

“A few interesting approaches,” Halden said. “There’s a European consortium led by Bechtel that’s proposing a more conservative timeline but very robust engineering. A joint venture between Japanese and Korean firms that’s promising some innovative materials science.” He paused. “And there’s one Brazilian company, Construtora Atlântico, that submitted a bid about thirty percent lower than anyone else.”

Bastos looked up sharply. “Are they credible?”

“I’m having our procurement team investigate them now. Their proposal is surprisingly detailed.”

Three thousand miles away, Tobias Raines called Henrique Cavalcanti, the CEO of Construtora Atlântico. “Henrique, have you put in the bid?”

“I have been working on that for weeks with my team. The RFP was written by academics, but we worked it out. When I calculated the cost, I cut fifteen billion from that, as you asked. It is a sealed bid process, but I am pretty sure we will be the standout for low price.”

“That’s good, we need to get the bid. What I need to be sure of is you can get this built quickly. Every day it is not built is costing me money.”

Henrique looked up from the structural calculations. “Our current design is eighteen months. As you know, projects like this face more delays than advances. This may take two years or more.”

“That won’t do. Let’s target fourteen months. You let me know what we can do to speed this along.”

“Yes, sir. Can I ask a question? They say their plans are speculative. Are you concerned it might not work well?”

“We build infrastructure for them. That will definitely be useful, whether the tower works as advertised or not,” Raines said. “Even if the pneumatic launch system is a complete failure, we’ll have built the first step we will need to support Jupiter operations. And if it fails, I believe they will figure out how to fix it.”

“Understood. We will continue to press for ideas to get this done faster.”

“Fantastic, Henrique. Let me give you a number of a metal supplier. Felipe runs a large mill in Mobile, Alabama called Planetary Metals. He can get you what you need fast. Let him know I sent you.”

“Will do.”

The call came three days later. Bastos was reviewing quarterly reports when Halden entered, with rolls of blueprints under his arm.

“Construtora Atlântico is a legitimate contractor. They have done a lot of work for the Brazilian government.”

Bastos set down her reports. “You’ve finished the background check?”

“Yes. They have a long and successful history. And a fascinating proposal.”

Halden spread out the Brazilian proposal on Bastos’s desk. “They made several changes. Twinned towers don’t just double launch cadence. It creates fault tolerance. If something goes wrong with tube A, you can use tube B until it is fixed. The tubes are designed to be built by jacking the top and inserting a new segment underneath. That process can also run in reverse, bringing it down for needed repairs or when there is a hurricane warning.”

“They are using the local limestone to run their own cement plant on site, which I think is smart, but they have also requested mineral rights to sell off the limestone they excavate. I wrote a note rejecting this idea, but I haven’t sent it yet. I wanted to talk to you.”

“Don’t look a gift horse in the mouth. Selling limestone may be why they are cheaper. The worst thing for us would be a huge pile of broken limestone and no place to send it.”

“OK, I will approve it. Another variance is the eliminated rail.”

“No, we need that to connect Baracoa to Punta de Maisi.”

“Instead, they want to use Mexican Modutram. The top speed on the intercity route will be about forty-five miles per hour, which is slower than the spec you and Webb put together, but they do much more. Yours had three stops: Baracoa, the airport, and the hangar. Theirs has unlimited stops and routes. They let it run all over the construction site, depending on where the work is, and all they have to do is paint some lines. Here this week, two hundred yards away next week. And the cars are very flexible. They can carry passengers, freight, and food trucks that can roll right up to the work sites. They even have yellow steel modules for scrapers and graders.”

“That’s close enough to rail for my needs,” Bastos said, scanning the timeline. “They are going to finish the smaller project in eighteen months? That’s half what the European consortium was promising.”

“And they’re guaranteeing performance. If the system doesn’t achieve the specified cadence, we don’t pay the final thirty percent.”

Bastos studied the technical drawings, trying to process the scale of what she was looking at. “I want this started now.”

“Yes, ma’am,” Halden answered. “I will get them started today.”

Nine months later, Bastos stood on the observation platform that had been built specifically for this moment. The concrete pad stretched for nearly a hundred yards, positioned a mile from the construction site.

Around her, a small crowd had gathered. Kim and Webb from JRI, and Carson from the board of directors. The hangar was still just a skeleton, but the twin towers were complete and guyed in place. They watched as the first pneumatic tube was jacked into place, a hundred meters at a time. They started with the crown, which looked like a pepper pot muzzle brake. In less than three hours, the tube was ready. They didn't yet have the entire pneumatic system ready, but they could run a low-pressure test. A stacked wooden slug, six meters across and ten meters high, stood where the rocket would go. Technicians began the countdown. It used a clock face with only a second hand. When it got close to the end, everyone on the platform was cheering, "Five, four, three, two, one." Nothing. It had launched, but it was still travelling. Six seconds later, they saw the slug exit at about six hundred miles per hour straight up. Five seconds after that, the report struck, shaking them right through their bodies. The slug continued up for another twenty seconds, before beginning its descent. Vanes on board directed it to land safely at sea.

"Linda," Henrique murmured.

Bastos looked out at the impossibly tall structure, the first piece of humanity's most ambitious attempt to reach space cheaply and routinely.

She thought about her desperate board meeting ten months ago, when she'd been scrambling for anything that might show progress. Now she had seen it launch a slug two and a half miles into the Cuban sky.

For the first time since coming to Baracoa, she felt like they were going to change the world.

Chapter 28: Sinoplex

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The tunnel stretched twenty kilometers beneath the East China Sea, its walls lined with gleaming superconducting cables thick as tree trunks. Dr. Elin Sarai walked beside Minister Chen Zhengrong and his technical team, their footsteps echoing in the cool air. Liquid nitrogen flowed through the outer jackets, keeping the cables cooled to 77 Kelvin.

“Ten megavolts DC,” Chen said, pride evident in his voice. “No AC/DC conversion losses. No AC at all. Direct from the tokaplex to all of eastern China on a double backbone mesh.”

Elin ran her hand along the insulated housing. Of course he wasn’t mentioning the energy consumption of cryogenic cooling, or accounting for takeoffs. There would be losses, but this was still an amazing achievement. She’d reviewed the specifications, but standing here made it real. This was a powerline scaled for a civilization that had outgrown fossil fuels and was reaching for something bigger.

“The takeoffs?” she asked.

Chen gestured to a junction ahead where smaller conduits branched upward. “Aerial grid at one megavolt. We’ve phased out AC transmission entirely. Every substation, every industrial complex, uses DC. Anyplace with legacy AC equipment runs their own inverters, but everyone knows AC is the past. DC transmission uses the conductor more efficiently and has no radiative or inductive loss. Your tokaplex will feed this grid a full terawatt, continuous. That will allow us to proceed with hyperelectrification.”

“If we can keep it fueled,” Elin thought to herself, trying to keep her expression neutral.

They emerged from the tunnel onto the artificial atoll, a C-shaped ring of reclaimed land and reinforced concrete two kilometers across. The opening faced east, toward the open sea. Enormous voltage converters lined the inner shore, their

cooling towers venting steam into the morning air. Connection points bristled with sensors and fail-safes. This was a fortress of infrastructure, designed to channel more energy than most countries could use.

“The platform arrives tomorrow,” Chen said. “Your team is ready?”

“We are,” Elin replied. “Gonçalo Silva will oversee certification. Your site general manager will have full operational control once we’re satisfied with the handoff.”

Chen nodded. “You will find General Manager Zhao to be quite capable. She managed the Tianwan Nuclear Power Plant for eight years and has just completed six months of training at the Monte Hermoso tokaplex. She is ready.”

Elin looked out at the empty water beyond the atoll’s mouth. Somewhere out there, the platform was crossing the Indian Ocean, pulled by a fleet of tugs. She had watched it leave Biloxi two months ago, a floating city of fusion technology, the magnetic containment vessel alone massing fifty thousand tons. It was far too big to cross any canal, so they had to route it around Africa.

“It will work,” she said, more to herself than to Chen.

The platform appeared on the horizon at dawn, a low dark shape flanked by eight ocean-going tugs. Elin stood on the atoll’s southern observation deck with Gonçalo and GM Zhao, watching through binoculars as the convoy approached.

“*Mãe de Deus*,” Gonçalo whispered. “I saw it in Biloxi. But out here...”

It was massive. The tokaplex core sat in the center, surrounded by heat exchangers, coolant systems, control buildings, and the power conversion equipment that would step the reactor’s output up to ten megavolts. The whole assembly rode on a stabilized floating platform designed to handle typhoon-force seas. Prefabricated in sections, welded together on artificial islands in Biloxi, the entire tokaplex was tested together for six months before the journey began.

The tugs positioned it carefully, fighting current and wind. Centimeter by centimeter, they nudged the platform toward the atoll’s opening. Harbor pilots called out adjustments over radio. Elin held her breath.

The platform’s bow crossed the threshold. She exhaled.

Three hours later, it was locked into place, connection booms extending to meet the platform’s docking hardpoints. The tugs released their lines and backed away, their job done. Now came Elin’s part.

“Zhao,” she said. “Let’s go see your reactor.”

Zhao walked them through the platform’s systems, describing the cooling loops, the magnetic coil assemblies, the plasma diagnostics without hesitation.

“I need to verify you know this unit’s startup sequence,” Gonalo said, pulling up the checklist on his tablet. “Walk me through it.”

Zhao nodded. “Let’s begin.”

They spent three days on certification. Zhao ran through normal startups, emergency shutdowns, partial failures, grid synchronization issues while Gonalo observed, checking her procedures against the design specs.

On the third evening, Gonalo found Elin. “Zhao’s ready,” he said, exhaustion in his voice. “Better than ready. This is her reactor now.”

The test fuel arrived quietly. It was a small bottle of ^3He , barely fifty grams, enough for a thirty-second burn. They loaded it into the fuel system at midnight, ran the startup sequence, and watched the plasma come alive. The magnetic fields held. The fusion reaction stabilized. For half a minute, the tokaplex poured a terawatt into the Chinese grid.

Then Gonalo shut it down, and they spent eight hours analyzing every data point.

“Perfect,” Zhao said, reviewing the thermal curves. “We’re ready for full operation.”

“Fantastic,” Elin said. “The production fuel load will arrive tomorrow.”

The delivery was scheduled for dawn, 6:02 AM, but Elin was on the atoll’s loading platform an hour early. A US Navy Destroyer had brought the ten-ton container from the Helion Dynamics warehouse in Dutch Harbor to a US Navy base in Okinawa. It was being delivered from there by tiltrotor to the terawatt tokaplex.

Raines had arranged it personally. Ten tons of ^3He , worth eight billion dollars, representing two months of operation. That was the total lunar output. Helion Dynamics would be dipping into their reserves to keep all the other reactors running for a while. Raines was betting lunar production would increase fast enough to take care of that. The entire supply chain depended on this working.

Elin watched the eastern horizon. At 6:01, the sun breached the sea, and for less than a second the top edge flashed brilliant green, then it was replaced by ordinary

gold. The tiltrotor appeared a minute later, on schedule, flanked by two escort craft. It descended slowly, rotors kicking up spray, and settled onto the platform. The rear ramp dropped, and a US Marine officer stepped out, followed by six armed Marines surrounding a cylindrical container the size of a small car.

Minister Chen was there, along with a PLA colonel and a squad of soldiers. The two groups eyed each other across the platform.

“Dr. Sarai?” the Marine lieutenant said.

“That’s me.”

“Captain Pak, US Marine Corps. I have a delivery for the Shanghai Terawatt Tokaplex Project. Transfer of custody requires your signature and authorization from the People’s Republic of China.”

The PLA colonel stepped forward. “Colonel Feng. I will be overseeing security of this facility and all future fuel deliveries.”

Pak handed Elin a slate. She scanned the manifest. 9.8 metric tons of ^3He , purity 99.97%, source Tycho Crater, shipped via Lunar Tug. She signed.

“Minister Chen?” she said.

Chen signed as well, transferring \$7,840,000,000 to Helion Dynamics. The transfer done, Pak and Colonel Feng added their authorizations. The Marines rolled the container down the ramp. The PLA soldiers came up, face to face with the US Marines. Captain Pak crisply saluted Colonel Feng. The PLA took custody and, with Chen’s technical team, moved the container into the tokaplex’s ^3He injection site.

Pak and his detail were still standing near the tiltrotor. He turned and saluted. “Good luck, Dr. Sarai, Minister Chen. We’ll return in two months with the next load.”

The tiltrotor lifted off, and Elin watched it disappear toward the horizon. Colonel Feng was already setting up a security perimeter around the injection site. This was his domain now. Eight billion dollars’ worth of fuel, the lifeblood of China’s energy future, sitting in a tank beneath the East China Sea.

Raines would love the optics. The US delivering fuel to China while both governments postured about energy independence. But Elin knew the truth: Li Yun Tan’s operation on the Moon was the real bottleneck. Even with the

Kagutsuchi Vortex online, even with production ramping up, the lunar supply couldn't keep pace with demand.

The Helion Dynamics reserves were currently over fifty tons, stockpiles Raines had been building since ^3He first flowed from the Moon. But reserves wouldn't last forever. The Moon needed to produce more, or the entire terawatt program would stall.

She pushed the thought aside. Today wasn't about supply chains. Today was about making history.

The activation ceremony was scheduled for noon. The observation deck was filled with ministers, corporate executives, and international press. Tobias Raines stood among them, shaking hands and making promises about second and third tokaplexes, about China leading the world into a fusion-powered future. Elin avoided the crowd, staying in the control room with Zhao and Gonalo.

"Are we ready?" Zhao asked.

Elin checked the status board. Fuel loaded. Cooling systems nominal. Grid synchronization confirmed. Beijing, Shanghai, and Guangzhou were standing by for the surge.

"We're ready."

Zhao settled into the operator's chair, her hands steady on the controls. Gonalo stood beside her, watching. Elin took a position at the monitoring station, eyes on the plasma diagnostics.

"Beginning startup sequence," Zhao said.

The magnetic coils energized, their hum vibrating through the platform. ^3He flowed into the reaction chamber. The plasma formed, a roiling ball of superheated gas trapped in invisible magnetic fields. Temperature climbed. Pressure stabilized. The fusion reaction ignited.

Power surged. One hundred gigawatts. Five hundred. Eight hundred. The numbers climbed smoothly, exactly as designed. At ninety-five percent, Zhao held for thirty seconds, confirming stability, then brought it to full output.

One terawatt. The grid absorbed it smoothly. Across eastern China, coal plants, no longer needed, began staged shutdowns. Gas generators went idle. The superconducting backbone carried the load that had been distributed across hundreds of smaller facilities.

On the grid display, Elin watched the power flow stabilize. This was just the beginning. The capacity was there now for the planned expansions in Shenzhen, for the western hydrogen pipeline, for the urban vertical agriculture complexes waiting for reliable power. For the hyperelectrification Chen kept talking about.

Elin allowed herself a small smile. They'd done it. The first terawatt tokaplex, operational and feeding the grid. Raines' vision, her design, Zhao's operation. It worked. But as she watched the ^3He injection metrics, she couldn't ignore the math. This tokaplex was burning as much as the Moon could produce by itself, forcing the smaller tokaplexes onto reserve fuel. And Raines was already negotiating for five more this size.

Outside, the crowd erupted in applause. Minister Chen was giving a speech about China's energy future, about leading the world, about the dawn of a new era. Raines was smiling, shaking hands, making deals.

Elin stayed in the control room, watching the numbers. Gonçalo stood beside her.

"It's working, Elin," he said quietly.

"For now."

Chapter 29: Reach

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Darrin loaded up a rover. Extra batteries, food, water, and a CO_2 cylinder. Li noticed him packing. “You’re packing the high-wheel? Planning on driving somewhere?” she asked.

“Not sure yet, but I am ready for it. I am getting new QR codes. These claims aren’t on our map. I don’t know where they are coming from. What I do know is they are sending their bags to Faraday Crater, which is at the easternmost edge of the cableway. I’m going there to take a look, see what happens to the bags. I am bringing the high-wheel because the terrain might be tough.”

She looked at his survey laser. Having it didn’t mean anything. He always had it. But it brought a question to mind. “Are you planning on stopping them?”

“Doubt it, but maybe. I want to see what is happening.”

Darrin scanned the QR code on the high-wheel rover and punched in the destination: Faraday Crater. He hooked up and was off. The uptake was smooth. The crater floor dropped away as he climbed over the rim.

The lunar highlands unfolded beneath him punctuated by bright crater scars, illuminated only by blue earthlight. The cable carried him at fifty miles per hour. Below he could see the mining camps, sweeping, bagging, sending the regolith to Tycho. Even though the power was capped, they could still sweep slowly straight through the lunar night.

An hour and a half out he passed Henderson & Associates, eight domes clustered around a central processing hub. Power lines gleamed on ceramic insulators. Nearly a dozen bags queued at their transfer station.

Thirty minutes from Faraday Crater he noticed a rover loaded with pressure bottles. A lone figure was at the tower, filling each bottle with oxygen. Darkside deaths had ended under the cableway.

After four hours, he came to the end of his cableway. There he found another cable, lighter, extending further east. Darrin had some of the new bags with him. He hooked them up to see where they would go. The platform switched them onto the minor line. Darrin hooked up and followed them out.

This line couldn't handle the thirty ton bags that had become the standard under the cableway. But three ten-ton bags held just as much. Darrin kept following the empties another fifty miles. Here, there were no more lines. The bags just sat on the platform, waiting to be claimed.

Darrin dropped down and parked his high-wheel a ways away where it wouldn't be seen, then walked back to watch. He buried himself in the regolith, leaving a gap so he could see the platform. He waited.

While he watched, other bags were coming to the end and staying. A man came up, hauling ten bags. He hooked them all on. Darrin pulled out his slate and took pictures of them. The QR code checked out. He knew that man. He stayed partly buried, watching the bags.

Another man drove up in a crawler, hauling a trailer with twenty bags. As he hung them on the cableway, Darrin noticed immediately that the bags had different codes, and none of them were people he knew. He watched the man take the empty bags and a few packages that had come in, toss them in the trailer, and leave the way he came.

As soon as he was over the horizon, Darrin got up and ran back to his crawler. He followed the crawler's tracks. They were easy to see. There were ruts from them being followed so many times. Darrin kept his distance. If the man was paranoid, he'd be seen.

The driver stopped. Darrin kept his high-wheel out of sight and crept up, staying low. As he watched, the man got off the crawler and put everything in a very large cup, then sealed that with another cup. The cup was huge compare to the load.

The man rolled the cup about ten feet. Then backed off. He walked to a table and tapped it a few times. Without warning, the cup started moving fast, then flew into the sky. Darrin recorded it on his slate. The slate estimated where it would land.

Far from the cableway, there was no way to get a message back to Li. Darrin got back to his high-wheel. He drove five miles north, staying out of sight, then let the slate take over driving. It was heading three hundred miles away. Darrin set a stop point fifty miles from the destination. He pressurized the interior and took his

helmet off. While the slate drove he boiled water and dropped in an algae biscuit. He ate it with a daily multivitamin.

He couldn't send it, but he compiled a note, anyway. "Li, found where the bags are coming from. Going to investigate."

Darrin was watching as he drove through. Camps looked different out here. Lights were off, since it was night. No one had power to burn. He saw K bottles stacked next to buried huts, nothing happening. He remembered living through lunar night, the biocore turned low, trying to rest and breathe less.

His lights were the only ones on, other than the Earthshine. Darrin started to doze when he saw a man running. He took control of the rover and turned towards the man. Looking at his right hand on the steering wheel, he was surprised to see the survey laser.

The man was waving his arms and running toward Darrin. There was a heterodyne squeal when the man got in range. "Hi, stranger! Can you help? My wife and I are going to darkside!"

Darrin sped up, driving straight toward the man. He got his helmet back on, and recompressed the air into storage. "On the way. Just two of you? Is she still alive?"

"She is in her suit in the hut. Regolith cut a seal on the airlock. We are leaking air. I saw your lights. I'm sorry, but we don't have any other hope."

"No worries." Darrin pulled up beside the man. "Get in."

The stranger got in. "My name's Bill. My wife is Helen."

"Plug in to the rover, and call your wife."

Bill plugged his suit into the electrical outlet. "Helen, get out here. Do you see the lights? Come this way."

Darrin kept driving to the pile of regolith the man had pointed out. Buried, it just looked like a hill. Then he saw a woman come around the corner walking slowly towards the rover.

"My air's thin. I don't have enough battery to recycle it."

Darrin pulled to a stop next to her and got her inside. Bill plugged her in, and Darrin gave her suit a charge of CO₂. The recycler started processing it, releasing pure oxygen into her helmet. He put a new destination on his slate, Faraday Crater.

He overrode the speed controls. When Bill and Helen had both recharged and brought their oxygen levels up, Darrin remembered to repressurized the rover.

Three minutes later, they reached four psi. He took off his own helmet. "You can take your helmets off if you want. We're OK in here. Are you hungry?"

They nodded in unison as they worked their helmets off. "My cup is dirty, but the water is hot."

He dropped another algae biscuit in the cup and added water. They used his fork to eat. "Since we couldn't take our suits off, all we could eat was tube paste. We ran out days ago."

"Help yourself to another biscuit." Darrin pointed to a cabinet with twenty biscuits, and some flavor packets.

The high-wheel handled terrain smoothly, even at speed. After a couple hours, the Faraday Crater tower showed. Darrin's slate chirped, surprising him. It had just sent his message to Li.

They pulled up next to the crater. "We need to suit up again. I'm going to open the door soon." He hit the compressor while they were getting their helmets on. Sound disappeared, except over Local Band. "I am going to recharge and refill my oxygen tanks. What are you two going to do?"

"We can't go back. Our hut is unlivable. I guess we just stay here at the tower, taking oxygen and power until dawn."

"Come dawn, what are you going to do?"

There was a long pause. Helen answered. "I have no idea."

"Have you heard of Tan's Cantina?"

They nodded.

"That's where you are going. She will feed you, and there is a hotel."

"We can pay. We have twenty grams back at our hut."

"You're not at your hut. You stay on this line for four hours, then you will see the cantina. Tan will take care of you. Can you wash dishes?"

Darrin waited until his rover was fully charged and their suits were charged. He programmed their destination and scanned their QR codes. He sent a note to Li to expect them.

They hooked on and waved as they left.

Darrin reset his destination to the package's expected impact site. Three hundred fifty miles. Three hundred to his stopping point. The rover started driving. He fell asleep.

Eight hours later, an alarm woke him. He had reached the stopping point. Darrin started driving himself towards the calculated impact point. Another one flashed overhead, coming toward him. His slate calculated the launch and impact points. The new calculations were with a half mile of his original calculations. He continued along slowly. The high-wheel was good for driving in rough areas, but not so good for hiding. Ten miles from the calculated site, he got out and started walking.

He walked about two miles when multiple spotlights lit up all around him. An open-top rover came up to him. The driver turned toward him. "You must be Darrin Brooks. We've been waiting for you. John Fletcher. I appreciate you making the trip. Get in."

Darrin got in the rover. John continued. "My guy watched you over in Faraday Crater, Watched you leave, watched you come back, watched you leave again. You aren't Tan, so he figured You must be Darrin Brooks."

"How did he tell you? Messages in the containers?"

"No, we've got a fiber optic cable under the regolith. It's for timing the launches, but it keeps us in touch. Do you want to see our setup?"

"Sure."

Fletcher drove over to the launch site. Four steel cups were nestled together. Another pair was out, and each half was partially loaded. The mass driver was a mile long. "This is Riccius, half a mile from an Odd Point Rendezvous site. Three hundred miles from Faraday Crater. Everyone within twenty miles of here trucks in their regolith, and I launch it to Faraday. The mass driver on that end catches it."

"How can you be so precise?"

“It’s the moon. No coriolis force, no wind. So we can get pretty close. And the vessels have some controls.”

“Where do you get the power?”

“We need less than you might think. Launches use a lot, but landings are regenerative. We have solar panels and batteries, and since we don’t sweep, it’s easier. Still, we send less during lunar night.”

He took out his slate and showed a diagram. “I’m thinking of building my own cableway. We would start with a fifty miles radius, move up as demand and income allows. I have not decided on a cross or an asterisk. Either way, it will make it easier for them to get to rendezvous, and, more importantly, it will serve as a collection system for their regolith, which we send you.”

Darrin brought out his own slate, showing their design. “I found that a loop out here with a single line coming in from the loop is efficient. It makes the cableway next to your operations less complex. Then an asterisk spreading out from that. Miners regularly add infill lines. It has worked pretty well for us.”

“That looks nice. I will tell the team on Earth. For us, though, it is a question of ROI. All those decisions are made downstairs,” Fletcher said, pointing at Earth.

Fletcher drove to where the container was being loaded. “Quarter inch steel cups. Cost a fortune to ship these up from Earth. They sent me six pairs. Tough. When we were starting, I put a few of them into the pebble berm. They came out slightly dented, but fine. We hammered them out with sledgehammers.”

Darrin ran his hand along the steel.

Fletcher brought him to a cup that was on its side, being loaded. “At each end is a guidance module. This small, tubular mass driver ejects regolith pellets at a thousand meters per second. The pellets are just sintered regolith. I put a ton in the front and in the back. They typically fire off a couple hundred pounds. The launch crushes them, so it is dust flying out. No matter. Probably safer. The pellets just make loading easier.”

“You see them loading in the bags. When both sides are full, they will join the cups and roll the vessel over to the launcher. My man on this side opens the vessel, takes the bags out and hangs them on your cableway. He picks up anything headed this way, such as empty bags, food, air, tools, parts, anything that can take high G forces. He then reseals the vessel and launches it back. Two-way flow of material. Regolith going toward you, material and supplies coming back.”

Darrin stared at Fletcher for a long moment. The concept was audacious. Ballistic cargo transport across three hundred miles of lunar surface but here it was, working. “Your mass driver must be amazing.”

“It’s great. Lets go look at the muzzle.” They got back in the rover and drove up to the end, which looked like a huge mound of pebbles.

“You can see our tube poking out through the hill there. It launches at a forty-five degree angle. You see that set of rings? They are packed with electromagnets allowing fine tuning on launch and final corrections on catch. The hill itself is pebbles. If a vessel misses, it runs straight into that pile, absorbing the energy quickly. The mass driver is only a mile long, and only starts curving to forty five degrees in the last third of the launch. Same on the other end.”

“Only a mile? That would need over twenty G. You would crush your cargo.”

“Twenty-five. We don’t ship cargo that would be crushed. Once a month we make the drive back and forth, carrying people and delicacies.”

As they drove around the site, Darrin saw a sign.

Send your regolith to Tycho
30%

“What’s that?”

“We never did any marketing or advertising except for those signs. We have six of them around here.”

“How is that working out?”

“Very well. We are making money and people keep bringing us their regolith.”

Darrin pulled up a chart showing mining growth under the cableway. “You can see it is growing everywhere. The fastest growth is under the six main lines. But lets just look at Faraday Crater.” He shifted the chart to show growth at that tower. “This includes the miners near Faraday Crater, those a little further out for whom Faraday is the closest tower, and you.”

“I can see the bump when we started shipping.”

“Me, too, but look at the growth. People near the tower are growing fast and are the wealthiest. Fifty percent growth since the tower was installed. Next are the people

east of the crater that need to drive a long ways to get to it. Solid growth. Thirty percent. But look at the unknowns. Those are your people. Essentially flat.”

“I should advertise more.”

“Your problem isn’t awareness. Your customers aren’t thriving.”

“They’re doing fine. Churn is under ten percent.”

“Do you know why the miners who are leaving are doing so?”

“We are not worried about those low numbers. All businesses have churn.”

“Not all. We don’t. Let’s look at miner growth.” Darrin brought up a chart showing the change over time of productivity by QR code.

“Miners under the cableway are making money fast. They’re buying better equipment, eating better, working safer. Every month they produce more. Your customers aren’t making enough to upgrade. The churn you see is miners darksiding or getting so scared of darksiding they can’t stay.”

“No one asked your opinion. We are doing alright.”

“When you send the regolith to Tycho, you asked my opinion. If we agree to keep processing these bags, it will be because we believe you are helping these miners thrive.”

Fletcher took a moment to breathe. He got his pulse back down. “Darrin, it’s not just about them. I have to thrive, too. We plan to extend this model all around your cableway and, eventually, all over the Moon. The more I make, the faster I can grow.”

“Your model can’t grow right now. You are hauling steel vessels up from Earth. You are trying to live on solar. And miners aren’t excited to work with you. Instead, you work with me.”

“What do you mean?”

“You need to build a cableway. I sell you what you need for that. Drop it off at Faraday Crater. I send you light titanium cable. Now you can make a cableway that doesn’t flex in the sun. You need vessels? I can stamp them. I can even run electric lines out here. I’ll charge you for it, but it will be steady power, day and night. My one requirement is you cut the miner’s fees to ten percent. Let them thrive, too. That is what I will agree to.”

“I’ll have to talk to the owners downstairs.”

“OK, you do that. And, when they sign, we will start processing your bags again.”

Fletcher jumped out of the rover. “Wait right here.”

He turned and ran into an office. When he took his helmet off, he didn’t turn off Local Band. Darrin couldn’t hear everything, just the shouted parts. “Then wake him up!” “He cut us off! Zero revenue!” “Ten Percent!” “No Choice!” The mumbling got fainter.

Darrin sat in the rover for hours. He wrote a note to Li, including all the details while they were fresh. He knew the message would not get sent until he was under the cableway.

Two hours later, Fletcher returned. “The board was delighted with your ideas. They are very excited about the possibilities of our cooperation. I am authorized to sign a contract with your terms.”

“I am less interested in you signing a contract than I am with you following the terms. There aren’t any lawyers up here, anyway. Start by changing your signs. Next, go talk to the miners. Talk to one every day, get to know them and their needs.”

Fletcher brought Darrin back to the high-wheeler. Darrin set the destination to Faraday Crater and went back to sleep. He slept through the entire trip until his slate chirped. His message to Li had been sent. He hooked his rover to the line and slept four more hours until he arrived in Tycho.

The terminal platform rose to meet him, friction dampers engaging smoothly. He unclipped and dropped onto familiar ground, feeling the vibration of the cableway system through his boots.

He cycled through the airlock, stripped off his pressure suit, and found Li in the cantina. “Welcome home, Darrin. Do you remember these two?” She pointed to Bill and Helen. “They didn’t just have a bad seal. All their equipment was cheap. I threw away their suits and sold them new ones. They could afford them. They had probably ten grams of ^3He .”

“Li, are you nuts? Those suits cost...”

Li looked at him sharply.

“I forget how much they cost. How much did you charge them?”

“A gram.”

“They look nice.”

“While you were off playing outside the cableway, Bill and Luke went back and pulled everything they had: helium, sweepers, solar panels, batteries, their rover, even their hut. Helen was a CPA before she came up here. She agreed to stay on. You know I need an accountant. Luke has Bill working on the ovens, for now. Point is, they agreed to stay.”

“I got your note. Thirty percent was too much. How did it go?” she asked without looking up.

“I got him down to ten percent.” Darrin pulled up a chair. “But we need to talk about our extraction fees.”

She looked up sharply. “What about my fee?”

“I told Fletcher we’d consider dropping to fifteen percent for his incoming cargo. From twenty.”

Li’s expression went flat. “You told him what?”

“Hear me out.”

“Fifteen percent? Darrin, do you understand what we provide?” She gestured at the window overlooking the extraction facility. “We maintain all of that. We staff all that. And now you want me to cut our fee by a quarter?”

“I know what we’ve built. But I also know what we haven’t built. Ninety-eight percent of the Moon is waiting for you. Bill and Helen? Luck. Darksides happen daily.”

“If they had stopped here first, I’d have checked their gear.”

“You can’t check everyone’s gear. But you can expand. Fletcher thinks he is making money in transportation. I think he is extending Tycho Crater across the Moon. Help him build it, for us.”

Li was quiet for a moment, studying the schematic. “We still lose five percent on extraction fees.”

“Do we?” Darrin zoomed out on the map, showing the full lunar sphere. “Right now we cover two percent of the Moon’s surface and generates sixty percent of

lunar ^3He . The other ninety eight percent of the Moon? It's just sitting there. Billions of tons of regolith that's barely being mined."

Li studied the map in silence. She heard him. This was huge. Still, she felt he hadn't heard her. "Darrin, the Moon is not London or DC. We don't have lawyers. We need simple agreements. Twenty percent is simple. I change that, everyone will start looking for a deal. My fee stands."

Darrin started to object, but she held up a hand.

"But," she continued, "I'll rebate five percent to Fletcher on all regolith that comes through his network during the growth phase. He charges the miners ten percent for transport, gets five percent rebate from us on their extraction fee, we charge twenty percent. Net cost to the miners is still thirty percent, Fletcher makes fifteen percent during construction, and we keep our nominal rate."

"You sure?"

"I'm sure." She turned back to her workstation. "Now go get some sleep. You look exhausted."

He started to protest, but she was right. The adrenaline was wearing off, leaving him hollow and tired.

"Wake me if anything comes up," he said.

"Nothing's going to come up. Go."

He made his way to the hut, stripped off his clothes, and collapsed into the bed. Sleep came immediately.

Chapter 30: Batidas

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

The design came together quietly, over late evenings and sketches on slates passed around the table. There was no formal announcement, just the slow coalescence of agreement between engineers who trusted one another's instincts. Daily there were seminars, small talk given by each researcher where they described their works in progress.

Rithika Patel stood confidently at the podium, surveying the audience of her peers. She cleared her throat and began her talk.

"Good morning, everyone. Today, I'd like to present our design for a helium separation system."

She clicked a button, and a diagram of the system appeared on the screen behind her.

"Our design uses a combination of compressors, membrane separators, and cryocoolers to achieve high-purity helium. The process starts with the ingestion of atmosphere at a steady rate, optimized for the density band around 2 atmospheres of pressure."

Rithika gestured to the diagram. "The hydrogen-helium mix is ingested by a large fan which blows the air across a zeolite membrane shield. The membrane passes only the hydrogen and helium."

A colleague in the audience raised his hand. "Rithika, can you explain the choice of membrane material for this shield?"

Rithika nodded. "The shield must block ammonia, sulfur compounds, and phosphines which would cause downstream problems. We are modeling this using a thin silicalite-1 zeolite with high porosity to minimize the pressure drop across the membrane. The relatively fragile zeolite requires strong backing to support it, which we provide with a glass honeycomb."

She clicked the button again, and a graph appeared on the screen.

“Once ingested, the gas is compressed in an axial flow turbine. The ambient pressure is two bar. We compress it to nearly three bar. Compression also heats the gas. Here it passes through a cone twenty meters long, ten meters in diameter. The cone is lined with a five-micron foil of a palladium-silver alloy over a high-strength inconel mesh. Hydrogen passes through this foil readily. By the end of the cone, the gas mixture contains less than one percent hydrogen.”

“That is an astonishing level of extraction.”

“Yes, palladium grabs the hydrogen quickly. Silver helps the hydrogen pass through the foil quickly and, of course, the high temperature helps.”

Rithika continued. “The hot hydrogen is passed back through a countercurrent flow with the incoming gas, heating up that gas before compression. The retentate is still hot. An intercooler brings it down to ambient temperature, 165K, then it gets refrigerated to about six Kelvin in a cryogenic recuperator. At that point, the helium passes through a nozzle, where a Joule-Thomson expansion drops the pressure and rapidly cools the gas for the final isotopic separation.”

“How cold does it come out?”

She clicked the button again, and a diagram of the fractional distillation tower appeared.

“A little less than five Kelvin. Ready to enter the fractional distillation column. The column is packed with aerogel, and the top is kept at 3.7K. The ^4He rains out and goes back to the cryogenic recuperator. The final gas phase is high-purity ^3He .”

Rithika concluded her talk. “Our combined design isolates high-purity ^3He efficiently. We’ve run the numbers, and we believe we can achieve a theoretical yield of approximately ten tons per day per engine, using about a hundred gigajoules per kilogram.”

The audience applauded, and afterwards, several colleagues approached her to ask questions and offer feedback.

Nikola Savic, a new grad student in orbital mechanics, was dazzled. He came up to talk to her. “Your system is the most efficient system I’ve seen. Off the record, do you have any idea how to get the ^3He off the dirigible?”

“As far as I know, that stage is theoretically impossible. Rockets can’t lift off from Jupiter’s atmosphere. I try not to think about it,” she said with a wink.

Savic's mind was spinning on the walk home. Real extraction, as much as you please, as long as you want. There just had to be a way to get the ^3He off. When he burst through the door, his roommate, Marco, looked up from his tablet, eyebrows raised at Savic's unusual energy.

"You should have been there," Savic said, dropping his bag and heading straight for the kitchen. "Rithika just solved helium extraction. Ten tons per day from a single engine. It's brilliant. They use membrane separation, fractional distillation, the whole thing integrated into a dirigible that can actually stay aloft in Jupiter's atmosphere."

Marco nodded politely. He was an atmospheric chemist, comfortable with theory but not particularly interested in engineering details.

"The problem," Savic continued, grabbing a drink from the fridge, "is that it's completely useless."

"Useless?"

"All that beautiful engineering, and the helium just sits there in the dirigible. There's no way to get it off Jupiter. The gravity well is too deep."

Savic slumped into the chair across from Marco, the excitement draining from his face. "She solved the hard part. And it doesn't matter at all because no one can solve the impossible part."

Marco looked at him for a moment. "You're an orbital mechanic, right? That's what you do. You figure out how to move things through space."

"This isn't moving things through space. This is escaping from Jupiter. Do you know what the escape velocity is? Sixty kilometers per second. That's not engineering, that's physics saying no."

"So figure it out," Marco said, returning to his tablet. "That's literally your job."

The challenge hung in the air. Savic stared at the ceiling, then pulled out his slate and started sketching orbital diagrams.

For the next week, he attacked the problem methodically. The tyranny of the rocket equation made chemical rockets hopeless. Planes were better. Electric ramjets. They could lift the helium high in the atmosphere, but there was no way to get it from a plane out of the Jovian gravity well.

Mass drivers came next. Electromagnetic launchers were used on Ceres, the Moon, and Mars. They were very efficient, both in energy and mass. No propellant, just electrical energy converted to kinetic energy through magnetic fields. A theoretical best case.

Savic spent three days modeling a mass driver capable of launching a ten-ton bottle of ^3He from the Jovian atmosphere to escape velocity. The numbers were staggering. The launcher would need to be hundreds of miles long to achieve the necessary acceleration without crushing a steel bottle. The power requirements were even worse. To launch one kilogram of ^3He would require the entire output of a large fission reactor.

He calculated the infrastructure required: massive superconducting coils, cryogenic cooling systems, power distribution networks that could handle gigawatt pulses. The platform would need to be the size of a small city just to house the equipment. That wouldn't fit on Rithika's elegant dirigible.

Savic pushed back from his desk in frustration. The mass driver was clearly impractical, but maybe it would be theoretically possible if Rithika's design didn't cruise at two atmospheres. That left an impossible amount of atmosphere above the rig. The drag crushed all hope. It would vaporize the bottle.

He had another idea, imagining a very light tethered balloon rising high with just this one cargo, and launching from far above all the clouds. But it made too small of a change.

Like everything else he'd tried, it ran up against the fundamental problem: Jupiter's gravity well was simply too deep for current technology to climb out of economically.

He was an orbital mechanic, supposedly. That's what his advisor said. But even orbital mechanics had limits.

At the end of the week, Marco found Savic hunched over calculations, barely moving. "Come with me, my man. You need food. And maybe some perspective. I found a Cuban place up in Baracoa, Café Havana. They have live music tonight. My treat."

Savic reluctantly agreed. He wasn't getting anywhere right now. "Alright. I can't even find a fresh way for the physics to say no. Let's go have fun. The problem will still be here when I get back."

“I invited you, not your slate,” Marco said, pulling the slate from Savic’s pocket and putting it face down on the desk.

They took a JRC shuttle bus to Baracoa and walked down to the cafe. The evening was warm, and they grabbed a table on the patio where a guitarist was setting up in the corner. The familiar comfort of black beans and rice, the smell of garlic and cumin, the gentle pulse of música criolla were exactly what Savic needed.

“So,” Marco said, attacking his ropa vieja, “still trying to escape Jupiter?”

“Not tonight,” Savic said, raising his mojito. “Tonight I’m officially not thinking about orbital mechanics.”

The guitarist began playing a slow and melodic tune that mixed with the harbor breeze. Savic let himself sink into the moment and enjoy the food, the music, and the simple pleasure of not calculating mass ratios or power requirements.

Across the patio, two women were seated together, drinking batidas, a thick frozen drink. He could only see the face of one of them, but it was captivating. She was clearly enjoying the music. A young man asked her friend to dance, and the woman he was watching was alone. “I’m gonna enjoy myself tonight,” he told Marco.

Marco followed Savic’s gaze. “Out of your league, Nick. Just enjoy the music.”

Savic stood and began walking toward her, marveling at the sparkle of her eyes, the bounce of her hair, and the way her cheeks looked when she drew in her batida through the straw. He watched absently as she worked the straw, pulling the drink from the bottom of the glass to her mouth. Such a simple mechanism. Direct connection. No intermediary steps, no complex machinery. Just suction and...

He stopped.

A straw.

He went straight back to the table and started talking to Marco. “You saw that woman drinking a batida through a straw? She isn’t really sucking it up, you know. She reduces the air pressure in her mouth, but it is the air pressure in the room that’s pushing it up through the straw. That is why her drink is flowing.”

Marco hardly knew what to do when Savic talked this way. Anything could suddenly turn into a lesson in physics, and Marco had no idea why it was so fascinating to Savic.

“You’re thinking about work again,” Marco said.

Savic was lost in thought. He couldn’t even hear Marco. His head was filled with material profiles and energy diagrams.

When they finished eating, Marco paid the bill. On the shuttle back to Jamal, Savic’s mouth kept running. Marco was tired. He was happy to see his friend so animated, but he felt overwhelmed by the discussion. He rolled his eyes and let the words wash over him. When they got home, he went to bed. Savic opened his slate and began doing calculations.

The next morning Savic called together some other grad students who had been at Rithika’s talk and his own advisor, Dr. Miriam Vance, to talk about the idea. “I think we all know that Rithika’s dirigibles are the best way to collect ^3He from Jupiter, but getting the ^3He out of the atmosphere, much less the gravity well, is completely impossible.” Rithika nodded as his friends murmured in agreement. “No ship can fly in, pick up a bottle of ^3He and then leave. The delta V is just too much. So I spent days looking at mass drivers, hoping to send cryocontainers of ^3He out.”

“The gravity well is too deep.”

“Tell me about it. I looked at a five-hundred-mile-long mass driver to get the power down to three hundred megawatts per ton. And it still needs terajoules, even ignoring atmospheric drag. I hoped to use the magnetosphere to help, but I was really just grasping at straws. Fundamentally, it is impossible.” A few heads nodded.

“Last night, I saw something new. A woman was drinking through a straw. And it struck me. The atmosphere already pushes itself up. With an engineered ‘straw’ the atmosphere would apply most of the force, and we would only need to apply a little extra to accelerate the ^3He and overcome friction, allowing it to rise up through the straw.”

“Do you want to make a ^3He fountain in the exoatmosphere?”

“At first I didn’t know what I wanted. An exoatmospheric fountain would have been a huge step forward. Then I remembered an old space travel concept, space elevators, which allowed you to launch heavy payloads into space without rockets.”

“Space elevators never worked.”

“They never made practical sense because by the time we could build a cable strong enough, LEO had way too many satellites. For a time, they talked about anchoring it on a ship, and letting the ship move around to avoid satellites, but you

could never dodge the satellites fast enough. It can't work on Earth. But we are talking about Jupiter. No satellites inside the orbit of Callisto, and the planet spins really fast."

"You realize, of course, that the gravity in the Jovian system is immense."

"Yes, that's true. Hence the original problem. So I did a calculation. If Rithika's dirigible is in the equatorial stream, we can calculate based on equatorial, or System I, rotation. But there is a band above that in which the equatorial jet is five hundred kilometers per hour faster. Geostationary from there is 88,000 km. We will be well inside the Galilean moons, even with runout for a counterbalance. I am calling these split rigs, since there is a dirigible and a space station working together, separated by an 88,000 kilometer straw."

"No material could possibly handle the stress."

"I am not a materials guy, but it looks like carbon nanotubes might. The tube would need a strong taper, maybe twenty or thirty times as thick at Jovian stationary orbit as at the dirigible. Then it thins out again as the cable proceeds to the counterbalance."

Hope seemed to glimmer. If they could pump the ^3He up, they could get around the impossible launch problem. The team grew cautiously excited. This was far from a solved problem, but some of the impossible parts had been converted to very difficult.

Savic's advisor, Dr. Vance, called a meeting with him a few days later. "Fascinating talk you gave, Nick. That sounded impromptu."

"Yes, well, Rithika's scheduled talk a week ago got me thinking, and last night I had a thought and wanted to run it by some people."

"It is an intriguing idea. I expect it will not work, but even chasing down a dead end can be a fascinating adventure. Do you think this is a topic you can write your thesis on?"

"Are you kidding? I'd love to."

"Here is what I will need you to do. I am going to get you a room for this. A small conference room with too many whiteboards. You and your friends are free to use that room whenever you want. What you need to do, though, is divide the problem into a set of smaller problems."

"That sounds great."

“Here is how you pay for it. Every week, someone from your group will give a talk on one aspect of this concept. I want clear titles, and deeply detailed talks.”

“Every week? What if I don’t have progress one week?”

“You can talk about any detail of the project that interests you. We learn as much from failure as success. More, actually, since it is so much more common and really makes us examine the research. On my part, I will see to it that there are always snacks at your talks.”

They began working on it, after hours, in locked rooms. The JRC didn’t pry as long as there were regular meetings with titles they could show the world to prove they were working on something. And Savic’s talks drew larger crowds each week. Savic let the others talk, too. Every week he would file the talk with a title, a transcript, the slides, and a count of the attendees. No one read it, but JRC tracked the attendee count.

There were numerous subproblems, and people picked them up as they could. Each problem got its own whiteboard, showing its current state.

The first was how to get the helium up the straw. Rithika herself came up with a plan for this. She proposed a series of simple booster pumps spaced five thousand kilometers apart, each adding about five psi. Not the only choice at all, and subject to optimization, but good enough. It was enough to keep the helium flowing. This was marked solved.

Jason, an electrical engineer from Michigan, attacked the energy problem. The total demand for extraction and pumping was around five hundred megawatts per engine. That would take eight thousand acres of solar panels near Jupiter. Not easy. Not possible. But Jason had an idea. Whisper-thin aluminized mylar formed to concentrate the light, bringing eight thousand acres down to eight hundred acres. Still enormous, and now there were new problems. Alignment had to be precise, and the mylar sheets were fragile.

Distribution was another issue. They needed power on both ends of the split rig. The mass of an aluminum cable was enormous, and the resistance was too much. Jason split it up. He put a large solar collector on the rig, another large one five thousand kilometers up at the first booster pump, and a small one at each booster pump after that. His ideas drew interest, but were not a solution. No one had improvements. The whiteboard stayed static.

A related problem was maintaining the verticality of the tube. As gas would flow up, it would need to be accelerated laterally. With a little, that was fine, but for tons

of gas, the tube would curve. Savic knew this was his problem. Tons of gas going up, all needing a gentle nudge to accelerate laterally. The force came from the sides of the straw in the cable. Savic added a second straw, but for this one he pumped in a propellant gas, and set up a series of ion emitters along the cable. Unprocessed atmosphere made sense and is very light, but it would also cause corrosion and hydrogen embrittlement. Then it struck him that Rithika's design created enormous quantities of purified ^4He . Using helium as the propellant worked well. It did not erode the tube, pumps or ion emitters and had a little more mass reducing the energy requirements. Respecting Rithika's booster pump plan, he put one ion emitter at each booster pump. Using this active system changed the energy consumption dramatically, requiring an extra four hundred megawatts just for the emitters, and requiring fifteen tons of ^4He to be pumped up per day to be used as a propellant, increasing the pumping expense another hundred megawatts. Savic finished his whiteboard and reluctantly restated the problem on Jason's board, which still had no closure.

Then there was material degradation. The tube would be bathed in Jupiter's radiation belts, high-energy protons and electrons eroding the material on timescales measured in months. The atmospheric end would face ammonia crystals and hydrogen sulfide aerosols, each gust stripping away microns. The tether would fray unless wrapped in protective layers that added prohibitive weight and complexity. Savic prevailed on Marco to address this. The obvious solution was a nice, solid coating of gold. The mass was far too high. But Marco looked at only protecting the last thousand miles. This helped enormously. Rather than gold, he put on a coating of boron nitride nanotubes (BNNT) over this region. They were far more stable, but didn't have quite the strength-to-weight ratio of CNTs. The induced weight penalty hurt everything else, and was made worse because it still eroded, so it had to be made overly thick. Marco outdid himself on that, though. He had crawlers moving up and down the tether inspecting it everywhere. When a defect was found, the crawler immediately did repairs using BNNTs. This allowed a thinner coating, because it would be repaired as it eroded. No one liked the weight penalty or the added complexity of this maintenance machine, but it seemed unavoidable. The team signed off on Marco's board.

The dirigible skin faced similar problems. Hydrogen seeped through most membranes over time and it embrittled metals. Jupiter's atmospheric chemistry was relentless. Materials that lasted years in Earth's stratosphere broke down in weeks here. Each dirigible would be operational for months at best before needing a complete overhaul. Marco worked a separate whiteboard for the dirigible, using aluminized coatings for the ballonets, BNNT coatings for the skin, and rigorous

crew inspection and maintenance. These were not solutions, per se, but tolerable degradation rates.

This was the furthest anyone had gotten. Better than anything else in the literature. The concept was sound; it was the execution that remained impossible. Materials never really got solved, and power was completely impossible. But every week, Savic or someone else in the group would give another talk, and Savic dutifully filed the reports. The split-rig stayed on whiteboards, a beautiful impossibility. At least it was an interesting topic.

Chapter 31: Tan's Cantina

Price of ^3He	
Current Production	14 tons / month
Current Consumption	
Time Since Start	

Li found Darrin working on the ovens. "Ready to go check it out?"

"Just a minute. I needed to replace the drive belt. I will be ready in five."

"Don't worry, Darrin, I've got it," Bill said.

Darrin passed the wrench to Bill. "Thanks. I am being summoned."

He then looked at Li. "Is it ready for a walk through?"

"It's overdue."

They hooked up to the cableway and glided across the crater floor, about twenty miles to the prominence. Darrin looked at the phase of the Earth. Ten days to dawn.

They got off the industrial cableway and hooked onto an ornate cable that went up a well-lit path.

"I didn't expect it to be so bright," Li said.

Darrin looked at the enormous dome. "They added titanium dioxide to the mix. It looks like marble."

The entrance was a large revolving door. There was a large emergency airlock beside it.

"These revolving doors are going to be a maintenance nightmare," Darrin said.

"Ellie told me they did this at Bounty. It's elegant."

Near the entrance, a simple line drawing of her face hung over a sign: "Welcome to Tan's Cantina."

They walked past the empty bar. Li checked sight lines, noted the mirrored wall behind the counter, tested the Earth window placement from different angles. The window was set into the dome's northern curve, two meters across. She looked at the full Earth, imagining the shaft of blue-white light it would cast across the floor when the lights were dimmed.

On the stage, a smooth jazz trio from Chicago was practicing.

"How long are they here?" Darrin asked.

"Three weeks here, then one week at Fletcher's Antipodes before they return here for a closing show."

"Fletcher built a club?" Darrin asked.

"I hear it's nice. Much smaller, of course. He should build more. Miners need someplace to visit."

"They need a haven."

Helen came in. "Tan, Amazon sent another message. They want a million square feet."

"I'll site them a couple miles from the depot. It will be fine. But I want you to word the response like it is a huge problem," Li said. "We are still negotiating the rent."

"What's up with Amazon?" Darrin asked.

"With Fletcher connecting the Moon, Amazon thinks they can get most deliveries to customers in a day. Helen, what would you have done if you and Bill could have had a delivery in a day?"

"We would have replaced the airlock and been fine. We would still be mining."

"The whole Moon needs that, but we don't need to tell them that, yet."

Helen left, and Li and Darrin kept walking around the cantina. They stepped out the east balcony.

"Do you like having Case out there? Without them it is an unspoiled view."

"I like having them. Easier to buy new equipment, and now there are mechanics."

"Komatsu and Sany Group also want to open dealerships."

“There’s plenty of space. We could handle ten manufacturers. My only concern is that we can kick them out if their equipment is no good.”

“It’s still my crater.”

They continued on the balcony until they were looking south. “The row houses look very comfortable. Helen loves hers. Things are growing so fast. Where are you hiring from?”

“I rarely get anyone under the cableway anymore. I have a deal with Andrei. Whenever someone is calling it quits on the Moon, he lets me know and I go talk to them. Some are happy to stay if they can be someplace like this: steady work, nice housing, power, hospitality, and community. For me, I love our hut. I think we chose well when we bought it, and it has grown on me since.”

Li glanced at him quickly. “I want to show you the hotel. Have you looked at it?”

“Only from the outside. Let’s see it.”

Li led him inside, and they walked through a few rooms. They had no windows, just small monitors in each room with a current view of Earth. She brought him to the top floor.

“Look at this room, Darrin. What do you think?”

Darrin went around the apartment. High ceilings. A real Earth window. The bedroom had a door and a full sized bed. It had a separate shower room with a bathtub as well. Kitchen, office, and a large room that was really just for nothing. You could sit there and talk.

“What is this room for? Visiting dignitaries? It is so extravagant. No one can afford this, Li.”

“We can. I built this for us. The area where our hut is interferes with the shopping mall I have planned. We could dig it out and move it, but I think this may be nicer.”

Darrin showed only his practical face. The tiny hut on the north edge of the prominence had been their home since they first arrived at Tycho crater. “If you like, Li.”

Li looked away. “Darrin, I need to stop by home before we go to the opening. I can’t be Tan tonight wearing a pressure suit.”

They took a rover to their home. Li asked Darrin to wait outside. Darrin felt alone when she entered the airlock. She was quick. Keeping her pressure suit and helmet on, she grabbed a large garment bag and came back out.

“OK, I’m ready,” Li said, getting back on the rover. Darrin drove them to a cableway station, and they took the cableway up to the Cantina.

Back in her office, Li opened the bag and took out a black silk qipao with red and gold embroidery and a simple suit. “Let’s get changed,” she said as she began undressing.

“Me, too?” Darrin asked, as he began changing. “I should comb my hair.”

When they were ready, they stepped out into the cantina. Lights were low. The full Earth shone brightly through the Earth window, casting a dazzling beam of bluish light on the floor. There was a table stacked with champagne from France and another with bubbly amaranth chichas from Mars. Li picked up the oversized scissors from behind the bar and they went to the elevator landing where the miners were waiting behind the red ribbon.

“Welcome to the all-new Tan’s Cantina!” she said cutting the ribbon, letting the miners in. They flooded through, whooping and cheering as the band began playing “Moonland”, a reworked version of “Birdland.”

Darrin was immediately caught up by various friends he had met as he worked on the cableway. He sipped a chicha while talking to Rook. “Didn’t expect to see you here, Rook. I figured you’d avoid the crowds and drop by a week later,” Darrin said with a wink.

“What, and miss my one chance of seeing you all gussied up in a suit? Ha!” Rook replied. “Nice tie.” Rook smirked and took two chichas and went to find some roast cavy.

Darrin blushed a little, but it was a nice suit and the dark red tie looked great. He talked to some guys from Ocean State Resources. Most of the miners were in their singlets they wore under their pressure suits, but Ocean State miners put on jumpsuits over that. A little more formal. They had almost completed their second claim, and were talking about starting a new claim beyond the cableway.

Li was in a different world. Several corporate representatives had come up from Earth for this event. Amazon had sent a representative, Case had, but it was Mark Davis from David Clark Company that caught her attention first, eager to get up here before Oceaneering. “Tan, have you seen our new pressure suits? They are

much more durable than what most people are using up here. I was thinking, what if we set up a storefront, nothing huge, out next to Case? Someplace miners could find replacements or upgrades.”

“How much space would you need?” she asked, playing a little more naive than she really was.

Mark pulled out his slate. “A showroom for a few different models and components, a couple dressing rooms, a workspace where we can refit and repair suits, and storage space. Total is about a thousand feet. If you had unpressurized warehouse space, we would probably like that much again.”

Li laughed when he mentioned dressing rooms. Earthers were so self conscious. “Yeah, I think we can accommodate that. Unpressurized space is easy. I will set that up in the north, near Amazon. Your retail space will be ready in two months, connected to the cantina. Two hundred dollars per foot, plus twenty percent of the gross. I will give you a premium spot near the entrance.”

Mark didn’t know what to expect on the space, but her taking a cut of the gross was not something his company would like. “How much would it be if we just paid upfront, no cut of the gross?”

“Oh, that would be fine. Tycho Crater is not for everyone. Do you have a claim?”

“No, I mean here. Here is where we want to be.”

“OK. Because here in my claim I can get you a thousand feet of premium mall space for two hundred dollars per foot plus twenty percent of the gross. Interested?”

“Say we were. Tell me about the space we would get.”

Tan led him to her office. She brought up the mall plans on a projector. “Here is the mall we are planning. Twenty thousand square feet on two floors. The main door will be facing north. First building from the depot and there is a cableway stop there. When they enter, you will be the first thing they see. I will let you stay there as long as you are the top performing business in the mall.”

“As long as we are the top performer?”

“Have you ever shopped in a grocery store? You notice that sometimes the canned food is on aisle one, but then you come back in a few months and it is on aisle two? I am not building a big mall here, and in that limited space I want companies that really make life better for miners. I will rearrange where the stores are to do that.

Soon Oceaneering will be here, and Teijin Frontier will want a space. And customers will decide which of you is most important.”

“Customers? A minute ago you said sales.”

“Yes, well, sales are how customers tell me which company is most important.”

“OK, Tan, you win. We have to be here. Just give me the papers, and I will sign the agreement.”

“Papers?” Tan looked through her Earth window. “Earth is adorable. On Earth, papers would make a lease enforceable in court. But this is the Moon. No court. No papers. We agree, and you can move in as soon as the structure is built.”

Mark and Li returned to the cantina. Mark took a glass of champagne, thinking about how this would go over when he called the home office. They were not going to like the terms. They would blame him, as if he ever had a choice. He downed the first glass and picked up another.

Darrin caught her eye. He motioned to the floor. “May I have this dance?”

Li blushed and demurred. “Let’s just stand here next to each other.”

As she watched the other miners in the shaft of earthlight, it struck her. Tycho Crater had become a city. The only real city off Earth.

Chapter 32: Corporate Meddling

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

After the heavy lift project, Halden was promoted to Senior System Analyst. Every day Halden's job devolved to making a report of daily launches, which he fully automated. He spent his days reading the titles of the scheduled talks to be given at the JRI. Everyone in JRC was allowed to attend any JRI seminar. None of his coworkers ever did.

The talks were often exercises in academic diplomacy, full of progress reports on projects that were progressing nowhere in particular. Halden skipped past the safe research. Then, three pages in, a title caught his eye.

“Atmospheric Tether Systems for Jovian Gas Extraction”

Atmospheric tethers. He'd never seen those words combined before. The core concept was a physical connection between Jupiter's atmosphere and orbital space. Direct resource extraction without the impossible energy costs of heavy lift vehicles.

Halden leaned back in his chair. This could be theoretical materials research dressed up with an exotic application. Or it could be the kind of systems-level thinking that actually solved problems instead of just studying them.

Either way, it deserved a closer look. He marked it in his calendar.

The day of the talk he took the JRC/JRI shuttle to Jamal. Halden got a small plate with grapes and cheese and sat to the side. Savic was speaking about tapers and diameters of the tether straw.

Halden was intrigued by the talk, but he was fascinated by how involved the JRI people were. The talk drew a good crowd, including several professors. Halden put the series on his recurring calendar.

He sat in the meetings quietly, absorbing who the people were. He read abstracts of the earlier talks, and started attending other talks by the key players, learning who

the people were as much as what the project was. He even began to chat with them after the talks.

During one such conversation, he followed them into the project room. He saw the whiteboards for the first time. The entire concept, broken into subproblems, each of which had been worked extensively. He read each one carefully, taking pictures. Back in his office, he recreated the whiteboards. Between the talks, he worked on them. Found inconsistencies. Wrote his own ideas.

About a month after he started attending, Rithika was giving a talk on ^3He purification. Halden raised a hand. “What is the pressure where your dirigible will be operating?”

She was caught off guard. “When I conceived of this, I was planning to operate at one to two Bar, but recently Savic changed the plan on orbital level. He noticed a high velocity equatorial band at fifty millibar. So I should adjust to use that pressure.”

By the second month, Halden had reframed the problem. He caught Savic and asked him to bring just a few to the project room. Halden was standing in front of a new whiteboard. “What is your project here?”

“I noticed a way to use Rithika’s extraction system with a straw to get ^3He into orbit. We thought this might be the solution to Jovian extraction, but there are too many problems, especially energy and durability.”

“Two brothers back in North Carolina thought people could fly. They didn’t know anything about it. They just made bicycles. But they made the worlds worst glider with an engine and it hopped one hundred twenty feet. Their invention didn’t teach anyone how to fly. Planes never looked much like theirs. Why did it matter?”

“The Wright brothers solved the control problem,” Rithika answered.

“They did, but the reason it mattered was, until that one hundred twenty foot hop, flight was all talk. After that, powered flight was real. Everyone got involved. Within a decade, aviation was an industry. I don’t need you to build an airliner. I just need the Orville Flyer. I want a prototype that can endure Jupiter long enough to deliver a single kilogram of ^3He to orbit. We can get that sample to Earth and I will set up a press conference and burn that kilogram of Jovian ^3He in Ilha Fusão. If we can do that, the world changes overnight.”

“And if it fails?” Savic asked.

“It fails quietly. No one will know,” Halden said.

Rithika was the first to catch on. “My design is for throughput. If you want just a sample, it shrinks down. A multi-stage centrifugal compressor, let the hydrogen bleed out. The dirigible was to support more equipment. We can just let the compressor hang from the tether. I need to simplify purification.” Halden handed her a marker and she put her new proposal on his board.

“Savic, is there something you can do to simplify this?” Halden asked.

Savic stood up and went to Halden’s board. “The work of a space elevator is not so much lifting the payload as it is accelerating the payload tangentially. Hence the string of emitters. But, for a short run, we don’t need the complication. We start the process moving too fast, and let the tether straw’s momentum speed it up. After a month, it will be strongly arcing, and maybe break. By then, we will be done.”

Step by step, the goal simplified, until they came to energy. “Making it smaller helps, but it is still a hard problem. I have to power real pumps, and we do not have the power for it,” Jason said.

Halden took the marker. “JRC is not bound by the fissile material ban. What about a KRUSTY reactor? Those are the size of a roll of paper towels.”

Jason’s eyes widened. “If we can use nukes, we can do this.” He began looking up compact fission designs on his slate.

Marco raised his hand. “Excuse me, Halden, sir. I understand your objective, and all the great ideas my colleagues are sharing, but this project is not mostly about helium collection. Mostly it is that straw. My advisor and I have worked out the mass of the straw under current taper models. We believe a production system would weigh twenty million tons, maybe more. The prototype you are proposing will be lighter, and move less gas. We can make the straw a hundred times thinner. Even so, it is still going to be at least two hundred thousand tons of extremely high grade CNT reinforced polymer. That doesn’t look like a prototype.”

Savic was still at the board. “We are looking at ninety thousand kilometers from upper atmosphere to stationary orbit. We are looking at stationary with respect to the upper equatorial band, because that is moving faster. The faster we are synchronized with, the closer synchronous orbit is. I want to put this in a nuclear ramjet, moving at five thousand meters per second. We can do that on Jupiter. The advantage is, at that speed, we can move synchronous orbit down thirty thousand kilometers. Rithika, you started with a dirigible, right? Why was that?”

“The collection and purification steps require a lot of heavy equipment. A dirigible lifts all that without burning energy to fight gravity.”

“Now we don’t have all that heavy equipment, and I don’t mind burning energy for a short proof of concept. The intake of your compressor could be bleed from the compressed intake.”

Marco stood up. “Your idea brings us a thousand new problems. We don’t know how to build the jet. But if we reduce the height to synchronous orbit like that, we can build this with less than five thousand tons of material. Exponentials are harsh when you ask more, but benevolent when you ask for less.”

The researchers were working together and didn’t notice when Halden left the meeting. On the shuttle, he reviewed the plan. The system would not last, but it would return material from Jupiter. JRC would celebrate the delivery. For a day. Then they would have nowhere left to hide.

Chapter 33: Triangulation

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Amira Bastos sat in her office looking out on Bahía de Miel, a perk of being the CEO. But right now she had a call to make.

“Hello, this is Gary Thompson, General Atomics space division.”

“Hi, Gary, this is Amira Bastos. What is this request for authorization to refuel one of your reactors in space? You know there is a fissile launch ban.”

“Yes, ma’am. However, JRC is exempt from that ban. We have an order here from Terrence Halden requesting a refueled nuclear reactor for use in orbit around Jupiter.”

“Halden? Who is Halden?”

Gary brought up the requestor line on the order form. “Well, ma’am, it says here he is a senior systems analyst in applied research.”

“Hold that order. I need to look into this.”

“Very well, ma’am. I await your call.”

Bastos walked out to the receptionist and asked to see Halden immediately. The nature of a large organization like JRC is that something being done immediately may take a week or so. In this case, Halden was in her office in less than three minutes.

“What is this order from General Atomics?”

“Yes, the order for a refueled reactor? Well within my budget authority. We are planning a total PR coup for JRC.”

“You might have mentioned it to me first.”

Halden was amused at the absurdity of her comment. As if he could just drop by and chat. “Yes, ma’am. I should have let you know.”

“OK, what is this PR project about?”

“I had an idea about extracting ^3He from Jupiter. I got the lab guys at JRI to work out the details. It looks like we can pull out a kilogram of ^3He from Jupiter and bring it to Earth. We would burn that at Ilha Fusão, providing one month’s power with real Jovian fuel. I think that will get some pressure off your back.”

“Are you certain this will work?”

“I am not. My plan is no press until the helium is on its way here. If we fail up there, no one need know about it except us.”

Bastos looked out on the bay again. People did not understand that this was not as simple as just flipping a switch. Raines had been publicly complaining about the lack of progress repeatedly, and others were starting to join in. This might be just the project to buy some breathing room. But Halden’s leash was clearly too long.

“OK, here is what we will do. You proceed, but keep me informed. I want a status report every week. Two things: no nuclear power, and don’t let Raines or anyone from Helion Dynamics find out about it.”

“Yes, ma’am, but one thing.”

“What?”

“Nothing can get us to Jupiter and back except a General Atomics ship.”

“You can take a Lunar Tug ship.”

“They are promising to have ships that can do it soon, but right now they are limited to Mars, and they are very slow. With our payload, it would take several trips, probably decades.”

She stared at him for over a minute, searching for a weakness. “I hate orbital mechanics. Fine. Nukes are authorized. Keep it low key.”

“Of course, ma’am.”

At JRI, the prototype was coming apart under its own complexity. The light ramjet design was not coming together and there was no easy way to separate ^3He from ^4He . At the current weight, the odds of surviving a full harvesting cycle were marginal. Marco floated a solution in the project chat. “Top off the ^3He with ^4He . Symbolic payload either way.”

Symbolic. Halden read the message and understood. Halden knew he needed experienced engineers to get through this phase. Helion Dynamics had the engineering expertise and the drive to push this project forward, but Bastos had left him with strict orders not to involve them.

Halden took a walk down the street. He stopped at his favorite taqueria, *La Esquina*, painted yellow with blue trim. He didn’t order anything, just asked to use the phone. The manager let Halden use his office. Calling Raines in Geneva was too risky. He called the operator in Unalaska and got the phone number for the Helion Dynamics storage facility. He navigated the directory to Raines’ extension and left a voicemail.

Halden gave the manager twenty bucks and his number, offering another twenty every time he relayed a message. The manager nodded. “Si, No problem.”

Raines got the voicemail. He called Elin. “Do you remember Terrence Halden, the heavy lift guy? I got a message from him. He’s taking a shot. The kids at JRI have worked out a plan, but they are going to need an adult to help.”

“You know Bastos won’t want me within a hundred miles of Baracoa. What can I do?”

“Can you talk to him? He left me a number.”

Halden just got back to the building when he received a message from the taqueria manager. “*Hola, Señor Halden, Doctora Elin Sarai has called. She left a number.*”

“I’m on my way.”

Halden didn’t even sit down. He sent her the designs as he walked back to *La Esquina*. At the restaurant, he called her back from their office. “Dr. Sarai?” he stumbled. He never thought he would talk to her.

A moment of silence. Elin was scanning through the files. “Interesting concept. I may be able to help build that. I have a shop that specializes in limited-run, hard-to-manufacture prototypes. How quickly can you get me solid specifications?”

“They aren’t making solid specs; they are sharing ideas. Recently, it has started to overwhelm them. People were even talking about bringing back a bottle of helium without purifying the isotope.”

“Maybe I can help.”

And that was it. He looked at the waves on his walk back. The sea hadn’t changed. The clouds were still dragging themselves over the edge of the horizon. Inside, everything was shifting.

A week later, Halden waited by the whiteboard, hands behind his back. The room was packed already. Several postdocs, engineers from the cryogenics group, and a bunch of interns were eager for the next talk. The project was frustrating, but it also felt like the single most important problem on Earth. The morning sun over Baracoa flooded in through the wide louvered windows, pooling light across the tiled floor, throwing long shadows under the chalk trays. People were chatting, arguing over details, talking about designs.

Then Dr. Elin Sarai stepped through the door, and everything stopped.

She wasn’t tall, but she moved like someone who had never been second-guessed. No lab coat. No prepared slides. Just the calm certainty of someone who had already done the thing the rest of them were still designing. Her presence sent a shock through the room. Not noise, not energy. Silence. Reverent, involuntary. A couple of them actually stood up. One nodded stiffly and muttered something reverent. Another bowed. Halden didn’t smile. He waited, then spoke her name as though it needed no title.

“This is Elin.”

That was all. He stepped aside. She looked around, scanning their faces like she was reading field data.

“I’ve read your plans,” she said. “Conceptually, it’s good work. You’ve shown flexibility from core concept to prototype which senior engineers still struggle with.”

They sat straighter.

“The thing you must stay true to, in the midst of this flexibility, is your purpose. If this vehicle goes up and brings back a bottle of ^4He with trace ^3He , the world will see a meaningless gesture. If it brings back one kilogram of high-purity, Jovian ^3He , we wake up the next day in a different world.”

She paused.

“You’re building the first example of an extraction supply chain for fusion-grade material. Do not think of this as a mock-up. This is every bit as important as Ilha Fusão.”

The tension in the room shifted. Before, it had been the strain of trying to live up to a concept. Now it was the recognition of obligation. Their work had consequences.

“The demand for isotope feedstock is set to jump. China’s terawatt reactor burns four tons per month. We have twenty more on order. Lunar extraction is slow and inefficient. You get one kilogram back from Jupiter, one real kilogram, measured and sealed, and it will no longer be a dream, it will be demand.”

Halden finally spoke again. His voice was level.

“If we design a system that can deliver one kilogram, we will surely fail. Too many things can go wrong. The system has to be designed to survive long enough to deliver ten kilograms. Plenty can go wrong once it gets there. If you design for ten, we have a much better chance of getting at least one.”

The earlier proposal to top the tank off with ^4He to “simulate delivery” had evaporated. That kind of thinking wouldn’t survive in a room where Dr. Elin Sarai expected action.

The room became a working room again. They weren’t guessing anymore. They were delivering specs to Dr. Sarai. In the quiet moments between tasks, a few still stole glances at her. That she had read their work and taken it seriously was an overwhelming validation.

Elin worked with them for an hour before leaving. Halden left a few moments later, and went to his office first.

Elin arrived first. She looked like someone waiting for a ferry, not the engineer behind the largest change in electric power since Edison.

The restaurant had once served construction crews during the early buildout of the JRC headquarters, and had never entirely closed. It served straightforward Cuban

fare: rice, beans, fried plantains, slow-cooked pork. More importantly, the people who still ate there didn't ask questions. It was a good place to be unseen.

Halden slipped in five minutes later. He ordered without looking at the menu and took the seat across from her.

Elin wasted no time. "This prototype is good," she said. "But I don't care about one kilogram or even a hundred. I want scale. I want to see if this pipe dream of yours can give me a hundred tons a month. I want production."

Halden leaned back. "That's not how the JRC management sees it. They need a headline. Something that photographs well and carries no risk."

Elin didn't bother to look surprised. "They don't want helium," she said. "They want relevance. I've met these people."

"I report to them. My hands are tied."

"So play along. Just don't forget the real work. You're not building a story. You're building an economy."

He watched her carefully. "I know the difference. But I can't bring this to them. Not directly. Not yet."

"That's fine," she said. "Bring it to me."

They ate slowly. The pork was heavy, the plantains sweet. No one at the other tables seemed to notice them.

When the check came, Elin paid in cash.

She stood, adjusting her satchel, and glanced back at him one last time. "You'll need allies, Halden. More than you've got now. When it's time to move, don't do it alone."

Then she left, walking to the edge of the plaza and stepping onto the bus headed for Santiago de Cuba. From there, a hopper would take her to Biloxi, where the next terawatt tokaplex was already under construction.

Halden stayed behind, finishing his coffee. The restaurant was quiet again, the world unchanged. But the message was clear. The fiction had a deadline. After that, it had to be real.

Chapter 34: Proof

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

JRI launched the prototype on a Wednesday without ceremony. The experiment itself had changed shape over the last six months. The dirigible had become a nuclear ramjet with a single centrifugal compressor and a palladium alloy membrane venting hydrogen. Isotopic separation was built into the first fifty miles of the straw. The Larmor resonance RF heater from a lunar helium extractor was used to keep the ^3He warm while the ^4He condensed on the sides of the straw. It would eventually plug up and need to be drained, but that was fine for a short term project.

Lunar Tug had their type I cargo ships ready by launch, and were happy to carry the gear to Jupiter. The new ships made the flight in six months.

Savic did not expect it to work. He had already explained that to his advisor, Dr. Miriam Vance. The atmosphere was hostile and the radiation could destroy it. The ramjet was operating too hot to survive. Dr. Vance scheduled his thesis defense, anyway.

It did work. The tether held. The helium climbed up the tube. It was slow, less than fifty grams per hour, but it was pure. By the fourth day, they had bottled five kilograms of ^3He and the ramjet was still running.

A week later, the tip of the intake cone melted. The change in aerodynamics destroyed the ramjet within a minute. Without thrust, the ramjet tore free of the tether and descended into Jupiter's depths. This was the expected end, the only possible end.

The helium in the tether continued up past synchronous, to the platform. There it was pumped into a simple K-bottle. When the tether was empty, the platform released it. Without the tether holding it back, the platform went into an elliptical orbit that brought it past Callisto's orbit. Lunar Tug had been waiting for it there, and moved to intercept. As they approached, the platform shot the K-bottle towards the Lunar Tug ship. It was caught in a net and brought into the cargo bay.

The platform continued on its elliptical orbit, going a little higher, then descending on an arc that intersected the atmosphere. It was no longer necessary.

Savic went over the process and showed slides of the catch at the end of his defense. Then his committee asked him to step out of the room. Behind closed doors, Dr. Vance asked a simple question. “Has any grad student at the JRI accomplished nearly as much?”

When Savic returned to the room, Dr. Vance popped the cork on a bottle of champagne.

Meanwhile, the Lunar Tug ship burned hard to get out of the Jovian gravity well and brought the bottle back to LEO. There it was transferred to a descender and dropped into Earth’s atmosphere. It was recovered at Punta de Maisi.

Twelve kilograms of high-purity ^3He from Jupiter was flown to Lisbon. A tiltrotor brought it to the Ilha Fusão tokaplex. Cameras swarmed. The Prime Minister of Portugal gave a speech that managed to include both Columbus and climate change in the same breath. Then came the moment, the ceremonial coupling of the feed line. The valve was turned, taking ^3He feed from Jovian helium.

Back in Jamal, JRI had a graduation ceremony. Nikola Savic, Rithica Patel, Marco Bianchi, and Jason Cole were all awarded their Ph.D.s. On a large screen behind the stage, the events at Ilha Fusão were playing.

Power output from Ilha Fusão remained stable. Sensors lit green across the board. For the first time, a fusion reactor ran on Jovian ^3He . The press reported that it was stable, self-sustaining, quiet. Elin laughed to herself. Of course the reactor didn’t care where the ^3He came from, but news had to make it sound dramatic.

It was exciting to her, too. This proved they could pull gas from Jupiter. No one had really believed that. Well, Raines had, but that was mostly because he needed it and didn’t know any better. He did not understand the gravity well problem, the radiation, the harsh atmosphere. Elin thought about the irony. The JRC, which seemed to Helion Dynamics fully intent on preventing mining on Jupiter, had actually found a solution that neither she nor anyone else at Helion Dynamics would ever have considered. It was because of JRC that this project would work.

Headlines followed in waves. “*Jupiter’s Fire.*” “*A New Helium Age.*” “*Lisbon Lights the Future.*” Financial analysts who had once called Jovian ^3He a vanity project now rebranded it as inevitable. Everyone wanted to know when the next bottle would arrive.

The world, it turned out, had been waiting for an excuse to believe.

The success hit JRC like a concussion. They'd managed the Jovian mining program as a political asset, expensive, photogenic, and safely in the distant future. The kind of initiative you could gesture toward in budget hearings without the burden of actual delivery. Now there was helium in a tank, power on a grid, and headlines naming them the architects of a new energy age.

Suddenly, they had to act like they meant it.

Internal memos pivoted overnight from risk language to rollout frameworks. Leadership scrambled for talking points. A new office was carved out of the executive floor, glass-walled and overlit, and handed to Halden with a congratulatory nod. The gesture was meant to look like reward, but it was containment, an attempt to box the fire they'd accidentally lit.

Halden knew what this meant. He took the office. He made no speeches. But the whiteboards at JRI started filling faster.

Publicly, JRC embraced the future. They released videos touting their long-term vision. They featured archival footage of internal meetings that had never taken place. Executives who'd once argued ${}^3\text{He}$ was impossible now recast themselves as visionaries. The whole thing had always been part of the plan, they insisted. Jovian ${}^3\text{He}$ was well understood by JRC, and they were just waiting for the engineers at JRI to finally address the problem.

But the pressure was enormous. With the headlines came demand. Production targets. Timelines. Questions from sovereign energy agencies. JRC, once the architects of risk assessment, now found themselves required to deliver. And they had no idea how to scale what had been, just months ago, a fringe experiment.

So they leaned on JRI. Hard. The message was clear. Commercialize this right now. Turn it into something that ships quarterly numbers. Halden, in his new office, nodded politely, and said he understood.

Halden sat alone in the new office, surrounded by silence and expensive materials. The glass was low-reflection and ballistic-rated. The walls were layered in real hardwood. A print of a tokamak torus hung on the wall. It was glossy, oversized, and meaningless. The desk was deep and shiny, the kind reserved for signatures and staged photos.

The office was not about status. It was about visibility. The open walls, the proximity to the boardroom, the curious placement of the security sensors, he had

stepped into a place where his privacy was no longer tolerated. They didn't expect him to disappear again into the back rooms of JRI or meet Elin at the café where the rice and plantains were still served from tin pots. Those days were over.

He now had an assistant. Her name was Liora. Young, competent, unreadable. She had been hired directly by the board and reported to them in parallel. Her role was framed as support for scheduling, outreach, travel, whatever Halden needed to enable him to deliver. It was clear from her first day that she had another task, to ensure the board knew Halden's decisions before Halden made them.

She asked questions lightly. She took careful notes. She had access to his calendar and kept it tightly managed. Her desk was directly outside his door. She offered coffee, then logged his answer.

Halden smiled through it all. He gave Liora every answer she asked for, then made sure to ask her questions back. Not to intimidate. He had no interest in a power game, and he knew he wouldn't win that, anyway. He just wanted to see how closely she was watching, and how deeply she understood.

She was good. He appreciated that. It made things easier.

The board had reasserted control. They would not be left behind again. So they had elevated him, encased him in transparency, and attached a quiet observer.

Halden smiled again. Not performatively. Not defiantly. It was simply the smile of a man who recognized the shape of the world he now lived in.

And was already planning his next move.

Halden's visit to JRI was announced a week in advance, scheduled down to the minute by Liora. She handled the logistics, vetted the agenda, and ensured every presentation aligned with the board's expectations. She vetted every attendee. The mood at the institute was respectful, but clearly cautious. JRI had once existed outside the orbit of executive attention. That era was over.

The conference room was sunlit and scrubbed clean of clutter. A small cadre of engineers and researchers presented their ideas. They spoke in turns, showing modular designs, projected scaling curves, and flexible deployment models. The slides were crisp. The numbers were plausible. They spoke the language of commercialization now.

Halden nodded throughout. He asked a few pointed questions, mostly about throughput assumptions and power budgets. He thanked them when they finished,

complimented their discipline. Then, turning to Liora, he said, “Let’s come again next week. Same team.”

She made a note.

As they left the room, Halden paused at the doorway, exchanged brief goodbyes, and stepped out into the corridor.

On the table behind him, unnoticed by his assistant, a single piece of paper remained.

It was not formatted. It bore no heading. Just a line of text in Halden’s careful handwriting:

Call Elin

Below that, a number.

Savic, still standing near the screen, picked it up and read it twice. No one else had noticed. The presentations had been pro forma, intended to look good to the board when they read Liora’s notes. But this note was for them.

By the time he looked toward the hallway, Halden was already gone.

Within a week, the vacations began.

Officially, the rest periods were both earned and necessary time off after the long push for the first tethered collector. Unofficially, no one asked questions. One engineer scheduled a trip to Madurai. Another took time in Biloxi. A materials specialist found herself rerouted through a research exchange program in Malta. One by one, they slipped out of the official workflow for a couple weeks. No one raised alarms. Liora logged the absences, but saw nothing particularly noteworthy. Their replacements were already trained, briefed, and eager.

No one of them went to the same place twice. But each met someone. Dr. Elin Sarai had a long memory and a short tolerance for bureaucracy. She knew who could build, and who was just waiting for permission. Her introductions were quiet and deliberate.

In São Paulo, a propulsion tech found himself seated next to Catherine Dahl at a private lunch. She listened, asked two questions, and pulled out a slate. Her sketches were brutal, elegant, and actionable.

From the base on Ceres, Dr. Knox briefed another JRI fellow on radiation shielding and passive cooling systems. The meeting was ten minutes long. The packet he handed over was three hundred pages.

Across these meetings, the design began to crystalize. Not by committee. Not through official channels. But through the steady cross-weaving of minds too tired to wait for permission.

By the end of the month, JRI was a different organization. Officially, they were still a white-paper institute. But beneath the paper trail, they were building something real.

Chapter 35: Split Rigs

Price of ^3He	
Current Production	
Current Consumption	
Time Since Start	

Halden knew better than to try to run the project. He became a translator. He put Dr. Savic in charge of design. After graduation, Savic took time off to see family. Only Halden knew he went to Biloxi.

Elin brought him to her Back Bay facility. “Manufacturing is not just mostly logistics. Logistics is at least a hundred twenty percent of the game. Fabrication is only a distraction.”

“JRC has no facilities.”

“So you find people who do. And don’t expect just manufacturing. They will deliver expertise. Lean on them.”

“You didn’t do that. You built everything right here.”

“At first I did, but when I started moving operations off site, we became much more productive. Half the land here used to be metal warehousing. Now I let Planetary Metals handle all that, and we are much more productive. Don’t do what I did. Copy my successes.”

When Savic returned, he subcontracted the orbital platform component to Navantia, a Spanish company with a long record in pressurized structures and naval-grade hulls. Their space division handled segment fabrication.

Each segment was as large as current launch systems could manage, just under the payload mass and diameter limits for a heavy-lift orbital insertion. The segments went by boat to JRC’s Punta de Maisí launch facility, where they were lifted individually to the Lunar Tug LEO hub.

The Split Rig program had no major failures. Engineering reviews passed. Procurement delays were minimal. Logistics bottlenecks seemed to resolve themselves. This, more than anything, made JRC uneasy. Nothing could run this smoothly.

Halden knew what made him look effective. Someone kept intercepting weak points before they failed. Vendors were replaced before delays turned critical. Material tolerances were corrected before fabrication. Load structures were rebalanced just ahead of final simulation. No one said how it happened. But the pattern was visible in metadata. Subtle edit trails. Changes logged at the last minute.

That someone was Elin. She was never listed, never named, but always present. Her corrections ran ahead of the project like a shock front. Quiet, systematic, and plausibly deniable.

He was not allowed to contact her directly. Any trace of correspondence would end the relationship. JRC would read it as confirmation that Halden wasn't in control, and control was his only real capital.

So he worked with it. If a vendor spec changed, he approved it without delay. If an engineer flagged a surprising correction, he noted it and moved on. He never asked where it came from. Never said thank you. He simply treated each intervention as luck. And his luck held.

By the time the platform neared completion, Halden's internal evaluations had begun to shift. No longer just competent. Now strategic, resilient, effective under pressure. JRC flagged him as a potential executive.

He read the memo, deleted it, and gave no indication he'd seen it.

He knew the truth; it wasn't his project. It was Elin's. He just kept the chair warm.

The tether weaving system sat on the space platform itself, eight machines, mounted like outriggers, continuously extruding structural cable, four down into Jupiter's upper atmosphere. Four up to the space platform. Three machines in each direction built solid braided tethers. The fourth was tubular, the straw that moved the helium.

The raw materials were shipped separately and sourced through different channels. The polymer came from Earth. It was a two-part epoxy, engineered for extreme tensile loading, rapid in-vacuum curing, and thermal resilience. JRC sourced it through duPont. Delivery was handled by Lunar Tug, which managed all outbound mass cargo from Earth to Callisto. The epoxy was packed into reinforced drums with RFID-linked seals. Punta de Maisí heavy lift moved them into the LEO hub, and Lunar Tug brought them to Callisto.

The yarn came from Mars. Lafevre had underbid Earth-based rivals for the carbon nanotube filament contract, partly on delivered price, mostly on quality. Their lines were cleaner, more uniform, and their facility had the vacuum extrusion capacity to scale without bottlenecks. The Martian yarn was packed into dense transport drums and launched from the Olympus Mons mass driver. The driver was extended and powered up to get it to the Jovian gravity well. From there, a navigating rocket took it on a transfer arc to Callisto, where tenders brought it in.

At Callisto, the drums were staged in orbital storage before being transferred to the tenders. Tenders could carry six drums at a time, one epoxy and five yarn. They brought the drums to the weaving platform. Automated mixers combined the epoxy on demand; filament feeders unspooled the yarn, feeding it into the braid channel with high-precision tension control. Each meter of tether contained Earth and Mars, resin and filament, locked in structural code, laid down strand by strand.

The light in the polymer lab was flat and violet, filtered through silica cladding to blunt the Martian sun. Dr. Marc Lafevre was watching the production numbers. Operations were fourteen percent ahead of schedule.

Across the room, one of his chemists, Inaya Cho, adjusted the alignment on a thin scaffold where a carbon nanotube thread had been stretched and exposed to cycles of heat, pressure, and trace reactive gases.

“Three thousand hours,” she said. “Then it starts to pit. At four thousand, you lose twelve percent of tensile strength. At six, the surface is flaking oxides.”

Lafevre didn’t answer immediately. The Jovian lower stratosphere, particularly near the water cloud band, was more chemically aggressive than JRC had modeled. Tethers made of carbon would not survive indefinite exposure.

He looked at the stress curve again, “What do you propose?”

Inaya stepped back from the rig. “Boron nitride nanotubes.”

He nodded. “BNNT has lower tensile strength, but with enough crosslinking, we can hit the load targets. It won’t oxidize the way carbon does in that soup.”

“Cost?”

“Maybe ten times higher. But not insane. And we only need it as a shell on the last thousand miles. I’ll send a note to Halden. He will understand.”

He drafted the memo himself. It was short, formal, with data appendices and a polite recommendation to coat or substitute the last thousand miles of the tether with boron filament. The overall budget impact was only about five percent.

No reply came that day. After a month, he began to wonder if it had ever been delivered. Then he got his answer.

JRC Internal Correspondence

Re. Proposal from Noctis Landing - Dr. Marc Lafevre

To. Halden, Split Rig Director

From. Anya Choudhury, Budget Oversight

Per our earlier exchange, Dr. Marc Lafevre has submitted an unsolicited design deviation regarding the tether material. He proposes replacing the contracted carbon nanotube yarn with a new boron nitride nanotube variant synthesized under controlled vacuum on Mars.

While it may sound like a safer path, it threatens the kind of budget and schedule overruns this project cannot afford. Switching to BNNT, regardless of potential performance gains, would introduce a new qualification cycle, trigger fresh regulatory review, and potentially nullify two binding procurement agreements with Earth-side partners. More critically, the substitution would open the door to renewed political scrutiny from stakeholders who already view the Martian contract as overreach.

Rather than risk the budget and timeline of this project, we will follow the original spec provided by JRI. We will not rebalance material sourcing mid-cycle. Every change feeds the perception that JRC lacks process discipline. That cannot happen here.

Please encourage Lafevre to focus on quality control and filament consistency, which will surely resolve the issue.

A.

Lafevre read the memo. He posted it on the wall and got back to work.

When Cho saw the memo, she found him immediately. "They rejected the proposal. So that's it?"

"This is a JRC project. We are just here to make the yarn."